

DevOps / SRE Interview Guide

Real interview questions from 85 companies - de-duplicated, grouped by topic and tool, with clear explanations you can say out loud.

3028	questions collected from 151 interview write-ups
574	unique, explained questions after removing repeats and merging near-duplicates
85	companies (plus interviews where the company was not named)
16	topics / tools, each split into sub-topics

Source: github.com/dhiman23/devops-interview-guide (community interview experiences, 2025-2026). Questions are quoted from candidates; explanations are written for learning and interview delivery. Always adapt examples to your own project experience.

How to use this guide

Part 1 - Topics and tools. Every unique question appears once, under the topic it belongs to (Linux, Git, Docker, Kubernetes, Terraform, AWS...). The line "Asked at" shows which companies asked it, so questions that many companies ask are the ones to master first. Several similar questions were merged into one answer.

Part 2 - Company index. Pick a company and see every question number that was asked there, each linked to its explained answer.

Part 3 - Most asked. A short revision list of the questions asked by the most companies.

How to explain an answer in an interview

1. **Define** it in one simple sentence (what it is).
2. **How it works** in 2-3 steps or a small diagram you can draw.
3. **Example** from a real project (tool names, numbers, results).
4. **Trade-off / when not to use it**, which shows senior thinking.
5. **Troubleshooting questions:** narrow down layer by layer (DNS, network, LB, app, DB), state what you would check first and the command, then fix and prevent.

Reading notes

- Answers marked with a shaded box are interview-delivery tips.
- Code is shown in monospace blocks; adjust names, regions and versions to your environment.
- Questions that were only a topic name, a greeting, or personal details (for example "introduce yourself" fragments, company names, test-input data) were removed or folded into the nearest real question.

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Part 1 - Questions by Topic / Tool

Questions inside each sub-topic are ordered by how many companies asked them.

Linux & OS

28 questions

Fundamentals

Q1. Linux file hierarchy; what do /var and /opt contain; is Linux an OS or kernel?

Process/machine states

Asked at (5): Arrise Solutions, EPAM, Morgan Stanley, Others (company not named), Verizon

Linux is the kernel; "Linux OS" (distribution) = kernel + GNU tools. Filesystem Hierarchy: / root, /etc configs, /var variable data (logs, spool, cache), /opt optional third-party software, /home users, /usr programs, /bin,/sbin binaries, /tmp, /dev, /proc, /boot, /mnt.

Process states: Running (R), Sleeping (S/D), Stopped (T), Zombie (Z). A zombie is a finished process whose parent has not yet read its exit status (`wait()`), still occupying a PID slot; it uses no CPU. Fix by fixing/restarting the parent (killing the parent lets init reap it). System runlevels/targets: multi-user, graphical, rescue.

Q2. Hard link vs soft (symbolic) link

Asked at (4): IBM, Morgan Stanley, Qentelli Solutions, Verizon

Hard link: another name for the same inode/data (`ln file hard`); survives deletion of the original, can't cross filesystems or link directories. Soft link: separate file pointing to a path (`ln -s file soft`); breaks if the target is removed, can cross filesystems and point to directories.

Shell & Commands

Q3. Difference between & and && in shell; cron expressions; scheduling every 15 minutes

Asked at (6): Deloitte, Infosys, Oracle, Sonata Software, Verizon, Wipro

`cmd &` runs the command in the background; `cmd1 && cmd2` runs `cmd2` only if `cmd1` succeeded (`||` runs on failure). Cron format: minute hour day-of-month month day-of-week command. Every 15 minutes: `* /15 * * * *` /path/script.sh (edit with `crontab -e`). On Windows PowerShell: `Register-ScheduledTask` with `New-ScheduledTaskTrigger -Once -At (Get-Date) -RepetitionInterval (New-TimeSpan -Minutes 15)`; or `schtasks /create /sc minute /mo 15 ...`

Q4. Explain commands like finger, comm, netstat, jq, yq, at, shuf, lsblk, less, last, nc, mtr, iftop, lsof, blkid, mkfs, nice, and check status of nginx/apache

Asked at (4): CTS, Netcracker, Others (company not named), Sigmoid

- `netstat/ss`: sockets and ports; `nc`: test ports/transfer data; `mtr`: traceroute+ping; `iftop`: live bandwidth per connection.
- `jq / yq`: parse JSON / YAML; `comm`: compare two sorted files; `shuf`: shuffle lines; `less`: page through files; `last`: login history; `finger`: user info; `at`: one-time scheduled job.
- `lsblk/blkid`: block devices and UUIDs; `mkfs`: create a filesystem; `lsof`: open files/ports; `nice`: set process priority.
- Service status: `systemctl status nginx` (or `apache2/httpd`), `curl -I localhost`, check logs in `/var/log/nginx`.

Q5. One-liners: remove first/last line with sed, count lines, find files > 1 GB, find ERROR case-insensitively; find vs sed

Asked at (4): Akamai, Morgan Stanley, Others (company not named), Synechron

```
sed '1d;$d' file          # remove first and last line
wc -l < file              # total lines
find / -type f -size +1G 2>/dev/null
grep -i "error" file.txt
```

`find` searches the filesystem for files by name/size/time; `sed` is a stream editor that transforms text (substitute, delete lines).

Q6. Shell scripting: store names of newly created files in a directory; assign/print variables; last 15 lines of a file; file size

Asked at (4): Deloitte, HCL, Others (company not named), TCS

```
name="devops"; echo "$name"           # variable
tail -n 15 file.log                  # last 15 lines
ls -lh file ; du -sh folder ; stat file # sizes
# watch a directory and record new files (needs inotify-tools)
inotifywait -m -e create /data | while read dir ev file; do echo "$file" >> created.txt; done
# or without inotify: ls -l /data > created.txt
```

Q7. Write shell scripts: count processes of a user, replace a word in a file, first match of a pattern, sum 1..100, triangular pattern, copy a directory to another VM over SSH

Asked at (4): Cisco, Nitor Infotech, Perfios, Sigmoid

```
ps -u ubuntu --no-headers | wc -l      # processes of user ubuntu
sed -i 's/oldname/newname/g' file.txt  # replace word
grep -m1 "pattern" file.txt            # first matching line
echo $(( (100*101)/2 ))                # 5050 | or: s=0; for i in {1..100}; do s=$((s+i)); done; echo $s
n=$1; for ((i=1;i<=n;i++)); do seq -s ' ' 1 $i; done # triangle of numbers
rsync -avz -e "ssh -i ~/.ssh/key.pem" /nobackup/ ubuntu@ubuntu2:/nobackup/
```

Show you can run and explain them; handle errors with `set -euo pipefail`.

Q8. Find and kill a process; check processes without ps or top

Asked at (3): Arrise Solutions, Others (company not named), Verizon

`ps aux | grep name`, `pgrep -f name`, then `kill <pid>` (SIGTERM) and `kill -9 <pid>` only if it ignores it; `pkill name`. Without `ps/top`: `read /proc (ls /proc | grep '^[0-9]', cat /proc/<pid>/cmdline)`, or `lsof`, `systemctl`.

Q9. Scripts: copy new files from a directory to a remote server via SCP; restart a service if it fails and log it

Asked at (3): Cisco, LTIMindtree, Others (company not named)

```
# restart if down + log
SVC=nginx
if ! systemctl is-active --quiet $SVC; then
    systemctl restart $SVC && echo "$(date) restarted $SVC" >> /var/log/svc_watch.log
fi
# copy new files every minute (cron) to remote
find /data -type f -mmin -1 -exec scp -i key.pem {} user@host:/backup/ \;
```

Run via cron or a systemd timer; better yet use systemd `Restart=on-failure` for services.

Q10. stdin, stdout, stderr; types of variables in Linux

Asked at (2): Morgan Stanley, Verizon

File descriptors 0 = stdin, 1 = stdout, 2 = stderr. Redirect: `cmd > out.txt 2> err.txt`, `cmd > all.txt 2>&1`, pipe `|`. Variables: local shell variables (`x=5`), environment variables exported to children (`export PATH`), special variables (`$?`, `$$`, `$0`, `$1`, `$#`, `$@`), and positional parameters.

Q11. What is iptables?

Asked at (1): IBM

The traditional Linux firewall front-end to netfilter, organised into tables (filter, nat, mangle) and chains (INPUT, OUTPUT, FORWARD). Rules match packets (port, IP, state) and ACCEPT/DROP/REJECT. Example: `iptables -A INPUT -p tcp --dport 22 -j ACCEPT`. kube-proxy also uses it for Services; newer systems use nftables/firewalld/ufw.

Troubleshooting

Q12. A Linux system goes down or is unreachable (SSH not working, private instances) - how do you troubleshoot?

Asked at (4): Akamai, CMT, CTS, Others (company not named)

Check reachability (`ping`, `nc -vz host 22`), console/serial output or instance status, security groups/firewall, then access via a console, SSM Session Manager, EC2 Instance Connect or bastion. On the box: `uptime`, `dmesg | tail`, `journalctl -xe`, `df -h` (full disk), `free -m` (OOM), `top`, network (`ip a`, `ss -tulpn`), and services (`systemctl --failed`). Fix, then document and add monitoring.

Q13. How do you check server system load?

Asked at (4): Arrise Solutions, Others (company not named), Synechron, UST

`uptime` / `top` / `htop` show load average (1,5,15 min); compare to CPU count (`nproc`) - load above core count means saturation. Check `vmstat 1`, `iostat -x`, `free -m`, `sar`, and `ps aux --sort=-%cpu` to find the culprit. High load with low CPU often means I/O wait.

Q14. What is Linux load average, how is it calculated and displayed? How do you check system load, CPU and memory?

Asked at (4): Arrise Solutions, Deloitte, Others (company not named), SquareOps

Load average is the average number of processes that are running or waiting (runnable, or in uninterruptible I/O wait) over the last 1, 5 and 15 minutes, shown by `uptime/top/w: load average: 0.50, 0.80, 1.20`. Interpret it against the CPU core count (`nproc`): on a 4-core machine, load 4 = fully utilised; above that, processes are queuing. Commands: `top/htop`, `vmstat 1`, `free -m`, `lscpu` / `cat /proc/cpuinfo` (CPU details), `iostat`, `sar`.

Q15. A user can't SSH in / you're locked out with no root access - what do you check and how do you recover?

Asked at (2): Others (company not named), Qentelli Solutions

Checklist: network reachability and port 22 (`nc -vz`), security group/firewall, `sshd` running (`systemctl status sshd`), correct key and permissions (700 ~/.ssh, 600 authorized_keys), user exists and isn't locked/expired, `sshd_config` (AllowUsers, PasswordAuthentication), disk full or `/etc/nologin`, logs `/var/log/auth.log` or `journalctl -u sshd`.

Locked out with no root: cloud - use SSM/serial console or stop the instance, mount its volume on another instance and fix `sshd_config/keys`; physical - boot to single-user/rescue mode (edit GRUB `init=/bin/bash`), reset the password or keys.

Q16. /var partition is 90% full - immediate action?

Asked at (1): Others (company not named)

`df -h` then `du -xh /var --max-depth=2 | sort -h | tail` to find the culprit (usually `/var/log`, `docker`, `cache`). Fix: compress/rotate and delete old logs (`journalctl --vacuum-size=200M`, `logrotate`), clear package cache, remove old docker images, check deleted-but-open files (`lsof +L1`), and expand the volume (LVM/EBS) if needed. Then add monitoring alerts at 80%.

Q17. A service on the server is leaking memory - how do you troubleshoot?

Asked at (1): IBM

Confirm with `free -m`, `ps aux --sort=-%mem`, track the process RSS over time (`pidstat -r`, Prometheus). Use language tools: heap dumps/jmap (Java), `tracemalloc` (Python), `pprof` (Go), `valgrind`. Check for unbounded caches, unclosed connections, and recent deploys. Short-term: restart/limit memory (cgroups, container limit with restart), scale out; long-term: fix the leak and add alerts.

Administration

Q18. File permission types; creating a user without SSH access; mount vs directory; limiting space per team and CPU/memory

Asked at (8): Accion Labs, Amadeus Labs, Amazon, Five9, Netcracker, Others (company not named), TCS, ZopSmart

- Permissions: read/write/execute for user/group/other (`chmod 755`, `chown`), special bits SUID, SGID, sticky.
- User without SSH: `useradd -s /sbin/nologin app` (or add `DenyUsers/Match` rules in `sshd_config`).

- A directory is just a folder; a mount point is a directory where a separate filesystem/device is attached (`mount /dev/xvdf /data`), so it has its own capacity.
- Space limits per team: separate filesystems/LVs, or disk quotas (`quota`, XFS project quotas).
- CPU/memory limits: cgroups/systemd (`systemctl set-property svc CPUQuota=50% MemoryMax=1G`), `ulimit`, or container limits.

Q19. What is logrotate? Script to compress logs older than 30 days, delete older than 90 days, back up the last 7 days; schedule via cron

Asked at (3): Infosys, Others (company not named), Qentelli Solutions

logrotate periodically rotates, compresses and removes logs per config in `/etc/logrotate.d/` (daily/weekly, rotate N, compress, postrotate). Script:

```
#!/bin/bash
LOGDIR=/var/log/app
find $LOGDIR -name "*.log" -mtime +30 -exec gzip {} \;
find $LOGDIR -name "*.gz" -mtime +90 -delete
find $LOGDIR -name "*.log" -mtime -7 | tar czf /backup/logs_$(date +%F).tgz -T -
# crontab: 0 2 * * * /opt/scripts/logmgmt.sh
```

Q20. What are ulimits? (and PIDs)

Asked at (2): CMT, Morgan Stanley

`ulimit` sets per-process resource limits (open files `-n`, processes `-u`, memory...). Check with `ulimit -a`; make permanent in `/etc/security/limits.conf` or systemd `LimitNOFILE`. "Too many open files" is a typical symptom. A PID is the unique number the kernel gives each process (`ps`, `kill <pid>`, PID 1 = init/systemd).

Q21. How do you securely decommission VMs including data shredding?

Asked at (1): Wipro

Automate with a workflow: remove from LB/monitoring, back up required data, revoke credentials, wipe data (cryptographic erase by destroying the encryption key, or `shred/wipefs` on disks), snapshot deletion, terminate/deprovision via IaC, release IPs, update CMDB, and log evidence for audit. In cloud with encrypted volumes, deleting the KMS key is the quickest proof of erasure.

Q22. How does Linux boot?

Asked at (1): Arrise Solutions

BIOS/UEFI performs POST -> loads the bootloader (GRUB) from disk -> GRUB loads the kernel and `initramfs` -> kernel initialises hardware, mounts the root filesystem -> starts PID 1 (systemd) -> systemd brings up services/targets (multi-user, graphical) -> login prompt. Troubleshoot with rescue/emergency mode and `journalctl -b`.

Q23. System date is behind the current date on a VM - how to fix it?

Asked at (1): Morgan Stanley

Time drift. Sync via NTP: `timedatectl set-ntp true`, or `chronyc makestep / ntpdate pool.ntp.org`, ensure `chronyd/ntpd` is running and firewall allows UDP 123; in AWS use the Amazon Time Sync Service (169.254.169.123). Wrong time breaks TLS, Kerberos, MFA and logs.

Storage & LVM

Q24. Filesystem shows 50% used but you cannot write; df -ih shows 100% - why? Disk I/O; mount point space

Asked at (4): Alphadyne, Netcracker, Others (company not named), TCS

Inodes are exhausted (`df -ih` shows 100% IUse%) - typically millions of tiny files (sessions, mail, logs). Find with `find / -xdev -type f | cut -d/ -f2 | sort | uniq -c`, delete or archive them. Disk space per mount point: `df -h`, `du -sh /path/*`. Disk I/O: `iostat -x 1`, `iotop`, `vmstat` (look for high `await/util`).

Q25. Add 50 GB to /opt using LVM without downtime

Asked at (1): Others (company not named)

1) Add a disk (attach EBS), `lsblk`. 2) `pvccreate /dev/xvdf`. 3) `vgextend vg_name /dev/xvdf`. 4) `lvextend -L +50G /dev/vg_name/lv_opt` (or `-l +100%FREE`). 5) Grow the filesystem online: `resize2fs /dev/vg_name/lv_opt` (ext4) or `xfs_growfs /opt` (xfs). Verify with `df -h`. `lvextend -r` does steps 4-5 together.

Security

Q26. Where are modern Linux user passwords stored? What are good password security practices?

Asked at (2): Alphadyne, Others (company not named)

Hashed passwords in `/etc/shadow` (readable only by root; `/etc/passwd` has account info). Practices: long passphrases, password manager, MFA, no reuse, rotate on compromise rather than forced frequent changes, logout/rate limiting, SSH keys instead of passwords, central identity (SSO/LDAP), secrets in a vault, and breached-password checks.

Q27. How do you disable root login on a server?

Asked at (1): IBM

In `/etc/ssh/sshd_config` set `PermitRootLogin no` (and `PasswordAuthentication no`), then `systemctl restart sshd`; use a sudo-enabled normal user. Optionally lock the root password: `passwd -l root`.

Q28. What are the top 5 infrastructure attacks and how do you mitigate them?

Asked at (1): Others (company not named)

1) DDoS - CDN/WAF, Shield, rate limiting, autoscaling. 2) Brute force/credential stuffing - MFA, lockouts, no password SSH. 3) Unpatched vulnerabilities/exploits - patching, vulnerability scans. 4) Misconfiguration/exposed storage or ports - IaC scanning, least privilege, CSPM. 5) Phishing/stolen credentials/insider - training, short-lived creds, PAM. Also malware/ransomware (EDR, backups) and supply-chain attacks (SBOM, signed artifacts).

Networking & Security

23 questions

Basics

Q29. OSI model layers; TCP vs UDP

Asked at (4): Akamai, Amazon, CMT, Others (company not named)

OSI: 1 Physical, 2 Data link (MAC, switches), 3 Network (IP, routers), 4 Transport (TCP/UDP, ports), 5 Session, 6 Presentation (TLS/encoding), 7 Application (HTTP, DNS). ALB works at L7, NLB at L4, security groups at L3/L4. TCP is connection-oriented, reliable, ordered (handshake, retransmit) - web, SSH; UDP is connectionless, fast, no guarantees - DNS, streaming, gaming.

Q30. Is 90.90.88.0/12 private or public? Is 90.0.0.9/0 public?

Asked at (3): Akamai, EPAM, SquareOps

Private ranges (RFC1918) are only 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16. 90.x.x.x is NOT in them, so it's public. A /0 mask means "all addresses" (0.0.0.0/0 represents the entire internet), not a single host. A /12 means 12 network bits and 20 host bits; check whether the address is the network address or a host.

Q31. IPv4 vs IPv6

Asked at (2): Akamai, CMT

IPv4 is 32-bit (about 4.3 billion addresses, written like 192.168.1.1); IPv6 is 128-bit (hex, e.g. 2001:db8::1) with a huge address space, no need for NAT, built-in IPsec support and simpler headers. Most clouds support dual-stack.

Q32. Is IP X in CIDR Y? Validate an IPv4 address

Asked at (2): Akamai, SquareOps

```
import ipaddress
print(ipaddress.ip_address("10.11.7.44") in ipaddress.ip_network("10.11.0.0/16")) # True
def valid_ipv4(s):
    try: ipaddress.IPv4Address(s); return True
    except ValueError: return False
```

By hand: compare the network bits - a /16 means the first two octets (10.11) must match.

Q33. Private vs public IP addresses; how many IPs does /32 allow?

Asked at (2): EPAM, SquareOps

Private IPs (10/8, 172.16/12, 192.168/16) are non-routable on the internet; public IPs are globally routable. NAT lets private hosts reach the internet. A /32 is exactly one IP address (used for host routes and security group rules such as 1.2.3.4/32); a /24 gives 256 (254 usable).

Q34. Which commands check network details and traffic flows on a system?

Asked at (2): CTS, Others (company not named)

`ip a` / `ifconfig` (interfaces), `ip route`, `ss -tulpn` / `netstat -tulpn` (listening ports), `ping`, `traceroute`/`mtr`, `dig`/`nslookup` (DNS), `curl -v`, `tcpdump -i eth0 port 80` / Wireshark (packet capture), `iftop` / `nload` (bandwidth), `iptables -L`, `nc -vz host port`.

DNS & Web

Q35. DNS: TTL, A/AAAA/CNAME records, custom resolvers, DNS failover, creating records

Asked at (7): CMT, Five9, One2N, Orion Innovation, Others (company not named), Sapient, Sigmoid

DNS maps names to IPs. A = IPv4, AAAA = IPv6, CNAME = alias to another name (not allowed at the zone apex; use Alias records in Route 53). TTL = how long resolvers cache a record; low TTL (60s) before migrations/failover, high TTL (hours) for stable records. Flow: client -> stub resolver -> recursive resolver (cache) -> root -> TLD -> authoritative. Failover: health-checked failover or weighted routing (Route 53) switches to a secondary IP when the primary is unhealthy; clients still honour TTL. Custom resolver: conditional forwarding / Route 53 Resolver endpoints for hybrid DNS.

Q36. What happens when you type **www.google.com** in the browser?

Asked at (6): Arrise Solutions, Commonwealth Bank, F5, Others (company not named), Perfios, Plansource ValueLabs

1) Browser/OS cache check. 2) DNS resolution: resolver -> root -> TLD (.com) -> authoritative server returns IP. 3) TCP 3-way handshake. 4) TLS handshake (certificate validation, session keys). 5) HTTP request sent; load balancer/CDN routes it to a server. 6) Server responds with HTML. 7) Browser renders, fetching CSS/JS/images.

Load Balancing & Proxies

Q37. Reverse proxy vs forward proxy; is a load balancer or Nginx required in this setup?

Asked at (5): IBM, Morgan Stanley, One2N, Oracle, Sapient

A forward proxy sits in front of clients and fetches the internet on their behalf (filtering, caching, anonymity). A reverse proxy sits in front of servers (Nginx, HAProxy, ALB): clients talk to it, and it forwards to backends - giving TLS termination, load balancing, caching, compression, path routing and hiding backend IPs. A load balancer is needed whenever there is more than one instance or you want HA and health checks; Nginx is useful as reverse proxy/static server/ingress when you need L7 features. Place the LB in public subnets (or internal) in front of the app tier.

Q38. What is a WAF (and the "application access firewall")? How do you customise it?

Asked at (3): Deloitte, EPAM, F5

A Web Application Firewall inspects HTTP traffic at L7 and blocks attacks such as SQL injection, XSS, bots using managed rule sets (OWASP CRS) and custom rules (rate limiting, geo, IP). Modes: detection vs prevention. Examples: AWS WAF on ALB/CloudFront, Azure WAF on App Gateway/Front Door. Tune to avoid false positives using exclusions and log analysis. (There's no standard "AAF" - clarify with the interviewer.)

Q39. Load balancer not accepting HTTPS; redirecting HTTP to HTTPS without installing a certificate; load balancer throttling

Asked at (2): SquareOps, ZS Associates

HTTPS troubleshooting: listener on 443 exists, certificate attached and valid, security group/NACL allows 443, target group health/protocol matches (HTTP vs HTTPS), DNS points to the LB, TLS policy supports client versions. Without installing a certificate you can only redirect: an HTTP:80 listener rule with a redirect action to HTTPS:443 (the HTTPS listener still needs a cert elsewhere, e.g. at CloudFront/ACM), or a redirect at the DNS/CDN layer. Throttling: check LB capacity (LCUs), 5xx/429 counts, connection limits, target capacity and pre-warm for spikes.

Q40. How does a GSLB (global server load balancer) work?

Asked at (1): Others (company not named)

GSLB uses DNS (or anycast) to direct users to the best data centre/region based on health checks, geography, latency or load. Examples: Route 53 latency/geo/failover policies, Azure Traffic Manager/Front Door, F5 GTM, Cloudflare. DNS TTL affects failover time.

TLS & Web

Q41. How do SSL/TLS certificates work; HTTP vs HTTPS; renew an expired App Gateway certificate; check a corrupt certificate; automate .csr/.cer/.pfx

Asked at (9): Arrise Solutions, CTS, F5, Five9, One2N, Orion Innovation, Others (company not named), Plansource ValueLabs, RelevantZ

A certificate binds a public key to a domain, signed by a CA. TLS handshake: client hello -> server sends cert chain -> client validates chain/expiry/hostname -> key exchange -> symmetric encryption. HTTPS = HTTP over TLS (encrypted, authenticated); HTTP is plaintext.

Formats: CSR (request you send to the CA), CER/CRT (public certificate), PFX/P12 (cert + private key bundle). Automate with OpenSSL (`openssl req -new -newkey rsa:2048 -nodes -keyout k.key -out r.csr, openssl pkcs12 -export ...`), Key Vault, cert-manager, ACM.

Expired App Gateway cert: renew at the CA, upload the new PFX (or Key Vault version) to the listener, test, and set expiry alerts. Check validity: `openssl x509 -in cert.pem -noout -dates -issuer` or `openssl s_client -connect host:443`.

Q42. How to create certificates for multiple sub-domains; Cross-Origin Resource Sharing (CORS)

Asked at (2): Morgan Stanley, SquareOps

Use a wildcard certificate (*.example.com) or SAN/multi-domain certificate; with ACM request both `example.com` and `*.example.com` and validate by DNS; cert-manager with Let's Encrypt works too.

CORS: browsers block scripts from one origin calling another unless the server responds with `Access-Control-Allow-Origin` (and allowed methods/headers); preflight OPTIONS requests check this. Configure on the API/Gateway/ingress with specific origins, not * in production.

Security Incident

Q43. A DDoS attack consumes 100% resources on cluster nodes/services - what do you do?

Asked at (1): Sigmoid

Contain: enable edge protection (CloudFront/WAF rate limiting, Shield Advanced, Cloudflare), block offending IPs/geos in WAF/NACL, scale out where possible, apply connection/rate limits at ingress, shed non-critical traffic, and use autoscaling with caps. Investigate traffic patterns in logs, restore service, then harden: CDN caching, rate limiting, bot protection, autoscaling policies, runbook and post-mortem.

Q44. A vulnerability is found in a production server/application (even if the fix takes months) - what do you do?

Asked at (1): IBM

Assess severity (CVSS, exploitability, exposure), contain/mitigate immediately without waiting for the full fix: WAF virtual patching rules, disable the vulnerable feature, restrict network access, isolate the host, add monitoring. Then plan the patch through change management (test in staging, schedule window, rollback plan), apply, rescan, and document. For a long-lead fix, keep compensating controls in place and track risk acceptance.

Q45. You find malware on a client's machine - how do you clean up and prevent it?

Asked at (1): Koerber Pharma

Isolate the machine from the network, preserve evidence (disk/memory image), identify the entry point, then rebuild from a known-good image rather than "cleaning" (immutable infra), rotate all credentials, patch the vulnerability, restore data from clean backups, and add EDR/antivirus, least privilege, network segmentation, and monitoring.

IAM & Identity

Q46. What authentication methods does HashiCorp Vault support?

Asked at (2): BMW TechWorks, Marsh McLennan

Auth methods: Token, AppRole (machines), Kubernetes (service account JWT), AWS/Azure/GCP IAM, LDAP, OIDC/JWT, GitHub, userpass, TLS certs. The Vault Agent injector sidecar authenticates and writes secrets into pods automatically.

Q47. Authentication vs authorization

Asked at (1): Others (company not named)

Authentication = proving who you are (password, MFA, certificate, token). Authorization = deciding what you may do (roles, policies, scopes). Example: logging in to AWS (authn) vs IAM policy permitting S3 read (authz). APIs use OAuth2/OIDC: ID token for identity, access token for permissions.

Q48. Explain Okta integration for identity and access management

Asked at (1): JPMorgan

Okta is the identity provider. Applications integrate through SAML or OIDC for SSO; AWS via IAM Identity Center/SAML federation, Kubernetes via OIDC; SCIM provisions/deprovisions users automatically; groups map to roles; MFA and conditional access enforce security. Benefits: central lifecycle, audit, no local passwords.

SSH & Access

Q49. How do you enable passwordless authentication between two servers?

Asked at (3): Alphadyne, IBM, Others (company not named)

Generate a key pair on the source: `ssh-keygen -t ed25519`, copy the public key: `ssh-copy-id user@server` (appends to `~/.ssh/authorized_keys`), ensure permissions (700 `~/.ssh`, 600 `authorized_keys`), then `ssh user@server` works without a password. Disable password auth in `sshd_config` for hardening.

HTTP

Q50. Common API/HTTP errors - what are 504, 501 and others?

Asked at (1): Others (company not named)

2xx success; 3xx redirects; 4xx client errors (400 bad request, 401 unauthenticated, 403 forbidden, 404 not found, 429 too many requests); 5xx server errors: 500 internal error, 501 not implemented (server doesn't support the method), 502 bad gateway (invalid response from upstream), 503 service unavailable (overloaded/down), 504 gateway timeout (upstream too slow).

Q51. HTTP request headers and HTTP methods

Asked at (1): F5

Methods: GET (read), POST (create), PUT (replace), PATCH (partial update), DELETE, HEAD, OPTIONS. Headers carry metadata: Host, User-Agent, Authorization, Content-Type, Accept, Cookie, X-Forwarded-For, Cache-Control, Origin. Responses have status codes and headers like Set-Cookie, Location, Content-Length.

Git & Version Control

14 questions

Basics

Q52. Git merge vs rebase (and squash)

Asked at (9): EXL Service, EY, Infosys, LTIMindtree, Marsh McLennan, NUOS INFO Systems, Others (company not named), Wikreate Media, ZopSmart

Merge joins branches with a merge commit, preserving the true history (`git checkout main; git merge feature`). Rebase replays your feature commits on top of the latest main, creating a linear history (`git checkout feature; git rebase main`) but rewrites commit SHAs. Rule: never rebase shared/public branches. Squash combines multiple commits into one (`git rebase -i` or `squash-merge`) for a clean history.

Q53. git fetch vs git pull; git pull vs git clone; push and pull

Asked at (6): Aspire, Capgemini, Deloitte, Others (company not named), Synechron, Wikreate Media

`git fetch` downloads new commits/refs from the remote into remote-tracking branches (origin/main) without changing your working files - safe to inspect first. `git pull` = fetch + merge (or rebase with `--rebase`) into your current branch. `git clone` copies a whole remote repo the first time. `git push` uploads your local commits to the remote branch. Use fetch to review before integrating; pull for quick updates.

Q54. Everyday git commands, push local changes to remote, checkout and clone

Asked at (5): Belcan, Infosys, L and T, LTIMindtree, ZopSmart

```
git clone <url>           # copy repo
git checkout -b feature   # (or git switch -c) new branch
git status / git diff
git add file.txt
git commit -m "message"
git pull --rebase origin main
git push origin feature
git log --oneline --graph
git stash / git stash pop
```

`git checkout <branch|commit|file>` switches branches or restores files. To push a change to one file: `add -> commit -> push`.

Q55. How do you handle merge conflicts (and resolve automatically)?

Asked at (4): Deloitte, HCL, IBM, L and T

A conflict happens when two branches change the same lines. Resolve: `git status`, open files and edit the `<<<<<< / ===== / >>>>>>` markers, choose/combine changes, `git add`, then `git commit` (or `git rebase --continue`). Options: manual edit, `git checkout --ours/--theirs file`, merge tools (`git mergetool`), abort (`git merge --abort`). Automatic strategies: `-X ours/theirs`, `git rerere` - use carefully. Check history on both sides (`git log --merge`). Reduce conflicts with small PRs and frequent rebases.

Q56. Git cherry-pick

Asked at (3): Deloitte, L and T, LTIMindtree

`git cherry-pick <sha>` applies one specific commit from another branch onto your current branch, e.g. porting a hotfix to a release branch without merging everything. Conflicts are resolved like a merge; `-x` records the source commit.

Q57. Git tags, stale branches, --force-with-lease vs --force, and what git add does internally

Asked at (3): Capgemini, Others (company not named), Persistent Systems

- Tags mark releases (`git tag -a v1.0 -m "release", git push --tags`); lightweight vs annotated.
- Stale branch: merged or long-inactive branch; prune with `git branch -d` and `git fetch --prune`.
- `--force-with-lease` only force-pushes if the remote hasn't changed since you last fetched (safe); `--force` overwrites blindly.
- `git add` hashes file contents into blob objects in `.git/objects` and records them in the index (staging area); commit then creates a tree and commit object pointing to it.

Q58. Extract commits of the last 3 days; compare two commits

Asked at (2): Publicis Global Delivery, Synechron

`git log --since="3 days ago" --oneline` (add `--author`, `--stat`). Compare: `git diff <sha1> <sha2>` (or `sha1..sha2`), `git show <sha>`, `git diff --stat`.

Q59. git stash and pop

Asked at (2): Aspire, Belcan

`git stash` saves uncommitted changes and gives you a clean working tree (to switch branches quickly); `git stash list`, `git stash pop` re-applies the latest and removes it, `git stash apply` re-applies and keeps it, `git stash drop` deletes. Use `-u` to include untracked files.

Q60. Git submodules - purpose

Asked at (1): SAP

A submodule embeds another repo at a pinned commit inside your repo (e.g. shared library), tracked in `.gitmodules`. Use `git clone --recurse-submodules` and `git submodule update --init`. Alternatives: package managers or monorepo.

Q61. How and from where do you clone a repo; local vs remote repo

Asked at (1): HCL

`git clone <URL>` (HTTPS with token or SSH key) from the central remote (GitHub/GitLab) creates a local repo with `origin` set. You work locally (commit), then `git push` to the remote. Mirrors/local bare repos are used for offline or internal replicas.

Q62. How do you migrate a Git repo from one platform to another (GitHub to GitLab) keeping history?

Asked at (1): Deloitte

```
git clone --mirror https://github.com/org/repo.git
cd repo.git
git remote set-url --push origin https://gitlab.com/org/repo.git
git push --mirror
```

`--mirror` copies all branches, tags and history. Then update CI configs, webhooks, secrets, branch protections and tell the team to re-point remotes. Large files need `git lfs migrate`.

Branching

Q63. Which branching strategy do you follow? How does it connect with CI to deploy to the right environment; feature branch design; avoiding breakage of the release branch; production bug approach

Asked at (16): Alphadyne, Deloitte, EXL Service, Encora, HCL, Hexaware, Infosys, L and T, LTIMindtree +7 more

Common models: GitFlow (develop/release/hotfix), trunk-based (short-lived feature branches + feature flags), GitHub Flow.

CI mapping: feature branch -> build + test + PR checks; merge to `develop` -> deploy Dev; `release/*` -> QA/UAT; `main/tag` -> Prod (with approval). In the pipeline use conditions (`when { branch 'main' }`), rules, `if: github.ref == ...`).

Protect main: PRs, required reviews, required status checks, no direct push. Production bug: branch hotfix from the prod tag, fix + test, deploy, merge back to main/develop.

Workflows

Q64. How do you trigger a pipeline only for a specific branch, ignore other branches, trigger on PRs; what validations run before/after commit?

Asked at (3): Others (company not named), Perfios, SquareOps

Use branch filters in the pipeline: GitHub Actions `on: push: branches: [main]` and `on: pull_request`; Jenkins `when { branch 'main' }` or `multibranch`; GitLab `rules: - if: $CI_COMMIT_BRANCH == "main"`; Azure trigger: `branches: include: [main]`. Before commit: local pre-commit hooks (lint, secret scan). After push/PR: CI runs build, tests, code quality and security scans; branch protection requires them to pass before merge.

Undo & Recovery

Q65. How do you remove a developer's commit, delete the last 2 commits, or recover when someone force-pushed over main?

Asked at (3): NUOS INFO Systems, Others (company not named), Turning

- Safe on shared branches: `git revert <sha>` creates a new commit that undoes it (history preserved) - the right answer for "remove developer 10's commit".
- Last 2 commits locally: `git reset --hard HEAD~2` (or `--soft` to keep changes); on a shared branch you must force-push, so prefer revert.
- Lost main after force push: find the old commit via `git reflog` on any clone that has it (or the CI/hosting's last-known SHA), `git branch main <sha>` or `git reset --hard <sha>`, then `git push --force-with-lease origin main`. Prevent with branch protection (no force-push).

Scripting & Coding

16 questions

Python

Q66. Which scripting languages have you used, and what automations/scripts have you written (cost optimisation, daily tasks, recent script)?

Asked at (9): Accion Labs, Deloitte, Hexaware, IBM, Infosys, Persistent Systems, Sigmoid, Synechron, Volkswagen Group Digital

Prepare real examples: Bash for log rotation, backups and deploy helpers; Python (boto3) for stopping idle EC2s outside business hours, cleaning unattached EBS/snapshots, tagging audits, Slack notifications, API health checks; Terraform/Ansible for provisioning. Format: problem -> script -> result (e.g. "saved ~\$1.2k/month"). Be ready to explain each line and error handling.

Q67. Python fundamentals: lists vs tuples vs sets, shallow vs deep copy, decorators, inheritance, GIL, libraries

Asked at (6): Accolite, Others (company not named), Sonata Software, TCS, Virtusa, Wipro

- List: mutable, ordered; tuple: immutable, ordered (hashable, faster); set: unordered, unique, fast membership.
- Shallow copy copies the outer container but shares nested objects; deep copy (`copy.deepcopy`) copies everything recursively.
- Decorator: a function that wraps another to add behaviour: `@timer def f(): ....`
- Inheritance: `class B(A):` reuses and overrides A's methods, `super()` calls the parent.
- GIL: CPython lets only one thread run Python bytecode at a time, so threads don't speed CPU-bound work; use multiprocessing, asyncio (I/O-bound), C extensions/NumPy, or another runtime (PyPy, Cython).
- Libraries: boto3, requests, os/subprocess, json, argparse, pytest, paramiko, pandas.

Q68. Coding: print a number pattern; find pairs summing to a target; longest substring without repeating characters; count word occurrences

Asked at (4): Five9, JPMorgan, Perfios, Sigmoid

```
# word count
s="Hello World Hello"; print(s.lower().split().count("hello")) # 2

# pairs with given sum
def pairs(a,t):
    seen,res=set(),[]
    for x in a:
        if t-x in seen: res.append((t-x,x))
        seen.add(x)
    return res

# longest substring without repeats (sliding window)
def longest(s):
    last,start,best={},0,0
    for i,c in enumerate(s):
        if c in last and last[c]>=start: start=last[c]+1
        last[c]=i; best=max(best,i-start+1)
    return best # "abcabcbb" -> 3

# triangle of increasing rows in reverse: n=4
n=1
for r in range(1,5):
    print(*range(n+r-1,n-1,-1)); n+=r
```

Explain complexity: pairs $O(n)$, longest substring $O(n)$. Talk through the approach before coding.

Q69. Exercises: divisibility by 3 and/or 5 with input validation; min/max/sum from a file; JSON ingestion into key format; check if an external API is reachable

Asked at (3): Others (company not named), Sigmoid, Turning

```
n = input("Enter number (2+ digits): ")
if not (n.isdigit() and len(n) >= 2): raise SystemExit("invalid")
n = int(n)
if n % 15 == 0: print("Divisible by both 3 and 5")
elif n % 3 == 0: print("Only 3")
elif n % 5 == 0: print("Only 5")
else: print("Neither")

import json, requests
data = json.load(open("data.json"))          # dict keyed access: data["key"]
try:
    requests.get("https://api.example.com/health", timeout=3).raise_for_status()
except requests.RequestException as e:
    print("API unreachable:", e)
```

Note that "both" must be checked first. For the file task, read numbers, convert with `int`, then `min()`, `max()`, `sum()` after validating the letter input.

Q70. Given a log with ERROR/INFO lines (timestamps, connection failed, retries), parse it and report errors; remove/count duplicates in a list

Asked at (3): JPMorgan, Others (company not named), Sigmoid

Python log parsing:

```
from collections import Counter
levels = Counter()
for line in open("app.log"):
    levels[line.split()[2].rstrip(":")] += 1  # ERROR / INFO
print(levels)
errors = [l for l in open("app.log") if " ERROR" in l]
```

Shell: `grep ERROR app.log | wc -l`, `awk '{print $3}' app.log | sort | uniq -c`. Duplicates: `list(set(items))` (or `dict.fromkeys` to keep order); counting: `Counter(items)`. In Java, streams: `list.stream().distinct()` and `Collectors.groupingBy(c -> c, Collectors.counting())`. Use a set to avoid duplicates (O(1) lookups, unique by definition).

Q71. Python string programs: slice 0 to 4, reverse a string, palindrome check, remove duplicate element

Asked at (3): Others (company not named), Sigmoid, Virtusa

```
s = "devops-engineer"
print(s[0:4])          # 'devo' (end index excluded)
print(s[::-1])         # reverse
rev = ""
for ch in s: rev = ch + rev  # reverse with a for loop
def is_pal(x): x=x.lower(); return x == x[::-1]
nums=[1,2,2,3,2]; nums.remove(2)  # removes only the FIRST 2
print(list(dict.fromkeys(nums)))  # unique, order kept
```

Q72. Python scripts: failed job IDs from job logs, unique IPs from a live log, monitor a directory for new files

Asked at (2): Accolite, Sigmoid

```
def failed_ids(logs):
    return [j["job_id"] for j in logs if j.get("status") == "FAILED"]

import re
ips = set(re.findall(r"\b\d{1,3}(\.?\d{1,3}){3}\b", open("app.log").read()))
print(ips, len(ips))

import os, time
seen = set(os.listdir("."))
while True:
    time.sleep(60)
    now = set(os.listdir("."))
    for f in now - seen: print("new:", f)
    seen = now
```

For a live log, tail the file (`for line in follow(f)`) and update the set.

Q73. A Python program fails due to memory issues - what could be the cause?

Asked at (1): UST

Loading large datasets fully into memory (use generators/chunking), memory leaks from unbounded caches/lists/global references, circular references holding objects, large string/dataframe copies, too many threads/processes, or container memory limits being too low. Diagnose with `tracemalloc`, `memory_profiler`, `objgraph`, and fix design or increase limits.

Q74. Create your own Python module that prints "Hello, World"

Asked at (1): Others (company not named)

Create `hello.py` with `def greet(): print("Hello, World")` and use it with `import hello; hello.greet();` running `python hello.py` works with `if __name__ == "__main__": greet()`. A package is a folder with `__init__.py`.

Q75. Python: separate list items by whether they start with 'a' or 'b'

Asked at (1): CTS

```
items = ['abc', 'vca', 'abc', 'bca']
a = [i for i in items if i.startswith('a')]
b = [i for i in items if i.startswith('b')]
print(a, b)      # ['abc', 'abc'] ['bca']
```

Q76. Python: what are 5/2 and 5//2? What does a=[0]; b={0}; a[0], b[0] give?

Asked at (1): Akamai

`5/2 = 2.5` (true division), `5//2 = 2` (floor division). `a[0] = 0` (list index). `b = {0}` is a set (not a dict), sets aren't subscriptable, so `b[0]` raises `TypeError`.

Q77. Reverse the words in a given list using Python

Asked at (1): Others (company not named)

```
words = ["devops", "cloud", "linux"]
print([w[::-1] for w in words])    # reverse each word
print(words[::-1])                # reverse list order
```

Clarify with the interviewer which they mean; both are one-liners using slicing.

Q78. What is monkey patching in Python?

Asked at (1): Synechron

Changing or extending a class/module at runtime, e.g. replacing a function with a fake in tests: `module.func = lambda: "mock"`. Useful for tests/hotfixes, but risky because it makes behaviour hard to follow.

Q79. Write a Python script to list running EC2 instances with the tag PROD

Asked at (1): Amazon

```
import boto3
ec2 = boto3.client("ec2")
resp = ec2.describe_instances(Filters=[
    {"Name": "instance-state-name", "Values": ["running"]},
    {"Name": "tag:Environment", "Values": ["PROD"]}])
for r in resp["Reservations"]:
    for i in r["Instances"]:
        print(i["InstanceId"], i["InstanceType"], i.get("PrivateIpAddress"))
```

Use a paginator for large accounts and loop over regions if needed.

Shell

Q80. File scripts: find a missing file, rename .txt files with date, delete files older than 10 days, find files modified >5 hours ago, "too many open files"

Asked at (4): IBM, Morgan Stanley, Others (company not named), Sigmoid

```
[ -f /path/file ] || echo "missing"                # missing file
for f in *.txt; do mv "$f" "${f%.txt}_${date +%F}.txt"; done    # rename with date
find /data -type f -mtime +10 -delete                # older than 10 days
find . -type f -mmin +300 -daystart -mtime -1         # modified >5h ago but today
ulimit -n 65535    # fix "Too many open files"; make permanent in /etc/security/limits.conf / systemd
LimitNOFILE; also close file handles in code
```

Q81. Find the 10th word in a file; shell one-liners

Asked at (1): Synechron

`tr -s ' \n' '\n' < file | sed -n '10p'` or `awk '{for(i=1;i<=NF;i++){c++; if(c==10){print $i; exit}}}'`
file. Alternative: `cat file | xargs -n1 | sed -n 10p`.

Docker & Containers

36 questions

Fundamentals

Q82. VM vs Docker; does Docker have its own kernel?

Asked at (3): Arrise Solutions, EY, L and T

VMs virtualise hardware with a hypervisor and each has a full guest OS (heavy, minutes to boot, strong isolation). Containers share the host kernel and isolate processes (MBs, seconds to start, less isolation). Docker does NOT have its own kernel (on Mac/Windows it runs a lightweight Linux VM).

Q83. Docker image vs container

Asked at (2): Flentas, Techdome

Image = read-only template (layers) containing app and dependencies, like a class or recipe. Container = a running (or stopped) instance of an image with a writable layer, like an object. One image can start many containers.

Q84. Can you run Docker inside a container (DinD)?

Asked at (1): LTIMindtree

Yes: either Docker-in-Docker (privileged `docker:dind` container with its own daemon, needs `--privileged`) or the safer option of mounting the host socket (`-v /var/run/docker.sock:/var/run/docker.sock`) so the container controls the host daemon. Both have security risk; for CI prefer Kaniko/Buildah/Podman rootless builds.

Q85. Docker architecture and how it runs on Linux

Asked at (1): Others (company not named)

Client (`docker` CLI) talks via REST to the Docker daemon (`dockerd`), which manages images, containers, networks and volumes and delegates to `containerd` and `runc` to create containers. Containers are normal Linux processes isolated by namespaces (`pid`, `net`, `mnt`, `uts`, `ipc`, `user`), limited by `cgroups`, and use layered filesystems (`overlay2`). Registry stores images.

Q86. How do Lambda containers work?

Asked at (1): EPAM

Lambda can run a container image (up to 10 GB) built on an AWS base image that includes the Runtime Interface Client. Push the image to ECR, create the function from it; Lambda pulls and runs it on invocation. Useful for large dependencies/ML.

Q87. How do you design a containerised environment for scalability and high availability?

Asked at (1): Nice

Stateless containers behind a load balancer, orchestrated by Kubernetes/ECS with multiple replicas spread across AZs, health checks (liveness/readiness), autoscaling (HPA + node autoscaler), resource requests/limits, rolling updates, externalised state (DB, Redis, object storage), centralised logs/metrics, and immutable images from a trusted registry.

Q88. How do you verify the integrity of a Docker image?

Asked at (1): Others (company not named)

Pull by digest (`image@sha256:...`) so content is exactly what was built; enable Docker Content Trust or sign with Cosign/Notary and verify on deploy; compare `docker inspect` digests; scan for tampering/vulnerabilities.

Q89. Useful Docker commands: list running/stopped containers, remove images/containers

Asked at (1): ZopSmart

`docker ps` (running), `docker ps -a` (all), `docker rm <id>`, `docker rm -f $(docker ps -aq)`, `docker rmi <image>`, `docker image prune`, `docker system prune -a`, `docker logs`, `docker exec -it`, `docker stop/start`.

Q90. What is privileged mode in Docker? Give an example

Asked at (1): ZS Associates

`docker run --privileged` gives the container almost all root capabilities on the host and access to all devices - it can mount filesystems, change network config, run Docker-in-Docker. Example: `docker run --privileged -it ubuntu` then `mount/ip link` operations work. Dangerous: a compromise gives host access, so use only when unavoidable; prefer specific `--cap-add`.

Dockerfile

Q91. Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in a Dockerfile?

Asked at (13): Accion Labs, CGI, Emphasis, HCL, Infosys, L and T, LTIMindtree, Nitor Infotech, Others (company not named) +4 more

```
FROM node:20-alpine
WORKDIR /app
COPY package*.json ./
RUN npm ci --omit=dev
COPY . .
EXPOSE 80
USER node
CMD ["node", "server.js"]
```

Instructions: FROM, WORKDIR, COPY/ADD, RUN, ENV, EXPOSE (documents the port), USER, ENTRYPOINT/CMD. Build `docker build -t app:1.0 .`, run `docker run -p 80:80 app:1.0`. Without a base image: FROM scratch (empty image) - used for static binaries. Order instructions for layer caching: dependencies first, source last.

Q92. CMD vs ENTRYPOINT

Asked at (9): Akamai, EPAM, Infosys, Koerber Pharma, Nice, Others (company not named), Sigmoid, Synechron, Virtusa

ENTRYPOINT defines the fixed executable; CMD gives default arguments (or a default command) and is easily overridden by `docker run image args`. Together: `ENTRYPOINT ["ping"] + CMD ["localhost"] -> docker run img google.com` replaces only CMD. Override ENTRYPOINT using `--entrypoint`. Use exec (JSON) form so signals reach the process.

Q93. How do you reduce Docker image size?

Asked at (9): CTS, Infosys, Intact Green Services, Marsh McLennan, Others (company not named), Qburst, Sony, TCS, Wipro

- Use small bases (alpine, slim, distroless).
- Multi-stage builds so compilers and build tools stay out.
- Combine RUN commands and clean caches (`apt-get clean && rm -rf /var/lib/apt/lists/*`).
- `.dockerignore` for node_modules, .git, tests.
- Install only production dependencies; don't copy unnecessary files.
- Inspect with `docker history` or `dive`.

Large images mean slower pulls and deploys, more storage, and a bigger attack surface.

Q94. Multistage Dockerfile and when it is useful (compiled languages?)

Asked at (9): CGI, Deloitte, HCL, Marsh McLennan, NUOS INFO Systems, Oracle, Others (company not named), Sigmoid, ZS Associates

Multiple FROM stages: build in a heavy image (with compilers), copy only the final artifact into a slim runtime image. Result: much smaller, more secure images.

```
FROM golang:1.22 AS build
WORKDIR /src
COPY . .
RUN CGO_ENABLED=0 go build -o app
FROM gcr.io/distroless/static
COPY --from=build /src/app /app
ENTRYPOINT ["/app"]
```

Ideal for compiled languages (Go, Java, .NET, C++) and also useful for Node front-ends (build -> nginx).

Q95. ADD vs COPY vs RUN

Asked at (8): EPAM, Nice, Nisum Technologies, Others (company not named), Qburst, Synechron, TCS, Virtusa

COPY copies files from build context into the image (preferred, predictable). ADD does the same plus can fetch URLs and auto-extract tar archives (less transparent, use only when needed). RUN executes a command at build time and creates a layer (e.g. `RUN apt-get install -y curl`), it doesn't copy files.

Q96. How do you pass values during docker build (ARG vs ENV)? How do you avoid leaking secrets in the build?

Asked at (2): Deloitte, Qburst

ARG is available at build time only: `docker build --build-arg VERSION=1.2 .` with ARG VERSION in the Dockerfile. ENV persists in the image at runtime. Don't pass secrets as ARG/ENV (visible in `docker history`); use BuildKit secrets: `RUN --mount=type=secret,id=npn ...` and `docker build --secret id=npn,src=.npnrc`, or multi-stage so the secret never reaches the final stage. At runtime pass `-e` or `--env-file`.

Q97. AWS EC2 Image Builder; Packer

Asked at (1): EPAM

EC2 Image Builder is a managed service that automates building, testing and distributing AMIs/container images through pipelines with recipes, components and schedules. Equivalent open-source: HashiCorp Packer. Used to bake patched golden images.

Q98. RUN vs exec (docker exec) commands

Asked at (1): EPAM

RUN is a Dockerfile instruction executed at image build time creating a layer. `docker exec` runs a command inside an already running container (debugging). Also `docker run` creates and starts a container from an image.

Q99. Write a Dockerfile for a Java application

Asked at (1): OPT IT

```
FROM maven:3.9-eclipse-temurin-17 AS build
WORKDIR /app
COPY pom.xml .
RUN mvn -q dependency:go-offline
COPY src ./src
RUN mvn -q package -DskipTests

FROM eclipse-temurin:17-jre
WORKDIR /app
COPY --from=build /app/target/*.jar app.jar
EXPOSE 8080
USER 1000
ENTRYPOINT ["java", "-jar", "app.jar"]
```

The first stage compiles; the final image only has the JRE and jar.

Registry & Delivery

Q100. Build an image in CI, tag and push it, and deploy updates (commands + GitHub Actions)

Asked at (6): Amazon, Nextturn, One2N, Others (company not named), Sigmoid, Virtusa

Commands:

```
docker build -t app:1.0 .
docker tag app:1.0 user/app:1.0
docker login
docker push user/app:1.0
```

GitHub Actions: checkout -> `docker/login-action` -> `docker/build-push-action` with tags `${{ github.sha }}`
-> scan -> update deployment (Helm/Kustomize/ArgoCD). In AWS, CodeBuild can run the build with `buildspec.yml`. Deploy by changing the image tag and rolling out; rollback = previous tag.

Q101. Where does docker pull fetch images from? How do you pull from a private registry, authenticate Jenkins, and what alternatives exist to Docker Hub? What if the push fails?

Asked at (4): Deloitte, Emphasis, Intact Green Services, Others (company not named)

Default is Docker Hub (docker.io); a full name like `registry.example.com/team/app:1.0` points elsewhere. Private registries need `docker login` (credentials in `~/.docker/config.json`); in K8s use `imagePullSecrets`. Jenkins authenticates with stored credentials (`withCredentials + docker login --password-stdin`) or an IAM role (`aws ecr get-login-password`). Alternatives: AWS ECR, Azure ACR, Google Artifact Registry, GitHub Container Registry, GitLab Registry, Harbor, Nexus, Artifactory.

Push failing: check login/token expiry, repository permissions, repo exists (ECR needs a repo created), image tag naming, network/proxy, rate limits, disk space.

Q102. How do you version/manage images in ECR, delete old or untagged ones, and push to ACR?

Asked at (3): Moodys, Others (company not named), Verizon

Version with immutable tags (git SHA, semantic version), never rely only on `latest`; enable tag immutability and scan on push. Clean up with ECR lifecycle policies (e.g. expire untagged images after 7 days, keep last 20 tagged). ACR: `az acr login -n myacr`, `docker tag app myacr.azurecr.io/app:1.0`, `docker push ...`, or `az acr build` to build in the cloud.

Networking & Storage

Q103. Bind mounts vs volumes; docker volume and prune; -p flag

Asked at (4): Koerber Pharma, TCS, Techdome, Verizon

- Volume: managed by Docker in `/var/lib/docker/volumes`, portable, preferred for persistent data (databases).
- Bind mount: maps a specific host path into the container, good for dev (live code reload) but depends on host structure.
- `docker system prune` removes stopped containers, unused networks, dangling images; `docker volume prune` removes unused volumes (careful with data).
- `-p host:container` means publish (port mapping), e.g. `-p 8080:80`; `-P` publishes all exposed ports.

Q104. Connect a backend container to RDS; isolate communication between two containers; Docker network types

Asked at (4): Arrise Solutions, IBM, Techdome, Zensar

Network drivers: bridge (default, per host), host (shares host network), none, overlay (multi-host/swarm), macvlan. User-defined bridge networks give DNS by container name and isolation - containers on different networks cannot talk. Backend -> RDS: pass the RDS endpoint and credentials via env/secrets, ensure RDS security group allows the EC2/container SG on 3306/5432, no special Docker config needed. To isolate: put only the two containers on a private user-defined network.

Q105. How do you expose an application in Docker, map ports (e.g. image on 8080, host on 9090), and what if the app port doesn't match?

Asked at (2): Deloitte, Others (company not named)

`EXPOSE 8080` documents the port; `-p 9090:8080` publishes host 9090 to container 8080 (`docker run -p 9090:8080 tomcat-app`). If the application listens on a port different from the one you map, connections fail/reset - the port mapping must target the port the app actually listens on (and bind to 0.0.0.0, not 127.0.0.1).

Q106. Storage drivers in Docker; container runtimes

Asked at (2): Arrise Solutions, Oracle

Storage drivers manage image layers and the writable layer: overlay2 (default, recommended), btrfs, zfs, devicemapper, vfs (testing). Container runtime: high-level (containerd, CRI-O) manages images/lifecycle; low-level (runc) creates the container process. Kubernetes uses CRI runtimes (containerd) - Docker engine isn't required.

Security

Q107. How do you secure Docker images and containers (scanning, base image maintenance, authorized images, non-root user)?

Asked at (9): EPAM, LTIMindtree, Nisum Technologies, SAP, Sigmoid, Syncortex, TCS, Verizon, Volkswagen Group Digital

- Build: minimal trusted base image, pin versions, non-root `USER` (`RUN adduser ... && USER app`), no secrets in layers, multi-stage.
- Scan: Trivy/Snyk in CI and registry; rebuild images regularly for base-image patches; fix by upgrading packages/base.
- Control: allow only signed/approved images from your private registry (Cosign + admission controller like Kyverno/OPA, ECR policies).
- Runtime: read-only filesystem, drop capabilities, no `--privileged`, resource limits, seccomp/AppArmor, network policies, Falco.

This is DevSecOps - security checks at every pipeline stage.

Q108. How do you scan Docker images (build time and registry level)?

Asked at (2): CGI, Deloitte

Tools: Trivy, Gype, Snyk, Docker Scout, Clair. In CI: `trivy image --exit-code 1 --severity HIGH,CRITICAL app:tag` to fail the build. At registry: ECR enhanced scanning (Inspector), Harbor, Artifactory Xray scan on push and continuously. Also use minimal base images, pin versions, non-root users, sign images (Cosign) and admission control.

Images & Layers

Q109. How are Docker images built - what are layers, how does layer caching work, and where does a rebuild start after a failed layer?

Asked at (7): Arrise Solutions, Marsh McLennan, Nextturn, Oracle, Sigmoid, TCS, ZS Associates

Each Dockerfile instruction (FROM, RUN, COPY, ADD...) creates a read-only layer; layers are stacked with a union filesystem and shared between images. The image config stores metadata (env, cmd, ports, history).

Caching: Docker reuses a cached layer if the instruction and its inputs are unchanged. If layer 6 of 10 fails and you fix it, the rebuild starts at layer 6 (layers 1-5 come from cache); once a layer changes, all layers after it rebuild. Hence put rarely-changing steps (install dependencies) before frequently changing ones (copy source). Intermediate layers aren't used at runtime except as shared cache; in multi-stage builds, earlier stages are discarded from the final image.

Troubleshooting

Q110. How do you debug inside a container and get its logs (also stream logs to S3)?

Asked at (3): NPCI, Others (company not named), Plansource ValueLabs

`docker exec -it <id> sh`, `docker logs -f --tail 100 <id>`, `docker inspect`, `docker stats`, `docker top`; for minimal images use `docker debug`/ephemeral debug containers (`kubectl debug`). Logging drivers (`json-file`, `awslogs`, `fluentd`) control where logs go. To send logs from a path to S3: mount the path as a volume and use a sidecar/Fluent Bit with the S3 output plugin (or a cron `aws s3 sync`).

Q111. Containers stop suddenly after starting - how to troubleshoot?

Asked at (2): Sony, Wipro

`docker ps -a` for exit code, `docker logs <id>`, `docker inspect`. Common causes: main process exits (no foreground process), crash/misconfig, missing env/secret, port conflict, OOM kill (exit 137, check `docker inspect` OOMKilled), permission errors, wrong CMD/ENTRYPOINT. Re-run interactively: `docker run -it --entrypoint sh image`.

Q112. A new image was deployed in production and fails immediately - what do you do?

Asked at (1): Others (company not named)

First restore service: roll back to the last known good image. Then investigate: logs, exit code, config/env/secret differences, health checks, resource limits, migration errors, image tag mistakes. Reproduce locally/staging, fix, add a test or smoke check, and use canary or blue-green next time.

Q113. App reachable from outside the container but packet loss inside it - how to troubleshoot?

Asked at (1): NPCI

Check DNS (`/etc/resolv.conf`), bridge/network config (`docker network inspect`), MTU mismatches, iptables/firewall rules, docker0 and overlay problems, conntrack table full, container CPU throttling, and test with `ping`, `traceroute`, `tcpdump` from a netshoot container.

Q114. Containers consume too much disk space - how do you fix it?

Asked at (1): Others (company not named)

`docker system df` to see usage; `docker system prune -a` (unused images/containers/networks), `docker volume prune`; limit log size (`--log-opt max-size=10m --log-opt max-file=3`), remove dangling layers, use smaller images, mount large data on separate volumes, and clean the build cache (`docker builder prune`).

Compose

Q115. Docker vs Docker Compose; startup order (DB before app) and depends_on

Asked at (6): AMEX, HCL, L and T, NPCI, Synchrotron, Zensar

Docker runs a single container; Compose defines and runs multi-container apps in a `docker-compose.yml` (services, networks, volumes) with one command `docker compose up`. `depends_on` controls start order only; it doesn't wait for the DB to be ready, so add `condition: service_healthy` with a healthcheck, or a retry/wait script.

```
services:
  db: { image: mysql:8, healthcheck: { test: ["CMD","mysqladmin","ping"] } }
  web: { image: app, depends_on: { db: { condition: service_healthy } } }
```

Orchestration

Q116. Docker Swarm vs Kubernetes - why K8s?

Asked at (3): One2N, Others (company not named), Rapidsoft

Swarm is simple and built into Docker but has a small ecosystem. Kubernetes has stronger scheduling, self-healing, autoscaling, rich ecosystem (Helm, operators, service mesh), cloud-managed offerings and industry adoption. Choose Swarm for tiny setups; Kubernetes for production at scale.

Q117. Disaster recovery for stateful apps in containers

Asked at (1): JPMorgan

Keep state outside containers in persistent volumes, back up volumes with snapshots (Velero for K8s), replicate databases across zones/regions, store images in a replicated registry, keep manifests in Git, and rehearse restore tests with defined RTO/RPO.

Kubernetes

132 questions

Architecture

Q118. Kubernetes architecture and workflow

Asked at (20): Accolite, Arrise Solutions, CMT, Capgemini, Deloitte, EY, Emphasis, Encora, F5 +11 more

Control plane: API server (front door), etcd (state), scheduler (chooses node), controller manager (reconciles desired vs actual). Worker nodes: kubelet (runs pods), kube-proxy (networking), container runtime. Flow: `kubectl apply` -> API server validates and stores in etcd -> scheduler assigns node -> kubelet starts containers -> controllers keep desired state.

Q119. What is a CustomResourceDefinition (CRD)? Desired state vs actual state

Asked at (5): EPAM, Intact Green Services, Netcracker, Oracle, Sigmoid

A CRD extends the Kubernetes API with your own resource types (e.g. `Certificate`, `PostgresCluster`); a controller/operator watches these and acts. Kubernetes works on desired state (what you declared) vs actual state (what exists); controllers continuously reconcile the difference (self-healing). "In-desired state" means they match.

Q120. How do you ensure high availability in Kubernetes (survive an AZ failure, multi-region, ECS Fargate/EKS)?

Asked at (4): Accion Labs, CMT, EPAM, Sony

Multiple control-plane nodes/managed control plane across AZs; worker nodes in at least 3 AZs; replicas ≥ 3 with `topologySpreadConstraints`/anti-affinity; PDBs; readiness probes; HPA + node autoscaling; multi-AZ storage (EFS, replicated databases or managed multi-AZ DBs; EBS is AZ-bound so use operators with replication); backups. Multi-region: separate clusters with GitOps, global DNS/accelerator failover and data replication, with staggered upgrades for zero downtime.

Q121. Is etcd SQL or NoSQL and why?

Asked at (3): HCL, Verizon, Zensar

NoSQL - a distributed, consistent key-value store using the Raft consensus algorithm. Kubernetes needs a small, strongly consistent store for cluster state with watch support, not relational queries. Run odd numbers (3/5) of members and back it up.

Q122. What are namespaces in Kubernetes/EKS?

Asked at (3): EPAM, Flentas, Others (company not named)

Virtual clusters inside a cluster that isolate names and resources (dev, qa, prod, team-a). They scope RBAC, ResourceQuota, NetworkPolicy and Service DNS. Default namespaces: default, kube-system, kube-public, kube-node-lease. Namespaces are not a hard security boundary on their own.

Q123. What happens in the backend when you run `kubectl get pods`?

Asked at (2): Capgemini, IBM

`kubectl` reads `kubeconfig`, authenticates (cert/token/IAM), sends an HTTPS GET to the API server `/api/v1/namespaces/<ns>/pods`; the API server authenticates, authorises (RBAC), runs admission, reads the pod objects from etcd (often through its watch cache) and returns JSON; `kubectl` formats it as a table. No node is contacted (kubelet reports status to the API server).

Q124. Which Kubernetes component does not run as a pod, and what is the agent managing the control-plane side?

Asked at (2): Others (company not named), Verizon

The kubelet (an OS service managed by systemd) and the container runtime run on each node outside pods. In `kubeadm` clusters the control-plane components (API server, scheduler, controller-manager, etcd) run as static pods created by the kubelet from manifests in `/etc/kubernetes/manifests` - so kubelet also manages the control plane there. In managed services (EKS/AKS) the control plane is hidden.

Q125. Argo CD architecture

Asked at (1): Others (company not named)

Components: API server (UI/CLI/API), repo server (renders Helm/Kustomize from Git), application controller (compares desired vs live state and syncs), Redis cache, Dex for SSO, ApplicationSet controller. Applications point to a Git path and a destination cluster/namespace; sync policy manual or automated with prune and self-heal.

Q126. Can you talk to the Kubernetes API with plain REST calls from an external app?

Asked at (1): Sigmoid

Yes. kubectl is just an HTTPS client of the API server. You can use `curl -H "Authorization: Bearer $TOKEN" --cacert ca.crt https://<api>/api/v1/namespaces/default/pods` with a service-account token or client cert, or client libraries (Python/Go). Access is governed by RBAC, so use a scoped service account.

Q127. Can you write directly to etcd manually?

Asked at (1): Others (company not named)

Technically yes using `etcdctl put`, but you must not: bypassing the API server skips validation, admission, auditing and can corrupt the cluster. Always change state through the API server. etcd access should be locked down with TLS and backed up regularly; secrets in etcd should be encrypted at rest.

Troubleshooting

Q128. Crashloopbackoff - possible reasons and troubleshooting steps

Asked at (12): AMEX, Cisco, Encora, IBM, Infosys, LTIMindtree, Nextturn, Rapidsoft, SquareOps +3 more

The container starts and keeps crashing, so kubelet backs off restarts. Causes: application error/exception, wrong command or args, missing env/secret/ConfigMap, failed dependency (DB), OOMKilled, failing liveness probe, permissions/volume mount issues, bad image. Steps: `kubectl logs pod --previous`, `kubectl describe pod` (exit code, events, last state), check probes, resources, config, run a debug container or override command to inspect.

Q129. A worker node is not joining the cluster - how do you debug? What if the master goes down? Can workloads run on master?

Asked at (9): Accion Labs, F5, Flentas, IBM, Nisum Technologies, Others (company not named), Sigmoid, Verizon, Zensar

Node not joining: check kubelet (`systemctl status kubelet`, `journalctl -u kubelet`), network connectivity to API server (6443), correct join token/certs (not expired), container runtime running, time sync, CNI installed, security groups, and node IAM role/aws-auth in EKS.

Master (control plane) down: existing pods keep running, but you can't schedule, scale or change anything; restore the API server/etcd, use multiple control-plane nodes (HA) and etcd backups.

Workloads on master: by default it is tainted (`NoSchedule`); remove the taint or add a toleration to schedule there (not recommended in production).

Q130. Are readiness probes failing? Can a container restart itself? What is the pause container?

Asked at (8): JPMorgan, Koerber Pharma, L and T, LTIMindtree, Others (company not named), Sigmoid, SquareOps, Verizon

Readiness probe failures remove the pod from Service endpoints (no traffic) but don't restart it; check `kubectl describe pod`, the probe path/port/timeout, app startup time (use `startupProbe`). Liveness probe failures make the kubelet restart the container; `restartPolicy` (Always/OnFailure) also restarts crashed containers. The pause container holds the pod's network namespace so other containers share the IP; deleting it kills the pod's networking.

Q131. Common Kubernetes errors and production issues you have faced (CrashLoopBackOff, ImagePullBackOff, OOMKilled...)

Asked at (8): Alphadyne, CTS, Deloitte, EY, IBM, Infosys, Others (company not named), Volkswagen Group Digital

Prepare 2-3 real stories in STAR form. Typical: ImagePullBackOff from a wrong tag/registry credentials; CrashLoopBackOff from missing config; OOMKilled due to limits too low; Pending due to no capacity; ingress 502 from readiness mismatch; node NotReady from disk pressure; DNS latency; certificate expiry. For each state symptom, command used (`describe`, `logs --previous`, `top`), root cause, fix and prevention (alerts, limits, policies).

Q132. How do you debug a failed deployment or a failed pod (even when you cannot exec into it)?

Asked at (7): Altimetrik, Infosys, Oracle, Others (company not named), SAP, Syncortex, Synechron

`kubectl get pods`, `describe` (events), `logs [--previous]`, `get events`, `rollout status`. If the container crashes and exec is impossible: `kubectl debug -it pod --image=busybox --target=app` (ephemeral container), `kubectl debug node/...`, or temporarily override the command with `sleep infinity`. Check image, config, secrets, probes, resources, scheduling and quotas, then fix and redeploy (or rollback).

Q133. How do you monitor pod health/performance and alert when pods go down?

Asked at (6): Infosys, JPMorgan, LTIMindtree, Persistent Systems, Sonata Software, Synechron

Metrics: Prometheus + kube-state-metrics + node_exporter + cAdvisor on Grafana dashboards (CPU, memory, restarts, readiness, pending). Logs: Fluent Bit -> ELK/Loki. Alerts:

`kube_pod_container_status_restarts_total` increases, `kube_deployment_status_replicas_available < desired`, pod not ready, node NotReady, OOMKills. Add liveness/readiness probes, Alertmanager routing to Slack/PagerDuty, and tracing for performance.

Q134. Ingress configured but the page doesn't load (or 404 from gateway) - what do you check?

Asked at (6): Amadeus Labs, LTIMindtree, Netcracker, Others (company not named), SquareOps, Wipro

Walk the path: DNS -> LB -> ingress controller pods and logs -> Ingress rule (host/path, ingressClass, annotations, TLS secret) -> Service selectors/ports -> endpoints (`kubectl get endpoints`) -> pod readiness. 404 usually means no matching host/path rule or backend; 502/503 means no healthy pods/endpoints. Verify security groups and the LB target group health.

Q135. A ReplicaSet has 3 pods and one isn't coming up / pods remain Pending even after HPA - why?

Asked at (5): Capgemini, Nisum Technologies, Others (company not named), TCS, ZS Associates

Pending pods: insufficient CPU/memory on nodes (no capacity - needs Cluster Autoscaler/Karpenter), unsatisfied nodeSelector/affinity/taints, PVC not bound, quota/LimitRange exceeded, image pull errors, IP exhaustion (VPC CNI), PDB or max nodes reached. Use `kubectl describe pod` and read the Events section; `kubectl get events`; check node `kubectl describe node` for allocatable vs requested.

Q136. Common reasons a node becomes NotReady (or an unhealthy node group) and how to identify the root cause

Asked at (5): IBM, Nextturn, Others (company not named), RelevantZ, Wipro

kubelet stopped/crashing, container runtime down, network/CNI failure or no route to API server, resource pressure (memory/disk/PID), expired certificates, node IAM role missing (EKS), security group blocking, cloud instance failure/spot reclaim, clock drift. Check `kubectl describe node` conditions, `journalctl -u kubelet`, `systemctl status containerd`, `dmesg`, `disk df -h`, and cloud console status checks; for EKS node groups check the node role, aws-auth mapping and AMI/bootstrap logs. Prevent with monitoring, autoscaling group health replacement, and drains.

Q137. A Service isn't mapped to a deployment - what could be wrong?

Asked at (2): Capgemini, Others (company not named)

Selector labels don't match the pod labels, targetPort/port mismatch, pods not Ready (excluded from endpoints), wrong namespace, or pods not running. Debug: `kubectl get endpoints svc`, `kubectl describe svc`, `kubectl get pods --show-labels`, and test with `kubectl port-forward`.

Q138. Database connection from a pod works for others but not for you - how to troubleshoot? How do you troubleshoot worker node issues beyond kubectl?

Asked at (2): CGI, Nextturn

Pod to DB: test with `kubectl exec` (nc/telnet/nslookup), check Service/DNS, NetworkPolicy, security groups, secrets/credentials, DB user host restrictions, and RBAC if the problem is access to the secret. "Only for you" suggests a difference in identity, namespace, or source IP allowlist.

Node issues: `kubectl describe node` (conditions: MemoryPressure/DiskPressure), SSH/SSM to the node, `journalctl -u kubelet`, container runtime logs, `dmesg` (OOM), disk/inodes, network/CNI, time sync, and cloud console status checks.

Q139. How do you call pod1 from pod2 without a Service; check events and resource limits?

Asked at (2): NPCI, SquareOps

Use the pod IP directly (`kubectl get pod -o wide`), or DNS of a headless service / StatefulSet pod name (`pod-0.svc.ns.svc.cluster.local`). It is fragile since pod IPs change. Events: `kubectl get events --sort-by=.lastTimestamp`, `kubectl describe pod`. Limits: `kubectl describe pod` (Requests/Limits), `kubectl top pod`, `kubectl get pod -o jsonpath='{.spec.containers[*].resources}'`.

Q140. A PVC is Pending (or pod pending due to disk) - how do you debug?

Asked at (1): Others (company not named)

`kubectl describe pvc` events: no matching StorageClass or PV, provisioner/CSI driver not installed or lacking IAM permissions, requested size/access mode unsupported, zone mismatch (volume in a different AZ than the node - use `WaitForFirstConsumer`), quota exhausted. For disk-related Pending pods check node ephemeral-storage and DiskPressure.

Q141. After a new deployment all pods (new and old) crashed - why?

Asked at (1): SAP

The new deployment exhausted node resources (no limits/requests), triggering OOM kills and evictions of neighbouring pods; or a shared dependency broke (config, DB migration, secret). Fix by setting requests/limits and ResourceQuota, PDBs, canary rollout, and a rollback.

Q142. During peak traffic the ingress controller routes slowly - debug? Increase replicas permanently or dynamically?

Asked at (1): SquareOps

Check controller CPU/memory throttling, worker connections/keepalive limits, upstream latency, number of rules/reloads, LB limits, logs and metrics (request duration, queue). Fix: tune worker settings, scale. Prefer dynamic scaling with HPA (CPU/requests) plus a sensible higher baseline `minReplicas` for known peaks, spread across nodes with anti-affinity.

Q143. HPA is configured but pod-level autoscaling isn't happening - how do you fix it?

Asked at (1): Amadeus Labs

Check: metrics-server installed (`kubectl top pods`), pods define resource `requests` (utilisation is relative to them), `kubectl describe hpa` for events/conditions (unknown metrics), min/max replicas, target too high, stabilisation windows, scheduler capacity (pods Pending - need node autoscaler), correct `scaleTargetRef`, and custom-metrics adapter health.

Q144. High CPU in one pod but clean logs - what next?

Asked at (1): JPMorgan

`kubectl top pod`, check for CPU limits/throttling (`container_cpu_cfs_throttled_seconds_total`), profile the process (`kubectl exec + top/jcmd/pprof/py-spy`), look at traffic patterns (hot key, retry storm, load imbalance), GC activity, infinite loops/background jobs, noisy neighbour on the node, and compare with replicas. Take thread dumps/flame graphs; then fix, scale or set limits.

Q145. If master and worker node firewalls break, what happens to running and new deployments? How do you communicate?

Asked at (1): Others (company not named)

Existing pods keep running (the container runtime works locally), but kubelets can't reach the API server so nodes eventually go NotReady, pod status updates stop, no new deployments, scaling or self-healing happen. Restore the firewall rules (6443, 10250, etc.). Communicate by declaring an incident: notify stakeholders of impact (no deploys), ETA, workaround, then post-incident review.

Q146. ImagePullBackOff - why does it occur and how do you troubleshoot?

Asked at (1): Others (company not named)

Kubernetes can't pull the image. Causes: wrong image name/tag, image doesn't exist, private registry without `imagePullSecrets`, expired credentials, registry/network/DNS issue, Docker Hub rate limit, node lacks IAM permission (ECR). Debug: `kubectl describe pod` (Events), verify manually `docker pull`, check secret, fix tag, add pull secret, or node role for ECR.

Q147. In a canary deployment half the traffic returns 502 - how do you troubleshoot?

Asked at (1): JPMorgan

502 appears for requests routed to the canary: check canary pod readiness and logs, port/targetPort mismatch, crashing container, wrong health check path, mismatch of service selector, protocol (HTTP vs HTTPS), keep-alive/timeouts, resource limits, and dependencies. Compare configs of stable and canary; abort and shift traffic back to stable immediately while investigating.

Q148. Intermittent DNS resolution issues during deployment - causes?

Asked at (1): JPMorgan

CoreDNS pods overloaded or restarting, ndots/search-domain amplification, conntrack/UDP race conditions, node-local DNS cache missing, network policy blocking port 53, upstream resolver throttling (e.g. VPC DNS limit of 1024 pps per ENI), pods starting before DNS is ready, or short TTL churn. Fix with NodeLocal DNSCache, scale CoreDNS, lower ndots, use FQDNs and retries.

Q149. Jenkins pipeline randomly fails at deployment to EKS with timeouts on kubectl apply

Asked at (1): IBM

Check the layers: network path from the agent to the EKS API endpoint (private endpoint, security groups, VPN), expired kubeconfig/token (`aws eks get-token`), API server throttling, large manifests / admission webhooks timing out, agent resource limits, DNS issues. Fix: use `--request-timeout`, retry logic, a stable agent inside the VPC, and check webhook availability.

Q150. Logs are incomplete - how do you troubleshoot across AKS, Ingress, app and infrastructure?

Asked at (1): NUOS INFO Systems

Check each hop: the app writes to stdout/stderr (not files), container log rotation/size limits, log agent (Fluent Bit/Azure Monitor agent) health and buffer/backpressure, node disk pressure, ingress access/error log settings, sampling or ingestion caps and retention in Log Analytics/ELK, clock skew, and network drops to the log backend.

Q151. Pods restart intermittently though liveness and readiness pass and the node kills them - how do you debug?

Asked at (1): Sony

Look at `kubectl describe pod` last state and reason: OOMKilled (cgroup limit), node eviction due to MemoryPressure/DiskPressure, preemption by higher priority pods, node drain/spot interruption, kernel OOM on the node, or ephemeral-storage limit. Check `kubectl get events`, node conditions, `dmesg`, kubelet logs, and tune requests/limits and priority classes.

Q152. The cluster feels slow - how do you start troubleshooting?

Asked at (1): Sony

Check API server latency and health (`kubectl get --raw /readyz`, metrics), etcd latency/disk, control-plane CPU/memory, node pressure (`kubectl top nodes`), noisy neighbours/throttling, CoreDNS latency, network/CNI, large numbers of objects or webhooks timing out, pending pods and scheduler queue, and recent changes. Use Prometheus dashboards and audit logs.

Q153. Users see a 10-second delay every 15 minutes in an app on AKS/Kubernetes, with no code change - how to start RCA?

Asked at (1): JPMorgan

Look for periodic causes: scheduled jobs/cron, GC pauses, HPA scale events or cold pods, CPU throttling, readiness/liveness restarts, DNS TTL/CoreDNS cache expiries, certificate/token refresh, node maintenance, connection pool recycle, backup jobs on the DB. Correlate timestamps in metrics/logs/traces (Prometheus, Application Insights) with the 15-min pattern and test by disabling suspects.

Q154. What is OOM and how do you resolve OOM issues?

Asked at (1): Verizon

Out Of Memory: the kernel OOM killer (or cgroup limit) kills a process using too much memory; in Kubernetes the container shows `OOMKilled` (exit code 137). Resolve: check `kubectl describe pod`, `kubectl top`, raise memory limit/requests, fix leaks (heap dumps), tune the runtime (JVM `-Xmx` within container limit), add HPA/more nodes, set right limits so the node isn't overcommitted.

Pods & Workloads

Q155. ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types

Asked at (19): Aspire, CTS, EPAM, EY, Encora, HCL, Hexaware, Infosys, LTIMindtree +10 more

- ReplicationController: legacy, equality-based selectors. ReplicaSet: keeps N identical pods (set-based selectors). Deployment: manages ReplicaSets and gives rolling updates and rollback - use this.
- DaemonSet: one pod per node (log agents, monitoring). StatefulSet: ordered, stable-identity pods with persistent storage (databases).
- In the manifest, Deployment has `strategy: RollingUpdate | Recreate`; canary/blue-green are done with extra Deployments/Services or Argo Rollouts.

Deleting a pod managed by a controller causes it to be recreated.

Q156. Compute reservation in manifests - requests vs limits

Asked at (6): Capgemini, Flentas, HCL, Netcracker, Sigmoid, Verizon

`resources.requests` is what the scheduler reserves for the pod (guaranteed minimum); `limits` is the maximum - exceeding memory limit gets the container `OOMKilled`, exceeding CPU gets throttled. QoS classes: Guaranteed (requests = limits), Burstable, BestEffort. Use LimitRange/ResourceQuota per namespace.

```
resources: { requests: { cpu: 250m, memory: 256Mi }, limits: { cpu: 500m, memory: 512Mi } }
```

Q157. Multi-container pods: can we run 1 container in 2 pods? Delete a pod and run multiple containers? Can a pod be used as an agent?

Asked at (6): Flentas, HCL, Infosys, Intact Green Services, L and T, Others (company not named)

A pod can hold several containers that share network and storage (sidecars: logging, proxy). A single container belongs to exactly one pod; to run more copies you create more pods (replicas). Using a pod as a long-running agent (e.g. Jenkins agent) is possible, but pods are ephemeral - use dynamic, per-job agent pods and persist artifacts elsewhere.

Q158. What is a StatefulSet and how is it different from a Deployment?

Asked at (5): HCL, One2N, Others (company not named), Publicis Global Delivery, TCS

StatefulSet manages stateful apps: pods get stable names (db-0, db-1), stable network identity via a headless service, ordered start/stop, and their own PersistentVolume per pod (volumeClaimTemplates) that follows the pod on reschedule. Deployment pods are interchangeable and stateless. Use StatefulSets for databases, Kafka, Elasticsearch.

Q159. What is an init container and why use it?

Asked at (5): Capgemini, HCL, Oracle, Others (company not named), Verizon

Init containers run to completion, sequentially, before the app containers start. Use them to wait for dependencies (DB ready), run migrations, download config, set permissions. If an init container fails, the pod restarts it and the main container doesn't start.

Q160. StatefulSet vs RDS for a database - how do you decide? MongoDB/MySQL on Kubernetes and operators

Asked at (2): Sigmoid, ZS Associates

Prefer a managed database (RDS/Cloud SQL) for backups, patching, HA and less ops; run a DB in Kubernetes only for portability, cost or special engines, using a StatefulSet (stable identity, volumeClaimTemplates) and a database operator (e.g. MongoDB/MySQL operator) that automates replication, failover, backups and upgrades through CRDs - unlike a plain Deployment pod, which just starts a container and knows nothing about replication or recovery. Consider storage class, anti-affinity across zones, PDBs, backups, resource limits.

Q161. What is a DaemonSet? How do you run Tomcat on each of 10 nodes? Which daemons run by default?

Asked at (2): HCL, Orion Innovation

A DaemonSet ensures exactly one pod runs on each (or selected) node, automatically added with new nodes - used for log collectors, monitoring agents, CNI, storage drivers. To run Tomcat on every node, define a DaemonSet with the Tomcat image (optionally node selectors). Default system DaemonSets: kube-proxy and the CNI plugin (aws-node/calico-node), plus CSI node drivers.

Q162. Deploying batch/ML jobs without impacting running jobs

Asked at (1): Moodys

Don't restart running jobs: deploy changes as new Job versions or use Argo Workflows/Airflow, keep the workers on separate node pools, use `terminationGracePeriodSeconds`/PDBs and drain nodes only after jobs finish, checkpoint long computations, and roll out new code for subsequent runs only.

Q163. How do you inspect environment variables and ConfigMaps in a pod; how do you update config safely?

Asked at (1): SquareOps

`kubectl exec pod -- env`, `kubectl describe pod`, `kubectl get cm <name> -o yaml`. Config is injected as env vars or mounted files; env vars need a pod restart, mounted ConfigMaps update eventually. Safe updates: version the ConfigMap (hash in name), roll out with `kubectl rollout restart deployment` (or Helm checksum annotation), validate in staging, keep Git history for rollback.

Q164. Labels vs annotations

Asked at (1): Capgemini

Labels are key/value tags used for selecting and grouping objects (Services, Deployments select pods by label). Annotations hold non-identifying metadata for tools (ingress rules, build info, prometheus.io/scrape) and can be larger; they can't be used in selectors.

Q165. Run one MySQL instance on each node using custom operators

Asked at (1): Sigmoid

Use a DaemonSet (one pod per node) or a StatefulSet with anti-affinity, backed by local/PVC storage; with a database operator (e.g. Percona/MySQL Operator) define a custom resource (CRD) and the operator controller handles provisioning, replication, backups and failover. Note that each node's DB is independent unless the operator configures replication.

Q166. Scheduling: StatefulSet pods with rolling updates; what is the pod name after mongo-0 dies?

Asked at (1): HCL

StatefulSets update pods in reverse ordinal order (mongo-2, then -1, -0) one at a time waiting for readiness (RollingUpdate with partition support), unlike Deployments which surge. When mongo-0 dies a new pod with the SAME name mongo-0 is created, re-attaches the same PVC and keeps its DNS identity. A Deployment pod gets a new random name.

Q167. Security context: can a pod be scheduled with runAsNonRoot: true and runAsUser: 0?

Asked at (1): Amadeus Labs

No - the settings conflict. `runAsNonRoot: true` makes the kubelet refuse to start a container that would run as UID 0, so the pod is created but fails with `CreateContainerConfigError`. Fix by running as a non-zero user (`runAsUser: 1000`) and building the image accordingly.

Q168. Service vs Deployment (and Deployment vs StatefulSet/DaemonSet basics)

Asked at (1): Others (company not named)

A Deployment manages the pods (replicas, updates); a Service gives them a stable virtual IP/DNS name and load balances to pods matching its selector. They complement each other: Deployment runs the app, Service exposes it.

Q169. What are operators? How does Elasticsearch run on Kubernetes (ECK)?

Asked at (1): Sigmoid

An operator is a custom controller plus CRDs that encodes operational knowledge (deploy, scale, backup, upgrade) for a stateful app. Elastic Cloud on Kubernetes (ECK) provides the `Elasticsearch` CRD; the operator creates StatefulSets, certificates, services and handles rolling upgrades and shard allocation. Without it you'd manage StatefulSets manually or use Helm charts.

Q170. What happens to data when a DB pod is down?

Asked at (1): Amazon

If data lives in the container's filesystem it is lost when the pod is deleted. With a PersistentVolume (EBS/EFS) mounted, the data survives and is re-attached when the pod is rescheduled (a StatefulSet keeps the same PVC). Still need backups and replication for node/AZ failures.

Q171. What should be in a production pod configuration?

Asked at (1): Wipro

Resource requests/limits, liveness/readiness/startup probes, non-root security context with read-only root filesystem, pinned immutable image tag, graceful termination settings, anti-affinity/topology spread, PodDisruptionBudget, config/secret from outside the image, logging to stdout, labels/annotations, and ServiceAccount with least privilege.

Q172. Why does a Deployment's pod name contain two sets of random characters?

Asked at (1): TCS

Name format: `<deployment>-<replicaset-hash>-<random>`. The first part is the hash of the pod template (identifies the ReplicaSet revision), the second is a random suffix identifying the individual pod.

Services & Networking

Q173. How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)?

Asked at (12): Amadeus Labs, EY, IBM, Infosys, JPMorgan, Netcracker, Nitor Infotech, Others (company not named), Persistent Systems +3 more

Every pod gets its own IP and can reach any other pod IP directly (flat network model provided by the CNI). Pods normally use Services: CoreDNS resolves `svc.namespace.svc.cluster.local` to the ClusterIP and kube-proxy load-balances to ready pods. Across namespaces use the FQDN (`svc.otherns`). To make it secure: NetworkPolicies to allow only required namespaces/labels, mTLS via service mesh, and RBAC separation.

Q174. Types of Kubernetes services

Asked at (9): Capgemini, EXL Service, EY, Intact Green Services, Oracle, Others (company not named), Perfios, Qburst, TCS

ClusterIP (internal virtual IP, default), NodePort (opens a port 30000-32767 on every node), LoadBalancer (cloud LB), ExternalName (DNS alias to an external name), and Headless (`clusterIP: None`, returns pod IPs - used by StatefulSets). Ingress/Gateway API sit above Services for HTTP routing.

Q175. Ingress vs Ingress controller vs Gateway API

Asked at (8): Emphasis, HCL, L and T, Netcracker, Others (company not named), Sigmoid, ZS Associates, Zensar

An Ingress is an API object with routing rules (host/path -> Service). It does nothing alone - an Ingress controller (NGINX, ALB, Traefik) implements it. Gateway API is the newer, more expressive and role-oriented successor (GatewayClass, Gateway, HTTPRoute/TCPRoute) with better multi-team, traffic splitting and protocol support.

Q176. How do you get a static IP for a Kubernetes service/ingress, and how do you expose the application?

Asked at (5): Capgemini, L and T, One2N, Orion Innovation, Qburst

Expose with a Service (ClusterIP internal, NodePort, LoadBalancer) or Ingress/Gateway API behind a load balancer. Static IP: reserve an Elastic IP/NLB with EIP allocation annotations (AWS), a reserved IP on GCP/Azure (`loadBalancerIP/annotation`), and point DNS (Route53/ExternalDNS) at it.

Q177. Load balancing in Kubernetes: ELB, Ingress, global load balancing, restricting public access to load balancers

Asked at (5): Deloitte, IBM, Infosys, Qburst, Sigmoid

Inside the cluster, a Service load-balances across pods (kube-proxy iptables/IPVS). Externally: Service type LoadBalancer (NLB/CLB) for L4, or an Ingress controller (ALB Ingress via AWS Load Balancer Controller, NGINX) for L7 host/path routing; ALB is not required for pod-to-pod balancing. Global LB (Route53, Global Accelerator, Cloud DNS, Front Door) steers users across regional clusters with health checks. Restrict public access with internal LBs (`internal` annotation / private subnets), security groups/source ranges, WAF, and private clusters.

Q178. Complete request flow from the user's browser to the backend pod (Ingress, DNS, LB, Services)

Asked at (3): EXL Service, ITC Infotech, Sigmoid

User types the URL -> DNS resolves it (Route53/CNAME) to the cloud load balancer -> traffic enters the VPC via IGW and reaches the LB in public subnets (TLS terminates here or at ingress) -> LB targets the ingress controller pods (via NodePort/instance or pod IP) -> Ingress rules match host/path -> routes to a Service (ClusterIP) -> kube-proxy/eBPF picks a pod endpoint -> CNI delivers to the pod IP -> container responds the same path back. Outbound traffic from pods to the internet goes through the node ENI and NAT gateway. Security groups, NACLs and NetworkPolicies filter at each hop.

Q179. How do you map SSL certificates to an Ingress? Rotate Key Vault certs for App Gateway with AKS ingress?

Asked at (3): Encora, Others (company not named), Sonata Software

Create a TLS secret (`kubectl create secret tls site-tls --cert=tls.crt --key=tls.key`) and reference it under `spec.tls.secretName` in the Ingress; or automate with cert-manager + Let's Encrypt (annotation `cert-manager.io/cluster-issuer`), or ACM ARN annotation on AWS ALB. For Azure: store the .pfx in Key Vault, let Application Gateway (AGIC) or the CSI driver fetch it, and rotate by adding a new certificate version (auto-rotation via polling or Event Grid triggers).

Q180. What is Container Network Interface (CNI)?

Asked at (3): Amazon, Arrise Solutions, NPCI

CNI is the standard plugin interface Kubernetes uses to configure pod networking: assign IPs, set up routes, and optionally enforce NetworkPolicy. Examples: AWS VPC CNI, Calico, Cilium, Flannel, Weave.

Q181. CoreDNS vs kube-proxy; why are kubelet and kube-proxy needed?

Asked at (2): ITC Infotech, Others (company not named)

CoreDNS provides name resolution (service names -> ClusterIPs). kube-proxy implements the Service networking on each node (iptables/IPVS rules that forward ClusterIP traffic to pod endpoints). kubelet is the node agent that talks to the API server, starts/stops containers through the runtime, runs probes and reports node/pod status. All three are needed on a working node.

Q182. How does kube-proxy communicate with nodes / work?

Asked at (2): Arrise Solutions, L and T

kube-proxy runs on every node, watches the API server for Services and Endpoints, and programs iptables or IPVS rules (or eBPF with Cilium) so traffic to a ClusterIP/NodePort is DNAT'd to one of the healthy backend pod IPs. It doesn't proxy traffic itself in modern modes.

Q183. What is Calico, and how does it differ from AWS VPC CNI?

Asked at (2): HCL, ZS Associates

Calico is a CNI plugin offering pod networking and powerful NetworkPolicy enforcement (BGP routing or overlay/VXLAN). AWS VPC CNI gives pods real VPC IP addresses (ENI secondary IPs) for native performance and security-group integration, but uses VPC IPs quickly. Common pattern: VPC CNI for networking + Calico for network policies. Pod IPs come from the VPC CIDR with VPC CNI, or from a separate pod CIDR with overlay networks.

Q184. A Service of type LoadBalancer is created but nodes are in private subnets - how does it provision a public load balancer?

Asked at (1): Sigmoid

The cloud controller manager (or AWS Load Balancer Controller), which runs in the control plane / with an IAM role, calls the AWS API to create the LB. The API call doesn't depend on node subnet; the LB is placed in subnets tagged `kubernetes.io/role/elb` (public) or `internal-elb`. It only needs the right IAM permissions and subnet tags.

Q185. CoreDNS in Kubernetes

Asked at (1): Arrise Solutions

CoreDNS is the cluster DNS server (a Deployment in kube-system) that resolves Service and pod names, e.g. `svc.namespace.svc.cluster.local`. Pods get it as nameserver via kubelet `/etc/resolv.conf`; configured through the `coredns` ConfigMap (forwarders, caching). DNS failures break service discovery - check its pods, logs, and `nslookup` from a debug pod.

Q186. Gateway API: request flow with multiple GatewayClasses across regions; preventing split-brain routing

Asked at (1): Sony

Gateway API separates roles: GatewayClass (which controller), Gateway (listeners/IPs), HTTPRoute (rules to Services). Flow: client -> DNS/global LB -> regional Gateway (controller of its GatewayClass) -> HTTPRoute match -> backend Service -> pod. To avoid split-brain routing: one authoritative route owner per hostname (route attachment via `allowedRoutes/namespace` selectors), health-checked global routing with a single active/weighted policy, consistent config from one GitOps source, and monitoring for conflicting routes.

Q187. How do you set up an ingress controller?

Asked at (1): Infinite Solutions

Install via Helm: `helm install ingress-nginx ingress-nginx/ingress-nginx -n ingress --create-namespace` (or AWS Load Balancer Controller with IRSA for ALB). It creates a LoadBalancer Service; point DNS at it; define Ingress resources with `ingressClassName`, host/path rules and TLS. Monitor the controller logs and metrics.

Q188. If TargetPort is not specified in a Service, what happens?

Asked at (1): Others (company not named)

`targetPort` defaults to the same number as `port`, so traffic goes to that port on the pod. If the container listens elsewhere, connections fail - set `targetPort` (number or named port) explicitly.

Q189. Write a NodePort service exposing port 8080

Asked at (1): Nitor Infotech

```
apiVersion: v1
kind: Service
metadata: { name: web-svc }
spec:
  type: NodePort
  selector: { app: web }
  ports:
    - port: 8080          # service port
      targetPort: 8080    # container port
      nodePort: 30080     # 30000-32767 on every node
```

If `targetPort` is omitted it defaults to the same value as `port`.

Deployments & Rollouts

Q190. Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS

Asked at (11): CTS, Capgemini, Hexaware, LTIMindtree, Nitor Infotech, OPT IT, Oracle, Orion Innovation, Others (company not named) +2 more

```
apiVersion: apps/v1
kind: Deployment
metadata: { name: web-app, labels: { app: web } }
spec:
  replicas: 3
  selector: { matchLabels: { app: web } }
  template:
    metadata: { labels: { app: web } }
    spec:
      containers:
        - name: httpd
          image: httpd:alpine
          ports: [{ containerPort: 80 }]
          resources: { requests: { cpu: 100m, memory: 128Mi } }
```

Quick command: `kubectl create deployment web-app --image=httpd:alpine --replicas=1`. For images in ECR/ACR: build and push each image, give the cluster pull permission (node role/ACR attach), and write one Deployment + Service per image.

Q191. Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines)

Asked at (10): Capgemini, EPAM, Infosys, JPMorgan, LTIMindtree, Others (company not named), SAP, Sigmoid, Turning +1 more

Run two Deployments `app-blue` and `app-green` (labels version: blue/green) in the same namespace (different labels) and one Service whose selector points to the active colour. Pipeline: deploy green, run smoke tests, switch the Service selector (`kubectl patch svc app -p '{"spec":{"selector":{"version":"green"}}}'`) or Ingress weight, monitor, and keep blue for instant rollback; then scale down. Automate with Argo Rollouts/Flagger, or on Azure with Terraform for infra + pipeline steps for traffic switch and rollback.

Q192. How do you achieve a zero-downtime application upgrade and redeploy?

Asked at (6): AMEX, IBM, Perfios, Sonata Software, Sony, SquareOps

Kubernetes rolling updates with `maxUnavailable: 0` and `maxSurge: 1`, working readiness probes, `preStop` hook/graceful shutdown (`terminationGracePeriodSeconds`), multiple replicas with a `PodDisruptionBudget`, and backward-compatible database changes. For stricter needs use blue-green or canary with a load balancer/Argo Rollouts.

Q193. PodDisruptionBudget, voluntary disruptions, minAvailable vs maxUnavailable

Asked at (6): EPAM, EY, Oracle, Others (company not named), Publicis Global Delivery, SquareOps

Voluntary disruptions are planned evictions (drain, upgrades, scale-down); involuntary are hardware failures. A `PodDisruptionBudget` limits how many pods of an app can be down during voluntary disruptions: `minAvailable: 2` (keep at least 2) or `maxUnavailable: 1`. Use `minAvailable` with absolute numbers for quorum systems; `maxUnavailable` (often percentage) for scalable apps. PDBs don't prevent involuntary failures.

Q194. How does kubectl rollout undo know which image to revert to? Is rollback handled by CI/CD or Kubernetes?

Asked at (4): Others (company not named), Sigmoid, SquareOps, TCS

A Deployment keeps old ReplicaSets (history, `revisionHistoryLimit`), each with the pod template (including the exact image tag) of its revision. `kubectl rollout undo deploy/app [--to-revision=N]` simply scales the previous ReplicaSet back up - it doesn't pull from the registry's "latest"; the old image tag must still exist in the registry (use immutable tags). Rollback can be triggered by Kubernetes (undo) or by CI/CD/GitOps (redploy or revert the previous version); best practice is automated, metric-driven rollback in the pipeline or Argo Rollouts.

Q195. During a canary of 10%, how do you verify health and what metrics before going to 100%? How do you implement canary with real-time monitoring and rollback?

Asked at (3): EPAM, EXL Service, SAP

Compare canary vs stable on the golden signals: error rate (5xx), latency (p95/p99), saturation (CPU/memory), business KPIs (conversions), restarts and logs. Use progressive steps (10% -> 25% -> 50% -> 100%) with a hold time at each. Automate with Argo Rollouts/Flagger analysis templates queried from Prometheus: if error rate or latency exceeds thresholds the rollout aborts and traffic shifts back to stable automatically.

Q196. With a single replica, does a rolling restart cause downtime? What happens step by step?

Asked at (3): JPMorgan, Sigmoid, SquareOps

With default strategy (`maxSurge 25%`, `maxUnavailable 25%`) a single replica rounds to surge 1 / unavailable 0, so a new pod is created first, waits for readiness, then the old pod is terminated: no downtime if readiness is correct. Sequence: new ReplicaSet scaled up -> pod scheduled, image pulled, init containers, container starts -> readiness passes -> added to endpoints -> old pod gets SIGTERM after `preStop` and is removed from endpoints, graceful termination -> old ReplicaSet scaled to 0. If `maxUnavailable: 1` and surge 0 there is downtime.

Q197. Deploy a Kubernetes app with Jenkins: plugins, and a Jenkinsfile for Node.js (build, push image, deploy)

Asked at (2): Persistent Systems, Synechron

Tools: Pipeline, Docker Pipeline, Kubernetes CLI/Kubernetes plugin (agents as pods), credentials binding; or call Helm/ArgoCD.

```
pipeline {
  agent any
  environment { IMG = "registry.example.com/node-app:${BUILD_NUMBER}" }
  stages {
    stage('Install & Test'){ steps{ sh 'npm ci && npm test' } }
    stage('Build Image'){ steps{ sh 'docker build -t $IMG .' } }
    stage('Push'){ steps{
      withCredentials([usernamePassword(credentialsId:'reg',usernameVariable:'U',passwordVariable:'P')
    ]){
      sh 'echo $P | docker login registry.example.com -u $U --password-stdin && docker push $IMG' }
    } }
    stage('Deploy'){ steps{ withKubeConfig(credentialsId:'kubeconfig'){
      sh 'kubectl set image deployment/node-app app=$IMG && kubectl rollout status deployment/node-app' } } }
  }
}
```

Explain: install/test, build immutable image tagged with build number, push using stored credentials, deploy by updating the image and waiting for the rollout (fails the build if unhealthy).

Q198. How do you manage deployment, scaling and rollback in Kubernetes? How to roll back a failed deployment?

Asked at (2): Nice, Others (company not named)

Deploy with a Deployment (rolling update), scale with `kubectl scale` or HPA, and roll back with `kubectl rollout status deploy/app`, `kubectl rollout history deploy/app`, `kubectl rollout undo deploy/app [--to-revision=2]`; with Helm `helm rollback app 1`; in GitOps revert the Git commit. For Docker, redeploy the previous image tag. Scenario: a bad image causes failed probes -> rolling update halts (maxUnavailable) -> undo restores the old ReplicaSet.

Q199. Post-deployment steps to confirm everything runs well

Asked at (2): Sigmoid, Synechron

`kubectl rollout status`, check pods Ready, logs, probes; smoke/health tests; check dashboards (error rate, latency, saturation); alerts quiet; verify DB migrations and external integrations; confirm autoscaling and resource usage; watch for a bake period; tag the release and update docs/notifications.

Q200. How do you implement feature toggles in deployment pipelines?

Asked at (1): Wipro

Decouple deployment from release: use a flag service (LaunchDarkly, Unleash, AWS AppConfig) or config maps/env vars; code checks the flag at runtime; enable per user/percentage/environment; combine with canary for gradual exposure; clean up flags after rollout. Flags allow instant disable without redeploy.

Cluster Management

Q201. Explain the full EKS/Kubernetes cluster upgrade process with zero downtime

Asked at (13): Amazon, Cisco, Deloitte, EPAM, Flentas, HCL, LTIMindtree, Netcracker, Others (company not named) +4 more

Plan: read release notes, check deprecated APIs (`kubent/pluto`), back up (etcd/Velero, IaC state), test in lower environments. Upgrade one minor version at a time: 1) control plane (EKS console/Terraform/eksctl), 2) add-ons (CoreDNS, kube-proxy, VPC CNI), 3) nodes - managed node groups rolling update or new node group/Karpenter AMI drift, cordon and drain old nodes honouring PDBs, 4) validate workloads and monitoring. Zero downtime relies on replicas > 1, PDBs, readiness probes, and surge capacity. Have a rollback plan (blue/green cluster for big jumps).

Q202. Basic Kubernetes commands (also Docker and Ansible commands)

Asked at (7): Deloitte, EY, Infosys, Oracle, Others (company not named), Qburst, ZopSmart

Kubernetes: `kubectl get pods|svc|deploy -A -o wide, describe, logs -f [--previous], exec -it pod -- sh, apply -f, delete, scale, rollout status|undo, top, port-forward, cordon/drain, config use-context, explain`.
Docker: `build, run, ps, logs, exec, images, rm, prune`. Ansible: `ansible all -m ping, ansible-playbook site.yml, ansible-vault encrypt, ansible-galaxy install`.

Q203. Have you created an EKS cluster? Explain the process and prerequisites (compute, networking, access)

Asked at (6): Deloitte, IBM, Persistent Systems, Sapient, SquareOps, ZopSmart

Prerequisites: AWS account/permissions, VPC with private subnets across 2-3 AZs (tagged for ELB), IAM roles for cluster and nodes, security groups, `kubectl/eksctl/Terraform`. Steps: create the cluster (`eksctl` or Terraform `aws_eks_cluster`), add managed node groups (or Karpenter/Fargate), enable OIDC/IRSA, install add-ons (VPC CNI, CoreDNS, kube-proxy, EBS CSI), configure access (aws-auth/access entries), install ingress controller, autoscaler, monitoring, then deploy workloads. Nodes in private subnets; API endpoint private or restricted; applications run on EKS.

Q204. How do you manage worker nodes, node groups, and how many clusters/add-ons do you manage?

Asked at (6): Deloitte, Flentas, Others (company not named), Sigmoid, SquareOps, ZS Associates

In EKS: managed node groups (AWS handles AMI/upgrades), self-managed groups, or Karpenter/Fargate. Configure instance types, min/max, labels/taints, launch template, and subnets. Manage nodes by `cordon/drain`, rolling AMI upgrades, autoscaling, and node health monitoring. Answer the numbers honestly: e.g. "4 clusters (dev/stage/prod/DR), ~15 add-ons: CoreDNS, kube-proxy, VPC CNI, metrics-server, ingress, cert-manager, ArgoCD, Prometheus."

Q205. How do you back up an entire Kubernetes cluster regularly?

Asked at (5): Amazon, HCL, Infinite Solutions, Sigmoid, Syncortex

Back up etcd snapshots (`etcdctl snapshot save`; managed clusters do this for you), plus workload manifests (kept in Git) and persistent volumes using Velero (schedules, backs up resources and PV snapshots to S3). Test restores regularly and store backups in another region.

Q206. Restore etcd from a backup onto a new VM; what happens if etcd stops or latency spikes?

Asked at (4): HCL, IBM, Others (company not named), Sony

Yes: install etcd, `etcdctl snapshot restore backup.db --data-dir /var/lib/etcd-new`, point the control plane at the new data dir and members, and restart the API server. If etcd stops, the API server can't read/write state: existing pods keep running but you can't create, update or schedule anything. High etcd latency (>500 ms) slows API requests, causes leader election timeouts, delays scheduler and controller reconciliation, and may cause cluster instability - use fast SSDs, dedicated hosts and defrag/compaction.

Q207. How do you handle node draining?

Asked at (3): Netcracker, Others (company not named), SquareOps

`kubectl cordon node` (stop scheduling), then `kubectl drain node --ignore-daemonsets --delete-emptydir-data` evicts pods gracefully respecting PodDisruptionBudgets; do maintenance/upgrade; `kubectl uncordon node`. Ensure replicas > 1 and PDBs so apps stay available.

Q208. How do you set up kubectl access, connect Jenkins to Kubernetes, and log in to pods?

Asked at (3): Others (company not named), RelevantZ, SquareOps

`kubectl` needs a kubeconfig (cluster endpoint, CA, credentials): EKS `aws eks update-kubeconfig --name c --region r`, AKS `az aks get-credentials`. Jenkins: store kubeconfig or service-account token as a credential, use the Kubernetes CLI plugin (`withKubeConfig`) or run agents inside the cluster with the Kubernetes plugin, with a least-privilege RBAC. Login to a pod: `kubectl exec -it pod -n ns -- /bin/sh`.

Q209. How do you connect to ECS containers and manage services like DBs, EC2, EKS, ECS?

Asked at (1): Deloitte

ECS Exec: `aws ecs execute-command --cluster c --task <id> --container app --interactive --command "/bin/sh"` (needs SSM permissions and `enableExecuteCommand`). Manage everything with IaC and connect to DBs through private networking, secrets from Secrets Manager, and SSM for EC2.

Q210. How do you find and remove orphan resources in Kubernetes?

Asked at (1): Aspire

Look for unattached PVCs/PVs, unused ConfigMaps/Secrets, Services with no endpoints, ReplicaSets with 0 replicas, completed Jobs, old namespaces, LoadBalancers and disks left in the cloud. Tools: `kubectl get all -A`, `kubectl get pv`, `kubectl-popeye`, `kor`, `kubectl get endpoints`; delete with labels/selectors and TTLs (`ttlSecondsAfterFinished`) after verifying.

Q211. How do you switch between clusters?

Asked at (1): Verizon

`kubectl config get-contexts`, `kubectl config use-context <name>`, `kubectl config current-context`; for EKS add contexts with `aws eks update-kubeconfig --name cluster --region r`. Tools: `kubectx/kubens`.

Q212. How to run a YAML manifest without creating a file?

Asked at (1): LTIMindtree

Pipe it to kubectl: `cat <EOF | kubectl apply -f - ... EOF`, or generate imperative objects: `kubectl run nginx --image=nginx`, `kubectl create deployment web --image=nginx --replicas=2`, `kubectl expose deployment web --port=80`. Use `--dry-run=client -o yaml` to generate a manifest.

Q213. If the cluster is in a private subnet, how do you access it with kubectl?

Asked at (1): Sapient

Options: a bastion/jump host or SSM in the VPC, VPN/Direct Connect into the VPC, a CI runner inside the VPC, a peered network, or enable the public endpoint restricted to specific CIDRs (EKS public+private mode). With a private endpoint, DNS resolves the API server to private IPs inside the VPC.

Security

Q214. How is access control implemented in Kubernetes (RBAC)?

Asked at (9): AMEX, EPAM, L and T, Netcracker, Oracle, Others (company not named), Sigmoid, SquareOps, ZopSmart

Authentication (certs, OIDC, tokens, IAM in EKS) identifies the user; Authorization via RBAC with Role/ClusterRole (permissions: verbs on resources) and RoleBinding/ClusterRoleBinding (assign to users, groups, service accounts). Add admission control (PSA, Kyverno/OPA), NetworkPolicies, and least privilege. Check with `kubectl auth can-i`.

Q215. Did you configure ingress and egress rules (NetworkPolicy)?

Asked at (5): CTS, EPAM, EXL Service, Morgan Stanley, Netcracker

Yes: default-deny, then allow specific traffic. Example allowing only frontend pods to reach backend on 8080:

```
kind: NetworkPolicy
spec:
  podSelector: { matchLabels: { app: backend } }
  policyTypes: [Ingress, Egress]
  ingress: [{ from: [{ podSelector: { matchLabels: { app: frontend } } }], ports: [{ port: 8080 } ] }]
```

Requires a CNI that enforces policies (Calico, Cilium).

Q216. Implement Kubernetes security on container and infrastructure sides; zero-trust networking without a service mesh; runtime threat detection with Falco

Asked at (4): EPAM, F5, Others (company not named), Sony

Container side: non-root, read-only filesystem, drop capabilities, Pod Security Standards, image scanning/signing, resource limits. Infra side: RBAC least privilege, private API endpoint, etcd encryption, audit logs, patched nodes, network policies, secrets encryption. Zero trust without mesh: default-deny NetworkPolicies/CiliumNetworkPolicies, workload identity, SPIFFE/SPIRE or Cilium mTLS, authz at the app/gateway. Runtime detection: deploy Falco (eBPF) with rules for shell in container, privilege escalation, unexpected network; send alerts to SIEM/Slack; Sysdig for forensics.

Q217. Where is aws-auth stored and what does it do?

Asked at (4): Accenture, Deloitte, Nitor Infotech, SquareOps

It's a ConfigMap named `aws-auth` in the `kube-system` namespace on EKS that maps IAM roles/users to Kubernetes RBAC groups (node roles need it to join). Newer clusters use EKS Access Entries instead. Edit it carefully - a mistake can lock you out.

Q218. How do you map an IAM role to a Kubernetes service account (IRSA) and how does the pod know which role to assume?

Asked at (3): Capgemini, Sapient, ZS Associates

EKS IRSA: 1) Associate an OIDC provider with the cluster. 2) Create an IAM role whose trust policy allows `sts:AssumeRoleWithWebIdentity` from that OIDC provider for `system:serviceaccount:<ns>:<sa>`. 3) Annotate the ServiceAccount: `eks.amazonaws.com/role-arn: arn:aws:iam::123:role/app`. 4) Pod uses that service account; the webhook injects a projected token and `AWS_ROLE_ARN`, and the SDK exchanges it for temporary credentials. Newer alternative: EKS Pod Identity.

Q219. How do you restrict everything except two services (NetworkPolicy)? How do you secure a web app from OWASP Top 10?

Asked at (3): F5, HCL, SquareOps

Apply a default-deny NetworkPolicy, then allow only specific ingress/egress to the two required services by label (or security groups/NSGs in cloud). OWASP Top 10: WAF with managed rules, input validation, parameterised queries, strong authn/authz, TLS, patching dependencies (SCA), secure headers, logging/monitoring, secrets management, and regular pen tests/DAST.

Q220. Limiting resource usage at namespace level (ResourceQuota and LimitRange)

Asked at (3): Alphadyne, Amazon, Capgemini

`ResourceQuota` caps total CPU/memory/pods/PVCs per namespace; `LimitRange` sets default and min/max requests/limits per container so pods without values still get limits. Apply them per team/environment namespace instead of editing every deployment.

Q221. Role vs RoleBinding; ClusterRole vs ClusterRoleBinding - when to use which?

Asked at (3): CTS, Netcracker, SquareOps

Role = set of permissions inside one namespace; RoleBinding = grants a Role (or ClusterRole) to users/groups/service accounts in that namespace. ClusterRole = permissions cluster-wide or on cluster-scoped resources (nodes); ClusterRoleBinding = grants that across the entire cluster. Use RoleBinding wherever possible for least privilege; even a ClusterRole can be bound in a single namespace via RoleBinding.

Q222. Design a multi-tenant Kubernetes platform where teams cannot impact each other

Asked at (1): Sony

Namespace per team with ResourceQuota/LimitRange (resource isolation), NetworkPolicies default-deny (network isolation), RBAC scoped to namespaces, Pod Security Standards, separate node pools via taints/affinity for noisy workloads, policy engine (Kyverno/OPA), per-team ArgoCD projects. For independent upgrade cycles or stronger isolation use separate clusters or vClusters per team. Add cost allocation and quotas.

Q223. Explain control plane components and how you would harden them for production

Asked at (1): EPAM

Components: kube-apiserver, etcd, scheduler, controller-manager (+ cloud-controller-manager).
Hardening: restrict API server access (private endpoint, IP allowlist), enable RBAC and audit logging, OIDC auth, disable anonymous auth, TLS everywhere, encrypt etcd at rest and restrict/backup it, admission controllers (PSA, policies), regular upgrades, and protect kubelet endpoints. Managed services (EKS) handle much of the control plane.

Q224. What are the security implications of Kubernetes Secrets in etcd without encryption?

Asked at (1): Wipro

Secrets are only base64-encoded, so anyone with etcd or API read access (or a backup) can read them in plain text. Mitigate: enable encryption at rest with KMS provider, strict RBAC, restrict etcd access, audit logs, use external secret managers (Vault, Secrets Manager via External Secrets/CSI), rotate secrets.

Helm & Kustomize

Q225. Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly

Asked at (15): Altimetrik, Deloitte, Encora, Flentas, HCL, Hexaware, Infosys, Nice, Nisum Technologies +6 more

Chart structure: `Chart.yaml` (name, version, appVersion, dependencies), `values.yaml` (default config), `templates/` (Go-templated manifests like `deployment.yaml`, `service.yaml`), `charts/` (subcharts). `helm upgrade` renders templates with new values, computes a diff against the last release and applies it, creating a new revision (`helm rollback` goes back). Don't edit a chart directly (especially third-party ones) - override values or use your own wrapper/umbrella chart so upgrades don't lose changes.

Q226. Helm commands, deploying via CI/CD, and rolling back a failed or partially deployed release

Asked at (6): EY, Encora, Marsh McLennan, Nextturn, Others (company not named), SquareOps

Commands: `helm repo add`, `helm install app ./chart -f values-prod.yaml`, `helm upgrade --install app ./chart --atomic --timeout 5m`, `helm list`, `helm history app`, `helm rollback app <rev>`, `helm uninstall`. In CI/CD: build and push image, then `helm upgrade --install --set image.tag=$SHA` (or update values in Git for ArgoCD). Use `--atomic` (auto rollback on failure) and `helm diff` in PRs. For a partially applied release: `helm history`, `helm rollback app <last good>`; check `kubectl describe` for failed resources, fix the chart (immutable fields, hooks, CRDs), and re-run.

Q227. How do you create a Kustomize file?

Asked at (1): Akamai

Create `kustomization.yaml` listing base resources and overlays:

```
resources: [deployment.yaml, service.yaml]
namePrefix: dev-
images: [{ name: app, newTag: "1.2" }]
patches: [{ path: replicas.yaml }]
```

Run `kubectl apply -k overlays/dev` (or `kubectl kustomize`). Use base + overlays per environment, no templating.

Q228. How do you extend a Helm chart with new templates (without breaking the original)?

Asked at (1): SquareOps

Add new YAML files under `templates/`, drive them from `values.yaml` (with `{{ if .Values.feature.enabled }}` toggles and defaults so existing releases are unaffected), use subcharts or umbrella charts for bigger changes, bump the chart version, run `helm lint` and `helm template --dry-run`, and test an upgrade.

Q229. What is Helm chart signing? Which tools sign charts?

Asked at (1): Deloitte

Signing proves a chart's integrity and origin. Helm supports provenance files (`helm package --sign --key name --keyring ~/.gnupg/secring.gpg`, verify with `helm verify`) using GPG; modern setups sign OCI charts/images with Cosign (Sigstore) and verify in admission policy.

Autoscaling

Q230. What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a misconfigured HPA cause cascading failure?

Asked at (11): Blue Yonder, IBM, Infosys, JPMorgan, One2N, Others (company not named), Sony, SquareOps, Syncortex +2 more

HPA scales a Deployment's replicas between min and max using metrics: CPU/memory utilisation (metrics-server), custom/external metrics (requests per second, queue length via Prometheus Adapter or KEDA). Festival spike: set higher `minReplicas`/scheduled scaling (KEDA cron) before the event plus metric-based scale-out, ensure nodes can scale (Cluster Autoscaler/Karpenter), and set requests/limits properly.

Misconfiguration risks: no resource requests (HPA can't compute), too-aggressive scale-down causing flapping, max too low, scaling on a lagging metric, or scaling out pods that overload a shared DB. Fix: stabilisation windows, sensible min/max, scale on user-facing metrics, load test, and protect dependencies with limits/queues.

Q231. Do you use Cluster Autoscaler or Karpenter? How do you provision Karpenter, how does it decide which node to provision, and how do you scale EKS on metrics?

Asked at (6): Flentas, Others (company not named), Persistent Systems, Plansource ValueLabs, SquareOps, ZS Associates

Cluster Autoscaler scales existing node groups when pods are Pending (slower, node-group bound). Karpenter watches unschedulable pods, reads their requests, constraints (nodeSelector, affinity, topology, taints) and picks the cheapest fitting instance types directly via the EC2 API, also consolidating/removing underused nodes.

Provision Karpenter: install the Helm chart, give it an IAM role (IRSA/Pod Identity) and an instance profile/node role, tag subnets and security groups for discovery, then create a `NodePool` (requirements, limits, disruption policy) and an `EC2NodeClass` (AMI family, subnets, SGs). Scale EKS on metrics: HPA for pods (CPU/RPS), Karpenter/CA for nodes, and KEDA for event-driven signals.

Q232. HPA with a StatefulSet; HPA vs VPA

Asked at (5): Flentas, HCL, One2N, Others (company not named), Persistent Systems

HPA can target a StatefulSet (`scaleTargetRef: kind: StatefulSet`), scaling pods one at a time in order; only sensible if the app can scale horizontally (needs sharding/replication). VPA adjusts a pod's CPU/memory requests (recommend/auto modes) - restarts pods to apply, and shouldn't be combined with HPA on the same CPU/memory metric. Use HPA for stateless scale-out, VPA for right-sizing.

Q233. Cluster Autoscaler vs Horizontal Pod Autoscaler (and Karpenter)

Asked at (4): Amazon, EPAM, Flentas, SquareOps

HPA adds/removes pod replicas based on metrics (CPU, memory, custom). Cluster Autoscaler adds/removes nodes when pods are Pending for lack of resources or nodes are underused. Karpenter is a faster, flexible alternative that launches the right-sized instance directly. HPA scales pods first, then the node autoscaler provides capacity for them.

Storage

Q234. ConfigMaps, PV and PVC

Asked at (5): Aspire, EY, HCL, Others (company not named), TCS

ConfigMap holds non-secret configuration, consumed as env vars or files. PersistentVolume (PV) is a piece of storage in the cluster (provisioned manually or dynamically); PersistentVolumeClaim (PVC) is a request for storage by a pod; Kubernetes binds a PVC to a matching PV (via StorageClass), giving pods durable storage independent of their lifecycle.

Q235. RWO vs RWX volumes

Asked at (4): Publicis Global Delivery, Sapient, SquareOps, Verizon

ReadWriteOnce: mounted read-write by a single node (EBS) - pods on that node can share it.
ReadWriteMany: many nodes read-write at once (EFS, NFS, CephFS). ReadOnlyMany: many nodes read-only. EBS can't be RWX, so shared storage needs EFS/NFS.

Q236. Auto volume expansion; What is CSI and CSI drivers?

Asked at (3): Capgemini, Encora, NPCI

CSI (Container Storage Interface) is the standard plugin API letting Kubernetes use any storage backend (EBS, EFS, Azure Disk, Ceph) via drivers deployed in the cluster. Volume expansion: set `allowVolumeExpansion: true` in the StorageClass, then edit the PVC's `resources.requests.storage` to a bigger value; the CSI driver (and filesystem) resize it, often online.

Q237. Storage classes in Kubernetes; what happens when a Deployment with one PVC is scaled from 1 to 3?

Asked at (3): Netcracker, SquareOps, TCS

A StorageClass defines how volumes are provisioned (provisioner, type gp3/io1, reclaim policy, volumeBindingMode, allowVolumeExpansion) enabling dynamic provisioning. If a Deployment uses a single PVC and scales to 3: with RWO (EBS) all pods must land on the same node (or the extra pods fail to attach and stay Pending/ContainerCreating); with RWX (EFS) all 3 share it. For per-replica storage use a StatefulSet's volumeClaimTemplates. EBS can be shared by multiple pods only on the same node, not across nodes.

Q238. Static vs dynamic volume provisioning

Asked at (2): Capgemini, NPCI

Static: an admin pre-creates PersistentVolumes and PVCs bind to them (good for existing disks/NFS).
Dynamic: a StorageClass with a provisioner creates the volume on demand when a PVC is created (EBS CSI, EFS) - standard in cloud. Use case: StatefulSet databases request storage via `volumeClaimTemplates` and get EBS volumes automatically.

Q239. How do you mount a PVC in a Deployment?

Asked at (1): SquareOps

```
spec:
  containers:
    - name: app
      volumeMounts: [{ name: data, mountPath: /data }]
  volumes:
    - name: data
      persistentVolumeClaim: { claimName: app-pvc }
```

The PVC must exist (and bind to a PV via StorageClass); for multiple replicas use RWX storage or a StatefulSet with `volumeClaimTemplates`.

Q240. What is the EFS CSI driver?

Asked at (1): SquareOps

A CSI driver that lets Kubernetes mount Amazon EFS as ReadWriteMany PersistentVolumes. Install it as an add-on, give it an IAM role (IRSA), create an EFS with mount targets, define a StorageClass and PVC. Use it where many pods across nodes share files.

Scheduling

Q241. How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity

Asked at (11): Capgemini, HCL, Infosys, Koerber Pharma, Netcracker, Nitor Infotech, Oracle, Others (company not named), Persistent Systems +2 more

Simplest: `nodeSelector` with a node label, or `nodeName`. More flexible: node affinity (`requiredDuringScheduling...` hard rule, `preferred...` soft rule) matching node labels. Pod anti-affinity spreads replicas away from each other (e.g. different nodes/zones). Taints on nodes + tolerations on pods control who may run there; `TopologySpreadConstraints` for even spreading.

Q242. Pod scheduling arithmetic: requests/limits vs node capacity (e.g. 4 vCPU/16 GB limits and 2 vCPU/10 GB requests; 8 vCPU/32 GB node; 30 of 32 GB used)

Asked at (2): Flentas, Others (company not named)

The scheduler uses REQUESTS, not limits. A pod requesting 2 vCPU / 10 GB fits on a 8 vCPU/32 GB node 3 times by memory (30 GB), and 4 by CPU, so 3 pods (minus system reserved resources, kubelet/daemonsets). If only 2 GB of 32 GB is free and the pod requests 500 MB it can be scheduled (request fits), even if its 4 GiB limit exceeds free memory - but it may be OOM-killed if it grows. HPA/VPA don't change this: a new pod needs a node with free requested capacity, otherwise it's Pending until the node autoscaler adds a node.

Secrets & Config

Q243. ConfigMap vs Secret

Asked at (8): EXL Service, EY, Intact Green Services, L and T, Marsh McLennan, Oracle, Others (company not named), Zensar

Both inject configuration into pods (env vars or mounted files). ConfigMap = non-sensitive data in plain text. Secret = sensitive data (passwords, tokens), base64-encoded (not encrypted by default), can be encrypted at rest, has stricter RBAC and is kept in tmpfs on nodes. Use external secret managers for real protection.

Q244. How do you manage secrets and config at scale in GitOps without leaking them?

Asked at (4): Accenture, EPAM, LTIMindtree, SquareOps

Keep secret values out of Git: External Secrets Operator or the Secrets Store CSI driver sync from Vault/AWS Secrets Manager/Key Vault; or commit encrypted secrets using Sealed Secrets or SOPS (age/KMS); restrict access with RBAC; rotate regularly; and only reference secret names in manifests.

Q245. What are Sealed Secrets?

Asked at (1): SquareOps

Bitnami Sealed Secrets lets you commit secrets to Git safely: you encrypt a Secret with the controller's public key using `kubeseal` into a SealedSecret; only the in-cluster controller can decrypt it into a normal Secret. It enables GitOps without exposing credentials.

Service Mesh

Q246. Design an Istio setup for your Kubernetes cluster

Asked at (3): EY, Oracle, Others (company not named)

Install Istio (istioctl/Helm) with istiod control plane and Envoy sidecars (or ambient mode); label namespaces for injection; define an Ingress Gateway, VirtualServices/DestinationRules for routing, canary and retries; enable mTLS (PeerAuthentication STRICT) and AuthorizationPolicies; integrate Kiali, Prometheus, Grafana, Jaeger for observability. Plan resource overhead and upgrades.

Q247. Is a service mesh always needed? Is the burden of Istio worth it?

Asked at (1): Wipro

No. A mesh adds sidecars, complexity and resource cost. Use it when you need mTLS everywhere, fine-grained traffic control (canary, retries, circuit breaking), uniform observability or zero-trust policy across many services. For smaller estates, simpler tools suffice: ingress controller, NetworkPolicies, library-level retries, Linkerd/Cilium (lighter) or ambient mode.

Cloud Kubernetes

Q248. Fargate vs EKS worker nodes; EKS vs ECS; EBS disadvantages in EKS

Asked at (7): Deloitte, EPAM, JPMorgan, Others (company not named), Persistent Systems, Sigmoid, SquareOps

Fargate on EKS runs each pod in its own micro-VM with no node management (no DaemonSets, no privileged pods, higher per-pod cost); managed node groups give full control, GPUs, DaemonSets, cheaper at scale. EKS = managed Kubernetes (portable, big ecosystem, more complexity); ECS = AWS-native simpler orchestrator (tighter integration, less portable). EBS volumes are AZ-bound and single-attach (RWO), so pods can't move across AZs without data and shared access isn't possible - use EFS for shared storage.

Q249. When would you prefer on-prem Kubernetes over EKS and vice versa? Why deploy frontend on EKS instead of S3+CloudFront?

Asked at (2): Flentas, Others (company not named)

On-prem: data residency/regulation, existing hardware, low latency to local systems, avoiding cloud cost at steady scale. EKS: managed control plane, elasticity, integration with IAM/ALB/EBS, less ops. Frontend: static SPAs are cheaper and faster on S3+CloudFront; running on EKS makes sense when the frontend does SSR, shares the same platform/pipeline, needs service-mesh features, or runtime config - otherwise S3+CloudFront is the better choice, and you should say so.

CI/CD

55 questions

Core Concepts

Q250. Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow, pre-build/build/post-build)

Asked at (25): Accolite, Amazon, BMW TechWorks, CTS, Deloitte, EXL Service, Emphasis, IBM, Infosys +16 more

CI (Continuous Integration) = every commit is built and tested automatically so integration bugs surface early. CD = Continuous Delivery (always deployable, manual approval to prod) or Continuous Deployment (auto-deploy).

Typical flow I describe:

1. Developer pushes code to Git -> webhook triggers pipeline.
2. Pre-build: checkout, restore dependencies/cache.
3. Build: compile (Maven/npm), unit tests, code quality (SonarQube), dependency/SAST scan.
4. Package: build Docker image, scan with Trivy, push to registry/Nexus with a version tag (commit SHA).
5. Deploy to Dev automatically (Helm/ArgoCD/Ansible), run smoke/integration tests.
6. Promote to QA/Staging, then prod after approval, using rolling/blue-green/canary.
7. Post-build: notifications (Slack), reports, monitoring and rollback if health checks fail.

Tools example: GitHub -> Jenkins/GitHub Actions -> SonarQube -> Docker/ECR -> Helm + ArgoCD -> EKS -> Prometheus/Grafana. Tailor to your project: name real tools, branch strategy and deploy frequency.

Q251. Canary vs blue-green (and other deployment strategies)

Asked at (14): Accion Labs, Capgemini, EXL Service, EY, HCL, IBM, Infosys, Koerber Pharma, Nisum Technologies +5 more

- Blue-green: two identical environments; deploy to idle one, test, then switch traffic instantly; rollback = switch back; cost is double infrastructure.
- Canary: release to a small percentage (5%), watch metrics, increase gradually; lowest risk, needs good observability and traffic splitting.
- Rolling: replace instances gradually (default in Kubernetes). Recreate: stop then start (downtime).

Q252. What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple environments, and scale to 1000+ clusters?

Asked at (10): EPAM, EXL Service, HCL, LTIMindtree, Nextturn, Nice, Nitor Infotech, SAP, Sony +1 more

GitOps = Git is the single source of truth for desired state; an in-cluster agent pulls and reconciles. Argo CD compares live state with Git; out-of-sync resources show as drift and can auto-sync with `selfHeal`. Multiple environments (E1, E2, E3): separate folders/overlays (Kustomize/Helm values) or branches, one Application (or ApplicationSet) per environment with different sync policies (auto for dev, manual for prod). At 1000+ clusters: ApplicationSets, hub-and-spoke or sharded controllers, progressive rollout waves, drift alerts, emergency hotfixes via an expedited PR path, manual overrides time-boxed and recorded.

Q253. How do you reduce a slow pipeline (40-60 min -> 15-20 min)? Caching in Jenkins/GitHub Actions

Asked at (9): CGI, IBM, JPMorgan, LTIMindtree, Nitor Infotech, Others (company not named), Persistent Systems, Sony, UST

Find the slowest stages first, then:

- Cache dependencies (Maven `~/.m2`, npm, pip; `actions/cache` in GitHub Actions; workspace/agent caches in Jenkins).
- Run independent stages/tests in parallel and shard tests.
- Use Docker layer caching and multi-stage builds; smaller base images.
- Build only what changed (path filters, monorepo tools), skip redundant stages on non-main branches.
- Use bigger/ephemeral agents, remove unnecessary sleeps and manual waits.
- Run quality gates asynchronously where safe.

Q254. How do you onboard a new project to multiple environments and handle releases/hotfixes?

Asked at (8): Alphadyne, Altimetrik, Elixir Labs, Hexaware, Sigmoid, SquareOps, TCS, Volkswagen Group Digital

Create repo with standard branching (trunk-based or GitFlow), pipeline template, environment configs (dev/QA/prod) in Helm values/tfvars, secrets in vault, approvals for prod, monitoring and access. Releases: tag from main/release branch; hotfix: branch from the prod tag, fix, test, deploy, merge back to main.

Q255. AWS CodePipeline/CodeBuild/CodeDeploy - how to set up, and CodePipeline vs Jenkins

Asked at (6): HCL, LTIMindtree, Persistent Systems, Synechron, TCS, ZopSmart

CodePipeline orchestrates stages: Source (CodeCommit/GitHub) -> CodeBuild (buildspec.yml compile/test/package) -> CodeDeploy (appspec.yml to EC2/ECS/Lambda) with approval gates. Pros: managed, native IAM. Jenkins: more plugins/flexibility, multi-cloud, self-managed. Choose CodePipeline for AWS-native, low-ops; Jenkins when you need complex logic or existing investment.

Q256. How do you handle parallel execution and jobs in CI/CD (Jenkins, GitHub Actions)?

Asked at (6): Deloitte, IBM, LTIMindtree, Nitor Infotech, Others (company not named), Zensar

Jenkins declarative: `parallel { stage('Unit'){...} stage('Lint'){...} }`. GitHub Actions: jobs run in parallel by default unless `needs:` is set; `strategy.matrix` runs combos in parallel. Make sure steps are independent, enough executors/runners exist, and use stash/artifacts to share results.

Q257. How do you implement rollback in a CI/CD pipeline? Describe a rollback you handled

Asked at (6): JPMorgan, LTIMindtree, Others (company not named), Sigmoid, SquareOps, TCS

Keep every artifact immutable and versioned so you can redeploy the previous version. Methods: `kubectl rollout undo` / `helm rollback`, ArgoCD revert commit, blue-green switch back, feature flags, database backward-compatible migrations. Add automated health checks/smoke tests with auto-rollback on failure. Story: describe cause (e.g. a config mismatch), how you detected (alerts), quick rollback time, and lessons (add pre-prod test, canary).

Q258. Maven lifecycle, Maven release, and build tools for Java

Asked at (6): EY, NatWest Group, Others (company not named), Synechron, Wipro, ZopSmart

Maven default lifecycle phases: validate, compile, test, package, verify, install, deploy (clean is a separate lifecycle). `mvn clean package` builds the jar/war. Maven release plugin automates versioning: `release:prepare` (remove SNAPSHOT, tag, bump) and `release:perform` (build and publish). Java builds use Maven or Gradle.

Q259. Design a CI/CD pipeline for a new app (Node.js, Angular/Java/.NET, developer gave only source code) and for 10 FE + 10 BE microservices with Jenkins

Asked at (5): Alphadyne, IBM, One2N, Others (company not named), TCS

Steps: Git repo + branch strategy, Dockerfile, pipeline: install/restore -> lint -> unit tests -> build (npm build / mvn package / dotnet publish) -> Sonar/SCA -> image build + scan -> push to registry -> deploy Dev -> integration tests -> QA -> approval -> Prod. For 20 microservices: one shared library/template, one multibranch pipeline per service (generated by Job DSL or an organization folder), path-based triggers, independent versioning, Helm charts per service, environment promotion via GitOps, and a parameterised shared deploy job.

Q260. Best practices to structure repos and pipelines in a large DevOps project; reusable templates for 50 applications

Asked at (4): EPAM, HCL, NUOS INFO Systems, Others (company not named)

Keep app code, infrastructure and pipeline library separate; use monorepo or polyrepo deliberately; shared pipeline templates (Jenkins shared libraries, GitHub reusable workflows, Azure YAML templates) so a fix is made once; parameterise per app/environment; versioned templates; standard naming and branching. For scale and HA: controller with ephemeral agents (Kubernetes pods), autoscaling runners (cost-efficient), config as code, backups.

Q261. CI/CD for ML models; ephemeral environments; SLA monitoring in a payments pipeline; iOS build automation

Asked at (4): EPAM, JPMorgan, Moodys, SquareOps

- ML: pipeline stages data validation -> training -> evaluation vs baseline -> register model (MLflow) -> package -> canary deploy -> monitor drift.
- Ephemeral envs: per-PR namespace/stack created by IaC/Helm, tested, then destroyed automatically with TTL.
- Payments SLA: define SLIs (success rate, latency) per hop, instrument with tracing/metrics, SLO alerts and synthetic transactions.
- iOS: needs macOS runners (Mac mini / cloud Mac); Fastlane for build/sign/test, certificates in secure storage, upload to TestFlight.

Q262. What challenges have you faced in CI/CD / a known critical production issue / prioritising multiple failures?

Asked at (4): Elixir Labs, Nice, Others (company not named), Synechron

Use STAR: Situation, Task, Action, Result. Example: "Our pipeline took 50 minutes and blocked releases. I profiled stages, added dependency caching, parallel test jobs and incremental Docker layers and cut it to 15 minutes (a ~70% improvement)." For multiple failures: triage by customer impact and blast radius, restore service first (rollback), then root cause, communicate, write a postmortem and add a preventive control.

Q263. Approvals in a CD pipeline

Asked at (3): Hexaware, SquareOps, TCS

Jenkins: `input message: 'Deploy to prod?'`, `submitter: 'release-team'`. Azure DevOps: Environments -> Approvals and checks. GitHub Actions: Environments with required reviewers. GitLab: protected environments / `when: manual`. Record who approved for audit.

Q264. Code quality vs code coverage

Asked at (3): JPMorgan, LTIMindtree, Sigmoid

Code coverage = percentage of code executed by tests (measured by JaCoCo, coverage.py). Code quality is broader: bugs, code smells, vulnerabilities, duplication, maintainability (SonarQube). High coverage doesn't guarantee quality. Manage coverage by setting a threshold in the quality gate (e.g. $\geq 80\%$ on new code), reports in CI, and fail the build below it.

Q265. GitHub Actions vs Argo CD; how does CI differ from GitOps CD?

Asked at (3): Nextturn, SAP, Wipro

GitHub Actions is a CI (and push-based CD) engine run on events. Argo CD is a pull-based GitOps controller inside the cluster that continuously syncs the cluster to what's in Git, detecting and healing drift. Typical combo: Actions builds/tests/pushes image and updates the manifest repo; Argo CD deploys.

Q266. Deploy to both cloud and on-premises (hybrid) and across multiple clouds - how do you design the pipeline?

Asked at (2): Rapidsoft, Synechron

Use one pipeline tool with self-hosted agents/runners inside each network (the agent in on-prem pulls jobs; no inbound firewall holes). Keep the artifact the same everywhere (container image), parameterise targets, use Terraform/Ansible for provisioning, and GitOps per environment. Manage credentials per target with federation/vault. Avoid per-cloud pipeline copies - use shared templates and per-cloud stages.

Q267. Do you commit tfvars to Git? How do you run Terraform in a CD pipeline?

Asked at (2): Flentas, Turning

Commit non-secret tfvars per environment; never commit secrets or state - supply them via the pipeline's secret store / TF_VAR_ variables. Pipeline: init (remote backend) -> fmt/validate -> tflint/checkov -> plan (saved to file, posted for review) -> manual approval -> apply of that exact plan file; use OIDC for cloud auth and state locking.

Q268. GitLab CI: experience, rules, pipelines

Asked at (2): EY, Sonata Software

.gitlab-ci.yml defines stages and jobs. rules: control when a job runs (replacing only/except), e.g. rules: - if: '\$CI_COMMIT_BRANCH == "main"'. Also: runners (shared/specific), artifacts, cache, environments, includes/templates, protected variables.

Q269. Use files/outputs of a previous stage in the current stage; refer to a variable from a previous stage

Asked at (2): Others (company not named), Sigmoid

Jenkins: stash name: 'build', includes: 'target/*.jar' then unstash 'build' (or archiveArtifacts), and set values via env.VAR = ... / returnStdout: true. GitHub Actions: upload-artifact/download-artifact, and echo "ver=1.2" >> \$GITHUB_OUTPUT then \${ steps.id.outputs.ver } or needs.job.outputs. Azure: ##vso[task.setvariable variable=x;isOutput=true] and stageDependencies.

Q270. A pipeline was deleted by a team member - how do you recreate it and prevent it in future?

Asked at (1): Others (company not named)

Recreate from pipeline-as-code (Jenkinsfile/YAML in Git) - this is why the pipeline definition must live in the repo, not only in the UI. Restore Jenkins from JENKINS_HOME backup if jobs were UI-created. Prevent: pipeline as code + Job DSL/CasC, restrict delete permissions (RBAC), branch protection/code review for pipeline changes, regular backups.

Q271. Automated pipeline to move a file from Azure Blob to GCP Cloud Storage

Asked at (1): Others (company not named)

Trigger (schedule or Event Grid). Pipeline step authenticates with both clouds (federated identities), copies via azcopy to GCS (azcopy copy supports S3, or use rclone/gsutil or Storage Transfer Service), verifies checksum, logs, and alerts on failure. Keep credentials in a secret store.

Q272. How do you remove human intervention when you need a platform to deploy a built package?

Asked at (1): Perfios

Infrastructure as Code: the pipeline (or a separate one) provisions the target using Terraform/CloudFormation, configures it with Ansible/cloud-init or deploys to managed platforms (Kubernetes/ECS) declaratively, so environments are created on demand and no one has to build servers manually.

Q273. How do you stop broken code from reaching production?

Asked at (1): Sony

Branch protection with required PR reviews and passing checks, automated tests and quality gates in CI, security scans, staging/UAT with smoke tests, approvals for production, progressive delivery (canary/blue-green) with automatic rollback, feature flags, and monitoring with alerts after release.

Q274. In a monorepo, how do you build and deploy only the relevant services?

Asked at (1): EPAM

Path filters (GitHub `paths:`, GitLab `rules:changes`, Jenkins `changeset`), compute changed dirs via `git diff --name-only origin/main...HEAD`, or use build tools like Nx/Bazel/Turborepo that understand dependency graphs. Build a matrix of only changed services.

Q275. Production pipeline is blocked on missing approvals and stakeholders are unreachable

Asked at (1): JPMorgan

Don't bypass controls silently. Follow the escalation path (backup approver / manager / on-call), define a delegated-approver policy ahead of time. For a genuine emergency, use a documented break-glass process with post-hoc approval and audit trail. Fix root cause by setting approval groups with multiple approvers and timeouts.

Jenkins

Q276. Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters

Asked at (10): Alphadyne, CGI, Cisco, Deloitte, EY, Intact Green Services, LTIMindtree, Nitor Infotech, Others (company not named) +1 more

Multibranch pipeline scans a repo and automatically creates a pipeline for each branch/PR having a Jenkinsfile. Declarative: structured `pipeline { agent stages }` syntax, simpler, validated. Scripted: free-form Groovy, more flexible but harder to maintain. Shared library: reusable Groovy in `vars/` (global steps) and `src/`, loaded with `@Library('lib')` `_`. Dynamic stages: generate in a `script {}` block using `for` over environment lists; use `when { expression { env.DEPLOY_ENV == 'prod' } }`. Variables are used as `${env.VAR}` / `params.NAME`; `parameters {}` lets downstream jobs know which environment to deploy.

Q277. Jenkins administration: standalone setup, memory issues, copying jobs, backup, concurrent builds, credentials, AWS auth, plugins

Asked at (9): Belcan, EY, Emphasis, Intact Green Services, Koerber Pharma, LTIMindtree, Nextturn, Others (company not named), Publicis Global Delivery

- Standalone controller works for small teams, but is a single point of failure: use agents for builds, backups, and HA.
- Memory problems: check JVM heap (`-Xmx`), GC logs, heap dump, plugins, huge build history; set `discard-old-builds` and run builds on agents.
- Copy a job: UI "New item -> Copy from", or export `config.xml`; move between nodes via `JENKINS_HOME` or Job DSL/CasC.
- Backup: `JENKINS_HOME` (jobs, `config.xml`, credentials, secrets) with ThinBackup plugin or snapshot; store off-box.
- Concurrency: limit executors, `disableConcurrentBuilds()`, lockable resources, autoscaled K8s agents.
- AWS auth: IAM role on agent or AWS Credentials plugin / OIDC. Useful plugins: Pipeline, Git, Docker, Kubernetes, AWS Steps, CloudBees AWS.
- Python app on AWS: build/test, package, deploy by SSH/Ansible/CodeDeploy/ECS.

Q278. Connect a new Jenkins to GitHub; triggers (webhook, poll SCM, after N commits)

Asked at (8): Belcan, CTS, Deloitte, Emphasis, IBM, Others (company not named), Sony, SquareOps

Steps: install Git and GitHub plugins, add credentials (PAT or SSH deploy key), create job/multibranch with repo URL. Trigger options: GitHub webhook (push event -> `http://jenkins/github-webhook/`) - best; Poll SCM (cron `H/5 * * * *`); periodic build; upstream job. "Build after N commits" has no native feature: use poll SCM with a script that counts commits since last build, or schedule builds and let Jenkins skip if no changes.

Q279. How do you troubleshoot failing, stuck or slow pipelines (build not starting, random artifact upload failures, fails only on Tuesdays, syntax error in stage 5)?

Asked at (7): Deloitte, JPMorgan, LTIMindtree, NUOS INFO Systems, Rapidsoft, SquareOps, Synechron

Method: read the console log from the first error, then check by layer.

- Not triggering/building: webhook delivery, SCM polling, branch filter, **when** conditions, no available agents/executors, disabled job.
- Stuck: waiting for executor, **input** step, hung process, network/DNS, resource limits - use thread dump / **top** on agent.
- Random artifact upload failure: network/proxy, repository (Nexus/S3) disk or quota, credentials expiry, timeouts, concurrent uploads, DNS.
- Only on Tuesdays: look for time-based causes - cron jobs competing for resources, scheduled backups/maintenance, certificate or token expiry, rate limits, time-zone logic, dependency updates.
- Syntax error in stage 5 of declarative pipeline: Jenkins parses the whole Jenkinsfile first, so the build fails immediately before stage 1 runs (in scripted pipelines earlier stages would run).

Ownership: the DevOps/platform team handles pipeline infra, developers handle their build/test failures.

Q280. Show a sample Jenkinsfile and explain stages, agent, post and environment blocks; what types of Jenkins pipelines/jobs exist

Asked at (6): Belcan, Emphasis, Intact Green Services, Others (company not named), SquareOps, ZopSmart

```
pipeline {
  agent any
  environment { APP = 'myapp'; REG = 'registry.example.com' }
  stages {
    stage('Checkout'){ steps{ checkout scm } }
    stage('Build'){ steps{ sh 'mvn -B clean package' } }
    stage('Test'){ steps{ sh 'mvn test' } }
    stage('Docker'){ steps{ sh "docker build -t $REG/$APP:$BUILD_NUMBER ." } }
    stage('Deploy'){ when{ branch 'main' } steps{ sh './deploy.sh' } }
  }
  post { success{ echo 'ok' } failure{ echo 'failed' } always{ cleanWs() } }
}
```

agent = where it runs, **environment** = variables, **post** = actions after stages by outcome. Types: Freestyle, Pipeline (declarative/scripted), Multibranch, Folder, Maven project.

Q281. What are executors and agents in Jenkins and how do they work?

Asked at (3): CGI, IBM, Sigmoid

An agent (node) is a machine that runs builds, connected to the controller by SSH or JNLP/WebSocket. An executor is a slot on a node that runs one build at a time; N executors = N concurrent builds. The controller schedules queued jobs onto free executors matching the job's label. Best practice: controller has 0 executors; use ephemeral agents (Kubernetes pods).

Q282. Give view-only access on 5 Jenkins jobs to other users

Asked at (2): Belcan, Others (company not named)

Use Role-Based Strategy or Matrix Authorization: create a role with only Overall/Read and Job/Read, scope it by job regex or folder, and assign it to users/groups. Put the jobs in a folder and apply folder-level permissions.

Q283. How do you configure a Flask application in Jenkins?

Asked at (1): Synechron

Create a pipeline job from the repo: stages **python** -m venv venv, **pip install -r requirements.txt**, **pytest**, flake8/Sonar, build a Docker image (gunicorn app), push, then deploy (SSH/Ansible/Kubernetes). Jenkins needs Python installed on the agent or runs inside a Python container agent.

Q284. Jenkins deployment works in Dev but fails in Production - how to troubleshoot?

Asked at (1): Wipro

Diff the environments: credentials/permissions, network/firewall/proxy, config and secrets values, resource quotas, tool and cluster versions, approval/branch conditions, feature differences, data volume and timeouts. Read the failing stage logs, reproduce in staging with prod-like config, and fix via parameterised config - not by editing prod manually.

Q285. Secure Jenkins pipelines

Asked at (1): Sony

Enable authentication/RBAC, use credentials store not env vars, restrict script approval and Groovy sandbox, run builds on ephemeral agents not the controller, keep plugins updated, enable audit logs, HTTPS, CSRF protection, limit who can configure jobs, and scan code/images in the pipeline.

Q286. What is Blue Ocean in Jenkins?

Asked at (1): Synechron

A modern Jenkins UI plugin that visualises pipelines (stages, parallel branches, logs), simplifies creating pipelines from Git, and shows branch/PR status. It is in maintenance mode now; the standard Pipeline Stage View/Graph plugins are used more.

Q287. Write a checkout stage with Git credentials

Asked at (1): CGI

```
stage('Checkout'){
  steps{
    git branch: 'main', credentialsId: 'github-creds', url: 'https://github.com/org/repo.git'
  }
}
```

Credentials are stored in Jenkins (username/token or SSH key), never in the Jenkinsfile.

DevSecOps

Q288. SonarQube and Snyk integration, quality gates, tokens and static analysis

Asked at (10): Blue Yonder, Deloitte, HCL, Hexaware, LTIMindtree, Nice, Others (company not named), Persistent Systems, Sony +1 more

SonarQube: install server, generate a token (My Account -> Security), add as Jenkins credential, install the SonarQube Scanner plugin, configure server URL under Manage Jenkins, add a stage:

```
stage('Sonar'){ steps{ withSonarQubeEnv('sonar'){ sh 'mvn sonar:sonar' } } }
stage('Gate'){ steps{ timeout(5){ waitForQualityGate abortPipeline:true } } }
```

Quality gate = pass/fail conditions (coverage > 80%, no new critical bugs/vulnerabilities, duplication < 3%). In Azure Pipelines use the SonarQube extension tasks Prepare/Analyze/Publish. Snyk scans open-source dependencies/containers via `snyk test` with an API token. Combine with unit/integration tests so code quality is enforced automatically.

Q289. Design an end-to-end DevSecOps pipeline (compliance, PCI-DSS, GDPR/HIPAA); handling vulnerabilities in a pipeline; security tool integrations

Asked at (5): EPAM, EY, IBM, Others (company not named), Wipro

Shift security left: pre-commit secret scanning, SAST (SonarQube/CodeQL), SCA (Snyk/Dependabot), container scan (Trivy), IaC scan (Checkov/tfsec), DAST (OWASP ZAP) on staging, signed images (Cosign) and SBOM, admission policies (OPA/Kyverno) at deploy, runtime monitoring. Compliance: approvals, immutable audit logs, separation of duties, evidence stored, data residency per region, encryption and access reviews. If a vulnerability is found: block the build, patch/upgrade the dependency, rotate exposed secrets, re-scan and document.

Q290. A developer asks you to remove the code-quality (Sonar) stage because it is slow. What do you do?

Asked at (1): Others (company not named)

Don't just remove the gate; fix the slowness: run analysis only on changed code (PR analysis), run it in parallel with tests, use incremental caching, size the scanner properly, and run full scans nightly. Explain the cost of skipped quality control (bugs in prod) and agree on a time budget.

GitHub Actions

Q291. GitHub Actions: workflow file, Java Maven build and artifact, manual trigger, concurrency, runners

Asked at (6): Marsh McLennan, Others (company not named), Persistent Systems, SquareOps, Synechron, TCS

A workflow is YAML in `.github/workflows/` with `on` (events), `jobs`, `runs-on`, and `steps`.

```
name: ci
on: [push, workflow_dispatch] # workflow_dispatch = manual trigger
concurrency: { group: ci-${{ github.ref }}, cancel-in-progress: true }
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-java@v4
        with: { distribution: temurin, java-version: 17, cache: maven }
      - run: mvn -B verify
      - uses: actions/upload-artifact@v4
        with: { name: app, path: target/*.jar }
```

`concurrency` prevents overlapping runs. Self-hosted runners: install runner agent on a VM/K8s (actions-runner-controller), register with a token, add labels, and keep them isolated/ephemeral.

Q292. needs keyword, concurrency, and matrix strategy in GitHub Actions

Asked at (3): Marsh McLennan, Others (company not named), Persistent Systems

- `needs`: `build` makes a job wait for another (dependencies/ordering); without it jobs run in parallel.
- `concurrency` controls simultaneous runs of the same group (cancel or queue duplicates) - different purpose from `needs`.
- `strategy.matrix` runs a job multiple times with combinations, e.g. node 18/20 x ubuntu/windows, in parallel:

```
strategy:
  matrix: { node: [18, 20], os: [ubuntu-latest, windows-latest] }
runs-on: ${{ matrix.os }}
```

Q293. Review a GitHub Actions workflow snippet: how do you spot incorrect or missing steps?

Asked at (2): Moodys, Others (company not named)

Check: pinned action versions (SHA), minimal `permissions`, secrets via `secrets.*` not echoed, caching, `needs` ordering, tests before deploy, environment protection, matrix coverage, timeouts, artifacts upload, failure notifications, concurrency settings and linting (actionlint). Suggest missing: security scan, test report, rollback and approval.

Q294. Structure a multi-stage GitHub Actions pipeline that builds, tests and deploys a containerised app to Kubernetes/EKS

Asked at (2): SquareOps, Wipro

Jobs: `test` (lint + unit tests) -> `build` (docker/build-push-action, tag with SHA, Trivy scan) -> `deploy-dev` (auto) -> `deploy-prod` (environment with required reviewers). Deploy step: configure AWS credentials via OIDC, `aws eks update-kubeconfig`, then `helm upgrade --install` or `kubectl set image`, with `kubectl rollout status` and rollback on failure. Use `needs`, caching, concurrency and environment secrets.

Q295. Workload identity federation between GitHub Actions and Azure/GCP securely

Asked at (1): EPAM

Use OIDC: GitHub issues a short-lived JWT per run; configure the cloud to trust GitHub's issuer and restrict by repo/branch/environment claims (federated credential on an Azure app / GCP workload identity pool). The workflow requests `permissions: id-token: write` and exchanges the token for temporary cloud credentials - no stored secrets.

Secrets in Pipelines

Q296. How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets Manager) and rotate them dynamically?

Asked at (13): EXL Service, Flentas, HCL, JPMorgan, Marsh McLennan, Nice, Orion Innovation, Others (company not named), Rapidsoft +4 more

Rule: never hard-code or echo secrets.

- Jenkins: Credentials plugin (secret text/ssh/username-password), injected via `withCredentials`, masked in logs; scoped per folder.
- Azure DevOps: secret variables / variable groups linked to Key Vault, or Key Vault task; pipeline uses a service connection (workload identity federation) with `get/list` permission on the vault.
- AWS: pipeline role calls `aws secretsmanager get-secret-value` or use OIDC federation, no static keys.
- Rotate automatically in the vault (Secrets Manager rotation / dynamic secrets in HashiCorp Vault), short-lived tokens, secret scanning (gitleaks) on commits.

Q297. Credentials appear in pipeline logs, or a developer hard-coded a password and pushed it - what do you do?

Asked at (4): EXL Service, Sigmoid, SquareOps, UST

Treat the secret as compromised: 1) revoke/rotate it immediately, 2) delete or mask the log and restrict access, 3) remove it from Git history (`git filter-repo` or BFG, force-push, and ask devs to re-clone) - but rotation is what matters, 4) find the cause (`set -x`, echo of env). Prevent: pre-commit hooks and CI secret scanning (gitleaks, trufflehog), masked credentials, secret managers, GitHub push protection.

Q298. What is CyberArk and how do you consume it in a pipeline?

Asked at (1): Others (company not named)

CyberArk is a privileged access / secrets management platform. Pipelines use Conjur or the Credential Provider/AAM: the job authenticates with its identity (host/JWT), requests the secret at runtime via REST API or Jenkins CyberArk plugin, uses it in memory and never stores it; access and usage are audited.

Azure DevOps

Q299. YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline; variables, parameters, schedules, logs

Asked at (12): Capgemini, HCL, Hexaware, Nextturn, Nice, Others (company not named), Persistent Systems, RelevantZ, Synechron +3 more

YAML pipelines are code in the repo: versioned, reviewable, branch-specific, reusable via templates and consistent across envs. Classic is GUI-based and harder to audit. Example skeleton:

```
trigger: [main]
stages:
- stage: Build
  jobs: [{ job: b, steps: [{ script: './build.sh' }] }]
- stage: DeployUAT
  dependsOn: Build
  jobs: [{ deployment: d, environment: uat, strategy: { runOnce: { deploy: { steps: [{ script:
    './deploy.sh' }] } } } } ]}]
```

Pass values via `variables:`, variable groups, runtime `parameters:`; schedule with `schedules:` cron; logs are stored per run in Azure DevOps (retention policy) and can be exported.

Q300. Azure DevOps: stakeholder vs admin; REST API policy on all repos; migration readiness

Asked at (2): NUOS INFO Systems, Others (company not named)

Stakeholder = free limited access (view boards, approve, no repo/pipeline authoring); Project/Collection Administrator can manage permissions and settings. REST API: loop over repos (GET `/_apis/git/repositories`) and POST branch policies (`/policy/configurations`) with a PAT. Migration readiness: inventory projects/pipelines, check size limits, identity mapping, extensions, run Azure DevOps Migration Tools validation, plan cut-over and training.

Q301. What is a PAT, a service connection / connection string, and self-hosted vs Microsoft-hosted agent?

Asked at (2): HCL, Zensar

PAT = Personal Access Token, used instead of a password to authenticate to Git/Azure DevOps APIs with limited scope and expiry. Service connection = stored, secured connection in Azure DevOps to an external service (Azure subscription, Docker registry, Kubernetes) that pipelines reference by name. Connection string = a text string with endpoint and credentials for a service such as a database or storage account (store in Key Vault). Self-hosted agents run on your own machines (custom tools, private network); Microsoft-hosted are managed, clean VMs per job.

Q302. What are deployment groups in Azure DevOps?

Asked at (1): TCS

A logical set of target machines with the deployment agent installed (classic release pipelines) to which you deploy in parallel/rolling, tagging servers (web, db). Modern YAML pipelines use Environments with resources (VMs, Kubernetes) instead.

Artifacts & Registries

Q303. Build artifacts vs pipeline artifacts; Nexus / Artifactory and storing dependencies

Asked at (5): EY, Others (company not named), RelevantZ, Synchrotron, TCS

In Azure DevOps, build artifacts are the older mechanism (stored in a container on the server); pipeline artifacts are newer, faster (deduplicated) and recommended. Artifact repositories like Nexus/Artifactory store versioned binaries (jars, npm, Docker images) and proxy public registries for caching, security scanning and reproducible builds.

Q304. JFrog Artifactory: repo types, use cases and Xray

Asked at (1): TCS

Artifactory is a universal artifact repository. Repo types: local (your own artifacts), remote (proxy/cache of public repos like Maven Central/Docker Hub), virtual (aggregates several under one URL). Use cases: store builds, Docker images, caching dependencies, promoting releases. Xray scans artifacts and dependencies for vulnerabilities and license issues and can block downloads.

Terraform & IaC

58 questions

Fundamentals

Q305. What is Terraform, how does it work, and what have you done with it?

Asked at (12): CTS, Capgemini, Deloitte, EPAM, EY, HCL, IBM, Infosys, Nice +3 more

Terraform is a declarative, cloud-agnostic Infrastructure-as-Code tool by HashiCorp. You describe desired infrastructure in HCL; Terraform builds a dependency graph, compares config with the state, and the `plan` shows changes, then `apply` calls provider APIs (AWS, Azure, Kubernetes) to reach the desired state. State records what it manages. Core workflow: write -> init -> plan -> apply -> destroy.

In your answer mention real use: VPC/subnets, EC2/ASG, EKS, RDS, S3, IAM with modules, remote state in S3 + locking, CI/CD pipelines, multiple environments, policy checks. Answer "console, CLI or Terraform?" with Terraform (IaC) for repeatability, review and audit.

Q306. What is CloudFormation and how does it compare with Terraform?

Asked at (6): Deloitte, IBM, Others (company not named), Persistent Systems, Synechron, Virtusa

CloudFormation is AWS's native IaC (JSON/YAML templates -> stacks, change sets, drift detection, rollback). Terraform (HCL) is multi-cloud with a larger ecosystem, modules and plan/apply workflow; CloudFormation has no state file to manage and supports new AWS services immediately. Choose CloudFormation for AWS-only shops wanting managed state; Terraform for multi-cloud and consistency.

Q307. Stateful vs stateless applications; how do you manage data for stateless apps?

Asked at (3): Blue Yonder, Koerber Pharma, Zensar

Stateless apps keep no client/session data in the instance, so any replica can serve any request - easy to scale and replace. Stateful apps (databases) keep data tied to specific instances and need persistent storage and stable identity. Stateless apps store data externally: databases, object storage, Redis for sessions. In Kubernetes: Deployment for stateless, StatefulSet for stateful.

Q308. Export existing Azure resources to Terraform

Asked at (1): NUOS INFO Systems

Use `aztfexport` (Azure Export for Terraform): `aztfexport resource-group my-rg` generates HCL and state; review, clean up and commit. Alternatively `terraform import` per resource or `import` blocks with `-generate-config-out` (TF 1.5+).

Q309. Terraform vs Ansible; when to use each and how they work together

Asked at (1): Nextturn

Terraform provisions infrastructure declaratively (VPC, servers, databases) with state; Ansible configures what runs on the machines (packages, files, services), procedural/idempotent and agentless. Together: Terraform creates the VMs and outputs IPs/inventory, Ansible configures them (or bake images with Packer). For containers/K8s, Terraform for the cluster, Helm/ArgoCD for apps.

Q310. Terraform vs Terragrunt

Asked at (1): Flentas

Terragrunt is a thin wrapper around Terraform that keeps configurations DRY: generates backend/provider config, handles dependencies between stacks (`dependency` blocks), runs commands across many modules (`run-all`) and supports per-environment inputs. Use it for large multi-account/multi-environment estates.

State Management

Q311. How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or deleted?

Asked at (15): CTS, Cisco, EPAM, F5, IBM, JPMorgan, Koerber Pharma, LTIMindtree, Oracle +6 more

Remote backend (S3 + locking, or Terraform Cloud) with encryption, versioning and tight IAM; separate state per environment; never commit tfstate to Git because it can contain secrets; mark sensitive variables.

Migrate: add a `backend "s3"` block and run `terraform init -migrate-state`.

Lost state: if S3 versioning is on, restore the previous version. Otherwise rebuild it using `terraform import` (or `import` blocks) for each existing resource so nothing gets recreated, then `terraform plan` should show no changes.

Q312. What is the tfstate file and what does it do?

Asked at (9): Belcan, Flentas, IBM, Others (company not named), Persistent Systems, Sapient, SquareOps, TCS, Techdome

JSON file mapping your configuration to real resources (IDs, attributes, dependencies, outputs). Terraform uses it to know what exists, compute diffs, track metadata and improve performance. It can contain sensitive data, so keep it in a secured remote backend (S3 + versioning + encryption + locking), not Git. Generated values (IDs, IPs) are stored there and exposed with outputs.

Q313. How do you remove a stuck state lock; how are state conflicts managed?

Asked at (8): Capgemini, EXL Service, Nitor Infotech, Others (company not named), RelevantZ, SquareOps, Virtusa, ZS Associates

Make sure no run is actually in progress, then `terraform force-unlock <LOCK_ID>` (or delete the lock item in DynamoDB / lock file in S3). Conflicts are avoided by remote state with locking, one pipeline run per state at a time (CI concurrency), and small state files split by component.

Q314. Pipeline fails because resources already exist - how do you handle it (Remove, Import, Plan)?

Asked at (7): Altimetrik, Aspire, Capgemini, Encora, NUOS INFO Systems, UST, Virtusa

Resource exists in the cloud but not in state. Import it: `terraform import aws_s3_bucket.b bucket-name` (or an `import {}` block), write the matching resource code, run `terraform plan` until it shows no changes. If it's a stale/wrong state entry, `terraform state rm` it. Otherwise remove the manually created duplicate if safe.

Q315. How do multiple teams deploy to the same AWS account without overwriting each other; how do you isolate state across environments and teams?

Asked at (4): EPAM, Sonata Software, Sony, SquareOps

Separate state per team/component/environment (different backend keys or workspaces), narrow IAM per team, naming/tag conventions and resource prefixes, shared infrastructure (network) in its own state exposed via outputs/`terraform_remote_state`, CODEOWNERS, SCP/permission boundaries, and CI pipelines per repo. Ideally one account per team/environment.

Q316. How do you make sure EC2 is not deleted when someone runs terraform destroy?

Asked at (3): CTS, Syncortex, Wipro

Add `lifecycle { prevent_destroy = true }` to the resource - Terraform errors out on destroy. Also enable EC2 termination protection (`disable_api_termination = true`), restrict IAM/CI permissions, and use `terraform plan` review before applying.

Q317. Terraform-generated RDS password was not saved - can you retrieve it? Changing an S3 bucket name?

Asked at (2): Sonata Software, SquareOps

If created with `random_password`, the value is stored in the state file: `terraform output -raw db_password` (if output) or `terraform state show random_password.db` / inspect state - secured state is therefore essential; otherwise reset the DB master password. S3 bucket names are immutable: changing `bucket` forces replacement (destroy + create), so copy data first or create a new bucket and migrate; do not rename in place.

Q318. terraform import vs data sources; bringing an unmanaged production environment under Terraform

Asked at (2): Capgemini, JPMorgan

`import` adopts an existing resource into state so Terraform manages (and can change) it. A `data` source only reads information about something managed elsewhere without owning it - use it for shared/foreign resources like an existing VPC or AMI. Plan to adopt a manual environment: inventory the resources, prioritise (network -> data -> compute), write HCL (use `import` blocks with `-generate-config-out` or tools like Terraformer/aztfexport), import in small batches, run `plan` until no diff, store state remotely, lock down console changes, add CI, and document. Don't change anything before the plan is clean.

Q319. backend.tf is in the repo but state doesn't appear in the storage account - why?

Asked at (1): CTS

Likely `terraform init` never ran/migrated with that backend, wrong storage account/container/key, auth failure, using a workspace key, or the apply used a local backend. Check `terraform init -reconfigure`, `.terraform/terraform.tfstate`, the backend block and container permissions.

Q320. terraform state mv / moved block; refactoring resources without destroying them

Asked at (1): Persistent Systems

`terraform state mv aws_instance.a module.app.aws_instance.a` renames/moves the address in state without recreating the resource; the declarative equivalent is a `moved { from = ... to = ... }` block. Use when renaming resources or moving them into modules; always run `plan` afterwards - it should show no changes.

Language Features

Q321. Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and user data; variables for instance_type and region; Lambda/App Service

Asked at (10): Altimetrik, Amazon, Cisco, Deloitte, Emphasis, Hexaware, LTIMindtree, Others (company not named), Qentelli Solutions +1 more

```
variable "region" { default = "ap-south-1" }
variable "instance_type" { default = "t3.micro" }
provider "aws" { region = var.region }

resource "aws_security_group" "ssh" {
  name = "ssh-only"
  ingress { from_port = 22 to_port = 22 protocol = "tcp" cidr_blocks = ["203.0.113.0/24"] }
  egress { from_port = 0 to_port = 0 protocol = "-1" cidr_blocks = ["0.0.0.0/0"] }
}

resource "aws_instance" "web" {
  ami = data.aws_ami.amzn.id
  instance_type = var.instance_type
  vpc_security_group_ids = [aws_security_group.ssh.id]
  user_data = file("setup.sh")
}

resource "aws_ebs_volume" "data" { availability_zone = aws_instance.web.availability_zone size = 20 }
resource "aws_volume_attachment" "att" { device_name = "/dev/sdf" volume_id = aws_ebs_volume.data.id
  instance_id = aws_instance.web.id }
resource "aws_s3_bucket" "b" { for_each = toset(["logs", "backups", "assets"]) bucket =
  "myorg-${each.key}" }
```

A 3-tier architecture needs: VPC, public/private subnets across AZs, IGW/NAT, route tables, security groups, ALB, ASG/EC2 (or EKS), RDS in DB subnets, and outputs.

Q322. What provisioners exist in Terraform?

Asked at (7): Aspire, Capgemini, Five9, Koerber Pharma, Optum, Others (company not named), Techdome

`local-exec` (run a command on the machine running Terraform), `remote-exec` (run on the created resource over SSH/WinRM), `file` (copy files), plus `connection` blocks. HashiCorp treats provisioners as a last resort; prefer user-data, cloud-init, Ansible/Packer. Failed provisioners taint the resource.

Q323. Terraform lifecycle (create_before_destroy, prevent_destroy, ignore_changes)

Asked at (6): Encora, Optum, Oracle, Others (company not named), Sigmoid, ZS Associates

The `lifecycle` block customises behaviour: `create_before_destroy` builds the replacement first (zero downtime), `prevent_destroy` blocks destruction, `ignore_changes` ignores drift on chosen attributes (e.g. tags set by autoscaling), `replace_triggered_by` forces replacement. Resources have a create -> update -> destroy lifecycle driven by plan/apply.

Q324. What is a provider and what is a data block in Terraform?

Asked at (5): EPAM, Sapient, Sigmoid, Sonata Software, SquareOps

A provider is a plugin that lets Terraform talk to an API (aws, azure, kubernetes), configured in `provider` blocks with versions pinned in `required_providers`. A `data` block reads existing information (an AMI, a VPC, a remote state) without creating anything: `data "aws_vpc" "main" { default = true }` and used as `data.aws_vpc.main.id`.

Q325. Explain Terraform dependency graph internals and resource dependencies; circular dependencies

Asked at (3): EPAM, Flentas, Sony

Terraform parses references (`aws_subnet.a.id`) to build a directed acyclic graph of resources, then walks it in parallel, creating dependencies first (destroy in reverse). Implicit dependencies come from references; explicit with `depends_on`. Cycles appear in real projects, e.g. two security groups referencing each other in inline rules, or modules passing outputs in both directions - fix by splitting into separate rule resources (`aws_security_group_rule`), introducing an intermediate resource, or restructuring modules. `terraform graph` visualises it.

Q326. Null resources in Terraform

Asked at (3): Others (company not named), Sapient, Sigmoid

`null_resource` runs provisioners or triggers actions without managing real infrastructure, re-running when `triggers` change. Used for local-exec scripts. Now `terraform_data` replaces it; prefer proper providers/config tools where possible.

Q327. Terraform locals vs variables; types (list, string, map of objects, tuple)

Asked at (3): Amazon, Nisum Technologies, Others (company not named)

Input variables are the module's parameters set from outside (tfvars, CLI, env). Locals are computed internal names to avoid repetition - cannot be overridden: `locals { name = "${var.env}-app" }`, used as `local.name`. Types: string, number, bool, list/tuple (ordered), set (unique), map/object (key-value; objects have typed attributes). A map of objects:

```
variable "servers" {
  type = map(object({ type = string, disk = number }))
  default = { web = { type = "t3.micro", disk = 20 }, db = { type = "t3.large", disk = 100 } }
}
```

A list is indexed by position, a string is a single text value (`split()` and `join()` convert).

Q328. count vs for_each; what happens to already-created instances if you remove an item from the list?

Asked at (3): Encora, LTIMindtree, ZS Associates

`count` indexes resources by position (`aws_instance.web[0]`); `for_each` keys them by map/set key (`aws_instance.web["app1"]`). With `count` over a list, removing the 2nd item shifts indexes, so Terraform destroys/recreates everything after it (e.g. instance 3 becomes index 2). With `for_each`, only the removed key's instance is destroyed; the others are untouched. Use `count` for simple on/off (`count = var.enabled ? 1 : 0`) and `for_each` for collections.

Q329. Create multiple EC2 instances with different configurations; create 50 instances at once

Asked at (2): Nextturn, UST

```
variable "servers" {
  default = { web = { type = "t3.micro", ami = "ami-1" }, db = { type = "m5.large", ami = "ami-2" } }
}
resource "aws_instance" "this" {
  for_each      = var.servers
  ami           = each.value.ami
  instance_type = each.value.type
  tags          = { Name = each.key }
}
```

For 50 identical ones use `count = 50` (or an Auto Scaling Group with desired capacity 50 - the cloud-native approach). Raise `-parallelism` if needed.

Q330. What is the Terraform lock file (.terraform.lock.hcl)?

Asked at (2): F5, Others (company not named)

Records the exact provider versions and hashes selected by `terraform init`, ensuring everyone and CI use the same provider builds. Commit it to Git; update with `terraform init -upgrade`. Not related to state locking.

Q331. Write a Terraform file to launch an EC2 in a public subnet in a VPC

Asked at (2): Infosys, Others (company not named)

```
resource "aws_vpc" "main" { cidr_block = "10.0.0.0/16" }
resource "aws_subnet" "public" {
  vpc_id = aws_vpc.main.id
  cidr_block = "10.0.1.0/24"
  map_public_ip_on_launch = true
}
resource "aws_internet_gateway" "igw" { vpc_id = aws_vpc.main.id }
resource "aws_route_table" "rt" {
  vpc_id = aws_vpc.main.id
  route { cidr_block = "0.0.0.0/0" gateway_id = aws_internet_gateway.igw.id }
}
resource "aws_route_table_association" "a" { subnet_id = aws_subnet.public.id route_table_id =
  aws_route_table.rt.id }
resource "aws_instance" "web" {
  ami = data.aws_ami.amzn.id
  instance_type = "t3.micro"
  subnet_id = aws_subnet.public.id
  vpc_security_group_ids = [aws_security_group.web.id]
}
```

Q332. Split 10.0.0.0/16 into /21 subnets using locals and cidrsubnet

Asked at (1): Encora

```
locals {
  vpc_cidr = "10.0.0.0/16"
  subnets = [for i in range(8) : cidrsubnet(local.vpc_cidr, 5, i)] # /16 + 5 bits = /21
}
resource "aws_subnet" "s" {
  count = length(local.subnets)
  vpc_id = aws_vpc.main.id
  cidr_block = local.subnets[count.index]
}
```

`cidrsubnet(prefix, newbits, netnum)`: `newbits = 21-16 = 5`, giving 32 possible /21 subnets.

Q333. What are main.tf, providers.tf, outputs.tf, variables.tf?

Asked at (1): SquareOps

Conventional file names (Terraform reads all .tf files in a folder): `main.tf` holds the main resources/modules, `providers.tf` configures providers (region, versions: `required_providers`), `variables.tf` declares inputs, `outputs.tf` declares values exposed after apply (and to other modules), plus `backend.tf/versions.tf/terraform.tfvars`.

Q334. What are meta-arguments in Terraform?

Asked at (1): Optum

Arguments available on every resource/module: `count`, `for_each`, `provider`, `depends_on`, and `lifecycle` (plus `providers` for modules). They change how Terraform creates and orders resources rather than configuring the resource itself.

Q335. What is the Terraform external data source / command and when to use it?

Asked at (1): Optum

The `external` data source runs an external program (script) that returns JSON, making its values available to Terraform. Use it when no provider exists for a lookup (custom API or script logic), but it hurts portability and determinism, so use as a last resort.

Q336. Why use dynamic blocks in Terraform? Show a skeleton for an Azure resource

Asked at (1): Others (company not named)

Dynamic blocks generate repeating nested blocks from a list/map, avoiding copy-paste.

```
resource "azurerm_network_security_group" "nsg" {
  name = "web-nsg"
  location = var.location
  resource_group_name = var.rg
  dynamic "security_rule" {
    for_each = var.rules
    content {
      name = security_rule.value.name
      priority = security_rule.value.priority
      direction = "Inbound"
      access = "Allow"
      protocol = "Tcp"
      source_port_range = "*"
      destination_port_range = security_rule.value.port
      source_address_prefix = "*"
      destination_address_prefix = "*"
    }
  }
}
```

Use sparingly; overuse hurts readability.

Workflow

Q337. Name basic Terraform commands and explain each

Asked at (9): Belcan, CTS, Deloitte, Nextturn, Nitor Infotech, Others (company not named), Sigmoid, Wipro, ZopSmart

`terraform init` (download providers/modules, set backend), `fmt` (format), `validate` (syntax check), `plan` (preview changes), `apply` (create/update), `destroy` (remove), `output`, `show`, `state list/rm/mv`, `import`, `workspace`, `taint/-replace`, `refresh`.

Q338. How do you manage multiple environments in Terraform?

Asked at (7): EPAM, Koerber Pharma, LTIMindtree, Nitor Infotech, Plansource ValueLabs, Qburst, RelevantZ

Options: separate directories/root modules per environment sharing reusable modules (clearest, recommended), workspaces (same code, separate state), or Terragrunt. Each environment has its own backend state key, `tfvars`, and cloud account. Promote changes dev -> stage -> prod through the pipeline.

Q339. What do terraform init, plan, apply, fmt do? Meaning of + and - in plan; auto-approve

Asked at (7): Accenture, Aspire, Emphasis, Hexaware, IBM, Others (company not named), ZopSmart

`init` downloads providers/modules and configures the backend; `plan` previews changes without applying; `apply` executes them (after confirmation, `-auto-approve` skips the prompt - use only in pipelines with a reviewed plan); `fmt` formats files to canonical style. Plan symbols: `+` create, `-` destroy, `~` update in place, `-/+` destroy and recreate (or `+/-` with `create_before_destroy`), `<=` read data source.

Q340. Do you run Terraform locally or through CI/CD? Explain the workflow; what if apply or the pipeline takes 25+ minutes?

Asked at (6): EXL Service, IBM, Koerber Pharma, NUOS INFO Systems, Others (company not named), TCS

Through CI/CD for consistency and audit. PR: fmt, validate, tfint, Checkov, `plan` posted to the PR; merge to main: approval, `apply` the saved plan using OIDC credentials, remote state with locking. Runners: self-hosted or managed (GitHub Actions, Jenkins, Atlantis, Terraform Cloud). Slow runs: split into smaller states/components, use `-parallelism`, `-target` only for emergencies, avoid refreshing huge states (`-refresh=false` cautiously), cache providers, pin provider versions, remove unnecessary data sources, split monorepo by layer (network, platform, apps).

Q341. Terraform workspaces - what, why, requirements, and how you manage them

Asked at (4): Accion Labs, Koerber Pharma, Optum, Sigmoid

Workspaces let one configuration have multiple named states (`terraform workspace new dev`, `select`, `show`; `terraform.workspace` in code). Requirement: a backend that supports multiple states (S3, remote). Good for similar short-lived/parallel environments; for prod vs non-prod with different accounts and settings, separate folders/backends are safer because workspaces share the same code and credentials.

Q342. Remove a resource from Terraform code without deleting it in the cloud; resources that must be outside Terraform's lifecycle; delete an LB after updates

Asked at (3): Flentas, Qburst, Sigmoid

Option 1: `terraform state rm aws_lb.x` then delete the block - the real resource stays untouched. Option 2: use a `removed { from = aws_lb.x lifecycle { destroy = false } }` block (TF 1.7+). For resources that must be fully manual: don't import them, reference them via `data` sources, or use `lifecycle { ignore_changes = all }`. To delete a resource Terraform manages, remove the block and apply (or `terraform destroy -target=...`).

Q343. Describe a real scenario where you used Terraform to provision scalable infrastructure

Asked at (1): Nice

Example answer: "I defined VPC, subnets, ALB, ASG with launch templates, RDS Multi-AZ and EKS using reusable modules and remote state per environment. A pipeline ran fmt/validate/tflint/checkov and plan on PR, and apply after approval. Autoscaling policies and ALB health checks gave scalability; the whole stack could be recreated in minutes for DR." Cover modules, state, CI/CD, and security.

Q344. How do you roll back changes in Terraform?

Asked at (1): Flentas

Terraform has no "undo": revert the code in Git to the last good commit and apply (plan first); restore a previous state version from S3 versioning only if state is wrong. Use small changes, plan review, `create_before_destroy`, `prevent_destroy`, targeted snapshots/backups for data, and blue-green infra for risky changes.

Q345. If a Terraform deployment fails midway, what is your approach?

Asked at (1): Sigmoid

Read the error and which resources succeeded; Terraform keeps state of what was created, so fix the cause (permissions, quota, dependency, typo) and re-run `terraform apply` - it continues from where it stopped. Check for a stuck state lock, partial/orphan resources (import or clean up), run `terraform plan` to confirm convergence, and avoid manual changes. Use `-target` sparingly and keep state backups/versions.

Modules

Q346. What is a Terraform module, what does it contain, how do you create, call, reference outputs and upgrade it?

Asked at (13): EPAM, Encora, Infosys, Nice, OPT IT, Others (company not named), Persistent Systems, Sapient, Sonata Software +4 more

A module is a folder of Terraform files used as a reusable component (VPC, EKS). Structure: `main.tf`, `variables.tf`, `outputs.tf`, `versions.tf`, README. Types: root module, child (local), registry/Git modules, published/private modules.

```
module "vpc" {
  source = "../modules/vpc"          # or "terraform-aws-modules/vpc/aws" with version = "~> 5.0"
  cidr   = "10.0.0.0/16"
}
module "vm" {
  source      = "../modules/vm"
  subnet_id = module.vpc.private_subnet_ids[0] # module output used as input to another module
}
```

Upgrade: change `version/ref`, run `terraform init -upgrade`, review `plan`, test in lower environments.

Benefits: DRY, consistency, versioning, separation of concerns.

Q347. How do you write a Terraform module for EKS; how do you create and manage clusters, and master/worker roles?

Asked at (7): Accion Labs, Altimetrik, Deloitte, IBM, ITC Infotech, Sonata Software, ZS Associates

Module structure: `variables.tf` (name, version, subnets, node groups), `main.tf` (IAM roles for cluster and nodes, `aws_eks_cluster`, `aws_eks_node_group`, OIDC provider, add-ons), `outputs.tf` (endpoint, CA data, OIDC ARN), `versions.tf`. Or use the community `terraform-aws-modules/eks` module. Master (control plane) is AWS-managed; worker nodes run the pods. Use remote state, separate environments with `tfvars`, and manage Kubernetes resources with the `helm/kubernetes` providers or `GitOps`.

Q348. Terraform repository/folder structure; how do dev/qa/prod consume shared modules; Terraform architecture

Asked at (7): Altimetrik, CMT, EXL Service, Flentas, Others (company not named), Sonata Software, Synechron

Typical layout:

```
modules/      # reusable: vpc, eks, rds
environments/
  dev/        main.tf backend.tf terraform.tfvars
  qa/         ...
  prod/       ...
```

Each environment root calls the same modules with different variables and its own state key/backend.

`provider.tf` holds provider/region, `versions.tf` pins versions. Architecture: Terraform core (CLI) + providers (plugins) + state + backend; core reads config, builds graph, plans, applies via provider APIs.

Q349. Refactor a legacy Terraform codebase / monorepo with hundreds of states into modules safely; how to avoid repeating 100 lines; slow growing codebase

Asked at (4): EPAM, Persistent Systems, RelevantZ, Sony

Move step by step: introduce modules for repeated patterns, use `moved` blocks / `state mv` so plans show no changes, split by layers and ownership (network, data, apps), version modules, use `for_each/locals/dynamic` blocks to remove duplication, add tests (`terratest`, `terraform test`) and run plan in CI at each step.

Blue/green state migration with no resource recreation gives zero downtime. Smaller states also fix slow plans.

AWS Resources

Q350. Multi-region Terraform: 3 regions, which region does an EC2 go to? Structure for EC2 in 3 regions

Asked at (3): Accenture, Syncortex, ZS Associates

Use provider aliases:

```
provider "aws" { alias = "usel" region = "us-east-1" }
provider "aws" { alias = "eu"    region = "eu-west-1" }
resource "aws_instance" "a" { provider = aws.usel ... }
resource "aws_instance" "b" { provider = aws.eu   ... }
```

A resource without `provider` = uses the default (non-aliased) provider's region - so with only aliased providers it errors; with a default provider, it deploys there. Better: a reusable module taking a `providers` map called once per region (or `for_each` over regions via module with providers), with one state per region.

Q351. Ensure a particular AMI exists using Terraform; start an EC2 when CPU > 80%

Asked at (2): Optum, Sony

Use the `aws_ami` data source with filters; a `precondition` fails the plan if none is found. For CPU-based scaling, EC2 shouldn't be started per alarm; use an Auto Scaling Group with a target-tracking policy (`ASGAverageCPUUtilization = 80`) and a launch template that pulls what it needs from S3 in user-data (via instance profile). Use a CloudWatch alarm + policy otherwise.

Q352. Create AWS EventBridge via Terraform

Asked at (1): Others (company not named)

```
resource "aws_cloudwatch_event_rule" "nightly" {
  name = "nightly"
  schedule_expression = "cron(0 2 * * ? *)"
}
resource "aws_cloudwatch_event_target" "t" {
  rule = aws_cloudwatch_event_rule.nightly.name
  arn = aws_lambda_function.job.arn
}
resource "aws_lambda_permission" "p" {
  action = "lambda:InvokeFunction"
  function_name = aws_lambda_function.job.name
  principal = "events.amazonaws.com"
  source_arn = aws_cloudwatch_event_rule.nightly.arn
}
```

Q353. Promote a read replica to primary in Terraform; AWS resources created with Terraform

Asked at (1): Emphasis

Terraform has no direct "promote" action; promote with the CLI/console (`aws rds promote-read-replica`), then update code: remove `replicate_source_db` from the resource and import/refresh so state matches. Alternatively use Aurora/global DB failover resources. Mention resources you've created: VPC, EC2/ASG, RDS, S3, IAM, EKS, Route 53, CloudWatch alarms.

Security & Policy

Q354. How do you prevent insecure infrastructure changes pushed through Terraform (like quality gates for code)?

Asked at (4): EPAM, JPMorgan, NUOS INFO Systems, Sony

Add policy-as-code gates in the pipeline: `tflint`, Checkov/tfsec/Trivy for misconfigurations, OPA/Conftest or Sentinel for policies (no public S3, required tags), mandatory plan review/PR approvals, restricted CI role permissions, and drift detection.

Q355. How do you handle secrets in Terraform?

Asked at (1): Flentas

Never hard-code them. Pass via environment variables (TF_VAR_), CI secret store, or fetch from Vault/Secrets Manager with data sources; mark variables `sensitive = true`; remember secrets still land in state, so encrypt the backend (S3 + KMS), restrict access, and prefer generating/rotating secrets outside Terraform or using write-only/ephemeral values (newer versions) and managed identities.

Q356. IAM users vs GitHub OIDC role vs Terraform Cloud role - which is more secure and when to use each?

Asked at (1): Others (company not named)

IAM user keys are long-lived secrets - least secure. GitHub OIDC role: the workflow gets a short-lived token and assumes an IAM role (trust restricted by repo/branch) - no stored secrets, best for GitHub Actions. Terraform Cloud/HCP dynamic credentials use the same OIDC approach for TFC runs. Use OIDC for CI, keep IAM users out of automation.

Q357. Least privilege for CloudFormation stacks and Terraform

Asked at (1): Others (company not named)

Give the deployment role only the permissions needed for the resources it manages (CFN service role per stack; Terraform role scoped via IAM policy with conditions/boundaries), separate roles per environment, no admin keys, OIDC federation, SCPs, and review roles with Access Analyzer / generate policies from CloudTrail.

Troubleshooting

Q358. Enable Terraform debug logs; investigate and validate Terraform errors

Asked at (3): Koerber Pharma, Rapidsoft, Sigmoid

`export TF_LOG=DEBUG; export TF_LOG_PATH=tf.log` (levels TRACE, DEBUG, INFO, WARN, ERROR). Validation flow: `terraform fmt -check`, `terraform validate` (syntax/schema), `tflint`, `terraform plan`, read the error line, check provider versions, credentials, quotas and dependencies. Typical real errors: state lock, resource already exists (import), provider auth, cycle, quota/IAM denied.

Q359. A resource created by Terraform failed during provisioning - what happens?

Asked at (1): Sigmoid

Terraform records the partially created resource as tainted if a provisioner fails (it will be destroyed and recreated next apply). If the create call fails, nothing is recorded in state for that resource; other independent resources continue unless they depend on it. Fix the error and re-run apply.

Q360. terraform apply is creating all resources again - possible problems?

Asked at (1): UST

State is missing/empty or pointing at the wrong backend/workspace/key, backend not initialised, resources renamed in code (address changed), a different provider region/account, or `count/for_each` keys changed. Check `terraform state list`, backend config, workspace, and use `moved` blocks or `state mv`.

Best Practices

Q361. Have you used Terraform Cloud / HCP? What does it provide?

Asked at (1): Sonata Software

Managed service for remote state, locking, remote runs, VCS-driven workflows, private module registry, team access control, Sentinel/OPA policy, cost estimation and dynamic credentials. Alternatives: Atlantis, Spacelift, env0, GitHub Actions + S3 backend.

Q362. Terraform best practices

Asked at (1): F5

Remote state with locking and encryption; small focused modules; version-pin providers and modules; separate environments; no secrets in code; code review + CI (fmt, validate, tflint, checkov, plan); least-privilege credentials via OIDC; consistent naming and tags; avoid manual changes and drift; `prevent_destroy` on critical resources; document with README/terraform-docs; test with terratest/`terraform test`.

Ansible & Config Management

12 questions

Basics & Playbooks

Q363. What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config management tools)

Asked at (12): Accion Labs, Alphadyne, CTS, Deloitte, IBM, Infosys, LTIMindtree, Nextturn, Nisum Technologies +3 more

Ansible is an open-source, agentless configuration management and automation tool. The control node connects to managed nodes over SSH (WinRM for Windows), pushes small Python modules, runs them and removes them. Playbooks are written in YAML and are idempotent: running again changes nothing unless the system differs from the desired state.

Typical uses: install and configure packages (Nginx, Java), patch servers, deploy apps, create cloud resources, harden OS.

Jenkins is a CI/CD orchestrator; Ansible is the configuration/deployment tool Jenkins often calls.

Alternatives: Puppet, Chef, Salt (agent-based).

Interview tip: always give a real example, e.g. "I patched 40 servers in rolling batches using a playbook with serial: 10."

Q364. What is a playbook? Write one to install/configure Apache/Nginx and ensure the service starts on boot

Asked at (9): Alphadyne, Capgemini, Cisco, IBM, L and T, Others (company not named), Rapidsoft, TCS, ZopSmart

A playbook is a YAML file mapping hosts to a list of tasks. Example:

```
- hosts: webservers
  become: yes
  tasks:
    - name: Install nginx
      apt: { name: nginx, state: present, update_cache: yes }
    - name: Start and enable
      service: { name: nginx, state: started, enabled: yes }
```

Run: `ansible-playbook -i inventory site.yml` (options: `--check` dry run, `--diff`, `-l host`, `-t tag`, `-e var=val`, `-v`). `become: yes` runs as root (sudo). An ad-hoc command without a playbook: `ansible all -i inventory -m ping` or `ansible web -m shell -a "uptime"`. Pass parameters with `vars`, `vars_files`, or `-e`. Gather client requirements (OS, servers, ports, secrets, rollback) before writing.

Q365. Deploy to 100 servers / patch 20+ servers with Ansible; install Ansible; Ansible Tower/AWX

Asked at (6): CTS, Capgemini, IBM, Others (company not named), TCS, Wipro

Put hosts in inventory, increase `forks`, and roll out in batches with `serial: 10` and `max_fail_percentage` so a bad change doesn't hit everything; use handlers to restart services. Install: `apt install ansible` or `pip install ansible` (Ubuntu/RedHat via `dnf/yum + EPEL`). Ansible Tower (AWX is the open-source upstream) adds a web UI, RBAC, credential storage, job scheduling, logs, REST API and workflows around playbooks.

Q366. Import vs include; block/rescue; passing output between tasks; templates

Asked at (5): Capgemini, Others (company not named), Sigmoid, Syncortex, TCS

- `import_*` is static (processed when the playbook is parsed, tags and conditions apply to everything); `include_*` is dynamic (evaluated at runtime, good for loops/conditions).
- `block` groups tasks with shared conditions; `rescue` handles errors; `always` always runs - like try/catch/finally.
- Pass output: `register: result` then use `result.stdout` in later tasks, or `set_fact` to share across blocks.
- Templates: Jinja2 `.j2` files rendered with variables using the `template` module (e.g. `nginx.conf` per environment).

Q367. Which Ansible modules have you used? (install packages, nginx, commands)

Asked at (5): Alphadyne, IBM, Infosys, Others (company not named), TCS

Common modules: `package/apt/yum/dnf` (install a specific package: `apt: name=nginx state=present`), `service/systemd`, `copy`, `template`, `file`, `lineinfile`, `user/group`, `git`, `command/shell`, `cron`, `uri`, `docker_container`, `debug`, `ping`, `setup` (facts), `unarchive`, `get_url`. "Command not found" for yum/apt usually means the module is run on a host with a different package manager (use `package` or `when: ansible_os_family`), or Python/the module isn't present on the target.

Q368. Ansible ad-hoc tasks: create 3 users mapped to groups, run only on DB servers excluding one, shut down all servers

Asked at (1): Capgemini

- Users and groups in one task with a loop:

```
- user: { name: "{{ item.name }}", groups: "{{ item.group }}", append: yes }
  loop: [ {name: u1, group: prod}, {name: u2, group: task}, {name: u3, group: qa} ]
```

(create groups first with the `group` module).

- Exclude a host: `ansible-playbook site.yml --limit 'dbservers:!db3'` (or `hosts: dbservers:!db3`).
- Ad-hoc shutdown: `ansible all -b -m shell -a "shutdown -h now"` (or `-m community.general.shutdown`). Use carefully.

Q369. Structure Ansible patch automation with pre-task checks, main tasks and post-task validation

Asked at (1): Wipro

Use a playbook with `pre_tasks` (check free disk, snapshot/backup, take node out of LB, verify service state), `tasks` (apply updates via `package/yum`, reboot if needed with `reboot` module), `post_tasks` (health check, put back in LB, verify versions) and handlers; run with `serial` for rolling batches and `block/rescue` to roll back on failure. Package as a role.

Q370. Write an Ansible playbook to install Docker on nodes

Asked at (1): ZopSmart

- ```
- hosts: nodes
 become: yes
 tasks:
 - apt: { name: [docker.io], state: present, update_cache: yes }
 - service: { name: docker, state: started, enabled: yes }
 - user: { name: ubuntu, groups: docker, append: yes }
```

Use the official `geerlingguy.docker` role for repository setup in production.

## Inventory, Roles & Vault

### Q371. What are Ansible roles? How do you create them, structure a three-tier app, publish to Galaxy, and manage shared plugins?

Asked at (6): Capgemini, L and T, Others (company not named), Synechron, TCS, Wipro

A role is a standard directory structure that packages tasks, handlers, templates, files, variables and defaults so they can be reused and shared. Create with `ansible-galaxy init myrole`. Layout: `tasks/`, `handlers/`, `templates/`, `defaults/`, `vars/`, `files/`, `meta/`.

For a three-tier app, create roles `web`, `app`, `db` and a site playbook that applies each to its group - clean, reusable, testable. Ansible Galaxy is the public hub for roles/collections: `ansible-galaxy init`, add metadata, push to GitHub, import into Galaxy; consumers use `ansible-galaxy install`. Shared custom plugins go into a common role or an Ansible collection that other roles depend on (declared in `meta/` or `requirements.yml`).

### Q372. What is an inventory file? How do you keep large inventories clean and store variables for different environments?

Asked at (6): Capgemini, IBM, L and T, Others (company not named), Syncortex, Wipro

Inventory lists the hosts Ansible manages, grouped (webservers, dbservers), static INI/YAML or dynamic (AWS EC2 plugin). Best practices: meaningful group names by role/environment/location, consistent host naming, separate inventory per environment (`inventories/prod`, `inventories/dev`), use `group_vars/` and `host_vars/` for variables, use dynamic inventory for cloud. Variable precedence: extra-vars (-e) > host\_vars > group\_vars > role defaults. For two VMs with different needs, use host/group variables and `when:` conditions.

### Q373. What is Ansible Vault? How do you encrypt files/playbooks and store credentials?

Asked at (5): Capgemini, IBM, L and T, Others (company not named), TCS

Vault encrypts sensitive data (passwords, keys, whole files or single variables).

- `ansible-vault create secrets.yml, encrypt, edit, decrypt, encrypt_string.`
- Run with `--ask-vault-pass` or `--vault-password-file`.
- Don't store the vault password in Git; take it from a CI secret.

For production consider HashiCorp Vault/AWS Secrets Manager lookups. For an encrypted MySQL dump: decrypt variables with Vault, use `community.mysql.mysql_db` with `state: dump`, then store the dump securely.

## Troubleshooting

### Q374. How do you debug a playbook? What if it times out on one host or runs for hours?

Asked at (4): CTS, IBM, Others (company not named), Syncortex

- Verbose: `-v` to `-vvvv`; `debug` module; `--check --diff; --step; --start-at-task; ANSIBLE_LOG_PATH` to see the log.
- One host times out: test SSH connectivity (`ansible host -m ping`), DNS, firewall, resource usage, slow tasks, set `timeout`, `async/poll`.
- Long running: find the stuck task with `-vvv`, use `async` for long tasks, increase `forks`, use `strategy: free, serial, gather_facts: no`, pipelining, mitogen/fact caching, and limit hosts.

## VPC & Networking

### Q375. NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need?

Asked at (12): Amazon, Arrise Solutions, CTS, Capgemini, Emphasis, HCL, Koerber Pharma, Others (company not named), Persistent Systems +3 more

- Internet Gateway: two-way door between VPC and internet for public subnets.
- NAT Gateway: managed service in a public subnet letting private instances go OUT to the internet, no inbound connections initiated from outside.
- NAT instance: self-managed EC2, cheaper but you handle scaling, HA and patching.

A NAT Gateway is AZ-scoped, so for HA use one per AZ. For 2 public + 2 private subnets: minimum 1 (cheap, single point of failure), recommended 2 (one per AZ). It cannot be placed in a private subnet to reach the internet.

### Q376. How is Route 53 configured and what routing policies exist?

Asked at (9): Akamai, Deloitte, Morgan Stanley, Orion Innovation, Others (company not named), Sapient, Sigmoid, Wikreata Media, Wipro

Create a hosted zone, point the domain's registrar to its name servers, then add records (A/AAAA/CNAME/Alias to ALB/CloudFront). Routing policies: simple, weighted (canary), latency, failover (with health checks), geolocation, geoproximity, multi-value. Prefer Alias over CNAME for the zone apex.

### Q377. Security Groups vs NACLs (and subnet vs NACL)?

Asked at (9): Aspire, CTS, EPAM, Emphasis, F5, Infosys, Others (company not named), Qentelli Solutions, SquareOps

- Security Group: attached to the instance/ENI, stateful (return traffic automatically allowed), allow rules only.
- NACL: attached to the subnet, stateless (must allow both directions), allow and deny rules, evaluated in numbered order.

Use SGs as the main control; use NACLs for coarse subnet-level blocking, e.g. block a malicious IP range. A subnet is a network segment; a NACL is the firewall around it.

### Q378. Describe your VPC / network architecture: subnets per tier, HA, sizing IP addresses.

Asked at (8): Accion Labs, Alphadyne, CTS, Deloitte, Others (company not named), Sapient, Sigmoid, SquareOps

Typical design: one VPC across 2-3 AZs, each AZ with a public subnet (ALB, NAT), private app subnets and isolated DB subnets.

- Route tables: public -> IGW; private -> NAT; DB -> no internet.
- Security groups chained: ALB -> app -> DB.
- Sizing: plan CIDR with growth in mind (e.g. /16 VPC, /20-/24 subnets); remember AWS reserves 5 IPs per subnet, and EKS/pods can consume many IPs.
- Add VPC endpoints for S3/ECR to avoid NAT cost.

For exposing an EC2 app to the internet you need VPC, public subnet, IGW, route table, security group, public/Elastic IP (and ideally ALB, Route53).

### Q379. An EC2 in a private subnet must receive inbound traffic (without NAT). How, and how does traffic flow from internet to private subnet?

Asked at (7): Accenture, Alphadyne, CTS, EPAM, Perfios, Sapient, Sigmoid

Put a load balancer (ALB/NLB) in the public subnets; it forwards to targets in private subnets. Security group of the instances allows traffic only from the LB's SG. Flow: Route53 -> IGW -> public ALB -> private target. Other options: SSM/VPN/Direct Connect for admin access, PrivateLink for service exposure.

### **Q380. VPC Peering: how does it work and what options connect VPC A and B (two-way or one-way)?**

Asked at (7): Deloitte, Intact Green Services, Nitor Infotech, Orion Innovation, Others (company not named), Zensar, ZopSmart  
Peering is a private one-to-one link between two VPCs (same or different region/account). Requirements: non-overlapping CIDRs, accepted request, route table entries on both sides, and security groups allowing the traffic. It is non-transitive.

Options: peering, Transit Gateway (many VPCs), PrivateLink/VPC endpoint service (expose only one service - ideal for "only A talks to B"), VPN. For one-way access use PrivateLink, or restrict with routes/SGs/NACLs, since routes are bidirectional but access can be limited to initiating from A.

### **Q381. How do you connect AWS to on-prem or to a client's environment (Direct Connect, VPN)?**

Asked at (6): Arrise Solutions, EPAM, Elixir Labs, ITC Infotech, Koerber Pharma, Others (company not named)

- Site-to-Site VPN: encrypted IPsec tunnels over the internet; quick and cheap.
- Direct Connect: dedicated private fibre; consistent latency/bandwidth, higher cost and lead time; often paired with VPN as backup.
- Use a Virtual Private Gateway or Transit Gateway; handle routing with BGP and avoid overlapping CIDRs.

For a third-party/client: VPN, peering, or PrivateLink to expose just one service.

### **Q382. Transit Gateway vs VPC peering; have you used Transit Gateway?**

Asked at (6): Deloitte, EPAM, Elixir Labs, F5, HCL, Others (company not named)

Peering is a one-to-one, non-transitive link. Transit Gateway is a regional hub: attach many VPCs/VPNs/Direct Connect once and route between them centrally (transitive). Attachments don't need a CIDR range; you control reachability with TGW route tables and each VPC's own route tables. CIDRs must not overlap.

### **Q383. An EC2 instance is unreachable (not a security group problem) / an app behind a load balancer is suddenly unavailable - how do you troubleshoot?**

Asked at (5): CTS, Deloitte, Others (company not named), UST, Wipro

Work from outside in:

1. DNS resolves correctly?
2. LB listener and target group health?
3. Security groups and NACL on both directions.
4. Route tables / IGW / NAT.
5. Instance status checks (system vs instance), CPU/memory, disk full.
6. App process and port listening (`ss -tulpn`), app logs.
7. OS firewall (iptables).
8. VPC Flow Logs to see rejected traffic, and use SSM or console output for access. Recent deployments/changes (CloudTrail) are the usual culprit.

### **Q384. How do you access an RDS in a private subnet securely?**

Asked at (5): Amazon, Aspire, Emphasis, OPT IT, RelevantZ

Connect through a bastion/jump host, SSM Session Manager port forwarding, VPN or Direct Connect. Security group allows 3306/5432 only from specific sources (app SG, bastion SG). Enforce SSL, IAM auth and per-user DB accounts, and never make the DB publicly accessible.

### **Q385. Purpose of VPC Endpoints and when to use them?**

Asked at (5): Deloitte, EPAM, HCL, Others (company not named), TCS

Endpoints let you reach AWS services privately without internet/NAT. Gateway endpoints (S3, DynamoDB) are free and use route table entries; Interface endpoints (PrivateLink) use ENIs for most other services. Benefits: security (traffic stays on AWS network), lower NAT cost. Example: private EC2 pulling from S3. Restrict the bucket policy to the endpoint (`aws:SourceVpce`).

### **Q386. How does SSL/TLS work with Certbot, Let's Encrypt and ACM? How is a certificate used with ALB/NLB, routing and security groups?**

Asked at (4): Alphadyne, Arrise Solutions, Others (company not named), Synechron

Flow: the client connects, the server presents a certificate signed by a trusted CA, the client verifies it, they agree session keys (TLS handshake), then traffic is encrypted. Let's Encrypt/Certbot proves domain ownership (HTTP-01/DNS-01) and issues free 90-day certs that must auto-renew. On AWS use ACM: free public certs that auto-renew and attach to ALB/CloudFront (cannot be exported). ALB terminates TLS on 443 and forwards to targets; security group must allow 443 from the internet.

### **Q387. Public vs private subnet - how can you tell, and what must be in place?**

Asked at (4): Arrise Solutions, Persistent Systems, Sigmoid, Wipro

A subnet is public if its route table has a route 0.0.0.0/0 to an Internet Gateway (and instances have public/Elastic IPs). It is private if there is no route to an IGW; outbound internet goes (if needed) via a NAT Gateway placed in a public subnet.

Check: look at the subnet's route table, not at its name.

### **Q388. How would you migrate an app running on a public EC2 to a private subnet following best practices?**

Asked at (3): Alphadyne, JPMorgan, Persistent Systems

1. Create private subnets in each AZ with routes to NAT (for outbound).
2. Place an ALB (public) in front, with an ACM certificate.
3. Launch the app via launch template/ASG in private subnets (or move the instance with an AMI).
4. Lock security groups so only the ALB can reach the app.
5. Replace SSH with SSM, shift DNS to the ALB, test, then remove the public IP.

Use blue/green to avoid downtime.

### **Q389. How many subnets can a VPC have, and can you extend a subnet's CIDR later?**

Asked at (2): Capgemini, Plansource ValueLabs

Default quota is 200 subnets per VPC (adjustable). A subnet's CIDR can NOT be changed after creation; add a new subnet (and new secondary VPC CIDR if needed) and migrate. So plan IP space ahead.

### **Q390. After connecting VPCs via Transit Gateway, how do you block A->B and B->C traffic?**

Asked at (1): EPAM

Use separate TGW route tables: associate each attachment with its own route table and only propagate/add routes for the destinations that should be reachable (or add blackhole routes for blocked CIDRs). Also tighten security groups/NACLs. Route-table based segmentation is the TGW way.

### **Q391. Can you create two CIDR blocks (172.x and 192.x) in the same VPC?**

Asked at (1): EPAM

Yes. A VPC supports secondary CIDR blocks (up to 5 by default, adjustable) as long as they don't overlap and fall within allowed ranges. Useful when you run out of IPs. Add subnets from the new block and update routes.

### **Q392. Does S3 need a VPC?**

Asked at (1): Accenture

No. S3 is a regional, public AWS service accessed via its endpoint. To keep traffic off the internet from a VPC, use a Gateway VPC Endpoint (free) and restrict the bucket policy to that endpoint.

### **Q393. EC2 should not talk to the internet but must talk to other instances. How?**

Asked at (1): Persistent Systems

Place in a private subnet with no route to IGW/NAT, and allow internal traffic using security groups referencing each other. For AWS services use VPC endpoints so nothing leaves the VPC.

### Q394. Elastic IP vs public IP?

Asked at (1): Belcan

A public IP is auto-assigned and released when the instance stops/terminates (changes on restart). An Elastic IP is a static public IPv4 address you own until you release it and can remap between instances. Unused Elastic IPs cost money.

### Q395. Major difference between AWS and GCP VPCs

Asked at (1): Qburst

AWS VPC is regional and subnets are AZ-scoped. GCP VPC is global; subnets are regional and span all zones, routes and firewall rules are applied at VPC level with tags/service accounts. Firewall in GCP is VPC-wide rules (no NACL); peering in GCP is simpler across regions.

### Q396. Private subnet changed to public by a developer - what do you do?

Asked at (1): Qentelli Solutions

Detect it (AWS Config rule / CloudTrail alert on route changes), then revert the route (remove IGW route), check for exposed resources, review who changed it. Prevent via IaC-only changes, restricted IAM (no ec2:CreateRoute for devs), SCP and Config auto-remediation.

### Q397. Someone from a partner without an AWS account must see a dashboard. How?

Asked at (1): Persistent Systems

Don't create IAM users. Options: Grafana/QuickSight dashboards with SSO or a viewer role/share link; CloudWatch dashboard sharing via SSO; or export a read-only dashboard behind an authenticating proxy. Apply read-only roles and time-limited access.

### Q398. Two VPCs have overlapping CIDRs and Transit Gateway isn't allowed. What do you do?

Asked at (1): EXL Service

Overlapping ranges can't be routed or peered. Use AWS PrivateLink: expose the target service behind an NLB as an endpoint service, and the other VPC creates an interface endpoint - no CIDR routing needed. Alternatives: NAT the overlapping range, add a secondary non-overlapping CIDR and re-IP, or longer-term renumber one VPC.

### Q399. Two apps in the same VPC, each behind a public ALB: how does App A reach App B?

Asked at (1): SquareOps

With a public ALB, A's request goes out through NAT/IGW to the internet, DNS resolves B's public ALB, and it re-enters. That's slow, costs NAT charges and exposes B. Better: use an internal ALB (or service discovery / Cloud Map) so traffic stays inside the VPC.

### Q400. What default components are created with a VPC?

Asked at (1): TCS

A main route table (local route), a default network ACL, a default security group, and DHCP options set. No subnets or internet gateway unless you use the default VPC/wizard.

## IAM & Security

### Q401. How do you enforce least-privilege access and restrict a user to only specific services (e.g. only EC2 and RDS)? What security best practices do you apply on AWS?

Asked at (15): Amazon, CGI, Capgemini, Cisco, EPAM, F5, Infosys, JPMorgan, Koerber Pharma +6 more

Least privilege = give only the actions on only the resources actually needed.

- Write a policy that allows only ec2:\* and rds:\* (ideally narrower) and attach it to a group/role; everything else is denied by default.
- Add permission boundaries or SCPs (Organizations) as a hard ceiling.

- Use conditions (MFA, source IP, region, tags).
- Use IAM Access Analyzer / last-accessed data to trim unused permissions.
- Best practices: MFA, no root usage, roles over keys, SSO, CloudTrail on, rotate and review regularly.

#### **Q402. How do you manage secrets in your project? (Secrets Manager, Parameter Store, Vault, encryption and protection)**

Asked at (9): Amazon, Capgemini, HCL, IBM, LTIMindtree, Marsh McLennan, Others (company not named), Sonata Software, SquareOps

Never put secrets in code, images or Git. Store them in a secrets service and fetch at runtime.

- AWS Secrets Manager: encrypted with KMS, automatic rotation, fine-grained IAM, audit through CloudTrail.
- SSM Parameter Store: cheaper, good for config and simple secrets.
- HashiCorp Vault: multi-cloud, dynamic short-lived credentials.
- In Kubernetes: External Secrets Operator / CSI driver, or Sealed Secrets, rather than plain base64 Secrets.
- Protect them with least-privilege IAM, KMS keys, rotation and access logging.

#### **Q403. How can you log in to an EC2 instance without a password, or if you lost the .pem key?**

Asked at (6): Amazon, CTS, Capgemini, Five9, Persistent Systems, Verizon

Normal login is SSH key pairs, not passwords. Alternatives: SSM Session Manager (no SSH port, IAM-controlled, logged) or EC2 Instance Connect (temporary key pushed).

Lost key:

1. Stop the instance, detach the root EBS volume.
2. Attach it to a helper instance, add a new public key to `~/.ssh/authorized_keys`, reattach.
3. Or, if SSM agent is running, just connect via Session Manager, or use user-data to inject a new key on restart.

Lesson: use SSM so keys do not matter.

#### **Q404. Types of IAM policies; explicit deny vs allow; why a user with a role/policy still cannot access S3; "admin without S3 access"**

Asked at (6): Amazon, HCL, Nitor Infotech, Others (company not named), Sonata Software, SquareOps

Policy types: identity-based (managed/inline), resource-based (bucket policy), permission boundaries, SCPs, session policies, ACLs.

Evaluation: default deny -> any explicit Deny always wins -> otherwise need an Allow.

If access fails check: explicit deny anywhere, bucket policy denying, SCP, permission boundary, KMS key policy for encrypted objects, wrong region/ARN, VPC endpoint policy, and Block Public Access. An "admin" can be blocked by a permission boundary or SCP, which is the answer interviewers expect.

#### **Q405. An S3 bucket was made public by mistake. How do you secure and audit it? How do you secure a bucket with sensitive client data?**

Asked at (4): IBM, Infosys, Others (company not named), Plansource ValueLabs

Immediate: turn on S3 Block Public Access (account and bucket), remove public ACLs/policies.

Audit: check CloudTrail data events and S3 server access logs to see who accessed what; use Access Analyzer and AWS Config.

Prevent: default SSE-KMS encryption, bucket policy denying non-TLS, versioning + MFA delete, VPC endpoints, least-privilege policies, Macie for sensitive data discovery, alarms on policy changes.

#### Q406. What is the difference between IAM Users and IAM Roles?

Asked at (4): Koerber Pharma, Others (company not named), SquareOps, ZopSmart

An IAM User is a permanent identity for one person or app, with long-lived credentials (password, access keys). An IAM Role has no credentials of its own; anyone or anything allowed to "assume" it gets temporary credentials from STS that expire automatically.

- Use Users only for humans who truly need them (better: SSO/Identity Center).
- Use Roles for EC2, Lambda, EKS pods, cross-account access and CI/CD.

*Interview line: "Users are for who you are, roles are for what you temporarily act as - I prefer roles because there are no secrets to leak or rotate."*

#### Q407. How do you protect and encrypt data on EC2/EBS?

Asked at (3): IBM, Others (company not named), Sapient

- Encrypt EBS volumes with KMS (can be enforced account-wide by default). Snapshots inherit encryption.
- Encrypt in transit with TLS.
- Restrict access: security groups, IAM roles, no public IPs.
- Back up with snapshots/AWS Backup; patch the OS; use IMDSv2.

To encrypt an existing unencrypted volume: snapshot -> copy snapshot with encryption -> create new volume -> swap.

#### Q408. How do you scan EC2 instances for vulnerabilities, and fix a vulnerable or compromised instance?

Asked at (3): BMW TechWorks, CGI, Others (company not named)

Scan with Amazon Inspector (agent-based, continuous), Trivy/Qualys/Nessus, and check OS patch compliance via SSM. Fix: patch via new AMI or Patch Manager, redeploy.

If compromised: isolate (swap to a deny-all security group, revoke the role's sessions), snapshot disks and memory for forensics, terminate and rebuild from a clean image, rotate credentials, review CloudTrail, find the entry point.

#### Q409. You have 10 AWS accounts and cannot use access keys. How do you log in securely?

Asked at (3): Accenture, Koerber Pharma, Others (company not named)

Use federation instead of keys.

- AWS IAM Identity Center (SSO) tied to your IdP (Okta/Azure AD); users sign in once and pick an account/role; CLI uses `aws sso login` and gets temporary credentials.
- Or one central identity account and cross-account roles with `sts assume-role`.
- Add MFA and Organizations for central governance.

Benefit: no long-lived keys, centrally revoked, fully audited.

#### Q410. A cross-account scenario: EC2 in Account A needs to read tokens (secrets) in Account B

Asked at (2): Capgemini, Others (company not named)

Create a role in Account B with a trust policy allowing Account A's EC2 role, and permissions to the secret. EC2's role calls `sts:AssumeRole` and uses the temporary credentials. Alternatively add a resource-based policy on the secret (and KMS key) granting Account A's role directly.

#### Q411. An EC2 instance has an IAM role - what can we do with it and how does it work?

Asked at (2): EPAM, HCL

The role is attached through an instance profile. The instance gets temporary credentials from the metadata service (169.254.169.254) and the AWS SDK/CLI uses them automatically, so no keys are stored on the server.

Use cases: reading S3, writing CloudWatch logs, SSM access, pulling from ECR. Tip: enforce IMDSv2 to prevent SSRF stealing credentials.

#### **Q412. Get the count of running EC2 instances and attached EBS volumes across 50-60 AWS accounts without logging in to each**

Asked at (2): Deloitte, Others (company not named)

Use AWS Organizations plus a cross-account role (e.g. OrganizationAccountAccessRole) and a script/Lambda that loops accounts: `sts assume-role`, then `describe-instances/describe-volumes` in each region. Or use AWS Resource Explorer / Config Aggregator / Systems Manager Explorer for a central multi-account inventory.

#### **Q413. How do you protect an API Gateway?**

Asked at (2): Cisco, Sonata Software

Layered controls: authentication (IAM, Cognito or Lambda authorizer), throttling and usage plans/API keys, AWS WAF for SQLi/XSS/bots, TLS with custom domain, resource policies (IP/VPC restrictions), request validation, and CloudWatch logging/alerts.

#### **Q414. How do you secure multiple microservices and websites?**

Asked at (2): Encora, JPMorgan

Defence in depth: WAF and TLS at the edge, authentication/authorization (OAuth/JWT at API gateway), network segmentation with security groups/NetworkPolicies, service mesh mTLS between services, secrets in a vault, least-privilege roles per service, image scanning, and centralised logging/alerting.

#### **Q415. Managed policy vs inline policy - what is the difference? How do you find and remove inline policies that contain "\*"?**

Asked at (2): Others (company not named), Sonata Software

A managed policy is a standalone, reusable policy (AWS-managed or customer-managed) that can be attached to many identities and is versioned. An inline policy is embedded in exactly one user/role/group and is deleted with it.

- Prefer customer-managed policies: reusable, auditable, versioned.
- Inline is for a strict one-to-one relationship.

To remove wildcards: loop over users/roles with boto3 or CLI (`list-users -> list-user-policies -> get-user-policy`), check whether Action/Resource equals "\*", then `delete-user-policy`. Always report first, then delete after approval.

#### **Q416. Secrets Manager vs Parameter Store - what is the difference?**

Asked at (2): Others (company not named), Verizon

Both store values encrypted with KMS.

- Secrets Manager: built for secrets, native automatic rotation (RDS etc.), cross-region replication, costs per secret per month.
- Parameter Store: general configuration plus secrets (SecureString), no native rotation, standard tier is free.

Rule of thumb: passwords/API keys that must rotate -> Secrets Manager; app config and non-rotating values -> Parameter Store.

#### **Q417. What best practices and security parameters do you follow when creating EC2 / RDS / MongoDB for production?**

Asked at (2): OPT IT, Others (company not named)

- Network: private subnet, no public IP, minimal security group rules, no 0.0.0.0/0 on SSH/DB ports.
- Access: IAM role instead of keys, SSM Session Manager instead of SSH, IMDSv2.
- Data: encrypted EBS/RDS with KMS, backups and snapshots, Multi-AZ for RDS.
- Operations: right-sized instances, tagging, monitoring and alarms, patching, golden AMIs, everything defined as IaC.

### Q418. What is CloudTrail and what do you use it for?

Asked at (2): Koerber Pharma, Persistent Systems

CloudTrail records every API call in your account (who, what, when, from where). Used for security investigations, compliance and change tracking. Send logs to S3 (with log-file validation), optionally to CloudWatch for alerts. Remember: management events are on by default, data events (S3 objects) must be enabled.

### Q419. What is an OIDC provider in AWS and how do you federate (Azure AD/GitHub) to AWS?

Asked at (2): EPAM, ITC Infotech

An IAM OIDC identity provider lets external systems authenticate with a signed token (JWT) and exchange it via STS `AssumeRoleWithWebIdentity` for temporary AWS credentials - no stored keys. Examples: GitHub Actions deploying to AWS, EKS service accounts (IRSA). For human users, federate your IdP (Azure AD/Okta) with IAM Identity Center via SAML/SCIM and map groups to roles.

### Q420. AWS Config vs CloudTrail

Asked at (1): Deloitte

CloudTrail records WHO did WHAT (API calls) and when. Config records the STATE of resources over time and checks compliance rules (what changed, and is it compliant). Together: Config says the bucket became public; CloudTrail says who did it.

### Q421. Bucket policy vs ACL?

Asked at (1): Others (company not named)

Bucket policy is a JSON resource policy that can grant or deny to accounts, roles, IPs, VPC endpoints with conditions - the recommended way. ACLs are older, coarse (read/write per grantee) and AWS recommends disabling them (Object Ownership: bucket owner enforced).

### Q422. Does a security group + WAF + DDoS protection stop bot attacks?

Asked at (1): F5

Security groups only filter ports/IPs. Shield protects against network/transport DDoS. Bots need WAF with rate-based rules, Bot Control managed rule groups and CAPTCHA. So you need all layers: Shield + WAF + SGs (+ CloudFront).

### Q423. How do you give a developer read-only access to RDS?

Asked at (1): Others (company not named)

Create an IAM policy allowing only describe/list actions on RDS (`rds:Describe*`, `rds:ListTagsForResource`), attach to the developer's group/role. For data access, create a read-only database user (`GRANT SELECT`) and connect through a read replica, preferably with IAM database authentication; keep the DB in a private subnet reachable through VPN/bastion/SSM.

### Q424. How do you handle certificate management and expiry in AWS?

Asked at (1): Koerber Pharma

Use ACM public certificates with DNS validation: they auto-renew as long as the validation CNAME stays. For imported certs ACM does not renew, so set EventBridge/CloudWatch alarms on `DaysToExpiry` and notify via SNS; for Let's Encrypt automate renewal with cert-manager/certbot timers. Monitor with a synthetic check.

### Q425. How do you prevent unauthorized users from spinning up expensive services (e.g. ML) in AWS?

Asked at (1): Moodys

Use Service Control Policies in AWS Organizations to deny those services (e.g. `sagemaker:*`) in all but approved accounts, plus IAM permission boundaries. Add budgets/alerts, restrict regions via SCP, and alarm on CloudTrail events for the service being created.

## Q426. How do you share an encrypted (KMS) AMI from Account 1 to Account 2?

Asked at (1): HCL

An AMI encrypted with the default AWS-managed key cannot be shared, so you need a customer-managed KMS key.

1. Re-copy the AMI encrypting it with a customer-managed key.
2. Edit the key policy to allow Account 2 (kms:Decrypt, DescribeKey, CreateGrant, etc.).
3. Share the AMI (and its snapshots) with Account 2's ID.
4. In Account 2, copy the AMI re-encrypting with their own key.

## Q427. IAM: group vs individual policies, onboarding new team members, and how do you secure an AWS account as an admin?

Asked at (1): HCL

Always attach policies to groups/roles (e.g. Developers, Admins) and add users to groups - far easier to audit and change than per-user policies. Onboarding: create the user in IAM Identity Center (or IAM) -> add to a group -> require MFA -> they log in via the SSO portal/console URL and assume a role. Admin hardening: lock the root user with MFA and no keys, use CloudTrail, GuardDuty, Config, Security Hub, SCPs, and billing alerts.

## Q428. List IAM users whose access keys have not been used in over 90 days

Asked at (1): Others (company not named)

```
import boto3, datetime as dt
iam = boto3.client("iam")
limit = dt.datetime.now(dt.timezone.utc) - dt.timedelta(days=90)
for u in iam.list_users()["Users"]:
 for k in iam.list_access_keys(UserName=u["UserName"])["AccessKeyMetadata"]:
 last =
 iam.get_access_key_last_used(AccessKeyId=k["AccessKeyId"])["AccessKeyLastUsed"].get("LastUse
dDate")
 if not last or last < limit:
 print(u["UserName"], k["AccessKeyId"])
```

Alternative: the IAM credential report. Then deactivate or rotate the keys.

## Q429. Recent AWS/Azure outages - what were your key takeaways?

Asked at (1): JPMorgan

Answer with principles: don't rely on one region or one control plane, design for multi-AZ and (for critical systems) multi-region failover, test DR regularly, avoid hidden single points of failure (DNS, auth, a single provider's console), set sensible timeouts/retries with back-off, and have a communication plan and runbooks. Show you learned from a real postmortem.

## Q430. Retrieve secrets from AWS Secrets Manager using Python (boto3)

Asked at (1): Sonata Software

```
import boto3, json
client = boto3.client("secretsmanager", region_name="ap-south-1")
resp = client.get_secret_value(SecretId="prod/db")
secret = json.loads(resp["SecretString"])
print(secret["username"])
```

The caller needs secretsmanager:GetSecretValue (and kms:Decrypt). Never print real secrets; cache them to avoid repeated calls.

## Q431. Secure a Lambda function

Asked at (1): LTIMindtree

Least-privilege execution role per function, secrets from Secrets Manager (not env vars in plain), put in VPC only if needed, validate inputs, keep dependencies patched/scanned, reserved concurrency to limit abuse, resource policy to restrict who can invoke, enable logging and X-Ray.

# Architecture & Migration

### Q432. Design a highly available, fault-tolerant backend on AWS

Asked at (16): Altimetrik, Amazon, Arrise Solutions, CMT, CTS, EXL Service, Emphasis, IBM, LTIMindtree + 7 more

Route53 -> CloudFront/WAF -> ALB across 3 AZs -> ASG/EKS/ECS in private subnets -> Multi-AZ RDS/Aurora + ElastiCache -> S3. Add health checks, auto healing, stateless services, backups, multi-region DR (pilot light/warm standby) where RTO/RPO require it, IaC and monitoring. Remove single points of failure at every layer.

### Q433. AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets)

Asked at (14): CTS, Deloitte, IBM, Koerber Pharma, Moodys, Others (company not named), Plansource ValueLabs, Qburst, Qentelli Solutions + 5 more

Find: Cost Explorer grouped by service/tag/account, check Budgets/Anomaly Detection, look at recent changes in CloudTrail.

Reduce: right-size instances (Compute Optimizer), Savings Plans/Reserved/Spot, shut down idle or non-prod at night, delete unattached EBS/old snapshots/unused Elastic IPs, S3 lifecycle to cheaper classes (IA/Glacier) and Intelligent-Tiering, VPC endpoints to cut NAT data charges, enforce tagging. For GCP buckets: lifecycle rules, Nearline/Coldline/Archive classes.

### Q434. Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A

Asked at (13): Amazon, CTS, Flentis, HCL, IBM, Intact Green Services, Koerber Pharma, NUOS INFO Systems, Others (company not named) + 4 more

Drift: run `terraform plan` to detect it, then either `terraform apply` to revert the resource to code or update the code to match (`terraform apply -refresh-only`). Prevent drift using restricted IAM and AWS Config drift detection. To create instance B without touching A, add a new resource block with a distinct name (or use `-target` rarely); Terraform only adds it.

### Q435. How do you design for an entire region going down (and keep data safe even if multi-cloud also fails)? How to redirect traffic cost-effectively?

Asked at (6): IBM, Infinite Solutions, JPMorgan, Sigmoid, Sony, ZS Associates

Pick a DR strategy by RTO/RPO and budget: backup & restore (cheapest), pilot light (data replicated, compute off), warm standby, active-active. Replicate data cross-region (S3 CRR, RDS cross-region replicas/Aurora Global), keep infra as code to recreate quickly, use Route53 health-check failover. To save cost, keep only the DB replica running and bring up app tier on demand from AMIs/IaC. For worst case keep immutable offline/third-party backups and test restores regularly.

### Q436. Which AWS storage/instance types exist (EC2 types, EBS access mode, EFS vs EBS, multi-attach, Storage Gateway)?

Asked at (6): Amazon, EPAM, EXL Service, Koerber Pharma, SquareOps, ZopSmart

- EC2 families: general purpose (t/m), compute (c), memory (r), storage (i/d), GPU (p/g).
- EBS: block storage; normally attached to one instance in the same AZ (io1/io2 supports Multi-Attach for specific clustered apps). It cannot be shared like a file system.
- EFS: managed NFS shared by many instances/AZs - choose for shared or elastic file needs (also the answer for a monolith with a local file system).
- Storage Gateway: hybrid bridge exposing S3/EBS as NFS/SMB/iSCSI to on-prem.

### Q437. How do you back up AWS services (EC2, RDS, S3, EBS) and how does Aurora back up?

Asked at (5): Deloitte, Elixir Labs, Koerber Pharma, RelevantZ, Sonata Software

Use AWS Backup with policies and retention across services, EBS snapshots (incremental, copyable), RDS automated backups + manual snapshots, S3 versioning + replication. Aurora continuously backs up to S3 and supports point-in-time restore within the retention window. To move EBS data to another server: snapshot -> create volume in target AZ -> attach -> mount.

### Q438. How did you migrate an application from on-premises to AWS?

Asked at (4): AMEX, BMW TechWorks, HCL, Others (company not named)

Phases: assess (inventory, dependencies, 6 Rs: rehost/replatform/refactor...), plan landing zone (accounts, VPC, IAM), migrate data (DMS for DB, DataSync/Snowball for files), migrate servers (Application Migration Service or containerisation), test, cut over with DNS switch and rollback plan, then optimise. Keep on-prem sync during cutover to minimise downtime.

### Q439. Event-driven architecture with S3, Lambda and SNS; and Lambda failures/retries (including external API issues)

Asked at (3): CMT, IBM, Turning

S3 upload event -> Lambda processes -> publishes to SNS (fan-out to email/SQS/other Lambdas). For reliability: configure retries, a dead-letter queue (SQS/SNS) or on-failure destination, idempotent code, timeouts and exponential back-off. For a payment Lambda with flaky external API: add retry with back-off, circuit breaker, idempotency keys to avoid double charges, VPC/NAT/DNS checks and CloudWatch alarms. SQS slowness: check consumer count/visibility timeout, queue depth and age, scale consumers, use DLQ and batch size tuning.

### Q440. Spot vs Reserved instances

Asked at (3): EPAM, Koerber Pharma, SquareOps

Spot = spare capacity at up to ~90% discount, can be interrupted with a 2-minute warning; good for stateless/batch/CI workloads. Reserved Instances / Savings Plans = 1-3 year commitment for ~40-70% discount, for steady baseline. On-Demand for unpredictable loads. Mix them, e.g. RI for base, Spot for burst.

### Q441. How do you deploy an application to AWS, and how do you provision AWS services?

Asked at (2): IBM, JPMorgan

Provision infrastructure with Terraform/CloudFormation (VPC, ASG/EKS, RDS), build and test in a CI pipeline (Jenkins/GitHub Actions), push artifacts/images to S3/ECR, deploy via CodeDeploy, ArgoCD or rolling/blue-green strategies, with secrets in Secrets Manager and monitoring through CloudWatch/Prometheus.

### Q442. What is a Region and an Availability Zone?

Asked at (2): Capgemini, JPMorgan

A Region is a geographic area with multiple isolated data-centre groups. An Availability Zone is one or more discrete data centres within a region with independent power, cooling and networking, connected by low-latency links (a few ms) but far enough apart (km) to avoid shared failure. Deploy across at least two AZs for high availability and across regions for DR.

### Q443. What is a cloud-agnostic strategy? How do you use conditionals in Terraform/Ansible for AWS, Azure and GCP?

Asked at (2): Others (company not named), Wipro

Cloud-agnostic = avoid lock-in by using portable tools (Terraform, Kubernetes, containers, open standards) behind thin provider-specific modules. In Terraform use variables such as `cloud = "aws"` with `count = var.cloud == "aws" ? 1 : 0` to switch modules; in Ansible use `when: ansible_facts...` or provider variables. Trade-off: portability vs losing powerful native services, so keep it pragmatic.

### Q444. AWS vs Azure - which would you choose and why?

Asked at (1): IBM

Show balance: AWS has the widest service range and maturity; Azure integrates best with Microsoft/AD/.NET shops. Choose by existing skills, enterprise agreements, compliance, specific services and cost. Say: "I pick the platform that fits the workload; I'm comfortable on both because the concepts (VPC/VNet, IAM/Entra, etc.) map one-to-one."

## Q445. EC2 User Data

Asked at (1): EPAM

A script (or cloud-init config) that runs once at first boot as root, used to install packages, join a cluster or fetch config. Limit 16 KB, logs in /var/log/cloud-init-output.log. Don't hide secrets in it. For repeated boots use `#cloud-boothook` or other tools.

## Q446. What is AWS CDK and its commands?

Asked at (1): Turning

CDK defines infrastructure using real programming languages (TypeScript/Python) that synthesise to CloudFormation. Commands: `cdk init`, `cdk synth` (generate template), `cdk diff`, `cdk deploy`, `cdk destroy`, `cdk bootstrap` (prepare account).

## Load Balancing & Scaling

### Q447. ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB?

Asked at (12): Arrise Solutions, Belcan, CMT, CTS, Deloitte, EPAM, HCL, Others (company not named), Perfios +3 more

- ALB: Layer 7 (HTTP/HTTPS), path/host routing, WebSockets, WAF, auth; good for web apps and microservices.
- NLB: Layer 4 (TCP/UDP/TLS), ultra-low latency, millions of requests, static IP/Elastic IP, preserves source IP.
- Classic ELB: legacy, avoid.

Cross-zone load balancing spreads traffic evenly across AZs (default on for ALB, off for NLB; may add inter-AZ cost). Internal (private) ALB costs the same as an internet-facing one; you pay hourly plus LCUs. Choose ALB for HTTP routing, NLB for raw TCP/static IP.

### Q448. How does AWS Auto Scaling work? (ASG, with ALB, scaling on memory/disk, ECS autoscaling)

Asked at (8): Capgemini, EPAM, Flentas, JPMorgan, Others (company not named), Plansource ValueLabs, Sonata Software, SquareOps

An Auto Scaling Group keeps a min/desired/max number of instances, built from a launch template, spread over AZs, registered to an ALB target group. Policies: target tracking (e.g. keep CPU at 50%), step, scheduled, predictive. Memory and disk are not default metrics: install the CloudWatch agent, publish custom metrics, then scale on them. ECS uses Application Auto Scaling on service desired count (and capacity providers to scale the EC2 cluster).

### Q449. Key CloudWatch metrics to focus on; how to monitor LB health, CPU spikes and web/app server availability?

Asked at (7): CMT, EPAM, Koerber Pharma, Others (company not named), Qentelli Solutions, Sigmoid, SquareOps

- EC2: CPUUtilization, StatusCheckFailed, NetworkIn/Out; memory & disk need the agent.
- ALB: HealthyHostCount/UnHealthyHostCount, TargetResponseTime, HTTPCode\_Target\_5XX, RequestCount.
- RDS: CPU, FreeableMemory, connections, ReadLatency/WriteLatency.

Set alarms to SNS. Logs are organized into log groups (collection) and log streams (per source); use Logs Insights for queries. For availability track health check status, error rates and latency.

### Q450. CloudFront - how do you use it to serve a frontend?

Asked at (5): Arrise Solutions, EPAM, Others (company not named), Persistent Systems, Sigmoid

CloudFront is a CDN caching content at edge locations. For a static UI: host the build in a private S3 bucket, create a CloudFront distribution with Origin Access Control, attach ACM cert and Route53 alias, set caching and invalidate on deploy. Benefits: low latency, DDoS protection with Shield/WAF, lower origin load.

### Q451. CloudWatch: what is it and how do you create custom metrics?

Asked at (3): EPAM, Persistent Systems, SquareOps

CloudWatch collects metrics, logs and events, supports alarms and dashboards. Custom metric: send values with `aws cloudwatch put-metric-data` or the CloudWatch agent (memory, disk, app KPIs), then build alarms and notify via SNS.

### Q452. Failover mechanism in load balancers; active-passive vs active-active

Asked at (2): JPMorgan, Sigmoid

LBs run health checks and stop routing to unhealthy targets; they are themselves multi-AZ. Active-active: all nodes serve traffic simultaneously - better utilisation, instant capacity, but needs state sync. Active-passive: standby takes over only on failure - simpler and cheaper to reason about, but idle capacity and a failover delay.

### Q453. How do you filter a specific IP in CloudWatch logs, S3 logs, or VPC Flow Logs? How to keep flow logs readable?

Asked at (2): Deloitte, EPAM

- CloudWatch Logs Insights: `fields @timestamp, @message | filter @message like /10.0.1.5/`.
- S3 access logs / ALB logs: query with Athena (`WHERE remote_ip = '...`).
- VPC Flow Logs: send to S3 or CloudWatch, query with Athena; filter on `srcaddr/dstaddr` and `action=REJECT`. Use custom formats to only capture needed fields, to separate critical info.

### Q454. The team changed the AMI to an ASG launch template - how do you ensure the new version deploys properly?

Asked at (2): EPAM, SquareOps

Use Instance Refresh with minimum healthy percentage so instances roll gradually; ALB health checks gate traffic; set a checkpoint/canary percentage and automatic rollback. Alternatively blue/green with a second ASG. Verify through metrics, target health and smoke tests.

### Q455. What is a sticky session in ALB? What problems can it cause?

Asked at (2): Amazon, Aspire

Session affinity makes the ALB send a client to the same target (using an ALB cookie) so in-memory session data stays valid. Problems: uneven load and lost sessions if the target dies. Better design: stateless apps with sessions in Redis/DynamoDB.

### Q456. Can ASG and load balancer be in different regions? Peak traffic 5-8 PM daily - how to configure?

Asked at (1): Sigmoid

No - an ASG and its ALB must be in the same region (ASG can span AZs). For multi-region use Route53 latency/failover routing or Global Accelerator in front of separate regional stacks.

For daily peaks, use scheduled scaling (raise min/desired before 5 PM, lower after 8 PM) plus target-tracking for surprises.

### Q457. During heavy load new instances take 2-3 minutes to become ready (and ASG is terminating instances). How do you speed up/avoid it?

Asked at (1): EPAM

- Bake a golden AMI so boot does not install software; keep user-data minimal.
- Use warm pools or pre-scaled buffer, and scale earlier (lower thresholds/predictive scaling).
- Tune health check grace period and instance warm-up; set scale-in protection / termination policies and cooldowns to stop flapping.
- Check scale-in vs scale-out thresholds so they don't conflict.

#### Q458. What is Amazon Athena?

Asked at (1): Persistent Systems

A serverless interactive query service that runs SQL directly on data in S3 (CSV, JSON, Parquet), billed per data scanned. Common DevOps use: querying ALB/CloudTrail/VPC Flow logs. Use partitioning and columnar formats to cut cost.

#### Q459. What is connection draining (deregistration delay)?

Asked at (1): CTS

When a target is removed or unhealthy, the load balancer stops sending new requests but lets in-flight requests finish for a set period (ALB deregistration delay, default 300s). Prevents dropped connections during deployments and scale-in.

### Serverless & Containers

#### Q460. AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions?

Asked at (14): Aspire, Capgemini, Deloitte, EXL Service, Emphasis, Flentas, JPMorgan, Koerber Pharma, LTIMindtree +5 more

Lambda runs code on events without managing servers; you pay per request/duration. Use cases: S3 triggers (image processing), API backends, cron jobs (EventBridge), automation (stop dev instances), stream processing.

Cold start = extra latency when a new execution environment is created. Reduce: lighter packages, faster runtimes (Node/Python/Go), avoid VPC where unneeded, provisioned concurrency, SnapStart for Java, keep init code outside the handler.

Step Functions orchestrate multiple Lambdas into workflows with retries and branching.

#### Q461. Fargate vs EC2 launch type; ECS Service vs Task; setting up ECS on EC2

Asked at (7): Deloitte, EPAM, IBM, Nisum Technologies, Others (company not named), Plansource ValueLabs, Virtusa

Fargate runs containers without managing servers (you pay per vCPU/memory). EC2 launch type gives you control of nodes (cheaper at scale, GPUs, daemons). A Task is a running instance of a task definition (like one pod); a Service keeps N tasks running, replaces failed ones and integrates with a load balancer. ECS on EC2: create cluster, ASG with ECS-optimized AMI and agent, task definition, service, ALB.

#### Q462. API Gateway: configuration, REST vs WebSocket APIs

Asked at (2): EPAM, Sonata Software

API Gateway fronts backends (Lambda, HTTP, VPC link). Configure resources/methods, integration, stages, authorizers, throttling, custom domain, logging. REST/HTTP APIs are request-response; WebSocket APIs keep a persistent two-way connection (chat, live updates) with connect/disconnect/message routes. HTTP APIs are cheaper and faster than REST APIs but have fewer features.

#### Q463. Lambda role is set up correctly but logs are not appearing in the CloudWatch log group

Asked at (1): Others (company not named)

Check that the execution role has logs:CreateLogGroup/CreateLogStream/PutLogEvents (AWSLambdaBasicExecutionRole), the function really runs (invocation errors, throttling), a VPC-attached Lambda has a route/endpoint to CloudWatch Logs, a log group retention/resource policy or SCP isn't blocking, and the function isn't being killed at init before logging.

#### Q464. What are the limitations of Lambda?

Asked at (1): EPAM

15-minute max runtime, memory 128 MB-10 GB, 10 GB ephemeral /tmp, 6 MB sync payload, concurrency limits (default 1,000 per region), cold starts, no persistent local state, package size limits (250 MB unzipped zip / 10 GB container), and vendor lock-in. Not ideal for long-running or steady heavy workloads.

#### Q465. When to use EC2 vs Lambda (scenarios)?

Asked at (1): Others (company not named)

Lambda: event-driven, short (<15 min), spiky or low traffic, minimal ops, pay per use. EC2: long-running, steady high utilisation, custom OS/kernel or GPU, large memory, predictable cost. Example: image thumbnailer -> Lambda; a high-traffic always-on API with licensed software -> EC2/containers.

### Storage (S3, EBS)

#### Q466. S3 lifecycle policies and storage classes; how do you manage and version them?

Asked at (7): Elixir Labs, HCL, Optum, Others (company not named), Persistent Systems, Sigmoid, SquareOps

Lifecycle rules automatically transition or expire objects. Classes: Standard, Intelligent-Tiering, Standard-IA, One Zone-IA, Glacier Instant/Flexible/Deep Archive. Example: logs move to IA after 30 days, Glacier after 90, expire after 365; noncurrent versions expire after 30 days; abort incomplete multipart uploads. Versioning keeps previous object versions so you can recover from deletes; combine with lifecycle to control cost.

#### Q467. Terraform state in S3: expire in 30 days? other options? S3 native locking command?

Asked at (4): Five9, LTIMindtree, Others (company not named), Wipro

State should NOT expire - losing it breaks Terraform. A lifecycle rule should only expire noncurrent versions (e.g. 30 days). Alternatives to S3: Terraform Cloud/HCP, GCS, Azure Blob, Consul. S3 now supports native locking (`use_lockfile = true` in the backend block, Terraform 1.10+) so DynamoDB is no longer required; previously you used a DynamoDB table with `dynamodb_table`.

#### Q468. S3 vs EBS (and EFS)?

Asked at (3): Koerber Pharma, Sapient, Wikreate Media

S3 is object storage accessed over HTTP, unlimited, cheap, regional. EBS is block storage attached to one EC2 (like a hard disk), low latency, AZ-bound. EFS is shared NFS file system for many instances. Use S3 for files/backups/static sites, EBS for OS/databases, EFS for shared files.

#### Q469. Sending EC2 log files to S3 on a schedule, checking CPU and alerting

Asked at (2): Amazon, ITC Infotech

Attach an IAM role with `s3:PutObject` to the instance, then a cron job: `aws s3 cp /var/log/app.log s3://bucket/$(date +%F)/`. For CPU, use CloudWatch metrics + alarm -> SNS email; or script checks with `aws cloudwatch get-metric-statistics`. Better: CloudWatch Agent for logs, or logrotate hooks.

#### Q470. AMI vs snapshots

Asked at (1): EPAM

A snapshot is a point-in-time backup of one EBS volume (stored in S3, incremental). An AMI is a full template to launch instances: it references one or more snapshots plus launch permissions and block-device mapping. Create an AMI from an instance to clone servers; use snapshots for volume backups.

#### Q471. Can a bucket in one region be accessed from another region?

Asked at (1): Amazon

Yes. S3 bucket names are global but the data lives in one region; any client with permission can access it over the internet or via cross-region VPC endpoint/peering. Latency and inter-region data transfer cost apply. Use the bucket's regional endpoint to avoid redirects.

#### Q472. EBS-backed vs instance store-backed instances

Asked at (1): Others (company not named)

EBS-backed: root volume is an EBS volume, persists after stop, can be stopped/started and resized. Instance store: ephemeral disk physically attached to the host, very fast but data is lost on stop/terminate/failure. Use instance store for caches/scratch data only.

### Q473. How do you speed up large S3 uploads and verify a partially failed 10 GB upload?

Asked at (1): Plansource ValueLabs

Use multipart upload (parts in parallel, retry only failed parts), Transfer Acceleration, or the CLI's automatic multipart. After failure, `aws s3api list-parts` shows which parts are stored, and you resume or abort; also set a lifecycle rule to abort incomplete multipart uploads. Verify integrity with ETag/checksums.

### Q474. What is Cross-Region Replication (CRR) in S3?

Asked at (1): Aspire

CRR automatically and asynchronously copies new objects from a bucket in one region to a bucket in another, for DR, compliance or lower latency. Requires versioning on both buckets and an IAM role; existing objects need Batch Replication. Use Replication Time Control for an SLA.

## Databases (RDS)

### Q475. RDS migration with minimal downtime, and upgrading MySQL 5.7 to 8.0

Asked at (6): Amazon, Capgemini, Others (company not named), Rapidsoft, RelevantZ, Sigmoid

Migration: use AWS DMS (full load + CDC) or a read replica, keep source running, validate data, then cut over briefly by pointing the app to the new endpoint. Upgrade: take a snapshot, test on a clone, review parameter/option groups, run pre-check (deprecated features), perform in-place upgrade in a maintenance window or use blue/green deployments for near-zero downtime, then analyze tables and monitor.

### Q476. App is slow and you suspect RDS - what do you check? Memory pressure but cannot resize?

Asked at (3): Capgemini, Sonata Software, SquareOps

Check CloudWatch/Performance Insights: CPU, FreeableMemory, connections, IOPS, replica lag, slow query log, locks. Quick no-downtime relief: add read replicas and move reads, enable connection pooling/RDS Proxy, kill long queries, add cache (ElastiCache), tune indexes, adjust parameter group dynamically, offload reports. Then plan a proper resize in Multi-AZ so failover keeps downtime tiny.

### Q477. How do you configure RDS and scale it (vertical/horizontal)?

Asked at (3): Others (company not named), Verizon, Wikreate Media

Choose engine, instance class, storage type/size (gp3/io1), Multi-AZ, subnet group in private subnets, security group, backups/retention, encryption, parameter group, monitoring. Vertical scaling: modify instance class (short downtime unless Multi-AZ). Horizontal: add read replicas, or shard; use storage autoscaling for disk.

### Q478. Multi-AZ vs Read Replica; secondary vs read-only RDS

Asked at (3): Capgemini, Others (company not named), Verizon

- Multi-AZ: synchronous standby in another AZ for high availability and automatic failover; standby cannot serve reads (in classic Multi-AZ).
- Read replica: asynchronous copy for scaling reads (and DR); manual promotion.

HA = Multi-AZ, performance = read replicas.

### Q479. Can you kill heavy queries? Handling DB failover while a user is writing

Asked at (2): SquareOps, Turning

Find them (`SHOW FULL PROCESSLIST`; `pg_stat_activity`), then `KILL <id>` / `pg_terminate_backend(pid)`; set query timeouts and indexes to prevent repeats. During failover (e.g. Multi-AZ takes 30-120s) writes fail: apps should use the cluster/DNS endpoint, retry with back-off, idempotent writes, connection pooling (RDS Proxy) and queue writes if necessary.

### Q480. RDS cross-region read replica (India -> London)?

Asked at (2): Orion Innovation, SquareOps

Create a cross-region read replica of the source DB (console/CLI), which replicates asynchronously. Needs backups enabled, encryption key in target region if encrypted, and network access. Used for DR and local reads. Can be promoted to standalone in a regional failure. Alternative: Aurora Global Database for lower lag.

### Q481. Make RDS accept only one user connection at a time

Asked at (1): Capgemini

Set the database-level limit: `ALTER USER 'app'@'%' WITH MAX_USER_CONNECTIONS 1;` (MySQL) or `ALTER ROLE app CONNECTION LIMIT 1;` (PostgreSQL), plus parameter group `max_connections` if the whole instance should have 1. Combine with security group and IAM auth.

### Q482. Production RDS storage is 95% full - how do you debug and prevent it in future?

Asked at (1): Others (company not named)

Debug: find what is growing (large tables, binary logs/WAL, temp tables, unpurged logs) using `information_schema` and CloudWatch `FreeStorageSpace`. Quick fix: increase allocated storage (online), purge/archive old data, drop unused indexes. Prevent: enable storage autoscaling, set CloudWatch alarms at 70/80%, data retention/archival jobs, partitioning, and capacity reviews.

### Q483. Traditional RDS vs Aurora?

Asked at (1): Sonata Software

Aurora is AWS's cloud-native engine (MySQL/PostgreSQL compatible) with a shared distributed storage layer replicated 6 ways across 3 AZs, up to 15 low-lag replicas, faster failover (~30s), auto-growing storage, serverless option; typically costs more but performs better.

### Q484. Why use RDS Proxy?

Asked at (1): Flentas

RDS Proxy pools and reuses database connections, protecting the DB from connection storms (especially from Lambda), speeds up failover (cuts failover time), and supports IAM auth and Secrets Manager integration.

## EC2 & Compute

### Q485. How do you patch EC2 instances?

Asked at (3): BMW TechWorks, Qburst, SAP

Best: immutable approach - build a new patched AMI (Packer/EC2 Image Builder) and roll it out through the ASG. For long-lived servers: SSM Patch Manager with patch baselines and maintenance windows, test in dev/stage first, and use rolling batches to keep the service up.

### Q486. How do you manage environment segregation in AWS and handle multiple accounts (and AWS + Azure with one DevOps process)?

Asked at (2): NUOS INFO Systems, Sigmoid

Use separate accounts per environment/workload under AWS Organizations (prod isolated from dev), SCP guardrails, centralised logging/security accounts, and per-account roles via SSO. Same Terraform code with different variables/state per account. For AWS+Azure: one CI/CD pipeline and IaC tool (Terraform), common tagging, central secrets/policy (OPA, Sentinel), unified monitoring and cost dashboards.

### Q487. Can we take AWS backups using shell scripting?

Asked at (1): Deloitte

Yes: use AWS CLI (`aws ec2 create-snapshot`, `aws backup start-backup-job`) in a script scheduled through cron or EventBridge, tag with retention date, and delete old snapshots. But managed AWS Backup or Data Lifecycle Manager policies are preferable because they are auditable and need less maintenance.

### **Q488. Can you separate disk space on EC2 (app on one partition, observability stack on another)? How?**

Asked at (1): BMW TechWorks

Yes. Attach a separate EBS volume, then `lsblk, mkfs.xfs /dev/xvdf`, mount to e.g. `/var/log` or `/opt/monitoring`, and add it to `/etc/fstab` (use UUID). Benefits: logs filling disk won't crash the app, separate size/IOPS/snapshots. Could also use LVM for resizing.

### **Q489. Error "IP address limit exceeded" while launching an EC2 - how to fix?**

Asked at (1): Capgemini

Either the subnet ran out of free IPs (use a bigger or another subnet, or release unused ENIs) or the account hit a limit (Elastic IPs: 5 per region, vCPU/instance quota). Check subnet available IPs, delete unused ENIs/EIPs, or request a Service Quotas increase.

### **Q490. How do you migrate an EC2 instance from one region to another?**

Asked at (1): Nextturn

Create an AMI of the instance, copy the AMI to the target region (`aws ec2 copy-image`), launch from it, then recreate networking (VPC, SG, key pair), re-attach/copy EBS or data, update DNS and test. Re-encrypt with a key in the new region. For minimal downtime use Application Migration Service/replicate data first.

### **Q491. You cannot SSH into an EC2 - how can you still install a package (e.g. tree)?**

Asked at (1): Nextturn

Use SSM Run Command / Session Manager (instance needs the SSM agent and role), EC2 Instance Connect, or user-data on a restart, or attach the root volume to another instance. With SSM: `aws ssm send-command --document-name AWS-RunShellScript --parameters 'commands=["yum install -y tree"]'`. Also check why SSH failed: SG, key, NACL, route, public IP.

# Azure & GCP

20 questions

## Azure Architecture

### Q492. Azure storage types, subscription types, Recovery Services Vault, backup choices and Azure Boards

Asked at (4): Elixr Labs, Hexaware, RelevantZ, Synechron

- Storage: Blob (objects), Files (SMB shares), Queue (messages), Table (NoSQL), Managed Disks.
- Subscriptions: Free, Pay-As-You-Go, Enterprise Agreement, CSP, Dev/Test; used to separate billing/limits/environments.
- Recovery Services Vault stores backups and Site Recovery replicas for VMs, SQL, Files; Azure Backup can also protect on-prem via MARS agent.
- Azure Boards (in Azure DevOps): work item tracking - epics, features, user stories, tasks, bugs, Kanban boards, sprints and backlogs.

### Q493. Design a highly available three-tier architecture in Azure (React frontend + Node backend); App Gateway with WAF for a banking app

Asked at (3): JPMorgan, Orion Innovation, Persistent Systems

Frontend: Azure Static Web Apps/Blob + Front Door/CDN. Backend: App Service/AKS across availability zones behind Application Gateway (WAF v2, TLS offload, OWASP rules). Data: Azure SQL/Cosmos DB with zone redundancy and geo-replication. Add Key Vault, private endpoints, NSGs, Azure Monitor/Log Analytics and autoscale. For banking: WAF in prevention mode, private networking, encryption at rest/in transit, DDoS protection, and audit logs.

### Q494. Scaling strategy for compute-intensive vs I/O-intensive workloads; throttled Azure Function; migrations

Asked at (3): Elixr Labs, JPMorgan, Qburst

CPU-bound: scale on CPU, use compute-optimised or GPU SKUs and scale out. I/O-bound: scale on queue length/latency, use premium disks, caching, async processing. A throttled Function: look at Application Insights (429s, host.json concurrency), raise plan limits (Premium plan / scale-out), add retry/back-off and queues. Migration: assess, replicate (Azure Migrate / Site Recovery), test, cut over; resource migration moves infrastructure, data migration moves databases/files. Moving an AKS/GKE node pool: create a new pool, cordon and drain old nodes, then delete the old pool.

### Q495. How do you implement DC/DR in Azure (and what is the purpose of DR)?

Asked at (2): RelevantZ, Synechron

DR exists so the business survives a site/region failure within agreed RTO (time to recover) and RPO (data loss). In Azure: paired regions, Azure Site Recovery for VM replication, geo-redundant storage (GRS), SQL failover groups/geo-replication, Traffic Manager or Front Door for failover, Azure Backup, and IaC to rebuild. Test failover drills regularly, and document runbooks.

### Q496. A .NET three-tier app uses App Service (Web API), Function Apps (background jobs) and Service Bus - how do you design and operate it?

Asked at (1): Elixr Labs

App Service hosts the API, Functions process Service Bus queue messages asynchronously (decoupling and retry), with Key Vault for secrets, Managed Identity for auth, Application Insights for tracing, VNet integration/private endpoints for security, deployment slots for zero-downtime releases, and autoscale rules on queue length/CPU.

### Q497. What is IAP (Identity-Aware Proxy) in GCP?

Asked at (1): Qburst

IAP verifies user identity and context before allowing access to apps or VMs, replacing VPNs/bastions. For SSH to private VMs: `gcloud compute ssh --tunnel-through-iap`. Access is granted with the IAP-secured Tunnel User role and enforced per request.

## Azure Networking

### Q498. Connect VMs in different VNets; define a VNet; Azure Load Balancer; ExpressRoute

Asked at (4): Alphadyne, Elixir Labs, Others (company not named), RelevantZ

- VNet: private address space (e.g. 10.0.0.0/16) split into subnets.
- Different VNets: use VNet Peering (regional or global, non-transitive) or VPN gateway/virtual WAN.
- Azure Load Balancer: layer-4; create frontend IP, backend pool (VMs), health probe and rule. Application Gateway is layer 7.
- ExpressRoute: private dedicated connection from on-prem to Azure via a provider; higher reliability and lower latency than VPN.

### Q499. Application Gateway vs Front Door (and L7/L4)

Asked at (2): CTS, Sonata Software

Application Gateway is a regional layer-7 load balancer with WAF, SSL offload and path routing within a VNet. Front Door is a global layer-7 entry point/CDN with edge acceleration, global failover and WAF across regions. Azure Load Balancer is layer 4. Use Front Door in front of multi-region apps and App Gateway per region.

### Q500. Firewall vs NSG

Asked at (2): LTIMindtree, Others (company not named)

NSG is a stateful L3/L4 allow/deny filter attached to subnets or NICs (IP, port, protocol). Azure Firewall is a centralised managed firewall with FQDN rules, threat intelligence, NAT and logging across VNets. Use NSGs for basic segmentation, Azure Firewall for central egress/ingress control.

### Q501. Block a specific domain in Azure Application Gateway

Asked at (1): Sonata Software

Use WAF custom rules on the Application Gateway WAF policy: match condition on `RequestHeaders` Host or `RequestUri` containing the domain, action Block, with a priority. For outbound domain blocking use Azure Firewall FQDN rules.

### Q502. Set up Azure Bastion for secure access to VMs

Asked at (1): Alphadyne

Azure Bastion gives browser-based RDP/SSH to VMs over TLS without public IPs. Create a subnet named `AzureBastionSubnet` (/26 or larger) in the VNet, deploy Bastion with a public IP, then connect from the portal to private VMs. Pair with NSGs; no open 22/3389 on the internet.

## AKS

### Q503. Use Azure Key Vault secrets in AKS; how does a pod get rotated secrets? AKS cost optimisation

Asked at (3): Blue Yonder, EXL Service, Flentas

Use the Secrets Store CSI Driver with the Azure Key Vault provider and a workload identity: define a `SecretProviderClass`, mount it as a volume (and optionally sync to a Kubernetes secret). Enable secret rotation (`--enable-secret-rotation`, poll interval) so mounted files update; apps must re-read files (env vars need restart, or use Reloader). Cost: right-size nodes, autoscaler, spot node pools, reserved instances, stop dev clusters at night, set requests/limits, use Karpenter-like node auto-provisioning, monitor with cost analysis.

## **Q504. Protect endpoints in AKS; give only one pod access to a storage account and block others**

Asked at (2): Others (company not named), Sonata Software

Protect: private cluster, internal load balancers, App Gateway + WAF, NetworkPolicies, Azure Firewall, authorized IP ranges. Storage per-pod access: use Azure AD Workload Identity - bind a Kubernetes ServiceAccount used only by that pod to a managed identity that has an RBAC role (Storage Blob Data Reader) on the storage account; disable shared key access and use a private endpoint plus firewall rules so other pods have no permission.

## **Q505. How do you integrate Entra ID with AKS, secure AKS, and make it highly available?**

Asked at (1): Blue Yonder

- Auth: enable AKS-managed Entra ID integration, map Entra groups to Kubernetes RBAC roles, disable local accounts.
- Security: private cluster, Azure Policy, network policies, Defender for Containers, managed identities, image scanning from ACR, Key Vault for secrets, restrict API server IPs.
- HA: use availability zones, multiple node pools (system + user), cluster autoscaler, at least 3 nodes, pod anti-affinity/PDBs, Standard-tier uptime SLA, multi-region with Front Door/Traffic Manager for DR.

## **Q506. Networking used in AKS (kubenet vs Azure CNI)**

Asked at (1): Sonata Software

kubenet: pods get IPs from an overlay range and traffic is NAT'd through the node (fewer VNet IPs, limited features). Azure CNI: pods get real VNet IPs (direct reachability, NSG integration, needs more IP planning); Azure CNI Overlay is a middle option; Cilium for policies. Choose Azure CNI for enterprise networking and private connectivity.

## **Azure Security**

### **Q507. Azure Key Vault: experience, access policies, and how MySQL can use it privately**

Asked at (2): Capgemini, Nice

Key Vault stores secrets, keys and certificates. Access is controlled by RBAC or access policies; apps authenticate with Managed Identity (no passwords). For private use: create a Private Endpoint into the VNet, disable public network access, and use a private DNS zone, so the MySQL/app talks to the vault over the Microsoft backbone only. Mention you defined policies yourself (least privilege: get/list only).

### **Q508. Managed Identity vs Service Principal**

Asked at (2): Encora, Others (company not named)

A Service Principal is an app identity in Entra ID with a secret/certificate you must create, store and rotate. A Managed Identity is a service principal whose credentials Azure manages for you automatically and that is attached to a resource (VM, Function, AKS). Use managed identity wherever possible (no secrets to leak); SPs for external systems such as CI/CD outside Azure.

### **Q509. How do you create an app registration in Azure?**

Asked at (1): Zensar

Entra ID -> App registrations -> New registration (name, supported account types, redirect URI). Then note Application (client) ID and Tenant ID, create a client secret or certificate (or federated credential), grant API permissions/role assignments. Creates the app identity and its service principal used for authentication.

## **Azure Monitoring**

## Q510. Log Analytics Workspace, alerts with action groups, KQL queries, and Dynatrace alert types

Asked at (3): Others (company not named), RelevantZ, Wipro

A Log Analytics workspace is the central store for Azure Monitor logs (VM, AKS, App Insights) queried with KQL. Alert flow: create alert rule -> scope (VM) -> condition (metric: Percentage CPU > 75% for 5 min, or KQL query) -> action group (email/SMS/webhook/runbook) -> severity -> save. Basic KQL: `Perf | where CounterName == "% Processor Time" | summarize avg(CounterValue) by bin(TimeGenerated, 5m)` or `ContainerLog | where LogEntry contains "error"`. Dynatrace alerts: static threshold, auto-adaptive baseline, and Davis AI anomaly detection problems.

## Q511. Monitor Azure VM memory and set alerts in Azure Monitor

Asked at (2): NUOS INFO Systems, Zensar

Install the Azure Monitor Agent (collects guest metrics like memory). Create an alert rule: select the VM -> condition "Available Memory" (or Percentage Memory > 80%) -> threshold and evaluation window -> action group (email/SMS/webhook) -> severity and name. Dashboards can live in Azure Monitor/Grafana.

# Monitoring & Observability

20 questions

## Concepts

### Q512. What monitoring tools and agents have you used and what did you do with them? What proactive monitoring have you implemented?

Asked at (10): Altimetrik, Cisco, Deloitte, EXL Service, EY, IBM, Infosys, Nice, OPT IT +1 more

Answer from experience with structure: Metrics (Prometheus/Grafana, CloudWatch, Datadog), Logs (ELK/Loki, CloudWatch Logs), Traces (Jaeger/X-Ray), Alerting (Alertmanager, PagerDuty/Opsgenie). Agents: node\_exporter, CloudWatch agent, Fluent Bit/Filebeat, Datadog agent. Describe your role: "I built dashboards for golden signals, created SLO-based alerts, reduced alert noise by 40%, added synthetic checks and capacity forecasts so we caught disk/cert expiry before users noticed."

### Q513. Monitoring vs observability; metrics, logs and traces and their tools; observability architecture

Asked at (6): Accion Labs, Amazon, Commonwealth Bank, Infosys, Koerber Pharma, Others (company not named)

Monitoring watches known signals and alerts when thresholds break ("is it working?"). Observability lets you ask new questions about system behaviour from its outputs ("why is it broken?").

Three pillars: Metrics (numeric time series - Prometheus, CloudWatch, Datadog), Logs (events - ELK/Loki/Splunk), Traces (request path across services - Jaeger, Tempo, OpenTelemetry, X-Ray). Logs tell what happened in one place; traces show where time was spent across services and connect them.

Architecture: instrumented apps (OpenTelemetry) -> collectors/agents -> storage backends -> dashboards (Grafana) and alerting (Alertmanager/PagerDuty) -> runbooks and automated remediation. It supports reliability by shortening detection and diagnosis time (MTTD/MTTR).

### Q514. What is Nagios and how do you integrate Jenkins with it?

Asked at (1): TCS

Nagios is a classic host/service monitoring and alerting tool using plugins (NRPE, SNMP) and check commands. Integration: install the Nagios plugin or use Jenkins' Nagios/webhook notifications and NRPE checks (e.g. `check_http/check_jenkins` against Jenkins' API) to monitor Jenkins health and job status.

## Prometheus

### Q515. Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and what you monitor

Asked at (10): EXL Service, EY, Elixir Labs, HCL, Infosys, JPMorgan, Netcracker, OPT IT, Others (company not named) +1 more

Prometheus is a pull-based metrics system with a time-series DB and PromQL; Grafana visualises data from sources (Prometheus, Loki, Elasticsearch, CloudWatch).

Setup: on Kubernetes use the kube-prometheus-stack Helm chart (Prometheus, Alertmanager, Grafana, node-exporter, kube-state-metrics). `prometheus.yml` defines scrape\_configs and rule files; Grafana: add Prometheus as data source (URL `http://prometheus:9090`), import/create dashboards, set variables, thresholds and alerts. Extensions = Grafana plugins (`grafana-cli plugins install`).

Components: node\_exporter (host metrics), kube-state-metrics (K8s objects), cAdvisor (container), blackbox\_exporter (probes), Alertmanager (routing), Pushgateway. Typical metrics: CPU, memory, disk, network, pod restarts, request rate, error rate, latency. Alerts: high CPU/memory, pod down, disk full, 5xx rate, latency SLO. Explain where you deployed it (in cluster for scraping with service discovery) and why.

### Q516. Have you created dashboards in Grafana?

Asked at (3): Altimetrik, EXL Service, Nice

Answer with a concrete example: add Prometheus as data source, build panels with PromQL (CPU, memory, request rate, error %, p95 latency), use variables (namespace, pod) for reuse, set thresholds and alerts, and store dashboards as code (JSON/Terraform/Helm) in Git. Follow RED (Rate, Errors, Duration) for services and USE for infrastructure.

## Q517. How does Prometheus fetch data?

Asked at (2): Infosys, Nisum Technologies

Pull model: Prometheus scrapes `/metrics` HTTP endpoints at intervals from targets found via static config or service discovery (Kubernetes). Short-lived jobs push to Pushgateway; exporters (node\_exporter, etc.) expose metrics for third-party systems. Data is stored in its time-series database and queried with PromQL.

## Q518. Alert if CPU stays above 90% for 5 minutes but only when running pods are below 5

Asked at (1): Sigmoid

```
- alert: HighCpuFewPods
 expr: |
 avg(rate(container_cpu_usage_seconds_total{namespace="app"}[5m])) /
 avg(kube_pod_container_resource_limits{resource="cpu"}) > 0.9
 and
 count(kube_pod_status_phase{namespace="app", phase="Running"} == 1) < 5
 for: 5m
```

`for: 5m` provides the duration and `and` combines both conditions.

## Q519. Prometheus memory keeps growing - how do you debug?

Asked at (1): Others (company not named)

Usually high cardinality (too many label values like `user_id/pod_ip`) or too many series. Check `prometheus_tsdb_head_series`, top series (`topk(10, count by (__name__)(__name__=~".+"))`), the TSDB status page, scrape intervals, long retention, and heavy queries. Fix: drop/relabel high-cardinality labels, increase scrape interval, shard (Thanos/Mimir/federation), set retention and limits, scale resources.

## Q520. Write a PromQL expression to alert if CPU usage is above 80% on any node

Asked at (1): Sigmoid

```
100 - (avg by (instance) (rate(node_cpu_seconds_total{mode="idle"}[5m])) * 100) > 80
```

In a rule: `alert: HighNodeCPU, for: 5m`, labels severity warning, annotation with `{{ $labels.instance }}` value.

## Logging & Alerting

### Q521. Extract errors from logs: fetch lines with 200 status, count ERR, split error and warning into separate files, log analysis script

Asked at (5): EPAM, IBM, Morgan Stanley, Others (company not named), Sigmoid

```
grep -i "error" app.log # fetch errors
grep -c "ERR" app.log # count of ERR lines
awk '$9==200' access.log # 200 status lines (nginx combined format)
grep -i error "$1" > errors.txt; grep -i warning "$1" > warnings.txt # ./split.sh test.log
awk '{print $9}' access.log | sort | uniq -c | sort -rn # status code summary
```

Use `zgrep` for rotated gz logs and `tail -f | grep` for live.

### Q522. How do you create custom alerts and handle CPU/disk alerts (e.g. CPU/memory beyond 75%)? How do you receive alerts?

Asked at (5): Infosys, LTIMindtree, One2N, Sonata Software, Synechron

Define the condition (metric + threshold + duration), severity and routing. CloudWatch: alarm on `CPUUtilization > 75%` for 5 minutes -> SNS -> email/Slack/PagerDuty. Prometheus: alert rule `expr: node_memory... > 0.75, for: 5m` -> Alertmanager receivers. Azure: alert rule + action group. For handling: acknowledge, check top consumers, scale or fix, runbook, and tune thresholds. Alert on symptoms and capacity trends, include runbook links.

### Q523. How do you increase debug log level? Application-level metrics and APM?

Asked at (3): Capgemini, Cisco, One2N

Raise the level via config or runtime flag without redeploying where possible (e.g. LOG\_LEVEL=DEBUG env var, logback/Log4j config, Spring Actuator `/actuator/loggers`, `kubectl set env`), and revert afterwards because debug logs are heavy. Application metrics: instrument code with libraries (Prometheus client, Micrometer, OpenTelemetry) exposing request rate, errors, latency histograms, queue sizes, business counters; APM tools (Datadog, New Relic, Dynatrace, App Insights) provide traces and performance.

### Q524. Which logs do you check (agent, console, database, slow query, kernel)?

Asked at (3): Arrise Solutions, CGI, SquareOps

System: `/var/log/syslog` or `messages`, `auth.log/secure`, kernel `dmesg` and `/var/log/kern.log`, `journalctl`.

App/web: application logs, nginx/apache access and error logs. Containers: `kubectl logs`, `docker logs`.

Database: error log, slow query log, general log, Performance Insights (RDS). Agents: Fluent

Bit/CloudWatch agent logs. If DB logs aren't being written check log settings in the parameter group, disk space, permissions, log destination (file vs CloudWatch) and flush.

### Q525. How would you build centralized logging across multiple clouds and environments?

Asked at (2): AMEX, EPAM

Ship logs from every source with agents (Fluent Bit/Filebeat/OTel Collector) to a central store - ELK/OpenSearch, Loki, Splunk or a cloud-native workspace. Normalise fields (service, env, cloud), tag with trace IDs, set retention tiers (hot/warm/cold in S3), control access with RBAC, and alert/dashboard on top. Use private connectivity and encryption in transit.

### Q526. Script to monitor disk usage and email/log an alert above 80%

Asked at (2): Nextturn, Others (company not named)

```
#!/bin/bash
LIMIT=80
df -hP | awk 'NR>1{gsub("%","", $5); if($5>'$LIMIT') print $6 " "$5"%"}' | while read mp use; do
 echo "$(date) ALERT: $mp at $use" >> /var/log/disk_alert.log
 echo "$mp is $use full" | mail -s "Disk alert $(hostname)" ops@example.com
done
```

Python version: `shutil.disk_usage('/')` and `smtplib`. Schedule with cron every 5 minutes, or better use Prometheus `node_exporter` alerts.

### Q527. Thousands of alerts arrive from a huge infrastructure with customer-defined thresholds - how do you handle it?

Asked at (2): Elixir Labs, Sony

Fight alert fatigue: alert on symptoms/SLOs (user impact), not every cause; group and deduplicate (Alertmanager), set severity and routing, use sensible durations (for: 5m), dynamic thresholds/anomaly detection, suppress during maintenance, and review noisy alerts regularly. Every alert must be actionable with a runbook.

### Q528. Design a highly available logging system for 100+ microservices across 3 regions

Asked at (1): JPMorgan

Agents (Fluent Bit DaemonSets) per cluster -> regional buffering (Kafka/Kinesis) -> regional OpenSearch/Loki clusters (multi-AZ) with replication or cross-cluster search, plus cold storage in S3 for long retention; central Grafana/Kibana querying all regions; structured JSON logs with correlation IDs; backpressure handling, retention tiers, access control, and monitoring of the pipeline itself.

### Q529. You can't install Filebeat on worker nodes - how else do you collect logs?

Asked at (1): Amazon

Use a sidecar container in each pod, run the collector as a Kubernetes DaemonSet (Fluent Bit), push logs directly from the application to a logging endpoint (HTTP/syslog/OpenTelemetry), use the platform's native logging (CloudWatch Container Insights, Azure Monitor, stdout via cloud agents), or have the app write to a shared volume harvested elsewhere.

## ELK / EFK

### Q530. How is the ELK stack set up end to end; how Kibana connects to Elasticsearch; how logs are segregated; indices

Asked at (4): Five9, Infosys, Others (company not named), Sigmoid

Agents (Filebeat/Fluent Bit DaemonSet) collect container/host logs -> Logstash or ingest pipelines parse/enrich -> Elasticsearch stores in indices (e.g. `app-prod-2025.01.01`) -> Kibana visualises. Segregation: separate indices or data streams per environment/app/team, index patterns (data views), roles and spaces for access, ILM for retention.

Kibana config (kibana.yml):

```
server.host: "0.0.0.0"
elasticsearch.hosts: ["https://elasticsearch:9200"]
elasticsearch.username: "kibana_system"
elasticsearch.password: "${KIBANA_PASSWORD}"
elasticsearch.ssl.certificateAuthorities: ["/certs/ca.crt"]
```

An index is like a database of related JSON documents divided into shards; Kibana index patterns select which indices to search.

### Q531. In an EFK/ELK stack, what does each component do? How does filtering and indexing work in Elasticsearch?

Asked at (1): JPMorgan

- Elasticsearch: distributed search/analytics engine; stores JSON documents in indices split into shards (primary + replica).
- Logstash (heavy, transformations) or Fluentd/Fluent Bit (lightweight collector): collect, parse (grok), filter and ship.
- Kibana: visualisation/search UI.
- Beats/Filebeat: lightweight shippers on hosts.

Flow: app logs -> Fluent Bit (DaemonSet) -> Elasticsearch -> Kibana. Indexing: text is analysed (tokenised) into an inverted index mapping terms to documents, so searches/filters are fast; use index templates, ILM (hot/warm/delete) and daily indices to control size.

# SRE Concepts

19 questions

## Incident Response

### **Q532. How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix? Describe a production issue you faced**

Asked at (10): AMEX, Accion Labs, Alphadyne, Altimetrik, CMT, Cisco, One2N, Others (company not named), Sony +1 more

Sev-1 process: acknowledge and declare an incident, assign roles (incident commander, comms, ops), stabilise first (rollback, failover, scale), communicate status regularly, then find root cause, fix, and write a blameless postmortem with action items.

RCA approach: check recent changes (deploys, config, IAM), dashboards (golden signals), logs/traces, dependency health, then isolate layer by layer (DNS, LB, app, DB, network).

Rollback vs hotfix: rollback if the fix is unknown or risk is high and the previous version is safe (fastest); hotfix if rollback is impossible (schema change) and the fix is small and tested. For your story, name the symptom, root cause, time to recover and prevention.

### **Q533. Sudden traffic surge makes the web application unresponsive - what steps do you take to mitigate?**

Asked at (3): Encora, IBM, Others (company not named)

1) Triage: check dashboards (CPU, latency, errors, DB connections). 2) Scale out fast: increase replicas/ASG, scale DB read capacity. 3) Protect: rate limiting, WAF/CDN caching, shed non-critical load, queue requests. 4) Fix bottlenecks (slow queries, connection pools). 5) Communicate, then post-mortem and set up autoscaling, load tests and capacity planning.

### **Q534. Blameless postmortems, reducing MTTR and preventing repeat incidents**

Asked at (1): Altimetrik

Blameless postmortem: timeline, impact, root causes (5 whys), what went well/poorly, action items with owners and due dates, no individual blame. Reduce recurrence: fix systemic causes, add tests/guardrails/alerts, runbooks, game days. Reduce MTTR: better alerting and dashboards, auto-remediation (self-healing scripts, auto-rollback), clear on-call runbooks, ChatOps, feature flags, quick rollback.

## Troubleshooting

### **Q535. A live app is slow/has latency for customers (e-commerce) - how do you troubleshoot? Which metrics do you check?**

Asked at (6): IBM, Koerber Pharma, Others (company not named), Persistent Systems, Sigmoid, SquareOps

Follow a layered method: scope the problem (all users? one region/endpoint? since when?) -> check recent changes -> dashboards for the golden signals: latency, traffic, errors, saturation (CPU, memory, IOPS, connections, disk queue depth) -> traces to find the slow service/DB query -> dependencies (DB slow queries, cache hit rate, external APIs) -> network/LB (target response time) -> scale or rollback -> fix the root cause (index, cache, N+1, pool size) and add alerts.

### **Q536. App is down and returning 503 - what steps? When do 503 errors occur?**

Asked at (2): Aspire, CTS

503 = service unavailable: no healthy backends or overloaded. Check LB target health, whether pods/instances are running and ready, deployments/rollouts, resource exhaustion (CPU/memory/connections), autoscaling limits, dependency failures (DB), rate limiting, maintenance mode, ingress/service selectors and endpoints. Fix by restoring healthy targets (rollback/scale), then root cause.

### Q537. Clients report 504 Gateway Timeout - how do you debug?

Asked at (2): Cisco, Wipro

504 = the gateway/LB did not get a timely response from the upstream. Check: LB idle timeout vs backend response time, target health, backend CPU/memory/thread pool, slow DB queries or downstream APIs, network/security groups, recent deployments, retries and connection pools. Look at LB access logs and APM traces; fix the slow dependency, scale, tune timeouts.

### Q538. Users complain about latency but monitoring says healthy - what do you do? How does monitoring help fix a latency problem?

Asked at (2): Infosys, Sony

Check what the monitoring doesn't cover: user-facing metrics (p95/p99, not averages), synthetic and real-user monitoring, regions/ISP, specific endpoints, third-party dependencies, DNS/CDN/TLS, client-side issues. Use traces to find the slow hop, compare saturation (CPU, DB, queue depth), check recent changes and logs. Close the gap by adding the missing SLI/dashboard/alert afterwards.

### Q539. A production app works for internal users but fails for external users with 403 - how to isolate?

Asked at (1): JPMorgan

Compare paths: internal (private/VPN) vs external (internet). Check WAF rules/geo or IP blocks, CDN/LB rules, security groups/NACLs, authentication/SSO or CORS settings, header/host rules in ingress, source IP allowlists, and WAF/LB logs for the 403 origin (which layer returned it). Reproduce with curl from outside.

## SLI, SLO & Error Budgets

### Q540. What are SLI, SLO, SLA and error budget? How do you decide and set them?

Asked at (7): AMEX, Accion Labs, Altimetrik, Commonwealth Bank, Infosys, Others (company not named), Synechron

SLI = a measured indicator of service quality (e.g. % of successful requests, p95 latency). SLO = internal target for an SLI over a window (99.9% availability per 30 days). SLA = contractual promise to customers with penalties (looser than SLO). Error budget = 100% - SLO (99.9% -> ~43 min downtime/month); while budget remains you ship features, when it's exhausted you focus on reliability.

Choosing SLIs: pick what users care about (availability, latency, correctness, freshness) for each user journey; set SLOs from historical data and business need, not 100%; review quarterly; alert on burn rate; use SLOs to drive release and prioritisation decisions.

### Q541. Have you configured SLO-based alerting?

Asked at (1): Altimetrik

Yes - define SLIs (availability = good requests/total, latency p95), SLO (99.9% over 30 days), error budget, and alert on burn rate with multi-window rules (fast burn pages; slow burn tickets). Example Prometheus multi-window burn-rate alerts; dashboards show budget remaining; deploys freeze when budget is exhausted.

### Q542. What is an ideal burn rate?

Asked at (1): Altimetrik

Burn rate = how fast you consume the error budget relative to the SLO window. Burn rate 1 means you'll exactly use your budget by the window end; ideal is  $\leq 1$ . Alerting: fast burn (e.g. 14.4x over 1 hour) pages; slow burn (e.g. 1x over 3 days) opens a ticket.

## Disaster Recovery

### **Q543. Explain your DR strategy: RTO, RPO, failover and traffic redirection; DR for DevOps infrastructure; what if users can't access the app during DR**

Asked at (6): EPAM, EXL Service, Elixir Labs, Rapidsoft, UST, Zensar

DR = restoring service after a major failure. RTO (how fast), RPO (how much data loss tolerated). Strategies by cost: backup & restore, pilot light, warm standby, multi-site active-active. Components: cross-region data replication (RDS replica, S3 CRR), IaC to rebuild, AMIs/images replicated, secrets/config replicated, DNS failover (Route 53 health checks / Traffic Manager), runbooks and regular DR drills. For DevOps tooling itself: back up Git, Jenkins config, Terraform state, registries and have a standby CI. If users can't access during DR: check DNS records/TTL, health of the standby, LB, certs, security groups, data sync status and roll the traffic switch.

### **Q544. How does AWS DRS (Elastic Disaster Recovery) work; failover and failback?**

Asked at (1): Others (company not named)

DRS installs an agent on source servers that continuously replicates block-level data to a low-cost staging area in the target region. On disaster, you launch recovery instances from the latest (or earlier point-in-time) snapshot within minutes (low RTO, seconds RPO). Failback: reverse replication from AWS to the original site, then cut back after data is synced. Drills can be run without impacting production.

## **Performance**

### **Q545. How do you reduce latency for users from abroad? Low-latency routing?**

Asked at (3): Koerber Pharma, Persistent Systems, Sigmoid

Put content close to users: CDN (CloudFront/Akamai) for static and cacheable content, Global Accelerator/anycast for dynamic traffic over the AWS backbone, Route 53 latency-based routing to multi-region deployments, regional read replicas and caches, compress and optimise assets, HTTP/2-3, connection reuse. For real-time apps (games, streaming) use regional servers, UDP/WebSocket, edge compute and sticky/sharded sessions with scalable stateless backends.

### **Q546. How do you perform load testing?**

Asked at (1): JPMorgan

Define goals (target RPS, p95 latency, error rate), use tools such as JMeter, k6, Locust or Gatling, run realistic scenarios against a production-like environment, ramp up gradually (load, stress, soak, spike tests), monitor CPU/memory/DB/latency, find bottlenecks and re-test. Automate in CI for regression.

### **Q547. Which metric tells me the application is up or down?**

Asked at (1): One2N

For Prometheus, the **up** metric (1 = target scrape succeeded); better, use blackbox probes and SLI metrics: success rate (non-5xx), latency, and health check status from the LB. Alert on user-facing symptoms, not only process up/down.

## **Reliability**

### **Q548. DevOps vs SRE**

Asked at (2): AMEX, Synechron

DevOps is a culture/practice set to break silos and speed delivery (CI/CD, IaC, collaboration). SRE is a specific implementation of those ideas from Google: apply software engineering to operations with SLOs, error budgets, toil reduction (<50%), blameless postmortems and automation. "SRE is a concrete way to do DevOps; DevOps asks what, SRE defines how with measurable reliability targets."

### **Q549. How does MongoDB high availability work (primary/secondary)?**

Asked at (1): Others (company not named)

A replica set has one primary (accepts writes) and secondaries that replicate the oplog; on primary failure an election chooses a new primary automatically (needs majority, so use an odd number of voting members or an arbiter). Read preferences allow reads from secondaries. Sharding adds horizontal scale.

### **Q550. What is chaos engineering?**

Asked at (1): Infosys

Deliberately injecting failures (kill pods/instances, add latency, drop network) in a controlled way to find weaknesses before they cause outages. Process: define steady state, hypothesis, small blast radius, run experiment, observe, fix. Tools: Chaos Monkey, Gremlin, LitmusChaos, AWS Fault Injection Service.

# Architecture & Design

10 questions

## Microservices

### Q551. Microservices vs monolith; what are microservices?

Asked at (2): Persistent Systems, TCS

Microservices split an application into small, independently deployable services, each owning one business capability and its data, communicating via APIs/events. Benefits: independent scaling/deploys, team autonomy, fault isolation. Costs: network complexity, distributed tracing, data consistency. A monolith is a single deployable unit - simpler to start, harder to scale and change.

### Q552. Payment succeeds but order/shipping service fails - how do you handle it? Optimistic locking in a REST update?

Asked at (2): JPMorgan, Wipro

Use the Saga pattern with compensating actions (refund or retry), outbox pattern and idempotency keys with an event queue so the order is eventually completed or payment reversed. Optimistic locking: include a version/ETag; PUT with **If-Match: version**, the DB updates **WHERE id=? AND version=?**; zero rows updated returns 409/412 so lost updates are avoided.

### Q553. Service discovery when moving from monolith to microservices; zero-downtime DB migrations; retry mechanism for failed API calls

Asked at (2): JPMorgan, Wipro

- Service discovery: Kubernetes DNS/Services, Consul, or a service mesh; strangler-fig pattern to extract services gradually.
- Zero-downtime DB migration: expand-contract (add new column, dual-write, backfill, switch reads, drop old later), backward-compatible changes, tools like Flyway/Liquibase, online schema change tools.
- Retries: exponential back-off with jitter, max attempts, timeouts, idempotent operations, circuit breaker; e.g. Python **tenacity**, or built-in retry in mesh/gateway.

### Q554. How do you enforce tenant isolation at the API layer in a multi-tenant application?

Asked at (1): Wipro

Identify the tenant from the auth token (claims), never from user input; enforce tenant ID in every query (row-level security or separate schemas/DBs), authorise per tenant in the API gateway, rate-limit per tenant, isolate resources (namespaces, quotas), encrypt per tenant, and audit logs.

### Q555. Why is PUT idempotent? What happens if I send the same PUT twice or change the location with the same name?

Asked at (1): F5

Idempotent means repeating the request gives the same end state. PUT replaces the resource at a known URI, so sending the same entry twice leaves one record (no duplicates). With the same name but a different location, the second PUT overwrites the first - the DB ends with the latest location. POST is not idempotent (creates a new record each time).

## Migration

### **Q556. Prechecks before migration; checks after an upgrade; risks of upgrading; what triggers a rollback**

Asked at (3): Cisco, Elixir Labs, SquareOps

Before: inventory and dependencies, backups/snapshots, compatibility and deprecation checks, maintenance window, test in staging, rollback plan, stakeholder approval. After: services healthy (pods ready, probes, error rate, latency), data integrity, logs and alerts clean, performance baseline, integrations working, smoke tests. Risks: downtime, data loss, breaking changes, version incompatibilities. Rollback triggers: failed health checks, error-rate or latency breaching thresholds, data corruption, failed smoke tests.

### **Q557. Approach to migrating a monolith to containers/microservices**

Asked at (1): L and T

Containerise the app as-is first (lift & shift in Docker), externalise config and state, add health checks, then split gradually (strangler pattern) by business domain, with CI/CD per service, service discovery, observability, and database decoupling. Deploy to Kubernetes/ECS with rolling updates.

## **Multi-Region**

### **Q558. Design a cost-optimised architecture for a nightly internal reporting app with 3-year log retention**

Asked at (1): JPMorgan

Run the job on demand: EventBridge schedule -> Lambda/Fargate/AWS Batch (spot) or an auto-stopped instance; no always-on servers. Data in S3 with lifecycle: Standard -> Glacier after 30-90 days, retained 3 years (Object Lock if compliance); query with Athena. Compress logs (Parquet), use small instances and spot/Savings Plans, alarms, tagging, budgets.

### **Q559. Do you create clusters in multiple regions and how do you manage them?**

Asked at (1): Amazon

Yes, for DR, latency or compliance. Manage with IaC per region, GitOps (ArgoCD ApplicationSets) from one repo, global traffic routing (Route53 latency/failover, Global Accelerator), replicated data layers (Aurora Global, S3 CRR), centralised observability and identity, and regular failover drills.

### **Q560. Landing zone, guardrails, SCP, Control Tower**

Asked at (1): HCL

A landing zone is a pre-configured, secure multi-account environment. AWS Control Tower sets it up with Organizations, SSO, centralised logging and audit accounts. Guardrails are governance rules - preventive ones implemented as SCPs, detective ones as AWS Config rules. SCPs set the maximum permissions for accounts in an OU.

# General DevOps & Concepts

6 questions

## Concepts

### Q561. Kafka / messaging services; AWS AppConfig; combine responses of multiple APIs

Asked at (2): JPMorgan, Sonata Software

Kafka is a distributed log for high-throughput event streaming (topics, partitions, consumer groups); alternatives: SQS/SNS, RabbitMQ, Kinesis. AppConfig (AWS) deploys application configuration/feature flags safely with validation and gradual rollout. Combining three API responses: call concurrently (async/parallel), set timeouts, handle partial failure, and merge in an aggregation layer (API gateway/BFF/GraphQL).

### Q562. What is MongoDB? Redis vs Memcached; which database would you use?

Asked at (2): Flentas, Others (company not named)

MongoDB is a document (NoSQL) database storing JSON-like documents with flexible schema, replica sets for HA and sharding for scale. Redis: in-memory store with data structures, persistence, replication, pub/sub; Memcached: simple multi-threaded key-value cache, no persistence. Choose Redis for richer features (sessions, queues), Memcached for pure simple caching.

### Q563. What is an API?

Asked at (2): Others (company not named), Volkswagen Group Digital

An Application Programming Interface is a contract that lets one program talk to another over defined requests/responses, usually REST over HTTP with JSON (GET, POST, PUT, DELETE). It hides internal implementation. Authentication uses tokens/API keys/OAuth.

### Q564. In Airflow, how do you debug a failed job?

Asked at (1): Others (company not named)

Open the DAG run in the UI, check the failed task's logs, task instance details (upstream/downstream states, retries), the scheduler/worker logs, connections and variables, resource issues; fix, then clear the task to re-run it. Add retries, alerts and SLAs.

### Q565. SQL: customers who placed more than 3 orders in the last 90 days

Asked at (1): Accolite

```
SELECT c.customer_name
FROM customers c
JOIN orders o ON o.customer_id = c.customer_id
WHERE o.order_date >= CURRENT_DATE - INTERVAL '90 days'
GROUP BY c.customer_id, c.customer_name
HAVING COUNT(*) > 3;
```

### Q566. Spring Boot Actuator

Asked at (1): JPMorgan

Add `spring-boot-starter-actuator` dependency and expose endpoints in `application.properties` (`management.endpoints.web.exposure.include=health,info,metrics,prometheus`). Provides `/actuator/health` for Kubernetes probes and metrics for Prometheus. Secure the endpoints.

# Behavioural & Project

8 questions

## Introduction & Experience

### **Q567. Introduce yourself / explain your current project, roles, day-to-day activities and previous experience**

Asked at (20): AMEX, Alphadyne, BMW TechWorks, CTS, EY, Emphasis, Flentas, IBM, Infosys +11 more

Use a 2-minute structure: Who (name, years of experience, core stack) -> What (project: purpose, users, architecture in one line) -> How (your responsibilities: CI/CD, Kubernetes, Terraform, monitoring) -> Impact (numbers: deployment time cut from 2h to 15 min, 99.9% uptime, 30% cost saving) -> Why this role.

Example project explanation: "We run a microservices platform on EKS. I own the CI/CD with GitHub Actions + ArgoCD, infrastructure in Terraform, and observability with Prometheus/Grafana; we deploy ~10 times a day with automatic rollback." Day-to-day: check alerts/dashboards, handle tickets/PRs, improve pipelines, patching, cost reviews, on-call. Be ready to draw the architecture.

### **Q568. What automation have you done / contributed? Do career achievements need to be big?**

Asked at (6): Amadeus Labs, CMT, Infosys, Perfios, TCS, Wipro

Give concrete examples with before/after: automated server patching with Ansible, self-service environment creation in Terraform, a Python script to stop idle dev instances (saved X\$/month), auto-rollback in pipelines. Achievements need not be big - small repeated-toil removals show ownership; always quantify (time saved, errors reduced).

## Teamwork & Communication

### **Q569. Working under tight deadlines / high pressure; managing a team; accountability and collaboration**

Asked at (3): Infosys, JPMorgan, Others (company not named)

STAR answers: prioritise by impact, break work into milestones, communicate risk early, ask for help, automate repeated steps, and avoid heroics that cause errors. Ownership: blameless culture - during a failure, fix first, then run a postmortem focused on process, not people. Introducing a new tool: show a small pilot with measurable gains, document, train, and get feedback to win buy-in.

### **Q570. How do you communicate technical issues to non-technical stakeholders, and avoid miscommunication with multilingual stakeholders?**

Asked at (2): Others (company not named), Wipro

Lead with business impact (who is affected, since when, workaround, ETA), avoid jargon, use analogies, give regular timed updates, and confirm understanding. For multilingual threads: short plain sentences, bullet points, written summaries with decisions/owners/dates, diagrams, and a recap call for critical items.

### **Q571. Do you have any questions for us?**

Asked at (1): JPMorgan

Always ask 2-3: "What does the on-call/incident process look like?", "What's the biggest reliability or delivery pain right now?", "How do you measure success for this role in 6 months?", "What is the tech stack and roadmap?" It shows interest and judgement.

### **Q572. How can AI assist developers in increasing productivity?**

Asked at (1): UST

Code completion and generation, test and documentation generation, faster debugging and log analysis, PR/code review assistance, writing IaC and scripts, and incident summarisation. Caveats: review everything, never paste secrets, check licensing and security. In DevOps: AIOps for anomaly detection and runbook automation.

### **Q573. Tell me about a successful DevOps transformation project and your role**

Asked at (1): Others (company not named)

STAR example: starting point (manual deployments, monthly releases), your actions (introduced Git workflow, CI/CD, IaC, monitoring, training), obstacles (resistance, skills) and how you won buy-in, results (deploy frequency from monthly to weekly/daily, lead time down, failure rate down) - cite DORA metrics.

## **Puzzles**

### **Q574. Puzzle: using two candles (each burns 60 min unevenly), measure 45 minutes**

Asked at (1): JPMorgan

Light candle A at both ends and candle B at one end at the same time. A burns out in 30 minutes; at that moment light the other end of B. B's remaining 30 minutes now burns in 15 minutes. Total = 45 minutes.

# Part 2 - Company Index

Questions asked at each company (click a number to jump to the answer). Duplicate variants were merged, so one entry can cover several differently-worded questions.

## Accenture - 7 questions

- Q217** Where is aws-auth stored and what does it do? [Kubernetes]
- Q244** How do you manage secrets and config at scale in GitOps without leaking them? [Kubernetes]
- Q339** What do terraform init, plan, apply, fmt do? Meaning of + and - in plan; auto-approve [Terraform & IaC]
- Q350** Multi-region Terraform: 3 regions, which region does an EC2 go to? Structure for EC2 in 3 regions [Terraform & IaC]
- Q379** An EC2 in a private subnet must receive inbound traffic (without NAT). How, and how does traffic flow... [AWS]
- Q392** Does S3 need a VPC? [AWS]
- Q409** You have 10 AWS accounts and cannot use access keys. How do you log in securely? [AWS]

## Accion Labs - 13 questions

- Q18** File permission types; creating a user without SSH access; mount vs directory; limiting space per... [Linux & OS]
- Q66** Which scripting languages have you used, and what automations/scripts have you written (cost... [Scripting & Coding]
- Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]
- Q120** How do you ensure high availability in Kubernetes (survive an AZ failure, multi-region, ECS Fargate/EKS)? [Kubernetes]
- Q129** A worker node is not joining the cluster - how do you debug? What if the master goes down? Can... [Kubernetes]
- Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]
- Q341** Terraform workspaces - what, why, requirements, and how you manage them [Terraform & IaC]
- Q347** How do you write a Terraform module for EKS; how do you create and manage clusters, and master/worker... [Terraform & IaC]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management]
- Q378** Describe your VPC / network architecture: subnets per tier, HA, sizing IP addresses. [AWS]
- Q513** Monitoring vs observability; metrics, logs and traces and their tools; observability architecture [Monitoring & Observability]
- Q532** How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix?... [SRE Concepts]
- Q540** What are SLI, SLO, SLA and error budget? How do you decide and set them? [SRE Concepts]

## Accolite - 5 questions

- Q67** Python fundamentals: lists vs tuples vs sets, shallow vs deep copy, decorators, inheritance, GIL,... [Scripting & Coding]
- Q72** Python scripts: failed job IDs from job logs, unique IPs from a live log, monitor a directory for new... [Scripting & Coding]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q565** SQL: customers who placed more than 3 orders in the last 90 days [General DevOps & Concepts]

## Akamai - 10 questions

- Q5** One-liners: remove first/last line with sed, count lines, find files > 1 GB, find ERROR... [Linux & OS]
- Q12** A Linux system goes down or is unreachable (SSH not working, private instances) - how do you... [Linux & OS]
- Q29** OSI model layers; TCP vs UDP [Networking & Security]
- Q30** Is 90.90.88.0/12 private or public? Is 90.0.0.9/0 public? [Networking & Security]
- Q31** IPv4 vs IPv6 [Networking & Security]
- Q32** Is IP X in CIDR Y? Validate an IPv4 address [Networking & Security]
- Q76** Python: what are 5/2 and 5//2? What does a=[0]; b={0}; a[0], b[0] give? [Scripting & Coding]
- Q92** CMD vs ENTRYPOINT [Docker & Containers]
- Q227** How do you create a Kustomize file? [Kubernetes]
- Q376** How is Route 53 configured and what routing policies exist? [AWS]

## Alphadyne - 20 questions

- Q24** Filesystem shows 50% used but you cannot write; df -ih shows 100% - why? Disk I/O; mount point space [Linux & OS]
- Q26** Where are modern Linux user passwords stored? What are good password security practices? [Linux & OS]
- Q49** How do you enable passwordless authentication between two servers? [Networking & Security]
- Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]
- Q131** Common Kubernetes errors and production issues you have faced (CrashLoopBackOff, ImagePullBackOff,... [Kubernetes])
- Q220** Limiting resource usage at namespace level (ResourceQuota and LimitRange) [Kubernetes]
- Q254** How do you onboard a new project to multiple environments and handle releases/hotfixes? [CI/CD]
- Q259** Design a CI/CD pipeline for a new app (Node.js, Angular/Java/.NET, developer gave only source code)... [CI/CD]
- Q276** Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters [CI/CD]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management])
- Q364** What is a playbook? Write one to install/configure Apache/Nginx and ensure the service starts on boot [Ansible & Config Management]
- Q367** Which Ansible modules have you used? (install packages, nginx, commands) [Ansible & Config Management]
- Q378** Describe your VPC / network architecture: subnets per tier, HA, sizing IP addresses. [AWS]
- Q379** An EC2 in a private subnet must receive inbound traffic (without NAT). How, and how does traffic flow... [AWS]
- Q386** How does SSL/TLS work with Certbot, Let's Encrypt and ACM? How is a certificate used with ALB/NLB,... [AWS]
- Q388** How would you migrate an app running on a public EC2 to a private subnet following best practices? [AWS]
- Q498** Connect VMs in different VNets; define a VNet; Azure Load Balancer; ExpressRoute [Azure & GCP]
- Q502** Set up Azure Bastion for secure access to VMs [Azure & GCP]
- Q532** How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix?... [SRE Concepts]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Altimetrik - 15 questions

- Q132** How do you debug a failed deployment or a failed pod (even when you cannot exec into it)? [Kubernetes]
- Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]
- Q254** How do you onboard a new project to multiple environments and handle releases/hotfixes? [CI/CD]
- Q314** Pipeline fails because resources already exist - how do you handle it (Remove, Import, Plan)? [Terraform & IaC]
- Q321** Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and... [Terraform & IaC]
- Q347** How do you write a Terraform module for EKS; how do you create and manage clusters, and master/worker... [Terraform & IaC]
- Q348** Terraform repository/folder structure; how do dev/qa/prod consume shared modules; Terraform architecture [Terraform & IaC]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q512** What monitoring tools and agents have you used and what did you do with them? What proactive... [Monitoring & Observability]
- Q516** Have you created dashboards in Grafana? [Monitoring & Observability]
- Q532** How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix?... [SRE Concepts]
- Q534** Blameless postmortems, reducing MTTR and preventing repeat incidents [SRE Concepts]
- Q540** What are SLI, SLO, SLA and error budget? How do you decide and set them? [SRE Concepts]
- Q541** Have you configured SLO-based alerting? [SRE Concepts]
- Q542** What is an ideal burn rate? [SRE Concepts]

## Amadeus Labs - 6 questions

- Q18** File permission types; creating a user without SSH access; mount vs directory; limiting space per... [Linux & OS]
- Q134** Ingress configured but the page doesn't load (or 404 from gateway) - what do you check? [Kubernetes]
- Q143** HPA is configured but pod-level autoscaling isn't happening - how do you fix it? [Kubernetes]
- Q167** Security context: can a pod be scheduled with runAsNonRoot: true and runAsUser: 0? [Kubernetes]
- Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]
- Q568** What automation have you done / contributed? Do career achievements need to be big? [Behavioural & Project]

## Amazon - 29 questions

- Q18** File permission types; creating a user without SSH access; mount vs directory; limiting space per... [Linux & OS]
- Q29** OSI model layers; TCP vs UDP [Networking & Security]
- Q79** Write a Python script to list running EC2 instances with the tag PROD [Scripting & Coding]
- Q100** Build an image in CI, tag and push it, and deploy updates (commands + GitHub Actions) [Docker & Containers]
- Q170** What happens to data when a DB pod is down? [Kubernetes]
- Q180** What is Container Network Interface (CNI)? [Kubernetes]
- Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]
- Q205** How do you back up an entire Kubernetes cluster regularly? [Kubernetes]
- Q220** Limiting resource usage at namespace level (ResourceQuota and LimitRange) [Kubernetes]
- Q233** Cluster Autoscaler vs Horizontal Pod Autoscaler (and Karpenter) [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q321** Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and... [Terraform & IaC]
- Q327** Terraform locals vs variables; types (list, string, map of objects, tuple) [Terraform & IaC]
- Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]
- Q384** How do you access an RDS in a private subnet securely? [AWS]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q402** How do you manage secrets in your project? (Secrets Manager, Parameter Store, Vault, encryption and... [AWS]
- Q403** How can you log in to an EC2 instance without a password, or if you lost the .pem key? [AWS]
- Q404** Types of IAM policies; explicit deny vs allow; why a user with a role/policy still cannot access S3;... [AWS]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]
- Q436** Which AWS storage/instance types exist (EC2 types, EBS access mode, EFS vs EBS, multi-attach, Storage... [AWS]
- Q455** What is a sticky session in ALB? What problems can it cause? [AWS]
- Q469** Sending EC2 log files to S3 on a schedule, checking CPU and alerting [AWS]
- Q471** Can a bucket in one region be accessed from another region? [AWS]
- Q475** RDS migration with minimal downtime, and upgrading MySQL 5.7 to 8.0 [AWS]
- Q513** Monitoring vs observability; metrics, logs and traces and their tools; observability architecture [Monitoring & Observability]
- Q529** You can't install Filebeat on worker nodes - how else do you collect logs? [Monitoring & Observability]
- Q559** Do you create clusters in multiple regions and how do you manage them? [Architecture & Design]

## AMEX - 10 questions

- Q115** Docker vs Docker Compose; startup order (DB before app) and depends\_on [Docker & Containers]
- Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]
- Q192** How do you achieve a zero-downtime application upgrade and redeploy? [Kubernetes]
- Q214** How is access control implemented in Kubernetes (RBAC)? [Kubernetes]
- Q438** How did you migrate an application from on-premises to AWS? [AWS]
- Q525** How would you build centralized logging across multiple clouds and environments? [Monitoring & Observability]
- Q532** How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix?... [SRE Concepts]
- Q540** What are SLI, SLO, SLA and error budget? How do you decide and set them? [SRE Concepts]
- Q548** DevOps vs SRE [SRE Concepts]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Arrise Solutions - 23 questions

- Q1** Linux file hierarchy; what do /var and /opt contain; is Linux an OS or kernel? Process/machine states [Linux & OS]
- Q8** Find and kill a process; check processes without ps or top [Linux & OS]
- Q13** How do you check server system load? [Linux & OS]
- Q14** What is Linux load average, how is it calculated and displayed? How do you check system load, CPU and... [Linux & OS]
- Q22** How does Linux boot? [Linux & OS]
- Q36** What happens when you type www.google.com in the browser? [Networking & Security]
- Q41** How do SSL/TLS certificates work; HTTP vs HTTPS; renew an expired App Gateway certificate; check a... [Networking & Security]
- Q82** VM vs Docker; does Docker have its own kernel? [Docker & Containers]
- Q104** Connect a backend container to RDS; isolate communication between two containers; Docker network types [Docker & Containers]
- Q106** Storage drivers in Docker; container runtimes [Docker & Containers]
- Q109** How are Docker images built - what are layers, how does layer caching work, and where does a rebuild... [Docker & Containers]

- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q180** What is Container Network Interface (CNI)? [Kubernetes]
- Q182** How does kube-proxy communicate with nodes / work? [Kubernetes]
- Q185** CoreDNS in Kubernetes [Kubernetes]
- Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]
- Q381** How do you connect AWS to on-prem or to a client's environment (Direct Connect, VPN)? [AWS]
- Q386** How does SSL/TLS work with Certbot, Let's Encrypt and ACM? How is a certificate used with ALB/NLB,... [AWS]
- Q387** Public vs private subnet - how can you tell, and what must be in place? [AWS]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]
- Q450** CloudFront - how do you use it to serve a frontend? [AWS]
- Q524** Which logs do you check (agent, console, database, slow query, kernel)? [Monitoring & Observability]

## Aspire - 14 questions

- Q53** git fetch vs git pull; git pull vs git clone; push and pull [Git & Version Control]
- Q59** git stash and pop [Git & Version Control]
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q210** How do you find and remove orphan resources in Kubernetes? [Kubernetes]
- Q234** ConfigMaps, PV and PVC [Kubernetes]
- Q314** Pipeline fails because resources already exist - how do you handle it (Remove, Import, Plan)? [Terraform & IaC]
- Q322** What provisioners exist in Terraform? [Terraform & IaC]
- Q339** What do terraform init, plan, apply, fmt do? Meaning of + and - in plan; auto-approve [Terraform & IaC]
- Q377** Security Groups vs NACLs (and subnet vs NACL)? [AWS]
- Q384** How do you access an RDS in a private subnet securely? [AWS]
- Q455** What is a sticky session in ALB? What problems can it cause? [AWS]
- Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]
- Q474** What is Cross-Region Replication (CRR) in S3? [AWS]
- Q536** App is down and returning 503 - what steps? When do 503 errors occur? [SRE Concepts]

## Belcan - 10 questions

- Q54** Everyday git commands, push local changes to remote, checkout and clone [Git & Version Control]
- Q59** git stash and pop [Git & Version Control]
- Q277** Jenkins administration: standalone setup, memory issues, copying jobs, backup, concurrent builds,... [CI/CD]
- Q278** Connect a new Jenkins to GitHub; triggers (webhook, poll SCM, after N commits) [CI/CD]
- Q280** Show a sample Jenkinsfile and explain stages, agent, post and environment blocks; what types of... [CI/CD]
- Q282** Give view-only access on 5 Jenkins jobs to other users [CI/CD]
- Q312** What is the tfstate file and what does it do? [Terraform & IaC]
- Q337** Name basic Terraform commands and explain each [Terraform & IaC]
- Q394** Elastic IP vs public IP? [AWS]
- Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]

## Blue Yonder - 5 questions

- Q230** What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a... [Kubernetes]
- Q288** SonarQube and Snyk integration, quality gates, tokens and static analysis [CI/CD]
- Q307** Stateful vs stateless applications; how do you manage data for stateless apps? [Terraform & IaC]
- Q503** Use Azure Key Vault secrets in AKS; how does a pod get rotated secrets? AKS cost optimisation [Azure & GCP]
- Q505** How do you integrate Entra ID with AKS, secure AKS, and make it highly available? [Azure & GCP]

## BMW TechWorks - 7 questions

- Q46** What authentication methods does HashiCorp Vault support? [Networking & Security]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q408** How do you scan EC2 instances for vulnerabilities, and fix a vulnerable or compromised instance? [AWS]
- Q438** How did you migrate an application from on-premises to AWS? [AWS]
- Q485** How do you patch EC2 instances? [AWS]
- Q488** Can you separate disk space on EC2 (app on one partition, observability stack on another)? How? [AWS]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Capgemini - 49 questions

- Q53** git fetch vs git pull; git pull vs git clone; push and pull [Git & Version Control]
- Q57** Git tags, stale branches, --force-with-lease vs --force, and what git add does internally [Git & Version Control]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q123** What happens in the backend when you run kubectl get pods? [Kubernetes]
- Q135** A ReplicaSet has 3 pods and one isn't coming up / pods remain Pending even after HPA - why? [Kubernetes]
- Q137** A Service isn't mapped to a deployment - what could be wrong? [Kubernetes]
- Q156** Compute reservation in manifests - requests vs limits [Kubernetes]
- Q159** What is an init container and why use it? [Kubernetes]
- Q164** Labels vs annotations [Kubernetes]
- Q174** Types of Kubernetes services [Kubernetes]
- Q176** How do you get a static IP for a Kubernetes service/ingress, and how do you expose the application? [Kubernetes]
- Q190** Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS [Kubernetes]
- Q191** Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines) [Kubernetes]
- Q218** How do you map an IAM role to a Kubernetes service account (IRSA) and how does the pod know which... [Kubernetes]
- Q220** Limiting resource usage at namespace level (ResourceQuota and LimitRange) [Kubernetes]
- Q236** Auto volume expansion; What is CSI and CSI drivers? [Kubernetes]
- Q238** Static vs dynamic volume provisioning [Kubernetes]
- Q241** How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity [Kubernetes]
- Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]
- Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]
- Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]
- Q313** How do you remove a stuck state lock; how are state conflicts managed? [Terraform & IaC]
- Q314** Pipeline fails because resources already exist - how do you handle it (Remove, Import, Plan)? [Terraform & IaC]
- Q318** terraform import vs data sources; bringing an unmanaged production environment under Terraform [Terraform & IaC]
- Q322** What provisioners exist in Terraform? [Terraform & IaC]
- Q364** What is a playbook? Write one to install/configure Apache/Nginx and ensure the service starts on boot [Ansible & Config Management]
- Q365** Deploy to 100 servers / patch 20+ servers with Ansible; install Ansible; Ansible Tower/AWX [Ansible & Config Management]
- Q366** Import vs include; block/rescue; passing output between tasks; templates [Ansible & Config Management]
- Q368** Ansible ad-hoc tasks: create 3 users mapped to groups, run only on DB servers excluding one, shut... [Ansible & Config Management]
- Q371** What are Ansible roles? How do you create them, structure a three-tier app, publish to Galaxy, and... [Ansible & Config Management]
- Q372** What is an inventory file? How do you keep large inventories clean and store variables for different... [Ansible & Config Management]
- Q373** What is Ansible Vault? How do you encrypt files/playbooks and store credentials? [Ansible & Config Management]
- Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]
- Q389** How many subnets can a VPC have, and can you extend a subnet's CIDR later? [AWS]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q402** How do you manage secrets in your project? (Secrets Manager, Parameter Store, Vault, encryption and... [AWS]
- Q403** How can you log in to an EC2 instance without a password, or if you lost the .pem key? [AWS]
- Q410** A cross-account scenario: EC2 in Account A needs to read tokens (secrets) in Account B [AWS]
- Q442** What is a Region and an Availability Zone? [AWS]
- Q448** How does AWS Auto Scaling work? (ASG, with ALB, scaling on memory/disk, ECS autoscaling) [AWS]
- Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]
- Q475** RDS migration with minimal downtime, and upgrading MySQL 5.7 to 8.0 [AWS]
- Q476** App is slow and you suspect RDS - what do you check? Memory pressure but cannot resize? [AWS]
- Q478** Multi-AZ vs Read Replica; secondary vs read-only RDS [AWS]
- Q481** Make RDS accept only one user connection at a time [AWS]
- Q489** Error "IP address limit exceeded" while launching an EC2 - how to fix? [AWS]
- Q490** How do you migrate an EC2 instance from one region to another? [AWS]
- Q507** Azure Key Vault: experience, access policies, and how MySQL can use it privately [Azure & GCP]
- Q523** How do you increase debug log level? Application-level metrics and APM? [Monitoring & Observability]

## CGI - 11 questions

- Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]
- Q94** Multistage Dockerfile and when it is useful (compiled languages?) [Docker & Containers]
- Q108** How do you scan Docker images (build time and registry level)? [Docker & Containers]
- Q138** Database connection from a pod works for others but not for you - how to troubleshoot? How do you... [Kubernetes]
- Q253** How do you reduce a slow pipeline (40-60 min -> 15-20 min)? Caching in Jenkins/GitHub Actions [CI/CD]
- Q276** Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters [CI/CD]
- Q281** What are executors and agents in Jenkins and how do they work? [CI/CD]
- Q287** Write a checkout stage with Git credentials [CI/CD]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q408** How do you scan EC2 instances for vulnerabilities, and fix a vulnerable or compromised instance? [AWS]
- Q524** Which logs do you check (agent, console, database, slow query, kernel)? [Monitoring & Observability]

## Cisco - 15 questions

- Q7** Write shell scripts: count processes of a user, replace a word in a file, first match of a pattern,... [Linux & OS]
- Q9** Scripts: copy new files from a directory to a remote server via SCP; restart a service if it fails... [Linux & OS]
- Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]
- Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]
- Q276** Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters [CI/CD]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q321** Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and... [Terraform & IaC]
- Q364** What is a playbook? Write one to install/configure Apache/Nginx and ensure the service starts on boot [Ansible & Config Management]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q413** How do you protect an API Gateway? [AWS]
- Q512** What monitoring tools and agents have you used and what did you do with them? What proactive... [Monitoring & Observability]
- Q523** How do you increase debug log level? Application-level metrics and APM? [Monitoring & Observability]
- Q532** How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix?... [SRE Concepts]
- Q537** Clients report 504 Gateway Timeout - how do you debug? [SRE Concepts]
- Q556** Prechecks before migration; checks after an upgrade; risks of upgrading; what triggers a rollback [Architecture & Design]

## CMT - 14 questions

- Q12** A Linux system goes down or is unreachable (SSH not working, private instances) - how do you... [Linux & OS]
- Q20** What are ulimits? (and PIDs) [Linux & OS]
- Q29** OSI model layers; TCP vs UDP [Networking & Security]
- Q31** IPv4 vs IPv6 [Networking & Security]
- Q35** DNS: TTL, A/AAAA/CNAME records, custom resolvers, DNS failover, creating records [Networking & Security]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q120** How do you ensure high availability in Kubernetes (survive an AZ failure, multi-region, ECS Fargate/EKS)? [Kubernetes]
- Q348** Terraform repository/folder structure; how do dev/qa/prod consume shared modules; Terraform architecture [Terraform & IaC]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q439** Event-driven architecture with S3, Lambda and SNS; and Lambda failures/retries (including external... [AWS]
- Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]
- Q449** Key CloudWatch metrics to focus on; how to monitor LB health, CPU spikes and web/app server availability? [AWS]
- Q532** How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix?... [SRE Concepts]
- Q568** What automation have you done / contributed? Do career achievements need to be big? [Behavioural & Project]

## Commonwealth Bank - 3 questions

- Q36** What happens when you type www.google.com in the browser? [Networking & Security]
- Q513** Monitoring vs observability; metrics, logs and traces and their tools; observability architecture [Monitoring & Observability]
- Q540** What are SLI, SLO, SLA and error budget? How do you decide and set them? [SRE Concepts]

## CTS - 35 questions

- Q4** Explain commands like finger, comm, netstat, jq, yq, at, shuf, lsblk, less, last, nc, mtr, iftop,... [Linux & OS]
- Q12** A Linux system goes down or is unreachable (SSH not working, private instances) - how do you... [Linux & OS]
- Q34** Which commands check network details and traffic flows on a system? [Networking & Security]
- Q41** How do SSL/TLS certificates work; HTTP vs HTTPS; renew an expired App Gateway certificate; check a... [Networking & Security]
- Q75** Python: separate list items by whether they start with 'a' or 'b' [Scripting & Coding]
- Q93** How do you reduce Docker image size? [Docker & Containers]
- Q131** Common Kubernetes errors and production issues you have faced (CrashLoopBackOff, ImagePullBackOff,... [Kubernetes])
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q190** Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS [Kubernetes]
- Q215** Did you configure ingress and egress rules (NetworkPolicy)? [Kubernetes]
- Q221** Role vs RoleBinding; ClusterRole vs ClusterRoleBinding - when to use which? [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD])
- Q278** Connect a new Jenkins to GitHub; triggers (webhook, poll SCM, after N commits) [CI/CD]
- Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q316** How do you make sure EC2 is not deleted when someone runs terraform destroy? [Terraform & IaC]
- Q319** backend.tf is in the repo but state doesn't appear in the storage account - why? [Terraform & IaC]
- Q337** Name basic Terraform commands and explain each [Terraform & IaC]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management])
- Q365** Deploy to 100 servers / patch 20+ servers with Ansible; install Ansible; Ansible Tower/AWX [Ansible & Config Management]
- Q374** How do you debug a playbook? What if it times out on one host or runs for hours? [Ansible & Config Management]
- Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]
- Q377** Security Groups vs NACLs (and subnet vs NACL)? [AWS]
- Q378** Describe your VPC / network architecture: subnets per tier, HA, sizing IP addresses. [AWS]
- Q379** An EC2 in a private subnet must receive inbound traffic (without NAT). How, and how does traffic flow... [AWS]
- Q383** An EC2 instance is unreachable (not a security group problem) / an app behind a load balancer is... [AWS]
- Q403** How can you log in to an EC2 instance without a password, or if you lost the .pem key? [AWS]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]
- Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]
- Q459** What is connection draining (deregistration delay)? [AWS]
- Q499** Application Gateway vs Front Door (and L7/L4) [Azure & GCP]
- Q536** App is down and returning 503 - what steps? When do 503 errors occur? [SRE Concepts]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Deloitte - 55 questions

- Q3** Difference between & and && in shell; cron expressions; scheduling every 15 minutes [Linux & OS]
- Q6** Shell scripting: store names of newly created files in a directory; assign/print variables; last 15... [Linux & OS]
- Q14** What is Linux load average, how is it calculated and displayed? How do you check system load, CPU and... [Linux & OS]
- Q38** What is a WAF (and the "application access firewall")? How do you customise it? [Networking & Security]
- Q53** git fetch vs git pull; git pull vs git clone; push and pull [Git & Version Control]
- Q55** How do you handle merge conflicts (and resolve automatically)? [Git & Version Control]
- Q56** Git cherry-pick [Git & Version Control]
- Q62** How do you migrate a Git repo from one platform to another (GitHub to GitLab) keeping history? [Git & Version Control]
- Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]
- Q66** Which scripting languages have you used, and what automations/scripts have you written (cost... [Scripting & Coding])
- Q94** Multistage Dockerfile and when it is useful (compiled languages?) [Docker & Containers]
- Q96** How do you pass values during docker build (ARG vs ENV)? How do you avoid leaking secrets in the build? [Docker & Containers]
- Q101** Where does docker pull fetch images from? How do you pull from a private registry, authenticate... [Docker & Containers]
- Q105** How do you expose an application in Docker, map ports (e.g. image on 8080, host on 9090), and what if... [Docker & Containers]
- Q108** How do you scan Docker images (build time and registry level)? [Docker & Containers]

- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q131** Common Kubernetes errors and production issues you have faced (CrashLoopBackOff, ImagePullBackOff,... [Kubernetes]) [Kubernetes]
- Q177** Load balancing in Kubernetes: ELB, Ingress, global load balancing, restricting public access to load... [Kubernetes]
- Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]
- Q202** Basic Kubernetes commands (also Docker and Ansible commands) [Kubernetes]
- Q203** Have you created an EKS cluster? Explain the process and prerequisites (compute, networking, access) [Kubernetes]
- Q204** How do you manage worker nodes, node groups, and how many clusters/add-ons do you manage? [Kubernetes]
- Q209** How do you connect to ECS containers and manage services like DBs, EC2, EKS, ECS? [Kubernetes]
- Q217** Where is aws-auth stored and what does it do? [Kubernetes]
- Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]
- Q229** What is Helm chart signing? Which tools sign charts? [Kubernetes]
- Q248** Fargate vs EKS worker nodes; EKS vs ECS; EBS disadvantages in EKS [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]) [CI/CD]
- Q256** How do you handle parallel execution and jobs in CI/CD (Jenkins, GitHub Actions)? [CI/CD]
- Q276** Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters [CI/CD]
- Q278** Connect a new Jenkins to GitHub; triggers (webhook, poll SCM, after N commits) [CI/CD]
- Q279** How do you troubleshoot failing, stuck or slow pipelines (build not starting, random artifact upload... [CI/CD]) [CI/CD]
- Q288** SonarQube and Snyk integration, quality gates, tokens and static analysis [CI/CD]
- Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]
- Q306** What is CloudFormation and how does it compare with Terraform? [Terraform & IaC]
- Q321** Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and... [Terraform & IaC]
- Q337** Name basic Terraform commands and explain each [Terraform & IaC]
- Q347** How do you write a Terraform module for EKS; how do you create and manage clusters, and master/worker... [Terraform & IaC]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management]) [Ansible & Config Management]
- Q376** How is Route 53 configured and what routing policies exist? [AWS]
- Q378** Describe your VPC / network architecture: subnets per tier, HA, sizing IP addresses. [AWS]
- Q380** VPC Peering: how does it work and what options connect VPC A and B (two-way or one-way)? [AWS]
- Q382** Transit Gateway vs VPC peering; have you used Transit Gateway? [AWS]
- Q383** An EC2 instance is unreachable (not a security group problem) / an app behind a load balancer is... [AWS]
- Q385** Purpose of VPC Endpoints and when to use them? [AWS]
- Q412** Get the count of running EC2 instances and attached EBS volumes across 50-60 AWS accounts without... [AWS]
- Q420** AWS Config vs CloudTrail [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q437** How do you back up AWS services (EC2, RDS, S3, EBS) and how does Aurora back up? [AWS]
- Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]
- Q453** How do you filter a specific IP in CloudWatch logs, S3 logs, or VPC Flow Logs? How to keep flow logs... [AWS]
- Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]
- Q461** Fargate vs EC2 launch type; ECS Service vs Task; setting up ECS on EC2 [AWS]
- Q487** Can we take AWS backups using shell scripting? [AWS]
- Q512** What monitoring tools and agents have you used and what did you do with them? What proactive... [Monitoring & Observability]

## Elixir Labs - 14 questions

- Q254** How do you onboard a new project to multiple environments and handle releases/hotfixes? [CI/CD]
- Q262** What challenges have you faced in CI/CD / a known critical production issue / prioritising multiple... [CI/CD]
- Q381** How do you connect AWS to on-prem or to a client's environment (Direct Connect, VPN)? [AWS]
- Q382** Transit Gateway vs VPC peering; have you used Transit Gateway? [AWS]
- Q437** How do you back up AWS services (EC2, RDS, S3, EBS) and how does Aurora back up? [AWS]
- Q466** S3 lifecycle policies and storage classes; how do you manage and version them? [AWS]
- Q492** Azure storage types, subscription types, Recovery Services Vault, backup choices and Azure Boards [Azure & GCP]
- Q494** Scaling strategy for compute-intensive vs I/O-intensive workloads; throttled Azure Function; migrations [Azure & GCP]
- Q496** A .NET three-tier app uses App Service (Web API), Function Apps (background jobs) and Service Bus -... [Azure & GCP]
- Q498** Connect VMs in different VNets; define a VNet; Azure Load Balancer; ExpressRoute [Azure & GCP]
- Q515** Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and... [Monitoring & Observability]
- Q527** Thousands of alerts arrive from a huge infrastructure with customer-defined thresholds - how do you... [Monitoring & Observability]

- Q543** Explain your DR strategy: RTO, RPO, failover and traffic redirection; DR for DevOps infrastructure;... [SRE Concepts]  
**Q556** Prechecks before migration; checks after an upgrade; risks of upgrading; what triggers a rollback [Architecture & Design]

## Emphasis - 17 questions

- Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]  
**Q101** Where does docker pull fetch images from? How do you pull from a private registry, authenticate... [Docker & Containers]  
**Q118** Kubernetes architecture and workflow [Kubernetes]  
**Q175** Ingress vs Ingress controller vs Gateway API [Kubernetes]  
**Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]  
**Q277** Jenkins administration: standalone setup, memory issues, copying jobs, backup, concurrent builds,... [CI/CD]  
**Q278** Connect a new Jenkins to GitHub; triggers (webhook, poll SCM, after N commits) [CI/CD]  
**Q280** Show a sample Jenkinsfile and explain stages, agent, post and environment blocks; what types of... [CI/CD]  
**Q321** Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and... [Terraform & IaC]  
**Q339** What do terraform init, plan, apply, fmt do? Meaning of + and - in plan; auto-approve [Terraform & IaC]  
**Q353** Promote a read replica to primary in Terraform; AWS resources created with Terraform [Terraform & IaC]  
**Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]  
**Q377** Security Groups vs NACLs (and subnet vs NACL)? [AWS]  
**Q384** How do you access an RDS in a private subnet securely? [AWS]  
**Q432** Design a highly available, fault-tolerant backend on AWS [AWS]  
**Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]  
**Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Encora - 16 questions

- Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]  
**Q118** Kubernetes architecture and workflow [Kubernetes]  
**Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]  
**Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]  
**Q179** How do you map SSL certificates to an Ingress? Rotate Key Vault certs for App Gateway with AKS ingress? [Kubernetes]  
**Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]  
**Q226** Helm commands, deploying via CI/CD, and rolling back a failed or partially deployed release [Kubernetes]  
**Q236** Auto volume expansion; What is CSI and CSI drivers? [Kubernetes]  
**Q314** Pipeline fails because resources already exist - how do you handle it (Remove, Import, Plan)? [Terraform & IaC]  
**Q323** Terraform lifecycle (create\_before\_destroy, prevent\_destroy, ignore\_changes) [Terraform & IaC]  
**Q328** count vs for\_each; what happens to already-created instances if you remove an item from the list? [Terraform & IaC]  
**Q332** Split 10.0.0.0/16 into /21 subnets using locals and cidrsubnet [Terraform & IaC]  
**Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]  
**Q414** How do you secure multiple microservices and websites? [AWS]  
**Q508** Managed Identity vs Service Principal [Azure & GCP]  
**Q533** Sudden traffic surge makes the web application unresponsive - what steps do you take to mitigate? [SRE Concepts]

## EPAM - 68 questions

- Q1** Linux file hierarchy; what do /var and /opt contain; is Linux an OS or kernel? Process/machine states [Linux & OS]  
**Q30** Is 90.90.88.0/12 private or public? Is 90.0.0.9/0 public? [Networking & Security]  
**Q33** Private vs public IP addresses; how many IPs does /32 allow? [Networking & Security]  
**Q38** What is a WAF (and the "application access firewall")? How do you customise it? [Networking & Security]  
**Q86** How do Lambda containers work? [Docker & Containers]  
**Q92** CMD vs ENTRYPOINT [Docker & Containers]  
**Q95** ADD vs COPY vs RUN [Docker & Containers]  
**Q97** AWS EC2 Image Builder; Packer [Docker & Containers]  
**Q98** RUN vs exec (docker exec) commands [Docker & Containers]  
**Q107** How do you secure Docker images and containers (scanning, base image maintenance, authorized images,... [Docker & Containers]  
**Q119** What is a CustomResourceDefinition (CRD)? Desired state vs actual state [Kubernetes]  
**Q120** How do you ensure high availability in Kubernetes (survive an AZ failure, multi-region, ECS Fargate/EKS)? [Kubernetes]  
**Q122** What are namespaces in Kubernetes/EKS? [Kubernetes]

- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q191** Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines) [Kubernetes]
- Q193** PodDisruptionBudget, voluntary disruptions, minAvailable vs maxUnavailable [Kubernetes]
- Q195** During a canary of 10%, how do you verify health and what metrics before going to 100%? How do you... [Kubernetes]
- Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]
- Q214** How is access control implemented in Kubernetes (RBAC)? [Kubernetes]
- Q215** Did you configure ingress and egress rules (NetworkPolicy)? [Kubernetes]
- Q216** Implement Kubernetes security on container and infrastructure sides; zero-trust networking without a... [Kubernetes]
- Q223** Explain control plane components and how you would harden them for production [Kubernetes]
- Q233** Cluster Autoscaler vs Horizontal Pod Autoscaler (and Karpenter) [Kubernetes]
- Q244** How do you manage secrets and config at scale in GitOps without leaking them? [Kubernetes]
- Q248** Fargate vs EKS worker nodes; EKS vs ECS; EBS disadvantages in EKS [Kubernetes]
- Q252** What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple... [CI/CD]
- Q260** Best practices to structure repos and pipelines in a large DevOps project; reusable templates for 50... [CI/CD]
- Q261** CI/CD for ML models; ephemeral environments; SLA monitoring in a payments pipeline; iOS build automation [CI/CD]
- Q274** In a monorepo, how do you build and deploy only the relevant services? [CI/CD]
- Q289** Design an end-to-end DevSecOps pipeline (compliance, PCI-DSS, GDPR/HIPAA); handling vulnerabilities... [CI/CD]
- Q295** Workload identity federation between GitHub Actions and Azure/GCP securely [CI/CD]
- Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q315** How do multiple teams deploy to the same AWS account without overwriting each other; how do you... [Terraform & IaC]
- Q324** What is a provider and what is a data block in Terraform? [Terraform & IaC]
- Q325** Explain Terraform dependency graph internals and resource dependencies; circular dependencies [Terraform & IaC]
- Q338** How do you manage multiple environments in Terraform? [Terraform & IaC]
- Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]
- Q349** Refactor a legacy Terraform codebase / monorepo with hundreds of states into modules safely; how to... [Terraform & IaC]
- Q354** How do you prevent insecure infrastructure changes pushed through Terraform (like quality gates for... [Terraform & IaC]
- Q377** Security Groups vs NACLs (and subnet vs NACL)? [AWS]
- Q379** An EC2 in a private subnet must receive inbound traffic (without NAT). How, and how does traffic flow... [AWS]
- Q381** How do you connect AWS to on-prem or to a client's environment (Direct Connect, VPN)? [AWS]
- Q382** Transit Gateway vs VPC peering; have you used Transit Gateway? [AWS]
- Q385** Purpose of VPC Endpoints and when to use them? [AWS]
- Q390** After connecting VPCs via Transit Gateway, how do you block A->B and B->C traffic? [AWS]
- Q391** Can you create two CIDR blocks (172.x and 192.x) in the same VPC? [AWS]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q411** An EC2 instance has an IAM role - what can we do with it and how does it work? [AWS]
- Q419** What is an OIDC provider in AWS and how do you federate (Azure AD/GitHub) to AWS? [AWS]
- Q436** Which AWS storage/instance types exist (EC2 types, EBS access mode, EFS vs EBS, multi-attach, Storage... [AWS]
- Q440** Spot vs Reserved instances [AWS]
- Q445** EC2 User Data [AWS]
- Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]
- Q448** How does AWS Auto Scaling work? (ASG, with ALB, scaling on memory/disk, ECS autoscaling) [AWS]
- Q449** Key CloudWatch metrics to focus on; how to monitor LB health, CPU spikes and web/app server availability? [AWS]
- Q450** CloudFront - how do you use it to serve a frontend? [AWS]
- Q451** CloudWatch: what is it and how do you create custom metrics? [AWS]
- Q453** How do you filter a specific IP in CloudWatch logs, S3 logs, or VPC Flow Logs? How to keep flow logs... [AWS]
- Q454** The team changed the AMI to an ASG launch template - how do you ensure the new version deploys properly? [AWS]
- Q457** During heavy load new instances take 2-3 minutes to become ready (and ASG is terminating instances)... [AWS]
- Q461** Fargate vs EC2 launch type; ECS Service vs Task; setting up ECS on EC2 [AWS]
- Q462** API Gateway: configuration, REST vs WebSocket APIs [AWS]
- Q464** What are the limitations of Lambda? [AWS]
- Q470** AMI vs snapshots [AWS]
- Q521** Extract errors from logs: fetch lines with 200 status, count ERR, split error and warning into... [Monitoring & Observability]
- Q525** How would you build centralized logging across multiple clouds and environments? [Monitoring & Observability]

**Q543** Explain your DR strategy: RTO, RPO, failover and traffic redirection; DR for DevOps infrastructure;... [SRE Concepts]

## EXL Service - 24 questions

**Q52** Git merge vs rebase (and squash) [Git & Version Control]

**Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]

**Q174** Types of Kubernetes services [Kubernetes]

**Q178** Complete request flow from the user's browser to the backend pod (Ingress, DNS, LB, Services) [Kubernetes]

**Q195** During a canary of 10%, how do you verify health and what metrics before going to 100%? How do you... [Kubernetes]

**Q215** Did you configure ingress and egress rules (NetworkPolicy)? [Kubernetes]

**Q243** ConfigMap vs Secret [Kubernetes]

**Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]

**Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]

**Q252** What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple... [CI/CD]

**Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]

**Q297** Credentials appear in pipeline logs, or a developer hard-coded a password and pushed it - what do you do? [CI/CD]

**Q313** How do you remove a stuck state lock; how are state conflicts managed? [Terraform & IaC]

**Q340** Do you run Terraform locally or through CI/CD? Explain the workflow; what if apply or the pipeline... [Terraform & IaC]

**Q348** Terraform repository/folder structure; how do dev/qa/prod consume shared modules; Terraform architecture [Terraform & IaC]

**Q398** Two VPCs have overlapping CIDRs and Transit Gateway isn't allowed. What do you do? [AWS]

**Q432** Design a highly available, fault-tolerant backend on AWS [AWS]

**Q436** Which AWS storage/instance types exist (EC2 types, EBS access mode, EFS vs EBS, multi-attach, Storage... [AWS]

**Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]

**Q503** Use Azure Key Vault secrets in AKS; how does a pod get rotated secrets? AKS cost optimisation [Azure & GCP]

**Q512** What monitoring tools and agents have you used and what did you do with them? What proactive... [Monitoring & Observability]

**Q515** Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and... [Monitoring & Observability]

**Q516** Have you created dashboards in Grafana? [Monitoring & Observability]

**Q543** Explain your DR strategy: RTO, RPO, failover and traffic redirection; DR for DevOps infrastructure;... [SRE Concepts]

## EY - 24 questions

**Q52** Git merge vs rebase (and squash) [Git & Version Control]

**Q82** VM vs Docker; does Docker have its own kernel? [Docker & Containers]

**Q118** Kubernetes architecture and workflow [Kubernetes]

**Q131** Common Kubernetes errors and production issues you have faced (CrashLoopBackOff, ImagePullBackOff,... [Kubernetes]

**Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]

**Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]

**Q174** Types of Kubernetes services [Kubernetes]

**Q193** PodDisruptionBudget, voluntary disruptions, minAvailable vs maxUnavailable [Kubernetes]

**Q202** Basic Kubernetes commands (also Docker and Ansible commands) [Kubernetes]

**Q226** Helm commands, deploying via CI/CD, and rolling back a failed or partially deployed release [Kubernetes]

**Q234** ConfigMaps, PV and PVC [Kubernetes]

**Q243** ConfigMap vs Secret [Kubernetes]

**Q246** Design an Istio setup for your Kubernetes cluster [Kubernetes]

**Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]

**Q258** Maven lifecycle, Maven release, and build tools for Java [CI/CD]

**Q268** GitLab CI: experience, rules, pipelines [CI/CD]

**Q276** Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters [CI/CD]

**Q277** Jenkins administration: standalone setup, memory issues, copying jobs, backup, concurrent builds,... [CI/CD]

**Q289** Design an end-to-end DevSecOps pipeline (compliance, PCI-DSS, GDPR/HIPAA); handling vulnerabilities... [CI/CD]

**Q303** Build artifacts vs pipeline artifacts; Nexus / Artifactory and storing dependencies [CI/CD]

**Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]

**Q512** What monitoring tools and agents have you used and what did you do with them? What proactive... [Monitoring & Observability]

**Q515** Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and... [Monitoring & Observability]

**Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## F5 - 16 questions

- Q36** What happens when you type `www.google.com` in the browser? [Networking & Security]
- Q38** What is a WAF (and the "application access firewall")? How do you customise it? [Networking & Security]
- Q41** How do SSL/TLS certificates work; HTTP vs HTTPS; renew an expired App Gateway certificate; check a... [Networking & Security]
- Q51** HTTP request headers and HTTP methods [Networking & Security]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q129** A worker node is not joining the cluster - how do you debug? What if the master goes down? Can... [Kubernetes]
- Q216** Implement Kubernetes security on container and infrastructure sides; zero-trust networking without a... [Kubernetes]
- Q219** How do you restrict everything except two services (NetworkPolicy)? How do you secure a web app from... [Kubernetes]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q330** What is the Terraform lock file (`.terraform.lock.hcl`)? [Terraform & IaC]
- Q362** Terraform best practices [Terraform & IaC]
- Q377** Security Groups vs NACLs (and subnet vs NACL)? [AWS]
- Q382** Transit Gateway vs VPC peering; have you used Transit Gateway? [AWS]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q422** Does a security group + WAF + DDoS protection stop bot attacks? [AWS]
- Q555** Why is PUT idempotent? What happens if I send the same PUT twice or change the location with the same... [Architecture & Design]

## Five9 - 8 questions

- Q18** File permission types; creating a user without SSH access; mount vs directory; limiting space per... [Linux & OS]
- Q35** DNS: TTL, A/AAAA/CNAME records, custom resolvers, DNS failover, creating records [Networking & Security]
- Q41** How do SSL/TLS certificates work; HTTP vs HTTPS; renew an expired App Gateway certificate; check a... [Networking & Security]
- Q68** Coding: print a number pattern; find pairs summing to a target; longest substring without repeating... [Scripting & Coding]
- Q322** What provisioners exist in Terraform? [Terraform & IaC]
- Q403** How can you log in to an EC2 instance without a password, or if you lost the `.pem` key? [AWS]
- Q467** Terraform state in S3: expire in 30 days? other options? S3 native locking command? [AWS]
- Q530** How is the ELK stack set up end to end; how Kibana connects to Elasticsearch; how logs are... [Monitoring & Observability]

## Flentas - 30 questions

- Q83** Docker image vs container [Docker & Containers]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q122** What are namespaces in Kubernetes/EKS? [Kubernetes]
- Q129** A worker node is not joining the cluster - how do you debug? What if the master goes down? Can... [Kubernetes]
- Q156** Compute reservation in manifests - requests vs limits [Kubernetes]
- Q157** Multi-container pods: can we run 1 container in 2 pods? Delete a pod and run multiple containers? Can... [Kubernetes]
- Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]
- Q204** How do you manage worker nodes, node groups, and how many clusters/add-ons do you manage? [Kubernetes]
- Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]
- Q231** Do you use Cluster Autoscaler or Karpenter? How do you provision Karpenter, how does it decide which... [Kubernetes]
- Q232** HPA with a StatefulSet; HPA vs VPA [Kubernetes]
- Q233** Cluster Autoscaler vs Horizontal Pod Autoscaler (and Karpenter) [Kubernetes]
- Q242** Pod scheduling arithmetic: requests/limits vs node capacity (e.g. 4 vCPU/16 GB limits and 2 vCPU/10... [Kubernetes]
- Q249** When would you prefer on-prem Kubernetes over EKS and vice versa? Why deploy frontend on EKS instead... [Kubernetes]
- Q267** Do you commit tfvars to Git? How do you run Terraform in a CD pipeline? [CI/CD]
- Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]
- Q310** Terraform vs Terragrunt [Terraform & IaC]
- Q312** What is the `tfstate` file and what does it do? [Terraform & IaC]
- Q325** Explain Terraform dependency graph internals and resource dependencies; circular dependencies [Terraform & IaC]
- Q342** Remove a resource from Terraform code without deleting it in the cloud; resources that must be... [Terraform & IaC]
- Q344** How do you roll back changes in Terraform? [Terraform & IaC]
- Q348** Terraform repository/folder structure; how do dev/qa/prod consume shared modules; Terraform architecture [Terraform & IaC]

**Q355** How do you handle secrets in Terraform? [Terraform & IaC]  
**Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]  
**Q448** How does AWS Auto Scaling work? (ASG, with ALB, scaling on memory/disk, ECS autoscaling) [AWS]  
**Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]  
**Q484** Why use RDS Proxy? [AWS]  
**Q503** Use Azure Key Vault secrets in AKS; how does a pod get rotated secrets? AKS cost optimisation [Azure & GCP]  
**Q562** What is MongoDB? Redis vs Memcached; which database would you use? [General DevOps & Concepts]  
**Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## HCL - 49 questions

**Q6** Shell scripting: store names of newly created files in a directory; assign/print variables; last 15... [Linux & OS]  
**Q55** How do you handle merge conflicts (and resolve automatically)? [Git & Version Control]  
**Q61** How and from where do you clone a repo; local vs remote repo [Git & Version Control]  
**Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]  
**Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]  
**Q94** Multistage Dockerfile and when it is useful (compiled languages?) [Docker & Containers]  
**Q115** Docker vs Docker Compose; startup order (DB before app) and depends\_on [Docker & Containers]  
**Q118** Kubernetes architecture and workflow [Kubernetes]  
**Q121** Is etcd SQL or NoSQL and why? [Kubernetes]  
**Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]  
**Q156** Compute reservation in manifests - requests vs limits [Kubernetes]  
**Q157** Multi-container pods: can we run 1 container in 2 pods? Delete a pod and run multiple containers? Can... [Kubernetes]  
**Q158** What is a StatefulSet and how is it different from a Deployment? [Kubernetes]  
**Q159** What is an init container and why use it? [Kubernetes]  
**Q161** What is a DaemonSet? How do you run Tomcat on each of 10 nodes? Which daemons run by default? [Kubernetes]  
**Q166** Scheduling: StatefulSet pods with rolling updates; what is the pod name after mongo-0 dies? [Kubernetes]  
**Q175** Ingress vs Ingress controller vs Gateway API [Kubernetes]  
**Q183** What is Calico, and how does it differ from AWS VPC CNI? [Kubernetes]  
**Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]  
**Q205** How do you back up an entire Kubernetes cluster regularly? [Kubernetes]  
**Q206** Restore etcd from a backup onto a new VM; what happens if etcd stops or latency spikes? [Kubernetes]  
**Q219** How do you restrict everything except two services (NetworkPolicy)? How do you secure a web app from... [Kubernetes]  
**Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]  
**Q232** HPA with a StatefulSet; HPA vs VPA [Kubernetes]  
**Q234** ConfigMaps, PV and PVC [Kubernetes]  
**Q241** How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity [Kubernetes]  
**Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]  
**Q252** What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple... [CI/CD]  
**Q255** AWS CodePipeline/CodeBuild/CodeDeploy - how to set up, and CodePipeline vs Jenkins [CI/CD]  
**Q260** Best practices to structure repos and pipelines in a large DevOps project; reusable templates for 50... [CI/CD]  
**Q288** SonarQube and Snyk integration, quality gates, tokens and static analysis [CI/CD]  
**Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]  
**Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]  
**Q301** What is a PAT, a service connection / connection string, and self-hosted vs Microsoft-hosted agent? [CI/CD]  
**Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]  
**Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]  
**Q382** Transit Gateway vs VPC peering; have you used Transit Gateway? [AWS]  
**Q385** Purpose of VPC Endpoints and when to use them? [AWS]  
**Q402** How do you manage secrets in your project? (Secrets Manager, Parameter Store, Vault, encryption and... [AWS]  
**Q404** Types of IAM policies; explicit deny vs allow; why a user with a role/policy still cannot access S3;... [AWS]  
**Q411** An EC2 instance has an IAM role - what can we do with it and how does it work? [AWS]  
**Q426** How do you share an encrypted (KMS) AMI from Account 1 to Account 2? [AWS]  
**Q427** IAM: group vs individual policies, onboarding new team members, and how do you secure an AWS account... [AWS]  
**Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]  
**Q438** How did you migrate an application from on-premises to AWS? [AWS]  
**Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]  
**Q466** S3 lifecycle policies and storage classes; how do you manage and version them? [AWS]

**Q515** Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and...

[Monitoring & Observability]

**Q560** Landing zone, guardrails, SCP, Control Tower [Architecture & Design]

## Hexaware - 12 questions

**Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]

**Q66** Which scripting languages have you used, and what automations/scripts have you written (cost... [Scripting & Coding]

**Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]

**Q190** Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS [Kubernetes]

**Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]

**Q254** How do you onboard a new project to multiple environments and handle releases/hotfixes? [CI/CD]

**Q263** Approvals in a CD pipeline [CI/CD]

**Q288** SonarQube and Snyk integration, quality gates, tokens and static analysis [CI/CD]

**Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]

**Q321** Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and...

[Terraform & IaC]

**Q339** What do terraform init, plan, apply, fmt do? Meaning of + and - in plan; auto-approve [Terraform & IaC]

**Q492** Azure storage types, subscription types, Recovery Services Vault, backup choices and Azure Boards [Azure & GCP]

## IBM - 61 questions

**Q2** Hard link vs soft (symbolic) link [Linux & OS]

**Q11** What is iptables? [Linux & OS]

**Q17** A service on the server is leaking memory - how do you troubleshoot? [Linux & OS]

**Q27** How do you disable root login on a server? [Linux & OS]

**Q37** Reverse proxy vs forward proxy; is a load balancer or Nginx required in this setup? [Networking & Security]

**Q44** A vulnerability is found in a production server/application (even if the fix takes months) - what do... [Networking & Security]

**Q49** How do you enable passwordless authentication between two servers? [Networking & Security]

**Q55** How do you handle merge conflicts (and resolve automatically)? [Git & Version Control]

**Q66** Which scripting languages have you used, and what automations/scripts have you written (cost... [Scripting & Coding]

**Q80** File scripts: find a missing file, rename .txt files with date, delete files older than 10 days, find... [Scripting & Coding]

**Q104** Connect a backend container to RDS; isolate communication between two containers; Docker network types [Docker & Containers]

**Q123** What happens in the backend when you run kubectl get pods? [Kubernetes]

**Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]

**Q129** A worker node is not joining the cluster - how do you debug? What if the master goes down? Can... [Kubernetes]

**Q131** Common Kubernetes errors and production issues you have faced (CrashLoopBackOff, ImagePullBackOff,... [Kubernetes]

**Q136** Common reasons a node becomes NotReady (or an unhealthy node group) and how to identify the root cause [Kubernetes]

**Q149** Jenkins pipeline randomly fails at deployment to EKS with timeouts on kubectl apply [Kubernetes]

**Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]

**Q177** Load balancing in Kubernetes: ELB, Ingress, global load balancing, restricting public access to load... [Kubernetes]

**Q192** How do you achieve a zero-downtime application upgrade and redeploy? [Kubernetes]

**Q203** Have you created an EKS cluster? Explain the process and prerequisites (compute, networking, access) [Kubernetes]

**Q206** Restore etcd from a backup onto a new VM; what happens if etcd stops or latency spikes? [Kubernetes]

**Q230** What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a... [Kubernetes]

**Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]

**Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]

**Q253** How do you reduce a slow pipeline (40-60 min -> 15-20 min)? Caching in Jenkins/GitHub Actions [CI/CD]

**Q256** How do you handle parallel execution and jobs in CI/CD (Jenkins, GitHub Actions)? [CI/CD]

**Q259** Design a CI/CD pipeline for a new app (Node.js, Angular/Java/.NET, developer gave only source code)... [CI/CD]

**Q278** Connect a new Jenkins to GitHub; triggers (webhook, poll SCM, after N commits) [CI/CD]

**Q281** What are executors and agents in Jenkins and how do they work? [CI/CD]

**Q289** Design an end-to-end DevSecOps pipeline (compliance, PCI-DSS, GDPR/HIPAA); handling vulnerabilities... [CI/CD]

**Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]

**Q306** What is CloudFormation and how does it compare with Terraform? [Terraform & IaC]

**Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]

**Q312** What is the tfstate file and what does it do? [Terraform & IaC]

**Q339** What do terraform init, plan, apply, fmt do? Meaning of + and - in plan; auto-approve [Terraform & IaC]

**Q340** Do you run Terraform locally or through CI/CD? Explain the workflow; what if apply or the pipeline... [Terraform & IaC]

- Q347** How do you write a Terraform module for EKS; how do you create and manage clusters, and master/worker... [Terraform & IaC]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management]
- Q364** What is a playbook? Write one to install/configure Apache/Nginx and ensure the service starts on boot [Ansible & Config Management]
- Q365** Deploy to 100 servers / patch 20+ servers with Ansible; install Ansible; Ansible Tower/AWX [Ansible & Config Management]
- Q367** Which Ansible modules have you used? (install packages, nginx, commands) [Ansible & Config Management]
- Q372** What is an inventory file? How do you keep large inventories clean and store variables for different... [Ansible & Config Management]
- Q373** What is Ansible Vault? How do you encrypt files/playbooks and store credentials? [Ansible & Config Management]
- Q374** How do you debug a playbook? What if it times out on one host or runs for hours? [Ansible & Config Management]
- Q402** How do you manage secrets in your project? (Secrets Manager, Parameter Store, Vault, encryption and... [AWS]
- Q405** An S3 bucket was made public by mistake. How do you secure and audit it? How do you secure a bucket... [AWS]
- Q407** How do you protect and encrypt data on EC2/EBS? [AWS]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]
- Q435** How do you design for an entire region going down (and keep data safe even if multi-cloud also... [AWS]
- Q439** Event-driven architecture with S3, Lambda and SNS; and Lambda failures/retries (including external... [AWS]
- Q441** How do you deploy an application to AWS, and how do you provision AWS services? [AWS]
- Q444** AWS vs Azure - which would you choose and why? [AWS]
- Q461** Fargate vs EC2 launch type; ECS Service vs Task; setting up ECS on EC2 [AWS]
- Q512** What monitoring tools and agents have you used and what did you do with them? What proactive... [Monitoring & Observability]
- Q521** Extract errors from logs: fetch lines with 200 status, count ERR, split error and warning into... [Monitoring & Observability]
- Q533** Sudden traffic surge makes the web application unresponsive - what steps do you take to mitigate? [SRE Concepts]
- Q535** A live app is slow/has latency for customers (e-commerce) - how do you troubleshoot? Which metrics do... [SRE Concepts]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Infinite Solutions - 3 questions

- Q187** How do you set up an ingress controller? [Kubernetes]
- Q205** How do you back up an entire Kubernetes cluster regularly? [Kubernetes]
- Q435** How do you design for an entire region going down (and keep data safe even if multi-cloud also... [AWS]

## Infosys - 45 questions

- Q3** Difference between & and && in shell; cron expressions; scheduling every 15 minutes [Linux & OS]
- Q19** What is logrotate? Script to compress logs older than 30 days, delete older than 90 days, back up the... [Linux & OS]
- Q52** Git merge vs rebase (and squash) [Git & Version Control]
- Q54** Everyday git commands, push local changes to remote, checkout and clone [Git & Version Control]
- Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]
- Q66** Which scripting languages have you used, and what automations/scripts have you written (cost... [Scripting & Coding]
- Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]
- Q92** CMD vs ENTRYPOINT [Docker & Containers]
- Q93** How do you reduce Docker image size? [Docker & Containers]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]
- Q131** Common Kubernetes errors and production issues you have faced (CrashLoopBackOff, ImagePullBackOff,... [Kubernetes]
- Q132** How do you debug a failed deployment or a failed pod (even when you cannot exec into it)? [Kubernetes]
- Q133** How do you monitor pod health/performance and alert when pods go down? [Kubernetes]
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q157** Multi-container pods: can we run 1 container in 2 pods? Delete a pod and run multiple containers? Can... [Kubernetes]
- Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]
- Q177** Load balancing in Kubernetes: ELB, Ingress, global load balancing, restricting public access to load... [Kubernetes]
- Q191** Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines) [Kubernetes]

- Q202** Basic Kubernetes commands (also Docker and Ansible commands) [Kubernetes]
- Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]
- Q230** What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a... [Kubernetes]
- Q241** How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD])
- Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]
- Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]
- Q331** Write a Terraform file to launch an EC2 in a public subnet in a VPC [Terraform & IaC]
- Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management])
- Q367** Which Ansible modules have you used? (install packages, nginx, commands) [Ansible & Config Management]
- Q377** Security Groups vs NACLs (and subnet vs NACL)? [AWS]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS])
- Q405** An S3 bucket was made public by mistake. How do you secure and audit it? How do you secure a bucket... [AWS]
- Q512** What monitoring tools and agents have you used and what did you do with them? What proactive... [Monitoring & Observability]
- Q513** Monitoring vs observability; metrics, logs and traces and their tools; observability architecture [Monitoring & Observability]
- Q515** Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and... [Monitoring & Observability]
- Q517** How does Prometheus fetch data? [Monitoring & Observability]
- Q522** How do you create custom alerts and handle CPU/disk alerts (e.g. CPU/memory beyond 75%)? How do you... [Monitoring & Observability]
- Q530** How is the ELK stack set up end to end; how Kibana connects to Elasticsearch; how logs are... [Monitoring & Observability]
- Q538** Users complain about latency but monitoring says healthy - what do you do? How does monitoring help... [SRE Concepts]
- Q540** What are SLI, SLO, SLA and error budget? How do you decide and set them? [SRE Concepts]
- Q550** What is chaos engineering? [SRE Concepts]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]
- Q568** What automation have you done / contributed? Do career achievements need to be big? [Behavioural & Project]
- Q569** Working under tight deadlines / high pressure; managing a team; accountability and collaboration [Behavioural & Project]

## Intact Green Services - 11 questions

- Q93** How do you reduce Docker image size? [Docker & Containers]
- Q101** Where does docker pull fetch images from? How do you pull from a private registry, authenticate... [Docker & Containers]
- Q119** What is a CustomResourceDefinition (CRD)? Desired state vs actual state [Kubernetes]
- Q157** Multi-container pods: can we run 1 container in 2 pods? Delete a pod and run multiple containers? Can... [Kubernetes]
- Q174** Types of Kubernetes services [Kubernetes]
- Q243** ConfigMap vs Secret [Kubernetes]
- Q276** Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters [CI/CD]
- Q277** Jenkins administration: standalone setup, memory issues, copying jobs, backup, concurrent builds,... [CI/CD]
- Q280** Show a sample Jenkinsfile and explain stages, agent, post and environment blocks; what types of... [CI/CD]
- Q380** VPC Peering: how does it work and what options connect VPC A and B (two-way or one-way)? [AWS]
- Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]

## ITC Infotech - 6 questions

- Q178** Complete request flow from the user's browser to the backend pod (Ingress, DNS, LB, Services) [Kubernetes]
- Q181** CoreDNS vs kube-proxy; why are kubelet and kube-proxy needed? [Kubernetes]
- Q347** How do you write a Terraform module for EKS; how do you create and manage clusters, and master/worker... [Terraform & IaC]
- Q381** How do you connect AWS to on-prem or to a client's environment (Direct Connect, VPN)? [AWS]
- Q419** What is an OIDC provider in AWS and how do you federate (Azure AD/GitHub) to AWS? [AWS]
- Q469** Sending EC2 log files to S3 on a schedule, checking CPU and alerting [AWS]

## JPMorgan - 51 questions

- Q48** Explain Okta integration for identity and access management [Networking & Security]
- Q68** Coding: print a number pattern; find pairs summing to a target; longest substring without repeating... [Scripting & Coding]
- Q70** Given a log with ERROR/INFO lines (timestamps, connection failed, retries), parse it and report... [Scripting & Coding]
- Q117** Disaster recovery for stateful apps in containers [Docker & Containers]
- Q130** Are readiness probes failing? Can a container restart itself? What is the pause container? [Kubernetes]
- Q133** How do you monitor pod health/performance and alert when pods go down? [Kubernetes]
- Q144** High CPU in one pod but clean logs - what next? [Kubernetes]
- Q147** In a canary deployment half the traffic returns 502 - how do you troubleshoot? [Kubernetes]
- Q148** Intermittent DNS resolution issues during deployment - causes? [Kubernetes]
- Q153** Users see a 10-second delay every 15 minutes in an app on AKS/Kubernetes, with no code change - how... [Kubernetes]
- Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]
- Q191** Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines) [Kubernetes]
- Q196** With a single replica, does a rolling restart cause downtime? What happens step by step? [Kubernetes]
- Q230** What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a... [Kubernetes]
- Q248** Fargate vs EKS worker nodes; EKS vs ECS; EBS disadvantages in EKS [Kubernetes]
- Q253** How do you reduce a slow pipeline (40-60 min -> 15-20 min)? Caching in Jenkins/GitHub Actions [CI/CD]
- Q257** How do you implement rollback in a CI/CD pipeline? Describe a rollback you handled [CI/CD]
- Q261** CI/CD for ML models; ephemeral environments; SLA monitoring in a payments pipeline; iOS build automation [CI/CD]
- Q264** Code quality vs code coverage [CI/CD]
- Q275** Production pipeline is blocked on missing approvals and stakeholders are unreachable [CI/CD]
- Q279** How do you troubleshoot failing, stuck or slow pipelines (build not starting, random artifact upload... [CI/CD]
- Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q318** terraform import vs data sources; bringing an unmanaged production environment under Terraform [Terraform & IaC]
- Q354** How do you prevent insecure infrastructure changes pushed through Terraform (like quality gates for... [Terraform & IaC]
- Q388** How would you migrate an app running on a public EC2 to a private subnet following best practices? [AWS]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q414** How do you secure multiple microservices and websites? [AWS]
- Q429** Recent AWS/Azure outages - what were your key takeaways? [AWS]
- Q435** How do you design for an entire region going down (and keep data safe even if multi-cloud also... [AWS]
- Q441** How do you deploy an application to AWS, and how do you provision AWS services? [AWS]
- Q442** What is a Region and an Availability Zone? [AWS]
- Q448** How does AWS Auto Scaling work? (ASG, with ALB, scaling on memory/disk, ECS autoscaling) [AWS]
- Q452** Failover mechanism in load balancers; active-passive vs active-active [AWS]
- Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]
- Q493** Design a highly available three-tier architecture in Azure (React frontend + Node backend); App... [Azure & GCP]
- Q494** Scaling strategy for compute-intensive vs I/O-intensive workloads; throttled Azure Function; migrations [Azure & GCP]
- Q515** Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and... [Monitoring & Observability]
- Q528** Design a highly available logging system for 100+ microservices across 3 regions [Monitoring & Observability]
- Q531** In an EFK/ELK stack, what does each component do? How does filtering and indexing work in Elasticsearch? [Monitoring & Observability]
- Q539** A production app works for internal users but fails for external users with 403 - how to isolate? [SRE Concepts]
- Q546** How do you perform load testing? [SRE Concepts]
- Q552** Payment succeeds but order/shipping service fails - how do you handle it? Optimistic locking in a... [Architecture & Design]
- Q553** Service discovery when moving from monolith to microservices; zero-downtime DB migrations; retry... [Architecture & Design]
- Q558** Design a cost-optimised architecture for a nightly internal reporting app with 3-year log retention [Architecture & Design]
- Q561** Kafka / messaging services; AWS AppConfig; combine responses of multiple APIs [General DevOps & Concepts]
- Q566** Spring Boot Actuator [General DevOps & Concepts]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

**Q569** Working under tight deadlines / high pressure; managing a team; accountability and collaboration [Behavioural & Project]

**Q571** Do you have any questions for us? [Behavioural & Project]

**Q574** Puzzle: using two candles (each burns 60 min unevenly), measure 45 minutes [Behavioural & Project]

## Koerber Pharma - 33 questions

**Q45** You find malware on a client's machine - how do you clean up and prevent it? [Networking & Security]

**Q92** CMD vs ENTRYPOINT [Docker & Containers]

**Q103** Bind mounts vs volumes; docker volume and prune; -p flag [Docker & Containers]

**Q118** Kubernetes architecture and workflow [Kubernetes]

**Q130** Are readiness probes failing? Can a container restart itself? What is the pause container? [Kubernetes]

**Q241** How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity [Kubernetes]

**Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]

**Q277** Jenkins administration: standalone setup, memory issues, copying jobs, backup, concurrent builds,... [CI/CD]

**Q307** Stateful vs stateless applications; how do you manage data for stateless apps? [Terraform & IaC]

**Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]

**Q322** What provisioners exist in Terraform? [Terraform & IaC]

**Q338** How do you manage multiple environments in Terraform? [Terraform & IaC]

**Q340** Do you run Terraform locally or through CI/CD? Explain the workflow; what if apply or the pipeline... [Terraform & IaC]

**Q341** Terraform workspaces - what, why, requirements, and how you manage them [Terraform & IaC]

**Q358** Enable Terraform debug logs; investigate and validate Terraform errors [Terraform & IaC]

**Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]

**Q381** How do you connect AWS to on-prem or to a client's environment (Direct Connect, VPN)? [AWS]

**Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]

**Q406** What is the difference between IAM Users and IAM Roles? [AWS]

**Q409** You have 10 AWS accounts and cannot use access keys. How do you log in securely? [AWS]

**Q418** What is CloudTrail and what do you use it for? [AWS]

**Q424** How do you handle certificate management and expiry in AWS? [AWS]

**Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]

**Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]

**Q436** Which AWS storage/instance types exist (EC2 types, EBS access mode, EFS vs EBS, multi-attach, Storage... [AWS]

**Q437** How do you back up AWS services (EC2, RDS, S3, EBS) and how does Aurora back up? [AWS]

**Q440** Spot vs Reserved instances [AWS]

**Q449** Key CloudWatch metrics to focus on; how to monitor LB health, CPU spikes and web/app server availability? [AWS]

**Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]

**Q468** S3 vs EBS (and EFS)? [AWS]

**Q513** Monitoring vs observability; metrics, logs and traces and their tools; observability architecture [Monitoring & Observability]

**Q535** A live app is slow/has latency for customers (e-commerce) - how do you troubleshoot? Which metrics do... [SRE Concepts]

**Q545** How do you reduce latency for users from abroad? Low-latency routing? [SRE Concepts]

## L and T - 20 questions

**Q54** Everyday git commands, push local changes to remote, checkout and clone [Git & Version Control]

**Q55** How do you handle merge conflicts (and resolve automatically)? [Git & Version Control]

**Q56** Git cherry-pick [Git & Version Control]

**Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]

**Q82** VM vs Docker; does Docker have its own kernel? [Docker & Containers]

**Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]

**Q115** Docker vs Docker Compose; startup order (DB before app) and depends\_on [Docker & Containers]

**Q118** Kubernetes architecture and workflow [Kubernetes]

**Q130** Are readiness probes failing? Can a container restart itself? What is the pause container? [Kubernetes]

**Q157** Multi-container pods: can we run 1 container in 2 pods? Delete a pod and run multiple containers? Can... [Kubernetes]

**Q175** Ingress vs Ingress controller vs Gateway API [Kubernetes]

**Q176** How do you get a static IP for a Kubernetes service/ingress, and how do you expose the application? [Kubernetes]

**Q182** How does kube-proxy communicate with nodes / work? [Kubernetes]

**Q214** How is access control implemented in Kubernetes (RBAC)? [Kubernetes]

**Q243** ConfigMap vs Secret [Kubernetes]

**Q364** What is a playbook? Write one to install/configure Apache/Nginx and ensure the service starts on boot [Ansible & Config Management]

- Q371** What are Ansible roles? How do you create them, structure a three-tier app, publish to Galaxy, and... [Ansible & Config Management]
- Q372** What is an inventory file? How do you keep large inventories clean and store variables for different... [Ansible & Config Management]
- Q373** What is Ansible Vault? How do you encrypt files/playbooks and store credentials? [Ansible & Config Management]
- Q557** Approach to migrating a monolith to containers/microservices [Architecture & Design]

## LTIMindtree - 42 questions

- Q9** Scripts: copy new files from a directory to a remote server via SCP; restart a service if it fails... [Linux & OS]
- Q52** Git merge vs rebase (and squash) [Git & Version Control]
- Q54** Everyday git commands, push local changes to remote, checkout and clone [Git & Version Control]
- Q56** Git cherry-pick [Git & Version Control]
- Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]
- Q84** Can you run Docker inside a container (DinD)? [Docker & Containers]
- Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]
- Q107** How do you secure Docker images and containers (scanning, base image maintenance, authorized images,... [Docker & Containers]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]
- Q130** Are readiness probes failing? Can a container restart itself? What is the pause container? [Kubernetes]
- Q133** How do you monitor pod health/performance and alert when pods go down? [Kubernetes]
- Q134** Ingress configured but the page doesn't load (or 404 from gateway) - what do you check? [Kubernetes]
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q190** Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS [Kubernetes]
- Q191** Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines) [Kubernetes]
- Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]
- Q212** How to run a YAML manifest without creating a file? [Kubernetes]
- Q244** How do you manage secrets and config at scale in GitOps without leaking them? [Kubernetes]
- Q252** What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple... [CI/CD]
- Q253** How do you reduce a slow pipeline (40-60 min -> 15-20 min)? Caching in Jenkins/GitHub Actions [CI/CD]
- Q255** AWS CodePipeline/CodeBuild/CodeDeploy - how to set up, and CodePipeline vs Jenkins [CI/CD]
- Q256** How do you handle parallel execution and jobs in CI/CD (Jenkins, GitHub Actions)? [CI/CD]
- Q257** How do you implement rollback in a CI/CD pipeline? Describe a rollback you handled [CI/CD]
- Q264** Code quality vs code coverage [CI/CD]
- Q276** Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters [CI/CD]
- Q277** Jenkins administration: standalone setup, memory issues, copying jobs, backup, concurrent builds,... [CI/CD]
- Q279** How do you troubleshoot failing, stuck or slow pipelines (build not starting, random artifact upload... [CI/CD]
- Q288** SonarQube and Snyk integration, quality gates, tokens and static analysis [CI/CD]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q321** Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and... [Terraform & IaC]
- Q328** count vs for\_each; what happens to already-created instances if you remove an item from the list? [Terraform & IaC]
- Q338** How do you manage multiple environments in Terraform? [Terraform & IaC]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management]
- Q402** How do you manage secrets in your project? (Secrets Manager, Parameter Store, Vault, encryption and... [AWS]
- Q431** Secure a Lambda function [AWS]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]
- Q467** Terraform state in S3: expire in 30 days? other options? S3 native locking command? [AWS]
- Q500** Firewall vs NSG [Azure & GCP]
- Q522** How do you create custom alerts and handle CPU/disk alerts (e.g. CPU/memory beyond 75%)? How do you... [Monitoring & Observability]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Marsh McLennan - 11 questions

- Q46** What authentication methods does HashiCorp Vault support? [Networking & Security]
- Q52** Git merge vs rebase (and squash) [Git & Version Control]
- Q93** How do you reduce Docker image size? [Docker & Containers]
- Q94** Multistage Dockerfile and when it is useful (compiled languages?) [Docker & Containers]
- Q109** How are Docker images built - what are layers, how does layer caching work, and where does a rebuild... [Docker & Containers]
- Q226** Helm commands, deploying via CI/CD, and rolling back a failed or partially deployed release [Kubernetes]
- Q243** ConfigMap vs Secret [Kubernetes]
- Q291** GitHub Actions: workflow file, Java Maven build and artifact, manual trigger, concurrency, runners [CI/CD]
- Q292** needs keyword, concurrency, and matrix strategy in GitHub Actions [CI/CD]
- Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]
- Q402** How do you manage secrets in your project? (Secrets Manager, Parameter Store, Vault, encryption and... [AWS]

## Moodys - 6 questions

- Q102** How do you version/manage images in ECR, delete old or untagged ones, and push to ACR? [Docker & Containers]
- Q162** Deploying batch/ML jobs without impacting running jobs [Kubernetes]
- Q261** CI/CD for ML models; ephemeral environments; SLA monitoring in a payments pipeline; iOS build automation [CI/CD]
- Q293** Review a GitHub Actions workflow snippet: how do you spot incorrect or missing steps? [CI/CD]
- Q425** How do you prevent unauthorized users from spinning up expensive services (e.g. ML) in AWS? [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]

## Morgan Stanley - 12 questions

- Q1** Linux file hierarchy; what do /var and /opt contain; is Linux an OS or kernel? Process/machine states [Linux & OS]
- Q2** Hard link vs soft (symbolic) link [Linux & OS]
- Q5** One-liners: remove first/last line with sed, count lines, find files > 1 GB, find ERROR... [Linux & OS]
- Q10** stdin, stdout, stderr; types of variables in Linux [Linux & OS]
- Q20** What are ulimits? (and PIDs) [Linux & OS]
- Q23** System date is behind the current date on a VM - how to fix it? [Linux & OS]
- Q37** Reverse proxy vs forward proxy; is a load balancer or Nginx required in this setup? [Networking & Security]
- Q42** How to create certificates for multiple sub-domains; Cross-Origin Resource Sharing (CORS) [Networking & Security]
- Q80** File scripts: find a missing file, rename .txt files with date, delete files older than 10 days, find... [Scripting & Coding]
- Q215** Did you configure ingress and egress rules (NetworkPolicy)? [Kubernetes]
- Q376** How is Route 53 configured and what routing policies exist? [AWS]
- Q521** Extract errors from logs: fetch lines with 200 status, count ERR, split error and warning into... [Monitoring & Observability]

## NatWest Group - 2 questions

- Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]
- Q258** Maven lifecycle, Maven release, and build tools for Java [CI/CD]

## Netcracker - 16 questions

- Q4** Explain commands like finger, comm, netstat, jq, yq, at, shuf, lsblk, less, last, nc, mtr, iftop,... [Linux & OS]
- Q18** File permission types; creating a user without SSH access; mount vs directory; limiting space per... [Linux & OS]
- Q24** Filesystem shows 50% used but you cannot write; df -ih shows 100% - why? Disk I/O; mount point space [Linux & OS]
- Q119** What is a CustomResourceDefinition (CRD)? Desired state vs actual state [Kubernetes]
- Q134** Ingress configured but the page doesn't load (or 404 from gateway) - what do you check? [Kubernetes]
- Q156** Compute reservation in manifests - requests vs limits [Kubernetes]
- Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]
- Q175** Ingress vs Ingress controller vs Gateway API [Kubernetes]
- Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]
- Q207** How do you handle node draining? [Kubernetes]
- Q214** How is access control implemented in Kubernetes (RBAC)? [Kubernetes]
- Q215** Did you configure ingress and egress rules (NetworkPolicy)? [Kubernetes]
- Q221** Role vs RoleBinding; ClusterRole vs ClusterRoleBinding - when to use which? [Kubernetes]
- Q237** Storage classes in Kubernetes; what happens when a Deployment with one PVC is scaled from 1 to 3? [Kubernetes]
- Q241** How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity [Kubernetes]
- Q515** Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and... [Monitoring & Observability]

## Nextturn - 17 questions

- Q100** Build an image in CI, tag and push it, and deploy updates (commands + GitHub Actions) [Docker & Containers]
- Q109** How are Docker images built - what are layers, how does layer caching work, and where does a rebuild... [Docker & Containers]
- Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]
- Q136** Common reasons a node becomes NotReady (or an unhealthy node group) and how to identify the root cause [Kubernetes]
- Q138** Database connection from a pod works for others but not for you - how to troubleshoot? How do you... [Kubernetes]
- Q226** Helm commands, deploying via CI/CD, and rolling back a failed or partially deployed release [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q252** What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple... [CI/CD]
- Q265** GitHub Actions vs Argo CD; how does CI differ from GitOps CD? [CI/CD]
- Q277** Jenkins administration: standalone setup, memory issues, copying jobs, backup, concurrent builds,... [CI/CD]
- Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]
- Q309** Terraform vs Ansible; when to use each and how they work together [Terraform & IaC]
- Q329** Create multiple EC2 instances with different configurations; create 50 instances at once [Terraform & IaC]
- Q337** Name basic Terraform commands and explain each [Terraform & IaC]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management]
- Q491** You cannot SSH into an EC2 - how can you still install a package (e.g. tree)? [AWS]
- Q526** Script to monitor disk usage and email/log an alert above 80% [Monitoring & Observability]

## Nice - 19 questions

- Q87** How do you design a containerised environment for scalability and high availability? [Docker & Containers]
- Q92** CMD vs ENTRYPOINT [Docker & Containers]
- Q95** ADD vs COPY vs RUN [Docker & Containers]
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q198** How do you manage deployment, scaling and rollback in Kubernetes? How to roll back a failed deployment? [Kubernetes]
- Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q252** What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple... [CI/CD]
- Q262** What challenges have you faced in CI/CD / a known critical production issue / prioritising multiple... [CI/CD]
- Q288** SonarQube and Snyk integration, quality gates, tokens and static analysis [CI/CD]
- Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]
- Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]
- Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]
- Q343** Describe a real scenario where you used Terraform to provision scalable infrastructure [Terraform & IaC]
- Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]
- Q507** Azure Key Vault: experience, access policies, and how MySQL can use it privately [Azure & GCP]
- Q512** What monitoring tools and agents have you used and what did you do with them? What proactive... [Monitoring & Observability]
- Q516** Have you created dashboards in Grafana? [Monitoring & Observability]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Nisum Technologies - 11 questions

- Q95** ADD vs COPY vs RUN [Docker & Containers]
- Q107** How do you secure Docker images and containers (scanning, base image maintenance, authorized images,... [Docker & Containers]
- Q129** A worker node is not joining the cluster - how do you debug? What if the master goes down? Can... [Kubernetes]
- Q135** A ReplicaSet has 3 pods and one isn't coming up / pods remain Pending even after HPA - why? [Kubernetes]
- Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]
- Q327** Terraform locals vs variables; types (list, string, map of objects, tuple) [Terraform & IaC]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management]
- Q461** Fargate vs EC2 launch type; ECS Service vs Task; setting up ECS on EC2 [AWS]
- Q517** How does Prometheus fetch data? [Monitoring & Observability]

## Nitor Infotech - 19 questions

- Q7** Write shell scripts: count processes of a user, replace a word in a file, first match of a pattern,... [Linux & OS]
- Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]
- Q189** Write a NodePort service exposing port 8080 [Kubernetes]
- Q190** Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS [Kubernetes]
- Q217** Where is aws-auth stored and what does it do? [Kubernetes]
- Q241** How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q252** What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple... [CI/CD]
- Q253** How do you reduce a slow pipeline (40-60 min -> 15-20 min)? Caching in Jenkins/GitHub Actions [CI/CD]
- Q256** How do you handle parallel execution and jobs in CI/CD (Jenkins, GitHub Actions)? [CI/CD]
- Q276** Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters [CI/CD]
- Q313** How do you remove a stuck state lock; how are state conflicts managed? [Terraform & IaC]
- Q337** Name basic Terraform commands and explain each [Terraform & IaC]
- Q338** How do you manage multiple environments in Terraform? [Terraform & IaC]
- Q380** VPC Peering: how does it work and what options connect VPC A and B (two-way or one-way)? [AWS]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q404** Types of IAM policies; explicit deny vs allow; why a user with a role/policy still cannot access S3;... [AWS]

## NPCI - 7 questions

- Q110** How do you debug inside a container and get its logs (also stream logs to S3)? [Docker & Containers]
- Q113** App reachable from outside the container but packet loss inside it - how to troubleshoot? [Docker & Containers]
- Q115** Docker vs Docker Compose; startup order (DB before app) and depends\_on [Docker & Containers]
- Q139** How do you call pod1 from pod2 without a Service; check events and resource limits? [Kubernetes]
- Q180** What is Container Network Interface (CNI)? [Kubernetes]
- Q236** Auto volume expansion; What is CSI and CSI drivers? [Kubernetes]
- Q238** Static vs dynamic volume provisioning [Kubernetes]

## NUOS INFO Systems - 14 questions

- Q52** Git merge vs rebase (and squash) [Git & Version Control]
- Q65** How do you remove a developer's commit, delete the last 2 commits, or recover when someone... [Git & Version Control]
- Q94** Multistage Dockerfile and when it is useful (compiled languages?) [Docker & Containers]
- Q150** Logs are incomplete - how do you troubleshoot across AKS, Ingress, app and infrastructure? [Kubernetes]
- Q260** Best practices to structure repos and pipelines in a large DevOps project; reusable templates for 50... [CI/CD]
- Q279** How do you troubleshoot failing, stuck or slow pipelines (build not starting, random artifact upload... [CI/CD]
- Q300** Azure DevOps: stakeholder vs admin; REST API policy on all repos; migration readiness [CI/CD]
- Q308** Export existing Azure resources to Terraform [Terraform & IaC]
- Q314** Pipeline fails because resources already exist - how do you handle it (Remove, Import, Plan)? [Terraform & IaC]
- Q340** Do you run Terraform locally or through CI/CD? Explain the workflow; what if apply or the pipeline... [Terraform & IaC]
- Q354** How do you prevent insecure infrastructure changes pushed through Terraform (like quality gates for... [Terraform & IaC]
- Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]
- Q486** How do you manage environment segregation in AWS and handle multiple accounts (and AWS + Azure with... [AWS]
- Q511** Monitor Azure VM memory and set alerts in Azure Monitor [Azure & GCP]

## One2N - 16 questions

- Q35** DNS: TTL, A/AAAA/CNAME records, custom resolvers, DNS failover, creating records [Networking & Security]
- Q37** Reverse proxy vs forward proxy; is a load balancer or Nginx required in this setup? [Networking & Security]
- Q41** How do SSL/TLS certificates work; HTTP vs HTTPS; renew an expired App Gateway certificate; check a... [Networking & Security]
- Q100** Build an image in CI, tag and push it, and deploy updates (commands + GitHub Actions) [Docker & Containers]
- Q116** Docker Swarm vs Kubernetes - why K8s? [Docker & Containers]
- Q158** What is a StatefulSet and how is it different from a Deployment? [Kubernetes]
- Q176** How do you get a static IP for a Kubernetes service/ingress, and how do you expose the application? [Kubernetes]
- Q230** What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a... [Kubernetes]
- Q232** HPA with a StatefulSet; HPA vs VPA [Kubernetes]

- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q259** Design a CI/CD pipeline for a new app (Node.js, Angular/Java/.NET, developer gave only source code)... [CI/CD]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q522** How do you create custom alerts and handle CPU/disk alerts (e.g. CPU/memory beyond 75%)? How do you... [Monitoring & Observability]
- Q523** How do you increase debug log level? Application-level metrics and APM? [Monitoring & Observability]
- Q532** How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix?... [SRE Concepts]
- Q547** Which metric tells me the application is up or down? [SRE Concepts]

## OPT IT - 8 questions

- Q99** Write a Dockerfile for a Java application [Docker & Containers]
- Q190** Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS [Kubernetes]
- Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]
- Q384** How do you access an RDS in a private subnet securely? [AWS]
- Q417** What best practices and security parameters do you follow when creating EC2 / RDS / MongoDB for... [AWS]
- Q512** What monitoring tools and agents have you used and what did you do with them? What proactive... [Monitoring & Observability]
- Q515** Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and... [Monitoring & Observability]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Optum - 9 questions

- Q322** What provisioners exist in Terraform? [Terraform & IaC]
- Q323** Terraform lifecycle (create\_before\_destroy, prevent\_destroy, ignore\_changes) [Terraform & IaC]
- Q334** What are meta-arguments in Terraform? [Terraform & IaC]
- Q335** What is the Terraform external data source / command and when to use it? [Terraform & IaC]
- Q341** Terraform workspaces - what, why, requirements, and how you manage them [Terraform & IaC]
- Q351** Ensure a particular AMI exists using Terraform; start an EC2 when CPU > 80% [Terraform & IaC]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]
- Q466** S3 lifecycle policies and storage classes; how do you manage and version them? [AWS]

## Oracle - 20 questions

- Q3** Difference between & and && in shell; cron expressions; scheduling every 15 minutes [Linux & OS]
- Q37** Reverse proxy vs forward proxy; is a load balancer or Nginx required in this setup? [Networking & Security]
- Q94** Multistage Dockerfile and when it is useful (compiled languages?) [Docker & Containers]
- Q106** Storage drivers in Docker; container runtimes [Docker & Containers]
- Q109** How are Docker images built - what are layers, how does layer caching work, and where does a rebuild... [Docker & Containers]
- Q119** What is a CustomResourceDefinition (CRD)? Desired state vs actual state [Kubernetes]
- Q132** How do you debug a failed deployment or a failed pod (even when you cannot exec into it)? [Kubernetes]
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q159** What is an init container and why use it? [Kubernetes]
- Q174** Types of Kubernetes services [Kubernetes]
- Q190** Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS [Kubernetes]
- Q193** PodDisruptionBudget, voluntary disruptions, minAvailable vs maxUnavailable [Kubernetes]
- Q202** Basic Kubernetes commands (also Docker and Ansible commands) [Kubernetes]
- Q214** How is access control implemented in Kubernetes (RBAC)? [Kubernetes]
- Q241** How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity [Kubernetes]
- Q243** ConfigMap vs Secret [Kubernetes]
- Q246** Design an Istio setup for your Kubernetes cluster [Kubernetes]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q323** Terraform lifecycle (create\_before\_destroy, prevent\_destroy, ignore\_changes) [Terraform & IaC]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]

## Orion Innovation - 10 questions

- Q35** DNS: TTL, A/AAAA/CNAME records, custom resolvers, DNS failover, creating records [Networking & Security]
- Q41** How do SSL/TLS certificates work; HTTP vs HTTPS; renew an expired App Gateway certificate; check a... [Networking & Security]
- Q161** What is a DaemonSet? How do you run Tomcat on each of 10 nodes? Which daemons run by default? [Kubernetes]
- Q176** How do you get a static IP for a Kubernetes service/ingress, and how do you expose the application? [Kubernetes]
- Q190** Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS [Kubernetes]
- Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]
- Q376** How is Route 53 configured and what routing policies exist? [AWS]
- Q380** VPC Peering: how does it work and what options connect VPC A and B (two-way or one-way)? [AWS]
- Q480** RDS cross-region read replica (India -> London)? [AWS]
- Q493** Design a highly available three-tier architecture in Azure (React frontend + Node backend); App... [Azure & GCP]

## Others (company not named) - 233 questions

- Q1** Linux file hierarchy; what do /var and /opt contain; is Linux an OS or kernel? Process/machine states [Linux & OS]
- Q4** Explain commands like finger, comm, netstat, jq, yq, at, shuf, lsblk, less, last, nc, mtr, iftop,... [Linux & OS]
- Q5** One-liners: remove first/last line with sed, count lines, find files > 1 GB, find ERROR... [Linux & OS]
- Q6** Shell scripting: store names of newly created files in a directory; assign/print variables; last 15... [Linux & OS]
- Q8** Find and kill a process; check processes without ps or top [Linux & OS]
- Q9** Scripts: copy new files from a directory to a remote server via SCP; restart a service if it fails... [Linux & OS]
- Q12** A Linux system goes down or is unreachable (SSH not working, private instances) - how do you... [Linux & OS]
- Q13** How do you check server system load? [Linux & OS]
- Q14** What is Linux load average, how is it calculated and displayed? How do you check system load, CPU and... [Linux & OS]
- Q15** A user can't SSH in / you're locked out with no root access - what do you check and how do you recover? [Linux & OS]
- Q16** /var partition is 90% full - immediate action? [Linux & OS]
- Q18** File permission types; creating a user without SSH access; mount vs directory; limiting space per... [Linux & OS]
- Q19** What is logrotate? Script to compress logs older than 30 days, delete older than 90 days, back up the... [Linux & OS]
- Q24** Filesystem shows 50% used but you cannot write; df -ih shows 100% - why? Disk I/O; mount point space [Linux & OS]
- Q25** Add 50 GB to /opt using LVM without downtime [Linux & OS]
- Q26** Where are modern Linux user passwords stored? What are good password security practices? [Linux & OS]
- Q28** What are the top 5 infrastructure attacks and how do you mitigate them? [Linux & OS]
- Q29** OSI model layers; TCP vs UDP [Networking & Security]
- Q34** Which commands check network details and traffic flows on a system? [Networking & Security]
- Q35** DNS: TTL, A/AAAA/CNAME records, custom resolvers, DNS failover, creating records [Networking & Security]
- Q36** What happens when you type www.google.com in the browser? [Networking & Security]
- Q40** How does a GSLB (global server load balancer) work? [Networking & Security]
- Q41** How do SSL/TLS certificates work; HTTP vs HTTPS; renew an expired App Gateway certificate; check a... [Networking & Security]
- Q47** Authentication vs authorization [Networking & Security]
- Q49** How do you enable passwordless authentication between two servers? [Networking & Security]
- Q50** Common API/HTTP errors - what are 504, 501 and others? [Networking & Security]
- Q52** Git merge vs rebase (and squash) [Git & Version Control]
- Q53** git fetch vs git pull; git pull vs git clone; push and pull [Git & Version Control]
- Q57** Git tags, stale branches, --force-with-lease vs --force, and what git add does internally [Git & Version Control]
- Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]
- Q64** How do you trigger a pipeline only for a specific branch, ignore other branches, trigger on PRs; what... [Git & Version Control]
- Q65** How do you remove a developer's commit, delete the last 2 commits, or recover when someone... [Git & Version Control]
- Q67** Python fundamentals: lists vs tuples vs sets, shallow vs deep copy, decorators, inheritance, GIL,... [Scripting & Coding]
- Q69** Exercises: divisibility by 3 and/or 5 with input validation; min/max/sum from a file; JSON ingestion... [Scripting & Coding]
- Q70** Given a log with ERROR/INFO lines (timestamps, connection failed, retries), parse it and report... [Scripting & Coding]
- Q71** Python string programs: slice 0 to 4, reverse a string, palindrome check, remove duplicate element [Scripting & Coding]
- Q74** Create your own Python module that prints "Hello, World" [Scripting & Coding]
- Q77** Reverse the words in a given list using Python [Scripting & Coding]
- Q80** File scripts: find a missing file, rename .txt files with date, delete files older than 10 days, find... [Scripting & Coding]
- Q85** Docker architecture and how it runs on Linux [Docker & Containers]
- Q88** How do you verify the integrity of a Docker image? [Docker & Containers]

- Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]
- Q92** CMD vs ENTRYPOINT [Docker & Containers]
- Q93** How do you reduce Docker image size? [Docker & Containers]
- Q94** Multistage Dockerfile and when it is useful (compiled languages?) [Docker & Containers]
- Q95** ADD vs COPY vs RUN [Docker & Containers]
- Q100** Build an image in CI, tag and push it, and deploy updates (commands + GitHub Actions) [Docker & Containers]
- Q101** Where does docker pull fetch images from? How do you pull from a private registry, authenticate... [Docker & Containers]
- Q102** How do you version/manage images in ECR, delete old or untagged ones, and push to ACR? [Docker & Containers]
- Q105** How do you expose an application in Docker, map ports (e.g. image on 8080, host on 9090), and what if... [Docker & Containers]
- Q110** How do you debug inside a container and get its logs (also stream logs to S3)? [Docker & Containers]
- Q112** A new image was deployed in production and fails immediately - what do you do? [Docker & Containers]
- Q114** Containers consume too much disk space - how do you fix it? [Docker & Containers]
- Q116** Docker Swarm vs Kubernetes - why K8s? [Docker & Containers]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q122** What are namespaces in Kubernetes/EKS? [Kubernetes]
- Q124** Which Kubernetes component does not run as a pod, and what is the agent managing the control-plane side? [Kubernetes]
- Q125** Argo CD architecture [Kubernetes]
- Q127** Can you write directly to etcd manually? [Kubernetes]
- Q129** A worker node is not joining the cluster - how do you debug? What if the master goes down? Can... [Kubernetes]
- Q130** Are readiness probes failing? Can a container restart itself? What is the pause container? [Kubernetes]
- Q131** Common Kubernetes errors and production issues you have faced (CrashLoopBackOff, ImagePullBackOff,... [Kubernetes]
- Q132** How do you debug a failed deployment or a failed pod (even when you cannot exec into it)? [Kubernetes]
- Q134** Ingress configured but the page doesn't load (or 404 from gateway) - what do you check? [Kubernetes]
- Q135** A ReplicaSet has 3 pods and one isn't coming up / pods remain Pending even after HPA - why? [Kubernetes]
- Q136** Common reasons a node becomes NotReady (or an unhealthy node group) and how to identify the root cause [Kubernetes]
- Q137** A Service isn't mapped to a deployment - what could be wrong? [Kubernetes]
- Q140** A PVC is Pending (or pod pending due to disk) - how do you debug? [Kubernetes]
- Q145** If master and worker node firewalls break, what happens to running and new deployments? How do you... [Kubernetes]
- Q146** ImagePullBackOff - why does it occur and how do you troubleshoot? [Kubernetes]
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q157** Multi-container pods: can we run 1 container in 2 pods? Delete a pod and run multiple containers? Can... [Kubernetes]
- Q158** What is a StatefulSet and how is it different from a Deployment? [Kubernetes]
- Q159** What is an init container and why use it? [Kubernetes]
- Q168** Service vs Deployment (and Deployment vs StatefulSet/DaemonSet basics) [Kubernetes]
- Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]
- Q174** Types of Kubernetes services [Kubernetes]
- Q175** Ingress vs Ingress controller vs Gateway API [Kubernetes]
- Q179** How do you map SSL certificates to an Ingress? Rotate Key Vault certs for App Gateway with AKS ingress? [Kubernetes]
- Q181** CoreDNS vs kube-proxy; why are kubelet and kube-proxy needed? [Kubernetes]
- Q188** If TargetPort is not specified in a Service, what happens? [Kubernetes]
- Q190** Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS [Kubernetes]
- Q191** Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines) [Kubernetes]
- Q193** PodDisruptionBudget, voluntary disruptions, minAvailable vs maxUnavailable [Kubernetes]
- Q194** How does kubectl rollout undo know which image to revert to? Is rollback handled by CI/CD or Kubernetes? [Kubernetes]
- Q198** How do you manage deployment, scaling and rollback in Kubernetes? How to roll back a failed deployment? [Kubernetes]
- Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]
- Q202** Basic Kubernetes commands (also Docker and Ansible commands) [Kubernetes]
- Q204** How do you manage worker nodes, node groups, and how many clusters/add-ons do you manage? [Kubernetes]
- Q206** Restore etcd from a backup onto a new VM; what happens if etcd stops or latency spikes? [Kubernetes]
- Q207** How do you handle node draining? [Kubernetes]
- Q208** How do you set up kubectl access, connect Jenkins to Kubernetes, and log in to pods? [Kubernetes]

- Q214** How is access control implemented in Kubernetes (RBAC)? [Kubernetes]
- Q216** Implement Kubernetes security on container and infrastructure sides; zero-trust networking without a... [Kubernetes]
- Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]
- Q226** Helm commands, deploying via CI/CD, and rolling back a failed or partially deployed release [Kubernetes]
- Q230** What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a... [Kubernetes]
- Q231** Do you use Cluster Autoscaler or Karpenter? How do you provision Karpenter, how does it decide which... [Kubernetes]
- Q232** HPA with a StatefulSet; HPA vs VPA [Kubernetes]
- Q234** ConfigMaps, PV and PVC [Kubernetes]
- Q241** How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity [Kubernetes]
- Q242** Pod scheduling arithmetic: requests/limits vs node capacity (e.g. 4 vCPU/16 GB limits and 2 vCPU/10... [Kubernetes]
- Q243** ConfigMap vs Secret [Kubernetes]
- Q246** Design an Istio setup for your Kubernetes cluster [Kubernetes]
- Q248** Fargate vs EKS worker nodes; EKS vs ECS; EBS disadvantages in EKS [Kubernetes]
- Q249** When would you prefer on-prem Kubernetes over EKS and vice versa? Why deploy frontend on EKS instead... [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]
- Q253** How do you reduce a slow pipeline (40-60 min -> 15-20 min)? Caching in Jenkins/GitHub Actions [CI/CD]
- Q256** How do you handle parallel execution and jobs in CI/CD (Jenkins, GitHub Actions)? [CI/CD]
- Q257** How do you implement rollback in a CI/CD pipeline? Describe a rollback you handled [CI/CD]
- Q258** Maven lifecycle, Maven release, and build tools for Java [CI/CD]
- Q259** Design a CI/CD pipeline for a new app (Node.js, Angular/Java/.NET, developer gave only source code)... [CI/CD]
- Q260** Best practices to structure repos and pipelines in a large DevOps project; reusable templates for 50... [CI/CD]
- Q262** What challenges have you faced in CI/CD / a known critical production issue / prioritising multiple... [CI/CD]
- Q269** Use files/outputs of a previous stage in the current stage; refer to a variable from a previous stage [CI/CD]
- Q270** A pipeline was deleted by a team member - how do you recreate it and prevent it in future? [CI/CD]
- Q271** Automated pipeline to move a file from Azure Blob to GCP Cloud Storage [CI/CD]
- Q276** Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters [CI/CD]
- Q277** Jenkins administration: standalone setup, memory issues, copying jobs, backup, concurrent builds,... [CI/CD]
- Q278** Connect a new Jenkins to GitHub; triggers (webhook, poll SCM, after N commits) [CI/CD]
- Q280** Show a sample Jenkinsfile and explain stages, agent, post and environment blocks; what types of... [CI/CD]
- Q282** Give view-only access on 5 Jenkins jobs to other users [CI/CD]
- Q288** SonarQube and Snyk integration, quality gates, tokens and static analysis [CI/CD]
- Q289** Design an end-to-end DevSecOps pipeline (compliance, PCI-DSS, GDPR/HIPAA); handling vulnerabilities... [CI/CD]
- Q290** A developer asks you to remove the code-quality (Sonar) stage because it is slow. What do you do? [CI/CD]
- Q291** GitHub Actions: workflow file, Java Maven build and artifact, manual trigger, concurrency, runners [CI/CD]
- Q292** needs keyword, concurrency, and matrix strategy in GitHub Actions [CI/CD]
- Q293** Review a GitHub Actions workflow snippet: how do you spot incorrect or missing steps? [CI/CD]
- Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]
- Q298** What is CyberArk and how do you consume it in a pipeline? [CI/CD]
- Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]
- Q300** Azure DevOps: stakeholder vs admin; REST API policy on all repos; migration readiness [CI/CD]
- Q303** Build artifacts vs pipeline artifacts; Nexus / Artifactory and storing dependencies [CI/CD]
- Q306** What is CloudFormation and how does it compare with Terraform? [Terraform & IaC]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q312** What is the tfstate file and what does it do? [Terraform & IaC]
- Q313** How do you remove a stuck state lock; how are state conflicts managed? [Terraform & IaC]
- Q321** Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and... [Terraform & IaC]
- Q322** What provisioners exist in Terraform? [Terraform & IaC]
- Q323** Terraform lifecycle (create\_before\_destroy, prevent\_destroy, ignore\_changes) [Terraform & IaC]
- Q326** Null resources in Terraform [Terraform & IaC]
- Q327** Terraform locals vs variables; types (list, string, map of objects, tuple) [Terraform & IaC]
- Q330** What is the Terraform lock file (.terraform.lock.hcl)? [Terraform & IaC]
- Q331** Write a Terraform file to launch an EC2 in a public subnet in a VPC [Terraform & IaC]
- Q336** Why use dynamic blocks in Terraform? Show a skeleton for an Azure resource [Terraform & IaC]
- Q337** Name basic Terraform commands and explain each [Terraform & IaC]
- Q339** What do terraform init, plan, apply, fmt do? Meaning of + and - in plan; auto-approve [Terraform & IaC]
- Q340** Do you run Terraform locally or through CI/CD? Explain the workflow; what if apply or the pipeline... [Terraform & IaC]
- Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]

- Q348** Terraform repository/folder structure; how do dev/qa/prod consume shared modules; Terraform architecture [Terraform & IaC]
- Q352** Create AWS EventBridge via Terraform [Terraform & IaC]
- Q356** IAM users vs GitHub OIDC role vs Terraform Cloud role - which is more secure and when to use each? [Terraform & IaC]
- Q357** Least privilege for CloudFormation stacks and Terraform [Terraform & IaC]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management]
- Q364** What is a playbook? Write one to install/configure Apache/Nginx and ensure the service starts on boot [Ansible & Config Management]
- Q365** Deploy to 100 servers / patch 20+ servers with Ansible; install Ansible; Ansible Tower/AWX [Ansible & Config Management]
- Q366** Import vs include; block/rescue; passing output between tasks; templates [Ansible & Config Management]
- Q367** Which Ansible modules have you used? (install packages, nginx, commands) [Ansible & Config Management]
- Q371** What are Ansible roles? How do you create them, structure a three-tier app, publish to Galaxy, and... [Ansible & Config Management]
- Q372** What is an inventory file? How do you keep large inventories clean and store variables for different... [Ansible & Config Management]
- Q373** What is Ansible Vault? How do you encrypt files/playbooks and store credentials? [Ansible & Config Management]
- Q374** How do you debug a playbook? What if it times out on one host or runs for hours? [Ansible & Config Management]
- Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]
- Q376** How is Route 53 configured and what routing policies exist? [AWS]
- Q377** Security Groups vs NACLs (and subnet vs NACL)? [AWS]
- Q378** Describe your VPC / network architecture: subnets per tier, HA, sizing IP addresses. [AWS]
- Q380** VPC Peering: how does it work and what options connect VPC A and B (two-way or one-way)? [AWS]
- Q381** How do you connect AWS to on-prem or to a client's environment (Direct Connect, VPN)? [AWS]
- Q382** Transit Gateway vs VPC peering; have you used Transit Gateway? [AWS]
- Q383** An EC2 instance is unreachable (not a security group problem) / an app behind a load balancer is... [AWS]
- Q385** Purpose of VPC Endpoints and when to use them? [AWS]
- Q386** How does SSL/TLS work with Certbot, Let's Encrypt and ACM? How is a certificate used with ALB/NLB,... [AWS]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q402** How do you manage secrets in your project? (Secrets Manager, Parameter Store, Vault, encryption and... [AWS]
- Q404** Types of IAM policies; explicit deny vs allow; why a user with a role/policy still cannot access S3;... [AWS]
- Q405** An S3 bucket was made public by mistake. How do you secure and audit it? How do you secure a bucket... [AWS]
- Q406** What is the difference between IAM Users and IAM Roles? [AWS]
- Q407** How do you protect and encrypt data on EC2/EBS? [AWS]
- Q408** How do you scan EC2 instances for vulnerabilities, and fix a vulnerable or compromised instance? [AWS]
- Q409** You have 10 AWS accounts and cannot use access keys. How do you log in securely? [AWS]
- Q410** A cross-account scenario: EC2 in Account A needs to read tokens (secrets) in Account B [AWS]
- Q412** Get the count of running EC2 instances and attached EBS volumes across 50-60 AWS accounts without... [AWS]
- Q415** Managed policy vs inline policy - what is the difference? How do you find and remove inline policies... [AWS]
- Q416** Secrets Manager vs Parameter Store - what is the difference? [AWS]
- Q417** What best practices and security parameters do you follow when creating EC2 / RDS / MongoDB for... [AWS]
- Q421** Bucket policy vs ACL? [AWS]
- Q423** How do you give a developer read-only access to RDS? [AWS]
- Q428** List IAM users whose access keys have not been used in over 90 days [AWS]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]
- Q438** How did you migrate an application from on-premises to AWS? [AWS]
- Q443** What is a cloud-agnostic strategy? How do you use conditionals in Terraform/Ansible for AWS, Azure... [AWS]
- Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]
- Q448** How does AWS Auto Scaling work? (ASG, with ALB, scaling on memory/disk, ECS autoscaling) [AWS]
- Q449** Key CloudWatch metrics to focus on; how to monitor LB health, CPU spikes and web/app server availability? [AWS]
- Q450** CloudFront - how do you use it to serve a frontend? [AWS]
- Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]
- Q461** Fargate vs EC2 launch type; ECS Service vs Task; setting up ECS on EC2 [AWS]
- Q463** Lambda role is set up correctly but logs are not appearing in the CloudWatch log group [AWS]
- Q465** When to use EC2 vs Lambda (scenarios)? [AWS]
- Q466** S3 lifecycle policies and storage classes; how do you manage and version them? [AWS]
- Q467** Terraform state in S3: expire in 30 days? other options? S3 native locking command? [AWS]
- Q472** EBS-backed vs instance store-backed instances [AWS]
- Q475** RDS migration with minimal downtime, and upgrading MySQL 5.7 to 8.0 [AWS]

- Q477** How do you configure RDS and scale it (vertical/horizontal)? [AWS]
- Q478** Multi-AZ vs Read Replica; secondary vs read-only RDS [AWS]
- Q482** Production RDS storage is 95% full - how do you debug and prevent it in future? [AWS]
- Q498** Connect VMs in different VNets; define a VNet; Azure Load Balancer; ExpressRoute [Azure & GCP]
- Q500** Firewall vs NSG [Azure & GCP]
- Q504** Protect endpoints in AKS; give only one pod access to a storage account and block others [Azure & GCP]
- Q508** Managed Identity vs Service Principal [Azure & GCP]
- Q510** Log Analytics Workspace, alerts with action groups, KQL queries, and Dynatrace alert types [Azure & GCP]
- Q513** Monitoring vs observability; metrics, logs and traces and their tools; observability architecture [Monitoring & Observability]
- Q515** Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and... [Monitoring & Observability]
- Q519** Prometheus memory keeps growing - how do you debug? [Monitoring & Observability]
- Q521** Extract errors from logs: fetch lines with 200 status, count ERR, split error and warning into... [Monitoring & Observability]
- Q526** Script to monitor disk usage and email/log an alert above 80% [Monitoring & Observability]
- Q530** How is the ELK stack set up end to end; how Kibana connects to Elasticsearch; how logs are... [Monitoring & Observability]
- Q532** How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix?... [SRE Concepts]
- Q533** Sudden traffic surge makes the web application unresponsive - what steps do you take to mitigate? [SRE Concepts]
- Q535** A live app is slow/has latency for customers (e-commerce) - how do you troubleshoot? Which metrics do... [SRE Concepts]
- Q540** What are SLI, SLO, SLA and error budget? How do you decide and set them? [SRE Concepts]
- Q544** How does AWS DRS (Elastic Disaster Recovery) work; failover and fallback? [SRE Concepts]
- Q549** How does MongoDB high availability work (primary/secondary)? [SRE Concepts]
- Q562** What is MongoDB? Redis vs Memcached; which database would you use? [General DevOps & Concepts]
- Q563** What is an API? [General DevOps & Concepts]
- Q564** In Airflow, how do you debug a failed job? [General DevOps & Concepts]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]
- Q569** Working under tight deadlines / high pressure; managing a team; accountability and collaboration [Behavioural & Project]
- Q570** How do you communicate technical issues to non-technical stakeholders, and avoid miscommunication... [Behavioural & Project]
- Q573** Tell me about a successful DevOps transformation project and your role [Behavioural & Project]

## Perfios - 13 questions

- Q7** Write shell scripts: count processes of a user, replace a word in a file, first match of a pattern,... [Linux & OS]
- Q36** What happens when you type www.google.com in the browser? [Networking & Security]
- Q64** How do you trigger a pipeline only for a specific branch, ignore other branches, trigger on PRs; what... [Git & Version Control]
- Q68** Coding: print a number pattern; find pairs summing to a target; longest substring without repeating... [Scripting & Coding]
- Q174** Types of Kubernetes services [Kubernetes]
- Q192** How do you achieve a zero-downtime application upgrade and redeploy? [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]
- Q272** How do you remove human intervention when you need a platform to deploy a built package? [CI/CD]
- Q379** An EC2 in a private subnet must receive inbound traffic (without NAT). How, and how does traffic flow... [AWS]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]
- Q568** What automation have you done / contributed? Do career achievements need to be big? [Behavioural & Project]

## Persistent Systems - 41 questions

- Q57** Git tags, stale branches, --force-with-lease vs --force, and what git add does internally [Git & Version Control]
- Q66** Which scripting languages have you used, and what automations/scripts have you written (cost... [Scripting & Coding]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q133** How do you monitor pod health/performance and alert when pods go down? [Kubernetes]
- Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]
- Q197** Deploy a Kubernetes app with Jenkins: plugins, and a Jenkinsfile for Node.js (build, push image, deploy) [Kubernetes]
- Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]
- Q203** Have you created an EKS cluster? Explain the process and prerequisites (compute, networking, access) [Kubernetes]

- Q231** Do you use Cluster Autoscaler or Karpenter? How do you provision Karpenter, how does it decide which... [Kubernetes]
- Q232** HPA with a StatefulSet; HPA vs VPA [Kubernetes]
- Q241** How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity [Kubernetes]
- Q248** Fargate vs EKS worker nodes; EKS vs ECS; EBS disadvantages in EKS [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q253** How do you reduce a slow pipeline (40-60 min -> 15-20 min)? Caching in Jenkins/GitHub Actions [CI/CD]
- Q255** AWS CodePipeline/CodeBuild/CodeDeploy - how to set up, and CodePipeline vs Jenkins [CI/CD]
- Q288** SonarQube and Snyk integration, quality gates, tokens and static analysis [CI/CD]
- Q291** GitHub Actions: workflow file, Java Maven build and artifact, manual trigger, concurrency, runners [CI/CD]
- Q292** needs keyword, concurrency, and matrix strategy in GitHub Actions [CI/CD]
- Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]
- Q306** What is CloudFormation and how does it compare with Terraform? [Terraform & IaC]
- Q312** What is the tfstate file and what does it do? [Terraform & IaC]
- Q320** terraform state mv / moved block; refactoring resources without destroying them [Terraform & IaC]
- Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]
- Q349** Refactor a legacy Terraform codebase / monorepo with hundreds of states into modules safely; how to... [Terraform & IaC]
- Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]
- Q387** Public vs private subnet - how can you tell, and what must be in place? [AWS]
- Q388** How would you migrate an app running on a public EC2 to a private subnet following best practices? [AWS]
- Q393** EC2 should not talk to the internet but must talk to other instances. How? [AWS]
- Q397** Someone from a partner without an AWS account must see a dashboard. How? [AWS]
- Q403** How can you log in to an EC2 instance without a password, or if you lost the .pem key? [AWS]
- Q418** What is CloudTrail and what do you use it for? [AWS]
- Q450** CloudFront - how do you use it to serve a frontend? [AWS]
- Q451** CloudWatch: what is it and how do you create custom metrics? [AWS]
- Q458** What is Amazon Athena? [AWS]
- Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]
- Q466** S3 lifecycle policies and storage classes; how do you manage and version them? [AWS]
- Q493** Design a highly available three-tier architecture in Azure (React frontend + Node backend); App... [Azure & GCP]
- Q535** A live app is slow/has latency for customers (e-commerce) - how do you troubleshoot? Which metrics do... [SRE Concepts]
- Q545** How do you reduce latency for users from abroad? Low-latency routing? [SRE Concepts]
- Q551** Microservices vs monolith; what are microservices? [Architecture & Design]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Plansource ValueLabs - 12 questions

- Q36** What happens when you type www.google.com in the browser? [Networking & Security]
- Q41** How do SSL/TLS certificates work; HTTP vs HTTPS; renew an expired App Gateway certificate; check a... [Networking & Security]
- Q110** How do you debug inside a container and get its logs (also stream logs to S3)? [Docker & Containers]
- Q231** Do you use Cluster Autoscaler or Karpenter? How do you provision Karpenter, how does it decide which... [Kubernetes]
- Q338** How do you manage multiple environments in Terraform? [Terraform & IaC]
- Q389** How many subnets can a VPC have, and can you extend a subnet's CIDR later? [AWS]
- Q405** An S3 bucket was made public by mistake. How do you secure and audit it? How do you secure a bucket... [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]
- Q448** How does AWS Auto Scaling work? (ASG, with ALB, scaling on memory/disk, ECS autoscaling) [AWS]
- Q461** Fargate vs EC2 launch type; ECS Service vs Task; setting up ECS on EC2 [AWS]
- Q473** How do you speed up large S3 uploads and verify a partially failed 10 GB upload? [AWS]

## Publicis Global Delivery - 6 questions

- Q58** Extract commits of the last 3 days; compare two commits [Git & Version Control]
- Q158** What is a StatefulSet and how is it different from a Deployment? [Kubernetes]
- Q193** PodDisruptionBudget, voluntary disruptions, minAvailable vs maxUnavailable [Kubernetes]
- Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]
- Q235** RWO vs RWX volumes [Kubernetes]
- Q277** Jenkins administration: standalone setup, memory issues, copying jobs, backup, concurrent builds,... [CI/CD]

## Qburst - 15 questions

- Q93** How do you reduce Docker image size? [Docker & Containers]
- Q95** ADD vs COPY vs RUN [Docker & Containers]
- Q96** How do you pass values during docker build (ARG vs ENV)? How do you avoid leaking secrets in the build? [Docker & Containers]
- Q174** Types of Kubernetes services [Kubernetes]
- Q176** How do you get a static IP for a Kubernetes service/ingress, and how do you expose the application? [Kubernetes]
- Q177** Load balancing in Kubernetes: ELB, Ingress, global load balancing, restricting public access to load... [Kubernetes]
- Q202** Basic Kubernetes commands (also Docker and Ansible commands) [Kubernetes]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q338** How do you manage multiple environments in Terraform? [Terraform & IaC]
- Q342** Remove a resource from Terraform code without deleting it in the cloud; resources that must be... [Terraform & IaC]
- Q395** Major difference between AWS and GCP VPCs [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q485** How do you patch EC2 instances? [AWS]
- Q494** Scaling strategy for compute-intensive vs I/O-intensive workloads; throttled Azure Function; migrations [Azure & GCP]
- Q497** What is IAP (Identity-Aware Proxy) in GCP? [Azure & GCP]

## Qentelli Solutions - 11 questions

- Q2** Hard link vs soft (symbolic) link [Linux & OS]
- Q15** A user can't SSH in / you're locked out with no root access - what do you check and how do you recover? [Linux & OS]
- Q19** What is logrotate? Script to compress logs older than 30 days, delete older than 90 days, back up the... [Linux & OS]
- Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]
- Q321** Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and... [Terraform & IaC]
- Q377** Security Groups vs NACLs (and subnet vs NACL)? [AWS]
- Q396** Private subnet changed to public by a developer - what do you do? [AWS]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]
- Q449** Key CloudWatch metrics to focus on; how to monitor LB health, CPU spikes and web/app server availability? [AWS]

## Rapidsoft - 12 questions

- Q116** Docker Swarm vs Kubernetes - why K8s? [Docker & Containers]
- Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q190** Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS [Kubernetes]
- Q266** Deploy to both cloud and on-premises (hybrid) and across multiple clouds - how do you design the... [CI/CD]
- Q279** How do you troubleshoot failing, stuck or slow pipelines (build not starting, random artifact upload... [CI/CD]
- Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]
- Q358** Enable Terraform debug logs; investigate and validate Terraform errors [Terraform & IaC]
- Q364** What is a playbook? Write one to install/configure Apache/Nginx and ensure the service starts on boot [Ansible & Config Management]
- Q475** RDS migration with minimal downtime, and upgrading MySQL 5.7 to 8.0 [AWS]
- Q543** Explain your DR strategy: RTO, RPO, failover and traffic redirection; DR for DevOps infrastructure;... [SRE Concepts]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## RelevantZ - 19 questions

- Q41** How do SSL/TLS certificates work; HTTP vs HTTPS; renew an expired App Gateway certificate; check a... [Networking & Security]
- Q136** Common reasons a node becomes NotReady (or an unhealthy node group) and how to identify the root cause [Kubernetes]
- Q208** How do you set up kubectl access, connect Jenkins to Kubernetes, and log in to pods? [Kubernetes]
- Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]
- Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]
- Q303** Build artifacts vs pipeline artifacts; Nexus / Artifactory and storing dependencies [CI/CD]
- Q313** How do you remove a stuck state lock; how are state conflicts managed? [Terraform & IaC]
- Q338** How do you manage multiple environments in Terraform? [Terraform & IaC]
- Q349** Refactor a legacy Terraform codebase / monorepo with hundreds of states into modules safely; how to... [Terraform & IaC]

- Q384** How do you access an RDS in a private subnet securely? [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]
- Q437** How do you back up AWS services (EC2, RDS, S3, EBS) and how does Aurora back up? [AWS]
- Q475** RDS migration with minimal downtime, and upgrading MySQL 5.7 to 8.0 [AWS]
- Q492** Azure storage types, subscription types, Recovery Services Vault, backup choices and Azure Boards [Azure & GCP]
- Q495** How do you implement DC/DR in Azure (and what is the purpose of DR)? [Azure & GCP]
- Q498** Connect VMs in different VNets; define a VNet; Azure Load Balancer; ExpressRoute [Azure & GCP]
- Q510** Log Analytics Workspace, alerts with action groups, KQL queries, and Dynatrace alert types [Azure & GCP]
- Q512** What monitoring tools and agents have you used and what did you do with them? What proactive... [Monitoring & Observability]

## SAP - 10 questions

- Q60** Git submodules - purpose [Git & Version Control]
- Q107** How do you secure Docker images and containers (scanning, base image maintenance, authorized images,...) [Docker & Containers]
- Q132** How do you debug a failed deployment or a failed pod (even when you cannot exec into it)? [Kubernetes]
- Q141** After a new deployment all pods (new and old) crashed - why? [Kubernetes]
- Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]
- Q191** Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines) [Kubernetes]
- Q195** During a canary of 10%, how do you verify health and what metrics before going to 100%? How do you... [Kubernetes]
- Q252** What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple... [CI/CD]
- Q265** GitHub Actions vs Argo CD; how does CI differ from GitOps CD? [CI/CD]
- Q485** How do you patch EC2 instances? [AWS]

## Sapient - 16 questions

- Q35** DNS: TTL, A/AAAA/CNAME records, custom resolvers, DNS failover, creating records [Networking & Security]
- Q37** Reverse proxy vs forward proxy; is a load balancer or Nginx required in this setup? [Networking & Security]
- Q203** Have you created an EKS cluster? Explain the process and prerequisites (compute, networking, access) [Kubernetes]
- Q213** If the cluster is in a private subnet, how do you access it with kubectl? [Kubernetes]
- Q218** How do you map an IAM role to a Kubernetes service account (IRSA) and how does the pod know which... [Kubernetes]
- Q235** RWO vs RWX volumes [Kubernetes]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q312** What is the tfstate file and what does it do? [Terraform & IaC]
- Q324** What is a provider and what is a data block in Terraform? [Terraform & IaC]
- Q326** Null resources in Terraform [Terraform & IaC]
- Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]
- Q376** How is Route 53 configured and what routing policies exist? [AWS]
- Q378** Describe your VPC / network architecture: subnets per tier, HA, sizing IP addresses. [AWS]
- Q379** An EC2 in a private subnet must receive inbound traffic (without NAT). How, and how does traffic flow... [AWS]
- Q407** How do you protect and encrypt data on EC2/EBS? [AWS]
- Q468** S3 vs EBS (and EFS)? [AWS]

## Sigmoid - 78 questions

- Q4** Explain commands like finger, comm, netstat, jq, yq, at, shuf, lsblk, less, last, nc, mtr, iftop,... [Linux & OS]
- Q7** Write shell scripts: count processes of a user, replace a word in a file, first match of a pattern,... [Linux & OS]
- Q35** DNS: TTL, A/AAAA/CNAME records, custom resolvers, DNS failover, creating records [Networking & Security]
- Q43** A DDoS attack consumes 100% resources on cluster nodes/services - what do you do? [Networking & Security]
- Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]
- Q66** Which scripting languages have you used, and what automations/scripts have you written (cost... [Scripting & Coding])
- Q68** Coding: print a number pattern; find pairs summing to a target; longest substring without repeating... [Scripting & Coding]
- Q69** Exercises: divisibility by 3 and/or 5 with input validation; min/max/sum from a file; JSON ingestion... [Scripting & Coding]
- Q70** Given a log with ERROR/INFO lines (timestamps, connection failed, retries), parse it and report... [Scripting & Coding]
- Q71** Python string programs: slice 0 to 4, reverse a string, palindrome check, remove duplicate element [Scripting & Coding]
- Q72** Python scripts: failed job IDs from job logs, unique IPs from a live log, monitor a directory for new... [Scripting & Coding]

**Q80** File scripts: find a missing file, rename .txt files with date, delete files older than 10 days, find... [Scripting & Coding]

**Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]

**Q92** CMD vs ENTRYPOINT [Docker & Containers]

**Q94** Multistage Dockerfile and when it is useful (compiled languages?) [Docker & Containers]

**Q100** Build an image in CI, tag and push it, and deploy updates (commands + GitHub Actions) [Docker & Containers]

**Q107** How do you secure Docker images and containers (scanning, base image maintenance, authorized images,... [Docker & Containers]

**Q109** How are Docker images built - what are layers, how does layer caching work, and where does a rebuild... [Docker & Containers]

**Q119** What is a CustomResourceDefinition (CRD)? Desired state vs actual state [Kubernetes]

**Q126** Can you talk to the Kubernetes API with plain REST calls from an external app? [Kubernetes]

**Q129** A worker node is not joining the cluster - how do you debug? What if the master goes down? Can... [Kubernetes]

**Q130** Are readiness probes failing? Can a container restart itself? What is the pause container? [Kubernetes]

**Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]

**Q156** Compute reservation in manifests - requests vs limits [Kubernetes]

**Q160** StatefulSet vs RDS for a database - how do you decide? MongoDB/MySQL on Kubernetes and operators [Kubernetes]

**Q165** Run one MySQL instance on each node using custom operators [Kubernetes]

**Q169** What are operators? How does Elasticsearch run on Kubernetes (ECK)? [Kubernetes]

**Q175** Ingress vs Ingress controller vs Gateway API [Kubernetes]

**Q177** Load balancing in Kubernetes: ELB, Ingress, global load balancing, restricting public access to load... [Kubernetes]

**Q178** Complete request flow from the user's browser to the backend pod (Ingress, DNS, LB, Services) [Kubernetes]

**Q184** A Service of type LoadBalancer is created but nodes are in private subnets - how does it provision a... [Kubernetes]

**Q190** Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS [Kubernetes]

**Q191** Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines) [Kubernetes]

**Q194** How does kubectl rollout undo know which image to revert to? Is rollback handled by CI/CD or Kubernetes? [Kubernetes]

**Q196** With a single replica, does a rolling restart cause downtime? What happens step by step? [Kubernetes]

**Q199** Post-deployment steps to confirm everything runs well [Kubernetes]

**Q204** How do you manage worker nodes, node groups, and how many clusters/add-ons do you manage? [Kubernetes]

**Q205** How do you back up an entire Kubernetes cluster regularly? [Kubernetes]

**Q214** How is access control implemented in Kubernetes (RBAC)? [Kubernetes]

**Q248** Fargate vs EKS worker nodes; EKS vs ECS; EBS disadvantages in EKS [Kubernetes]

**Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]

**Q254** How do you onboard a new project to multiple environments and handle releases/hotfixes? [CI/CD]

**Q257** How do you implement rollback in a CI/CD pipeline? Describe a rollback you handled [CI/CD]

**Q264** Code quality vs code coverage [CI/CD]

**Q269** Use files/outputs of a previous stage in the current stage; refer to a variable from a previous stage [CI/CD]

**Q281** What are executors and agents in Jenkins and how do they work? [CI/CD]

**Q297** Credentials appear in pipeline logs, or a developer hard-coded a password and pushed it - what do you do? [CI/CD]

**Q323** Terraform lifecycle (create\_before\_destroy, prevent\_destroy, ignore\_changes) [Terraform & IaC]

**Q324** What is a provider and what is a data block in Terraform? [Terraform & IaC]

**Q326** Null resources in Terraform [Terraform & IaC]

**Q337** Name basic Terraform commands and explain each [Terraform & IaC]

**Q341** Terraform workspaces - what, why, requirements, and how you manage them [Terraform & IaC]

**Q342** Remove a resource from Terraform code without deleting it in the cloud; resources that must be... [Terraform & IaC]

**Q345** If a Terraform deployment fails midway, what is your approach? [Terraform & IaC]

**Q358** Enable Terraform debug logs; investigate and validate Terraform errors [Terraform & IaC]

**Q359** A resource created by Terraform failed during provisioning - what happens? [Terraform & IaC]

**Q366** Import vs include; block/rescue; passing output between tasks; templates [Ansible & Config Management]

**Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]

**Q376** How is Route 53 configured and what routing policies exist? [AWS]

**Q378** Describe your VPC / network architecture: subnets per tier, HA, sizing IP addresses. [AWS]

**Q379** An EC2 in a private subnet must receive inbound traffic (without NAT). How, and how does traffic flow... [AWS]

**Q387** Public vs private subnet - how can you tell, and what must be in place? [AWS]

**Q432** Design a highly available, fault-tolerant backend on AWS [AWS]

**Q435** How do you design for an entire region going down (and keep data safe even if multi-cloud also... [AWS]

**Q449** Key CloudWatch metrics to focus on; how to monitor LB health, CPU spikes and web/app server availability? [AWS]

**Q450** CloudFront - how do you use it to serve a frontend? [AWS]

**Q452** Failover mechanism in load balancers; active-passive vs active-active [AWS]

**Q456** Can ASG and load balancer be in different regions? Peak traffic 5-8 PM daily - how to configure? [AWS]

**Q466** S3 lifecycle policies and storage classes; how do you manage and version them? [AWS]

- Q475** RDS migration with minimal downtime, and upgrading MySQL 5.7 to 8.0 [AWS]
- Q486** How do you manage environment segregation in AWS and handle multiple accounts (and AWS + Azure with... [AWS]
- Q518** Alert if CPU stays above 90% for 5 minutes but only when running pods are below 5 [Monitoring & Observability]
- Q520** Write a PromQL expression to alert if CPU usage is above 80% on any node [Monitoring & Observability]
- Q521** Extract errors from logs: fetch lines with 200 status, count ERR, split error and warning into... [Monitoring & Observability]
- Q530** How is the ELK stack set up end to end; how Kibana connects to Elasticsearch; how logs are... [Monitoring & Observability]
- Q535** A live app is slow/has latency for customers (e-commerce) - how do you troubleshoot? Which metrics do... [SRE Concepts]
- Q545** How do you reduce latency for users from abroad? Low-latency routing? [SRE Concepts]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Sonata Software - 35 questions

- Q3** Difference between & and && in shell; cron expressions; scheduling every 15 minutes [Linux & OS]
- Q67** Python fundamentals: lists vs tuples vs sets, shallow vs deep copy, decorators, inheritance, GIL,... [Scripting & Coding]
- Q133** How do you monitor pod health/performance and alert when pods go down? [Kubernetes]
- Q179** How do you map SSL certificates to an Ingress? Rotate Key Vault certs for App Gateway with AKS ingress? [Kubernetes]
- Q192** How do you achieve a zero-downtime application upgrade and redeploy? [Kubernetes]
- Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q268** GitLab CI: experience, rules, pipelines [CI/CD]
- Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]
- Q315** How do multiple teams deploy to the same AWS account without overwriting each other; how do you... [Terraform & IaC]
- Q317** Terraform-generated RDS password was not saved - can you retrieve it? Changing an S3 bucket name? [Terraform & IaC]
- Q324** What is a provider and what is a data block in Terraform? [Terraform & IaC]
- Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]
- Q347** How do you write a Terraform module for EKS; how do you create and manage clusters, and master/worker... [Terraform & IaC]
- Q348** Terraform repository/folder structure; how do dev/qa/prod consume shared modules; Terraform architecture [Terraform & IaC]
- Q361** Have you used Terraform Cloud / HCP? What does it provide? [Terraform & IaC]
- Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]
- Q402** How do you manage secrets in your project? (Secrets Manager, Parameter Store, Vault, encryption and... [AWS]
- Q404** Types of IAM policies; explicit deny vs allow; why a user with a role/policy still cannot access S3;... [AWS]
- Q413** How do you protect an API Gateway? [AWS]
- Q415** Managed policy vs inline policy - what is the difference? How do you find and remove inline policies... [AWS]
- Q430** Retrieve secrets from AWS Secrets Manager using Python (boto3) [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q437** How do you back up AWS services (EC2, RDS, S3, EBS) and how does Aurora back up? [AWS]
- Q448** How does AWS Auto Scaling work? (ASG, with ALB, scaling on memory/disk, ECS autoscaling) [AWS]
- Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]
- Q462** API Gateway: configuration, REST vs WebSocket APIs [AWS]
- Q476** App is slow and you suspect RDS - what do you check? Memory pressure but cannot resize? [AWS]
- Q483** Traditional RDS vs Aurora? [AWS]
- Q499** Application Gateway vs Front Door (and L7/L4) [Azure & GCP]
- Q501** Block a specific domain in Azure Application Gateway [Azure & GCP]
- Q504** Protect endpoints in AKS; give only one pod access to a storage account and block others [Azure & GCP]
- Q506** Networking used in AKS (kubenet vs Azure CNI) [Azure & GCP]
- Q522** How do you create custom alerts and handle CPU/disk alerts (e.g. CPU/memory beyond 75%)? How do you... [Monitoring & Observability]
- Q561** Kafka / messaging services; AWS AppConfig; combine responses of multiple APIs [General DevOps & Concepts]

## Sony - 27 questions

- Q93** How do you reduce Docker image size? [Docker & Containers]
- Q111** Containers stop suddenly after starting - how to troubleshoot? [Docker & Containers]
- Q120** How do you ensure high availability in Kubernetes (survive an AZ failure, multi-region, ECS Fargate/EKS)? [Kubernetes]
- Q151** Pods restart intermittently though liveness and readiness pass and the node kills them - how do you... [Kubernetes]
- Q152** The cluster feels slow - how do you start troubleshooting? [Kubernetes]
- Q186** Gateway API: request flow with multiple GatewayClasses across regions; preventing split-brain routing [Kubernetes]
- Q192** How do you achieve a zero-downtime application upgrade and redeploy? [Kubernetes]
- Q206** Restore etcd from a backup onto a new VM; what happens if etcd stops or latency spikes? [Kubernetes]
- Q216** Implement Kubernetes security on container and infrastructure sides; zero-trust networking without a... [Kubernetes]
- Q222** Design a multi-tenant Kubernetes platform where teams cannot impact each other [Kubernetes]
- Q230** What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a... [Kubernetes]
- Q252** What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple... [CI/CD]
- Q253** How do you reduce a slow pipeline (40-60 min -> 15-20 min)? Caching in Jenkins/GitHub Actions [CI/CD]
- Q273** How do you stop broken code from reaching production? [CI/CD]
- Q278** Connect a new Jenkins to GitHub; triggers (webhook, poll SCM, after N commits) [CI/CD]
- Q285** Secure Jenkins pipelines [CI/CD]
- Q288** SonarQube and Snyk integration, quality gates, tokens and static analysis [CI/CD]
- Q315** How do multiple teams deploy to the same AWS account without overwriting each other; how do you... [Terraform & IaC]
- Q325** Explain Terraform dependency graph internals and resource dependencies; circular dependencies [Terraform & IaC]
- Q349** Refactor a legacy Terraform codebase / monorepo with hundreds of states into modules safely; how to... [Terraform & IaC]
- Q351** Ensure a particular AMI exists using Terraform; start an EC2 when CPU > 80% [Terraform & IaC]
- Q354** How do you prevent insecure infrastructure changes pushed through Terraform (like quality gates for... [Terraform & IaC]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q435** How do you design for an entire region going down (and keep data safe even if multi-cloud also... [AWS]
- Q527** Thousands of alerts arrive from a huge infrastructure with customer-defined thresholds - how do you... [Monitoring & Observability]
- Q532** How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix?... [SRE Concepts]
- Q538** Users complain about latency but monitoring says healthy - what do you do? How does monitoring help... [SRE Concepts]

## SquareOps - 87 questions

- Q14** What is Linux load average, how is it calculated and displayed? How do you check system load, CPU and... [Linux & OS]
- Q30** Is 90.90.88.0/12 private or public? Is 90.0.0.9/0 public? [Networking & Security]
- Q32** Is IP X in CIDR Y? Validate an IPv4 address [Networking & Security]
- Q33** Private vs public IP addresses; how many IPs does /32 allow? [Networking & Security]
- Q39** Load balancer not accepting HTTPS; redirecting HTTP to HTTPS without installing a certificate; load... [Networking & Security]
- Q42** How to create certificates for multiple sub-domains; Cross-Origin Resource Sharing (CORS) [Networking & Security]
- Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]
- Q64** How do you trigger a pipeline only for a specific branch, ignore other branches, trigger on PRs; what... [Git & Version Control]
- Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]
- Q130** Are readiness probes failing? Can a container restart itself? What is the pause container? [Kubernetes]
- Q134** Ingress configured but the page doesn't load (or 404 from gateway) - what do you check? [Kubernetes]
- Q139** How do you call pod1 from pod2 without a Service; check events and resource limits? [Kubernetes]
- Q142** During peak traffic the ingress controller routes slowly - debug? Increase replicas permanently or... [Kubernetes]
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q163** How do you inspect environment variables and ConfigMaps in a pod; how do you update config safely? [Kubernetes]
- Q192** How do you achieve a zero-downtime application upgrade and redeploy? [Kubernetes]
- Q193** PodDisruptionBudget, voluntary disruptions, minAvailable vs maxUnavailable [Kubernetes]
- Q194** How does kubectl rollout undo know which image to revert to? Is rollback handled by CI/CD or Kubernetes? [Kubernetes]
- Q196** With a single replica, does a rolling restart cause downtime? What happens step by step? [Kubernetes]
- Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]
- Q203** Have you created an EKS cluster? Explain the process and prerequisites (compute, networking, access) [Kubernetes]

- Q204** How do you manage worker nodes, node groups, and how many clusters/add-ons do you manage? [Kubernetes]
- Q207** How do you handle node draining? [Kubernetes]
- Q208** How do you set up kubectl access, connect Jenkins to Kubernetes, and log in to pods? [Kubernetes]
- Q214** How is access control implemented in Kubernetes (RBAC)? [Kubernetes]
- Q217** Where is aws-auth stored and what does it do? [Kubernetes]
- Q219** How do you restrict everything except two services (NetworkPolicy)? How do you secure a web app from... [Kubernetes]
- Q221** Role vs RoleBinding; ClusterRole vs ClusterRoleBinding - when to use which? [Kubernetes]
- Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]
- Q226** Helm commands, deploying via CI/CD, and rolling back a failed or partially deployed release [Kubernetes]
- Q228** How do you extend a Helm chart with new templates (without breaking the original)? [Kubernetes]
- Q230** What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a... [Kubernetes]
- Q231** Do you use Cluster Autoscaler or Karpenter? How do you provision Karpenter, how does it decide which... [Kubernetes]
- Q233** Cluster Autoscaler vs Horizontal Pod Autoscaler (and Karpenter) [Kubernetes]
- Q235** RWO vs RWX volumes [Kubernetes]
- Q237** Storage classes in Kubernetes; what happens when a Deployment with one PVC is scaled from 1 to 3? [Kubernetes]
- Q239** How do you mount a PVC in a Deployment? [Kubernetes]
- Q240** What is the EFS CSI driver? [Kubernetes]
- Q244** How do you manage secrets and config at scale in GitOps without leaking them? [Kubernetes]
- Q245** What are Sealed Secrets? [Kubernetes]
- Q248** Fargate vs EKS worker nodes; EKS vs ECS; EBS disadvantages in EKS [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,...) [CI/CD]
- Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]
- Q254** How do you onboard a new project to multiple environments and handle releases/hotfixes? [CI/CD]
- Q257** How do you implement rollback in a CI/CD pipeline? Describe a rollback you handled [CI/CD]
- Q261** CI/CD for ML models; ephemeral environments; SLA monitoring in a payments pipeline; iOS build automation [CI/CD]
- Q263** Approvals in a CD pipeline [CI/CD]
- Q278** Connect a new Jenkins to GitHub; triggers (webhook, poll SCM, after N commits) [CI/CD]
- Q279** How do you troubleshoot failing, stuck or slow pipelines (build not starting, random artifact upload... [CI/CD]
- Q280** Show a sample Jenkinsfile and explain stages, agent, post and environment blocks; what types of... [CI/CD]
- Q288** SonarQube and Snyk integration, quality gates, tokens and static analysis [CI/CD]
- Q291** GitHub Actions: workflow file, Java Maven build and artifact, manual trigger, concurrency, runners [CI/CD]
- Q294** Structure a multi-stage GitHub Actions pipeline that builds, tests and deploys a containerised app to... [CI/CD]
- Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]
- Q297** Credentials appear in pipeline logs, or a developer hard-coded a password and pushed it - what do you do? [CI/CD]
- Q305** What is Terraform, how does it work, and what have you done with it? [Terraform & IaC]
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- Q312** What is the tfstate file and what does it do? [Terraform & IaC]
- Q313** How do you remove a stuck state lock; how are state conflicts managed? [Terraform & IaC]
- Q315** How do multiple teams deploy to the same AWS account without overwriting each other; how do you... [Terraform & IaC]
- Q317** Terraform-generated RDS password was not saved - can you retrieve it? Changing an S3 bucket name? [Terraform & IaC]
- Q324** What is a provider and what is a data block in Terraform? [Terraform & IaC]
- Q333** What are main.tf, providers.tf, outputs.tf, variables.tf? [Terraform & IaC]
- Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]
- Q377** Security Groups vs NACLs (and subnet vs NACL)? [AWS]
- Q378** Describe your VPC / network architecture: subnets per tier, HA, sizing IP addresses. [AWS]
- Q399** Two apps in the same VPC, each behind a public ALB: how does App A reach App B? [AWS]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q402** How do you manage secrets in your project? (Secrets Manager, Parameter Store, Vault, encryption and... [AWS]
- Q404** Types of IAM policies; explicit deny vs allow; why a user with a role/policy still cannot access S3;... [AWS]
- Q406** What is the difference between IAM Users and IAM Roles? [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q436** Which AWS storage/instance types exist (EC2 types, EBS access mode, EFS vs EBS, multi-attach, Storage... [AWS]
- Q440** Spot vs Reserved instances [AWS]
- Q447** ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB? [AWS]
- Q448** How does AWS Auto Scaling work? (ASG, with ALB, scaling on memory/disk, ECS autoscaling) [AWS]
- Q449** Key CloudWatch metrics to focus on; how to monitor LB health, CPU spikes and web/app server availability? [AWS]
- Q451** CloudWatch: what is it and how do you create custom metrics? [AWS]

- Q454** The team changed the AMI to an ASG launch template - how do you ensure the new version deploys properly? [AWS]
- Q466** S3 lifecycle policies and storage classes; how do you manage and version them? [AWS]
- Q476** App is slow and you suspect RDS - what do you check? Memory pressure but cannot resize? [AWS]
- Q479** Can you kill heavy queries? Handling DB failover while a user is writing [AWS]
- Q480** RDS cross-region read replica (India -> London)? [AWS]
- Q524** Which logs do you check (agent, console, database, slow query, kernel)? [Monitoring & Observability]
- Q535** A live app is slow/has latency for customers (e-commerce) - how do you troubleshoot? Which metrics do... [SRE Concepts]
- Q556** Prechecks before migration; checks after an upgrade; risks of upgrading; what triggers a rollback [Architecture & Design]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]

## Syncortex - 10 questions

- Q107** How do you secure Docker images and containers (scanning, base image maintenance, authorized images,... [Docker & Containers]
- Q132** How do you debug a failed deployment or a failed pod (even when you cannot exec into it)? [Kubernetes]
- Q205** How do you back up an entire Kubernetes cluster regularly? [Kubernetes]
- Q230** What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a... [Kubernetes]
- Q241** How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity [Kubernetes]
- Q316** How do you make sure EC2 is not deleted when someone runs terraform destroy? [Terraform & IaC]
- Q350** Multi-region Terraform: 3 regions, which region does an EC2 go to? Structure for EC2 in 3 regions [Terraform & IaC]
- Q366** Import vs include; block/rescue; passing output between tasks; templates [Ansible & Config Management]
- Q372** What is an inventory file? How do you keep large inventories clean and store variables for different... [Ansible & Config Management]
- Q374** How do you debug a playbook? What if it times out on one host or runs for hours? [Ansible & Config Management]

## Synechron - 41 questions

- Q5** One-liners: remove first/last line with sed, count lines, find files > 1 GB, find ERROR... [Linux & OS]
- Q13** How do you check server system load? [Linux & OS]
- Q53** git fetch vs git pull; git pull vs git clone; push and pull [Git & Version Control]
- Q58** Extract commits of the last 3 days; compare two commits [Git & Version Control]
- Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]
- Q66** Which scripting languages have you used, and what automations/scripts have you written (cost... [Scripting & Coding]
- Q78** What is monkey patching in Python? [Scripting & Coding]
- Q81** Find the 10th word in a file; shell one-liners [Scripting & Coding]
- Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]
- Q92** CMD vs ENTRYPOINT [Docker & Containers]
- Q95** ADD vs COPY vs RUN [Docker & Containers]
- Q115** Docker vs Docker Compose; startup order (DB before app) and depends\_on [Docker & Containers]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q132** How do you debug a failed deployment or a failed pod (even when you cannot exec into it)? [Kubernetes]
- Q133** How do you monitor pod health/performance and alert when pods go down? [Kubernetes]
- Q197** Deploy a Kubernetes app with Jenkins: plugins, and a Jenkinsfile for Node.js (build, push image, deploy) [Kubernetes]
- Q199** Post-deployment steps to confirm everything runs well [Kubernetes]
- Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q255** AWS CodePipeline/CodeBuild/CodeDeploy - how to set up, and CodePipeline vs Jenkins [CI/CD]
- Q258** Maven lifecycle, Maven release, and build tools for Java [CI/CD]
- Q262** What challenges have you faced in CI/CD / a known critical production issue / prioritising multiple... [CI/CD]
- Q266** Deploy to both cloud and on-premises (hybrid) and across multiple clouds - how do you design the... [CI/CD]
- Q279** How do you troubleshoot failing, stuck or slow pipelines (build not starting, random artifact upload... [CI/CD]
- Q283** How do you configure a Flask application in Jenkins? [CI/CD]
- Q286** What is Blue Ocean in Jenkins? [CI/CD]
- Q291** GitHub Actions: workflow file, Java Maven build and artifact, manual trigger, concurrency, runners [CI/CD]
- Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]
- Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]
- Q303** Build artifacts vs pipeline artifacts; Nexus / Artifactory and storing dependencies [CI/CD]
- Q306** What is CloudFormation and how does it compare with Terraform? [Terraform & IaC]

- Q348** Terraform repository/folder structure; how do dev/qa/prod consume shared modules; Terraform architecture [Terraform & IaC]
- Q371** What are Ansible roles? How do you create them, structure a three-tier app, publish to Galaxy, and... [Ansible & Config Management]
- Q386** How does SSL/TLS work with Certbot, Let's Encrypt and ACM? How is a certificate used with ALB/NLB,... [AWS]
- Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]
- Q492** Azure storage types, subscription types, Recovery Services Vault, backup choices and Azure Boards [Azure & GCP]
- Q495** How do you implement DC/DR in Azure (and what is the purpose of DR)? [Azure & GCP]
- Q515** Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and... [Monitoring & Observability]
- Q522** How do you create custom alerts and handle CPU/disk alerts (e.g. CPU/memory beyond 75%)? How do you... [Monitoring & Observability]
- Q540** What are SLI, SLO, SLA and error budget? How do you decide and set them? [SRE Concepts]
- Q548** DevOps vs SRE [SRE Concepts]

## TCS - 47 questions

- Q6** Shell scripting: store names of newly created files in a directory; assign/print variables; last 15... [Linux & OS]
- Q18** File permission types; creating a user without SSH access; mount vs directory; limiting space per... [Linux & OS]
- Q24** Filesystem shows 50% used but you cannot write; df -ih shows 100% - why? Disk I/O; mount point space [Linux & OS]
- Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]
- Q67** Python fundamentals: lists vs tuples vs sets, shallow vs deep copy, decorators, inheritance, GIL,... [Scripting & Coding]
- Q93** How do you reduce Docker image size? [Docker & Containers]
- Q95** ADD vs COPY vs RUN [Docker & Containers]
- Q103** Bind mounts vs volumes; docker volume and prune; -p flag [Docker & Containers]
- Q107** How do you secure Docker images and containers (scanning, base image maintenance, authorized images,... [Docker & Containers]
- Q109** How are Docker images built - what are layers, how does layer caching work, and where does a rebuild... [Docker & Containers]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]
- Q135** A ReplicaSet has 3 pods and one isn't coming up / pods remain Pending even after HPA - why? [Kubernetes]
- Q158** What is a StatefulSet and how is it different from a Deployment? [Kubernetes]
- Q172** Why does a Deployment's pod name contain two sets of random characters? [Kubernetes]
- Q174** Types of Kubernetes services [Kubernetes]
- Q194** How does kubectl rollout undo know which image to revert to? Is rollback handled by CI/CD or Kubernetes? [Kubernetes]
- Q234** ConfigMaps, PV and PVC [Kubernetes]
- Q237** Storage classes in Kubernetes; what happens when a Deployment with one PVC is scaled from 1 to 3? [Kubernetes]
- Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]
- Q254** How do you onboard a new project to multiple environments and handle releases/hotfixes? [CI/CD]
- Q255** AWS CodePipeline/CodeBuild/CodeDeploy - how to set up, and CodePipeline vs Jenkins [CI/CD]
- Q257** How do you implement rollback in a CI/CD pipeline? Describe a rollback you handled [CI/CD]
- Q259** Design a CI/CD pipeline for a new app (Node.js, Angular/Java/.NET, developer gave only source code)... [CI/CD]
- Q263** Approvals in a CD pipeline [CI/CD]
- Q291** GitHub Actions: workflow file, Java Maven build and artifact, manual trigger, concurrency, runners [CI/CD]
- Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]
- Q302** What are deployment groups in Azure DevOps? [CI/CD]
- Q303** Build artifacts vs pipeline artifacts; Nexus / Artifactory and storing dependencies [CI/CD]
- Q304** JFrog Artifactory: repo types, use cases and Xray [CI/CD]
- Q312** What is the tfstate file and what does it do? [Terraform & IaC]
- Q340** Do you run Terraform locally or through CI/CD? Explain the workflow; what if apply or the pipeline... [Terraform & IaC]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management]
- Q364** What is a playbook? Write one to install/configure Apache/Nginx and ensure the service starts on boot [Ansible & Config Management]
- Q365** Deploy to 100 servers / patch 20+ servers with Ansible; install Ansible; Ansible Tower/AWX [Ansible & Config Management]
- Q366** Import vs include; block/rescue; passing output between tasks; templates [Ansible & Config Management]
- Q367** Which Ansible modules have you used? (install packages, nginx, commands) [Ansible & Config Management]
- Q371** What are Ansible roles? How do you create them, structure a three-tier app, publish to Galaxy, and... [Ansible & Config Management]
- Q373** What is Ansible Vault? How do you encrypt files/playbooks and store credentials? [Ansible & Config Management]
- Q385** Purpose of VPC Endpoints and when to use them? [AWS]

- Q400** What default components are created with a VPC? [AWS]
- Q401** How do you enforce least-privilege access and restrict a user to only specific services (e.g. only... [AWS]
- Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]
- Q514** What is Nagios and how do you integrate Jenkins with it? [Monitoring & Observability]
- Q551** Microservices vs monolith; what are microservices? [Architecture & Design]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]
- Q568** What automation have you done / contributed? Do career achievements need to be big? [Behavioural & Project]

## Techdome - 7 questions

- Q83** Docker image vs container [Docker & Containers]
- Q103** Bind mounts vs volumes; docker volume and prune; -p flag [Docker & Containers]
- Q104** Connect a backend container to RDS; isolate communication between two containers; Docker network types [Docker & Containers]
- Q118** Kubernetes architecture and workflow [Kubernetes]
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q312** What is the tfstate file and what does it do? [Terraform & IaC]
- Q322** What provisioners exist in Terraform? [Terraform & IaC]

## Turning - 9 questions

- Q65** How do you remove a developer's commit, delete the last 2 commits, or recover when someone... [Git & Version Control]
- Q69** Exercises: divisibility by 3 and/or 5 with input validation; min/max/sum from a file; JSON ingestion... [Scripting & Coding]
- Q191** Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines) [Kubernetes]
- Q267** Do you commit tfvars to Git? How do you run Terraform in a CD pipeline? [CI/CD]
- Q434** Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A [AWS]
- Q439** Event-driven architecture with S3, Lambda and SNS; and Lambda failures/retries (including external... [AWS]
- Q446** What is AWS CDK and its commands? [AWS]
- Q460** AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions? [AWS]
- Q479** Can you kill heavy queries? Handling DB failover while a user is writing [AWS]

## UST - 13 questions

- Q13** How do you check server system load? [Linux & OS]
- Q73** A Python program fails due to memory issues - what could be the cause? [Scripting & Coding]
- Q191** Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines) [Kubernetes]
- Q253** How do you reduce a slow pipeline (40-60 min -> 15-20 min)? Caching in Jenkins/GitHub Actions [CI/CD]
- Q297** Credentials appear in pipeline logs, or a developer hard-coded a password and pushed it - what do you do? [CI/CD]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q314** Pipeline fails because resources already exist - how do you handle it (Remove, Import, Plan)? [Terraform & IaC]
- Q321** Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and... [Terraform & IaC]
- Q329** Create multiple EC2 instances with different configurations; create 50 instances at once [Terraform & IaC]
- Q360** terraform apply is creating all resources again - possible problems? [Terraform & IaC]
- Q383** An EC2 instance is unreachable (not a security group problem) / an app behind a load balancer is... [AWS]
- Q543** Explain your DR strategy: RTO, RPO, failover and traffic redirection; DR for DevOps infrastructure;... [SRE Concepts]
- Q572** How can AI assist developers in increasing productivity? [Behavioural & Project]

## Verizon - 23 questions

- Q1** Linux file hierarchy; what do /var and /opt contain; is Linux an OS or kernel? Process/machine states [Linux & OS]
- Q2** Hard link vs soft (symbolic) link [Linux & OS]
- Q3** Difference between & and && in shell; cron expressions; scheduling every 15 minutes [Linux & OS]
- Q8** Find and kill a process; check processes without ps or top [Linux & OS]
- Q10** stdin, stdout, stderr; types of variables in Linux [Linux & OS]
- Q102** How do you version/manage images in ECR, delete old or untagged ones, and push to ACR? [Docker & Containers]
- Q103** Bind mounts vs volumes; docker volume and prune; -p flag [Docker & Containers]
- Q107** How do you secure Docker images and containers (scanning, base image maintenance, authorized images,... [Docker & Containers]
- Q121** Is etcd SQL or NoSQL and why? [Kubernetes]

**Q124** Which Kubernetes component does not run as a pod, and what is the agent managing the control-plane side? [Kubernetes]

**Q129** A worker node is not joining the cluster - how do you debug? What if the master goes down? Can... [Kubernetes]

**Q130** Are readiness probes failing? Can a container restart itself? What is the pause container? [Kubernetes]

**Q154** What is OOM and how do you resolve OOM issues? [Kubernetes]

**Q156** Compute reservation in manifests - requests vs limits [Kubernetes]

**Q159** What is an init container and why use it? [Kubernetes]

**Q201** Explain the full EKS/Kubernetes cluster upgrade process with zero downtime [Kubernetes]

**Q211** How do you switch between clusters? [Kubernetes]

**Q235** RWO vs RWX volumes [Kubernetes]

**Q251** Canary vs blue-green (and other deployment strategies) [CI/CD]

**Q403** How can you log in to an EC2 instance without a password, or if you lost the .pem key? [AWS]

**Q416** Secrets Manager vs Parameter Store - what is the difference? [AWS]

**Q477** How do you configure RDS and scale it (vertical/horizontal)? [AWS]

**Q478** Multi-AZ vs Read Replica; secondary vs read-only RDS [AWS]

## **Virtusa - 14 questions**

**Q67** Python fundamentals: lists vs tuples vs sets, shallow vs deep copy, decorators, inheritance, GIL,... [Scripting & Coding]

**Q71** Python string programs: slice 0 to 4, reverse a string, palindrome check, remove duplicate element [Scripting & Coding]

**Q92** CMD vs ENTRYPOINT [Docker & Containers]

**Q95** ADD vs COPY vs RUN [Docker & Containers]

**Q100** Build an image in CI, tag and push it, and deploy updates (commands + GitHub Actions) [Docker & Containers]

**Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]

**Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]

**Q230** What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a... [Kubernetes]

**Q241** How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity [Kubernetes]

**Q306** What is CloudFormation and how does it compare with Terraform? [Terraform & IaC]

**Q313** How do you remove a stuck state lock; how are state conflicts managed? [Terraform & IaC]

**Q314** Pipeline fails because resources already exist - how do you handle it (Remove, Import, Plan)? [Terraform & IaC]

**Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]

**Q461** Fargate vs EC2 launch type; ECS Service vs Task; setting up ECS on EC2 [AWS]

## **Volkswagen Group Digital - 6 questions**

**Q66** Which scripting languages have you used, and what automations/scripts have you written (cost... [Scripting & Coding]

**Q107** How do you secure Docker images and containers (scanning, base image maintenance, authorized images,... [Docker & Containers]

**Q131** Common Kubernetes errors and production issues you have faced (CrashLoopBackOff, ImagePullBackOff,... [Kubernetes]

**Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]

**Q254** How do you onboard a new project to multiple environments and handle releases/hotfixes? [CI/CD]

**Q563** What is an API? [General DevOps & Concepts]

## **Wikreate Media - 6 questions**

**Q52** Git merge vs rebase (and squash) [Git & Version Control]

**Q53** git fetch vs git pull; git pull vs git clone; push and pull [Git & Version Control]

**Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]

**Q376** How is Route 53 configured and what routing policies exist? [AWS]

**Q468** S3 vs EBS (and EFS)? [AWS]

**Q477** How do you configure RDS and scale it (vertical/horizontal)? [AWS]

## **Wipro - 50 questions**

**Q3** Difference between & and && in shell; cron expressions; scheduling every 15 minutes [Linux & OS]

**Q21** How do you securely decommission VMs including data shredding? [Linux & OS]

**Q63** Which branching strategy do you follow? How does it connect with CI to deploy to the right... [Git & Version Control]

**Q67** Python fundamentals: lists vs tuples vs sets, shallow vs deep copy, decorators, inheritance, GIL,... [Scripting & Coding]

**Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]

**Q93** How do you reduce Docker image size? [Docker & Containers]

**Q111** Containers stop suddenly after starting - how to troubleshoot? [Docker & Containers]

**Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]

**Q134** Ingress configured but the page doesn't load (or 404 from gateway) - what do you check? [Kubernetes]

- Q136** Common reasons a node becomes NotReady (or an unhealthy node group) and how to identify the root cause [Kubernetes]
- Q171** What should be in a production pod configuration? [Kubernetes]
- Q200** How do you implement feature toggles in deployment pipelines? [Kubernetes]
- Q224** What are the security implications of Kubernetes Secrets in etcd without encryption? [Kubernetes]
- Q225** Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly [Kubernetes]
- Q247** Is a service mesh always needed? Is the burden of Istio worth it? [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q252** What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple... [CI/CD]
- Q258** Maven lifecycle, Maven release, and build tools for Java [CI/CD]
- Q265** GitHub Actions vs Argo CD; how does CI differ from GitOps CD? [CI/CD]
- Q276** Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters [CI/CD]
- Q284** Jenkins deployment works in Dev but fails in Production - how to troubleshoot? [CI/CD]
- Q289** Design an end-to-end DevSecOps pipeline (compliance, PCI-DSS, GDPR/HIPAA); handling vulnerabilities... [CI/CD]
- Q294** Structure a multi-stage GitHub Actions pipeline that builds, tests and deploys a containerised app to... [CI/CD]
- Q296** How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets... [CI/CD]
- Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]
- Q311** How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or... [Terraform & IaC]
- Q316** How do you make sure EC2 is not deleted when someone runs terraform destroy? [Terraform & IaC]
- Q337** Name basic Terraform commands and explain each [Terraform & IaC]
- Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]
- Q365** Deploy to 100 servers / patch 20+ servers with Ansible; install Ansible; Ansible Tower/AWX [Ansible & Config Management]
- Q369** Structure Ansible patch automation with pre-task checks, main tasks and post-task validation [Ansible & Config Management]
- Q371** What are Ansible roles? How do you create them, structure a three-tier app, publish to Galaxy, and... [Ansible & Config Management]
- Q372** What is an inventory file? How do you keep large inventories clean and store variables for different... [Ansible & Config Management]
- Q375** NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need? [AWS]
- Q376** How is Route 53 configured and what routing policies exist? [AWS]
- Q383** An EC2 instance is unreachable (not a security group problem) / an app behind a load balancer is... [AWS]
- Q387** Public vs private subnet - how can you tell, and what must be in place? [AWS]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q433** AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets) [AWS]
- Q443** What is a cloud-agnostic strategy? How do you use conditionals in Terraform/Ansible for AWS, Azure... [AWS]
- Q467** Terraform state in S3: expire in 30 days? other options? S3 native locking command? [AWS]
- Q510** Log Analytics Workspace, alerts with action groups, KQL queries, and Dynatrace alert types [Azure & GCP]
- Q532** How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix?... [SRE Concepts]
- Q537** Clients report 504 Gateway Timeout - how do you debug? [SRE Concepts]
- Q552** Payment succeeds but order/shipping service fails - how do you handle it? Optimistic locking in a... [Architecture & Design]
- Q553** Service discovery when moving from monolith to microservices; zero-downtime DB migrations; retry... [Architecture & Design]
- Q554** How do you enforce tenant isolation at the API layer in a multi-tenant application? [Architecture & Design]
- Q567** Introduce yourself / explain your current project, roles, day-to-day activities and previous experience [Behavioural & Project]
- Q568** What automation have you done / contributed? Do career achievements need to be big? [Behavioural & Project]
- Q570** How do you communicate technical issues to non-technical stakeholders, and avoid miscommunication... [Behavioural & Project]

## Zensar - 16 questions

- Q104** Connect a backend container to RDS; isolate communication between two containers; Docker network types [Docker & Containers]
- Q115** Docker vs Docker Compose; startup order (DB before app) and depends\_on [Docker & Containers]
- Q121** Is etcd SQL or NoSQL and why? [Kubernetes]
- Q129** A worker node is not joining the cluster - how do you debug? What if the master goes down? Can... [Kubernetes]
- Q155** ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types [Kubernetes]
- Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]
- Q175** Ingress vs Ingress controller vs Gateway API [Kubernetes]
- Q243** ConfigMap vs Secret [Kubernetes]
- Q256** How do you handle parallel execution and jobs in CI/CD (Jenkins, GitHub Actions)? [CI/CD]
- Q299** YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline;... [CI/CD]

- Q301** What is a PAT, a service connection / connection string, and self-hosted vs Microsoft-hosted agent? [CI/CD]
- Q307** Stateful vs stateless applications; how do you manage data for stateless apps? [Terraform & IaC]
- Q380** VPC Peering: how does it work and what options connect VPC A and B (two-way or one-way)? [AWS]
- Q509** How do you create an app registration in Azure? [Azure & GCP]
- Q511** Monitor Azure VM memory and set alerts in Azure Monitor [Azure & GCP]
- Q543** Explain your DR strategy: RTO, RPO, failover and traffic redirection; DR for DevOps infrastructure;... [SRE Concepts]

## ZopSmart - 20 questions

- Q18** File permission types; creating a user without SSH access; mount vs directory; limiting space per... [Linux & OS]
- Q52** Git merge vs rebase (and squash) [Git & Version Control]
- Q54** Everyday git commands, push local changes to remote, checkout and clone [Git & Version Control]
- Q89** Useful Docker commands: list running/stopped containers, remove images/containers [Docker & Containers]
- Q91** Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in... [Docker & Containers]
- Q202** Basic Kubernetes commands (also Docker and Ansible commands) [Kubernetes]
- Q203** Have you created an EKS cluster? Explain the process and prerequisites (compute, networking, access) [Kubernetes]
- Q214** How is access control implemented in Kubernetes (RBAC)? [Kubernetes]
- Q250** Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,... [CI/CD]
- Q255** AWS CodePipeline/CodeBuild/CodeDeploy - how to set up, and CodePipeline vs Jenkins [CI/CD]
- Q258** Maven lifecycle, Maven release, and build tools for Java [CI/CD]
- Q280** Show a sample Jenkinsfile and explain stages, agent, post and environment blocks; what types of... [CI/CD]
- Q337** Name basic Terraform commands and explain each [Terraform & IaC]
- Q339** What do terraform init, plan, apply, fmt do? Meaning of + and - in plan; auto-approve [Terraform & IaC]
- Q363** What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config... [Ansible & Config Management]
- Q364** What is a playbook? Write one to install/configure Apache/Nginx and ensure the service starts on boot [Ansible & Config Management]
- Q370** Write an Ansible playbook to install Docker on nodes [Ansible & Config Management]
- Q380** VPC Peering: how does it work and what options connect VPC A and B (two-way or one-way)? [AWS]
- Q406** What is the difference between IAM Users and IAM Roles? [AWS]
- Q436** Which AWS storage/instance types exist (EC2 types, EBS access mode, EFS vs EBS, multi-attach, Storage... [AWS]

## ZS Associates - 22 questions

- Q39** Load balancer not accepting HTTPS; redirecting HTTP to HTTPS without installing a certificate; load... [Networking & Security]
- Q90** What is privileged mode in Docker? Give an example [Docker & Containers]
- Q94** Multistage Dockerfile and when it is useful (compiled languages?) [Docker & Containers]
- Q109** How are Docker images built - what are layers, how does layer caching work, and where does a rebuild... [Docker & Containers]
- Q128** Crashloopbackoff - possible reasons and troubleshooting steps [Kubernetes]
- Q135** A ReplicaSet has 3 pods and one isn't coming up / pods remain Pending even after HPA - why? [Kubernetes]
- Q160** StatefulSet vs RDS for a database - how do you decide? MongoDB/MySQL on Kubernetes and operators [Kubernetes]
- Q173** How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)? [Kubernetes]
- Q175** Ingress vs Ingress controller vs Gateway API [Kubernetes]
- Q183** What is Calico, and how does it differ from AWS VPC CNI? [Kubernetes]
- Q204** How do you manage worker nodes, node groups, and how many clusters/add-ons do you manage? [Kubernetes]
- Q218** How do you map an IAM role to a Kubernetes service account (IRSA) and how does the pod know which... [Kubernetes]
- Q230** What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a... [Kubernetes]
- Q231** Do you use Cluster Autoscaler or Karpenter? How do you provision Karpenter, how does it decide which... [Kubernetes]
- Q313** How do you remove a stuck state lock; how are state conflicts managed? [Terraform & IaC]
- Q323** Terraform lifecycle (create\_before\_destroy, prevent\_destroy, ignore\_changes) [Terraform & IaC]
- Q328** count vs for\_each; what happens to already-created instances if you remove an item from the list? [Terraform & IaC]
- Q346** What is a Terraform module, what does it contain, how do you create, call, reference outputs and... [Terraform & IaC]
- Q347** How do you write a Terraform module for EKS; how do you create and manage clusters, and master/worker... [Terraform & IaC]
- Q350** Multi-region Terraform: 3 regions, which region does an EC2 go to? Structure for EC2 in 3 regions [Terraform & IaC]
- Q432** Design a highly available, fault-tolerant backend on AWS [AWS]
- Q435** How do you design for an entire region going down (and keep data safe even if multi-cloud also... [AWS]

## Part 3 - Most Asked Questions (revision list)

| #    | Question                                                                                                       | Topic                       | Cos. |
|------|----------------------------------------------------------------------------------------------------------------|-----------------------------|------|
| Q250 | Explain CI/CD and walk through an end-to-end pipeline you built (stages, tools, flow,...)                      | CI/CD                       | 25   |
| Q118 | Kubernetes architecture and workflow                                                                           | Kubernetes                  | 20   |
| Q567 | Introduce yourself / explain your current project, roles, day-to-day activities and previous experience        | Behavioural & Project       | 20   |
| Q155 | ReplicaSet vs ReplicationController vs Deployment; DaemonSet vs StatefulSet; deployment types                  | Kubernetes                  | 19   |
| Q63  | Which branching strategy do you follow? How does it connect with CI to deploy to the right environment;...     | Git & Version Control       | 16   |
| Q432 | Design a highly available, fault-tolerant backend on AWS                                                       | AWS                         | 16   |
| Q225 | Helm: Chart.yaml, values.yaml, templates folder, upgrades and why not edit a chart directly                    | Kubernetes                  | 15   |
| Q311 | How do you manage Terraform state securely, migrate local state to S3, and recover if it is lost or deleted?   | Terraform & IaC             | 15   |
| Q401 | How do you enforce least-privilege access and restrict a user to only specific services (e.g. only EC2 and...) | AWS                         | 15   |
| Q251 | Canary vs blue-green (and other deployment strategies)                                                         | CI/CD                       | 14   |
| Q433 | AWS bill increased suddenly - how do you find and reduce costs? (EC2, S3, GCP buckets)                         | AWS                         | 14   |
| Q460 | AWS Lambda: what is it, use cases, cold start, and how to reduce it? Step Functions?                           | AWS                         | 14   |
| Q91  | Write a basic Dockerfile; create custom images; can you write one without a base image? What goes in a...      | Docker & Containers         | 13   |
| Q201 | Explain the full EKS/Kubernetes cluster upgrade process with zero downtime                                     | Kubernetes                  | 13   |
| Q296 | How do you handle secrets in pipelines (Jenkins credentials, Azure DevOps / Key Vault, AWS Secrets...)         | CI/CD                       | 13   |
| Q346 | What is a Terraform module, what does it contain, how do you create, call, reference outputs and upgrade it?   | Terraform & IaC             | 13   |
| Q434 | Fixing a Terraform-created EC2 that someone changed manually; creating a new EC2 B without destroying A        | AWS                         | 13   |
| Q128 | Crashloopbackoff - possible reasons and troubleshooting steps                                                  | Kubernetes                  | 12   |
| Q173 | How do pods communicate (pod-to-pod, across namespaces, service discovery, traffic routing)?                   | Kubernetes                  | 12   |
| Q299 | YAML pipelines vs classic pipelines in Azure DevOps; how to build and structure a pipeline; variables,...      | CI/CD                       | 12   |
| Q305 | What is Terraform, how does it work, and what have you done with it?                                           | Terraform & IaC             | 12   |
| Q363 | What is Ansible, how does it work and what have you used it for? (Ansible vs Jenkins, config management tools) | Ansible & Config Management | 12   |
| Q375 | NAT Gateway vs Internet Gateway vs NAT Instance; where does NAT Gateway live; how many do you need?            | AWS                         | 12   |
| Q447 | ALB vs NLB (and ELB); layer, when to choose; cross-zone communication; cost of private vs public ALB?          | AWS                         | 12   |
| Q190 | Write a Deployment YAML (httpd/nginx with 3 replicas) and deploy 3 images from ECR/ACR to EKS/AKS              | Kubernetes                  | 11   |
| Q230 | What is HPA, its triggers, how do you scale pods for a festival-day spike, and how can a misconfigured HPA...  | Kubernetes                  | 11   |

| #    | Question                                                                                                      | Topic                       | Cos. |
|------|---------------------------------------------------------------------------------------------------------------|-----------------------------|------|
| Q241 | How do you deploy an app on a specific node (worker node 2)? Node affinity and anti-affinity                  | Kubernetes                  | 11   |
| Q191 | Blue-green deployment in Kubernetes (same namespace, fully automated, rollback via Terraform/pipelines)       | Kubernetes                  | 10   |
| Q252 | What is GitOps (Argo CD, Flux)? How does the tool detect and manage drift, manage multiple environments,...   | CI/CD                       | 10   |
| Q276 | Multibranch pipeline; declarative vs scripted; shared libraries; dynamic stages; variables and parameters     | CI/CD                       | 10   |
| Q288 | SonarQube and Snyk integration, quality gates, tokens and static analysis                                     | CI/CD                       | 10   |
| Q321 | Write Terraform code: multiple S3 buckets; EC2 with EBS volume; EC2 with SSH-only security group and user...  | Terraform & IaC             | 10   |
| Q512 | What monitoring tools and agents have you used and what did you do with them? What proactive monitoring...    | Monitoring & Observability  | 10   |
| Q515 | Explain Prometheus and Grafana: installation, configuration, dashboards, data sources, components and what... | Monitoring & Observability  | 10   |
| Q532 | How do you handle a Sev-1? Root cause when an app goes down in the cloud? Roll back or hotfix? Describe a...  | SRE Concepts                | 10   |
| Q41  | How do SSL/TLS certificates work; HTTP vs HTTPS; renew an expired App Gateway certificate; check a corrupt... | Networking & Security       | 9    |
| Q52  | Git merge vs rebase (and squash)                                                                              | Git & Version Control       | 9    |
| Q66  | Which scripting languages have you used, and what automations/scripts have you written (cost optimisation,... | Scripting & Coding          | 9    |
| Q92  | CMD vs ENTRYPOINT                                                                                             | Docker & Containers         | 9    |
| Q93  | How do you reduce Docker image size?                                                                          | Docker & Containers         | 9    |
| Q94  | Multistage Dockerfile and when it is useful (compiled languages?)                                             | Docker & Containers         | 9    |
| Q107 | How do you secure Docker images and containers (scanning, base image maintenance, authorized images,...       | Docker & Containers         | 9    |
| Q129 | A worker node is not joining the cluster - how do you debug? What if the master goes down? Can workloads...   | Kubernetes                  | 9    |
| Q174 | Types of Kubernetes services                                                                                  | Kubernetes                  | 9    |
| Q214 | How is access control implemented in Kubernetes (RBAC)?                                                       | Kubernetes                  | 9    |
| Q253 | How do you reduce a slow pipeline (40-60 min -> 15-20 min)? Caching in Jenkins/GitHub Actions                 | CI/CD                       | 9    |
| Q277 | Jenkins administration: standalone setup, memory issues, copying jobs, backup, concurrent builds,...          | CI/CD                       | 9    |
| Q312 | What is the tfstate file and what does it do?                                                                 | Terraform & IaC             | 9    |
| Q337 | Name basic Terraform commands and explain each                                                                | Terraform & IaC             | 9    |
| Q364 | What is a playbook? Write one to install/configure Apache/Nginx and ensure the service starts on boot         | Ansible & Config Management | 9    |
| Q376 | How is Route 53 configured and what routing policies exist?                                                   | AWS                         | 9    |
| Q377 | Security Groups vs NACLs (and subnet vs NACL)?                                                                | AWS                         | 9    |
| Q402 | How do you manage secrets in your project? (Secrets Manager, Parameter Store, Vault, encryption and...        | AWS                         | 9    |
| Q18  | File permission types; creating a user without SSH access; mount vs directory; limiting space per team and... | Linux & OS                  | 8    |
| Q95  | ADD vs COPY vs RUN                                                                                            | Docker & Containers         | 8    |
| Q130 | Are readiness probes failing? Can a container restart itself? What is the pause container?                    | Kubernetes                  | 8    |

| #    | Question                                                                                              | Topic      | Cos. |
|------|-------------------------------------------------------------------------------------------------------|------------|------|
| Q131 | Common Kubernetes errors and production issues you have faced (CrashLoopBackOff, ImagePullBackOff,... | Kubernetes | 8    |
| Q175 | Ingress vs Ingress controller vs Gateway API                                                          | Kubernetes | 8    |
| Q243 | ConfigMap vs Secret                                                                                   | Kubernetes | 8    |
| Q254 | How do you onboard a new project to multiple environments and handle releases/hotfixes?               | CI/CD      | 8    |